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Key Points:
- Paleocene orogenesis took place across most of Sundaland including south Indochina
- Kilometers of orogenic exhumation and structuring affected southern Cambodia and Vietnam
- Plate collision along the Sundaland margin most likely drove deformation

Supporting Information:
- Supporting Information S1

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Cenozoic deformation and exhumation of the Kampot Fold Belt and implications for south Indochina tectonics

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Abstract Latest Mesozoic to earliest Cenozoic deformation affected SE Asia's Sundaland core. The deformation event bridges the Mesozoic SE Asian fusion with the Cenozoic era of rifting, translation, basin formation, and the creation of modern SE Asian oceans. Southern Cambodia and Vietnam are central to this shift, but geological investigations of the region are in their infancy. Based on apatite and zircon fission track analyses (AFTA and ZFTA), stratigraphic and structural observations, seismic data, thermal maturity, and igneous rock dating, the geological evolution of southern Cambodia and Vietnam is investigated. Diverse depositional styles, igneous activity, structural deformation and subsurface unconformities testify to a highly variable Phanerozoic tectonic setting. Major latest Cretaceous to Paleocene thrusting and uplift affected the Kampot Fold Belt and surrounding regions and the associated up to ~11 km exhumation probably exceeds earlier denudation events since at least Permian time. The present relief of the Bokor Mountains rising high above the Kampot Fold Belt represents an artifact after differential erosion and only 2.5–4.5 km of erosion affected this area. The latest Cretaceous to Paleocene orogenesis affected much of greater Indochina probably owing to plate collision along eastern Sundaland or a combination of collisions along both east and west Sundaland. AFTA and ZFTA data document protracted cooling of Cretaceous granites and locally elevated thermal gradients persisting a few tens of million years after their emplacement. The thermal gradient had stabilized by early Miocene time, and Miocene cooling probably reflects a renewed denudation pulse driven by either regional tectonism or climate-enhanced erosion.

1. Introduction

Since the geological mapping by the Geological Survey of Indochina concluded in the 1970s [*Bureau de Recherches Géologiques et Minières (BRGM)*, 1971a, 1971b, 1973a, 1973b, 1973c; *Fontaine*, 1971], geoscientific research activities in Cambodia have been limited, which is largely also the case for SW Vietnam. Consequently, information on the geology of Cambodia and southwestern Vietnam is sparse and include the release of 1:200,000 scale geological maps (1:50,000 for the Phu Quouc Island) by the Geological Survey of Vietnam and topical studies [*Saurin*, 1935; *Fromaget*, 1941; *Tien*, 1991; *Vysotsky et al.*, 1994; *Hoa*, 1996a, 1996b, 1996c, 1996d, 1996e; *Den*, 1998; *Linh*, 2004; *Fyhn et al.*, 2010a; *Ridd and Morley*, 2011; *Tri and Khuc*, 2011; *Tsuchiyama et al.*, 2016]. We investigate the geology and tectonic development in the logistically difficult southern Cambodia and Vietnam based on fieldwork, seismic interpretation, and laboratory analyses of rock samples and assess the implications to regional SE Asian development through time (Figure 1).

The outline of central SE Asia has generally been attributed to mainly Permo-Triassic and middle to late Cenozoic tectonism (Figure 1) [*Metcalfe*, 2009; *Hall*, 2002; *Morley*, 2002]. More recently, the importance of Cretaceous and early Cenozoic tectonism was emphasized [*Hall*, 2012; *Morley*, 2012]. Focus on the Cretaceous to early Cenozoic development has mainly involved the plate convergence along western Sundaland [*Morley*, 2012; *Watkinson et al.*, 2008, 2011]. Even so, in southeastern Indochina, a major plate tectonic reconfiguration took place between latest Cretaceous and middle to late Eocene time [*Fyhn et al.*, 2010a]. Subduction and collisional arc-related magmatism testifies to middle and Late Cretaceous paleo-Pacific oceanic convergence along the southeastern margin of Indochina [*Nguyen et al.*, 2004]. After magmatism ceased, much of the region underwent massive intracontinental rifting, basin formation and translation linked

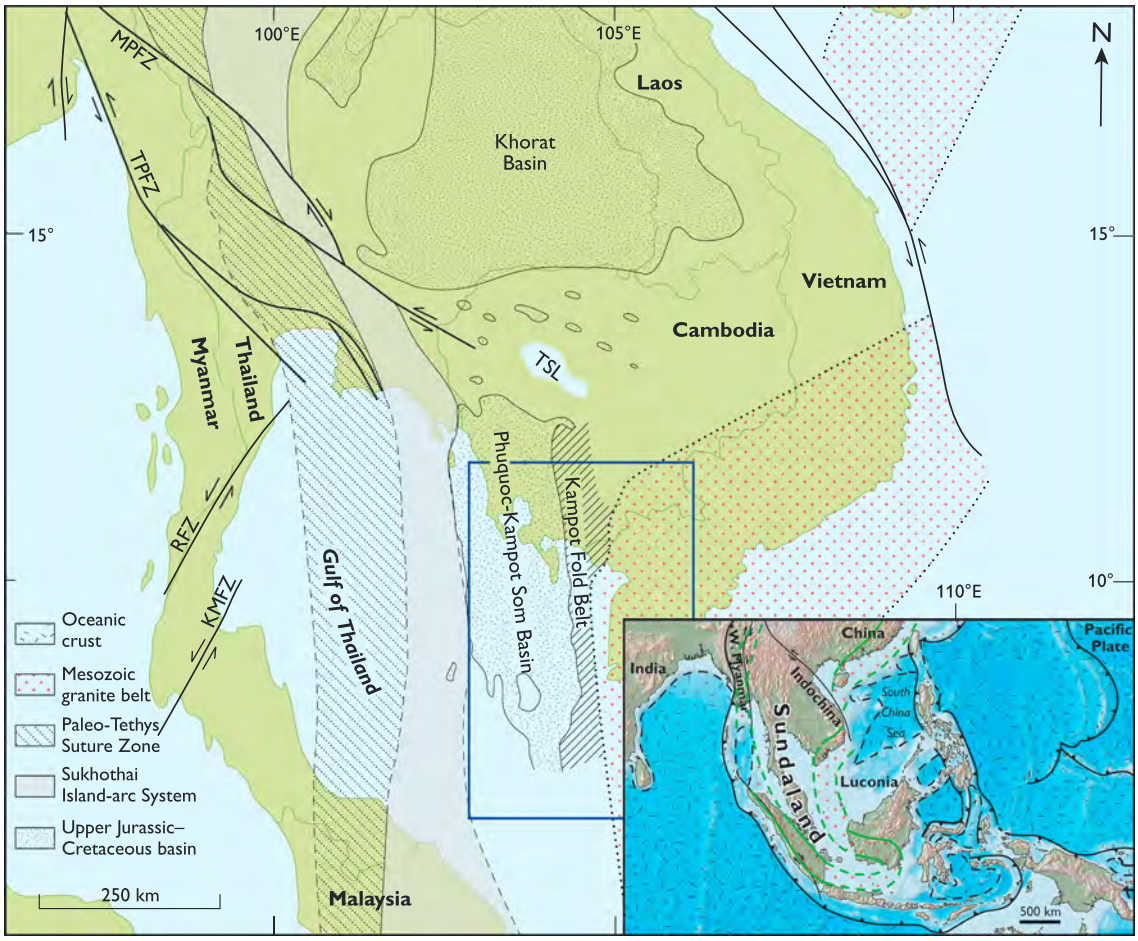


Figure 1. Outline of major structural and volcanic belts and Jurassic-Cretaceous basins. The SE Asian Sundaland core is delineated in insert map. Boxed study area outlines Figure 2. Sukhothai Island-Arc System and Paleo Tethys Suture Zone modified after *Metcalfe* [2013a, 2013b]. KMFZ = Khlong Marui Fault Zone, MPFZ = Mai Ping Fault Zone, RFZ = Ranong Fault Zone, TPFZ = Three Pagodas Fault Zone, and TSL = Tonle Sap Lake.

with the collision of the Indian plate with Eurasia from after the early Paleogene [*Molnar and Tapponnier*, 1975; *Tapponnier et al*., 1990; *Matthews et al*., 1997; *Lee and Watkins*, 1998; *Lee et al*., 2001; *Fyhn et al*., 2009a, 2009b, 2010b; *Pubellier and Morley*, 2013]. This important tectonic shift is poorly understood as Paleocene and lower Eocene strata and crystalline rocks representing the period are lacking. Based on analyses of fission tracks in apatite and zircon, vitrinite reflectance, igneous rock dating, and investigation of outcrop and subsurface stratigraphy and structures, this study addresses this important tectonic reconfiguration.

Located in southeastern Indochina, the Kampot Fold Belt provides important insight to a poorly studied part of Sundaland and records the tectonic forces acting on SE Asia through Paleozoic to Cenozoic time (Figure 1). The Kampot Fold Belt crops out in isolated parts of southeastern Cambodia and on remote southwest Vietnamese islands and seems comparable to better studied fold belts in Thailand and Laos (Figure 2). The fold belt is situated between the Early Cretaceous Phuquoc-Kampot Som Basin in the west and the Late Mesozoic paleo-Pacific magmatic arc in the east and is separated from the latter by a folded and intruded transition zone (Figures 1 and 2). Until recently, the fold belt was considered to be Permo-Triassic in age associated with Indosinian deformation or even older [*Dien et al*., 2008; *Sone et al*., 2012]. However, more recent investigations have suggested a younger deformation history associated with inversion of the Phuquoc-Kampot Som Basin to the west [*Fyhn et al*., 2010a]. The Kampot Fold Belt is composed by strongly folded and thrusted Paleozoic and Mesozoic rocks. It is not clear whether deformation primarily formed during a single tectonic event or reflects different generations of deformation as is the case in other fold belts in SE Asia [*Helmcke*, 1986; *Upton*, 1999; *Ahrendt et al*., 1993; *Stokes et al*., 1996; *Lepvrier et al*., 2004; *Morley*, 2004;

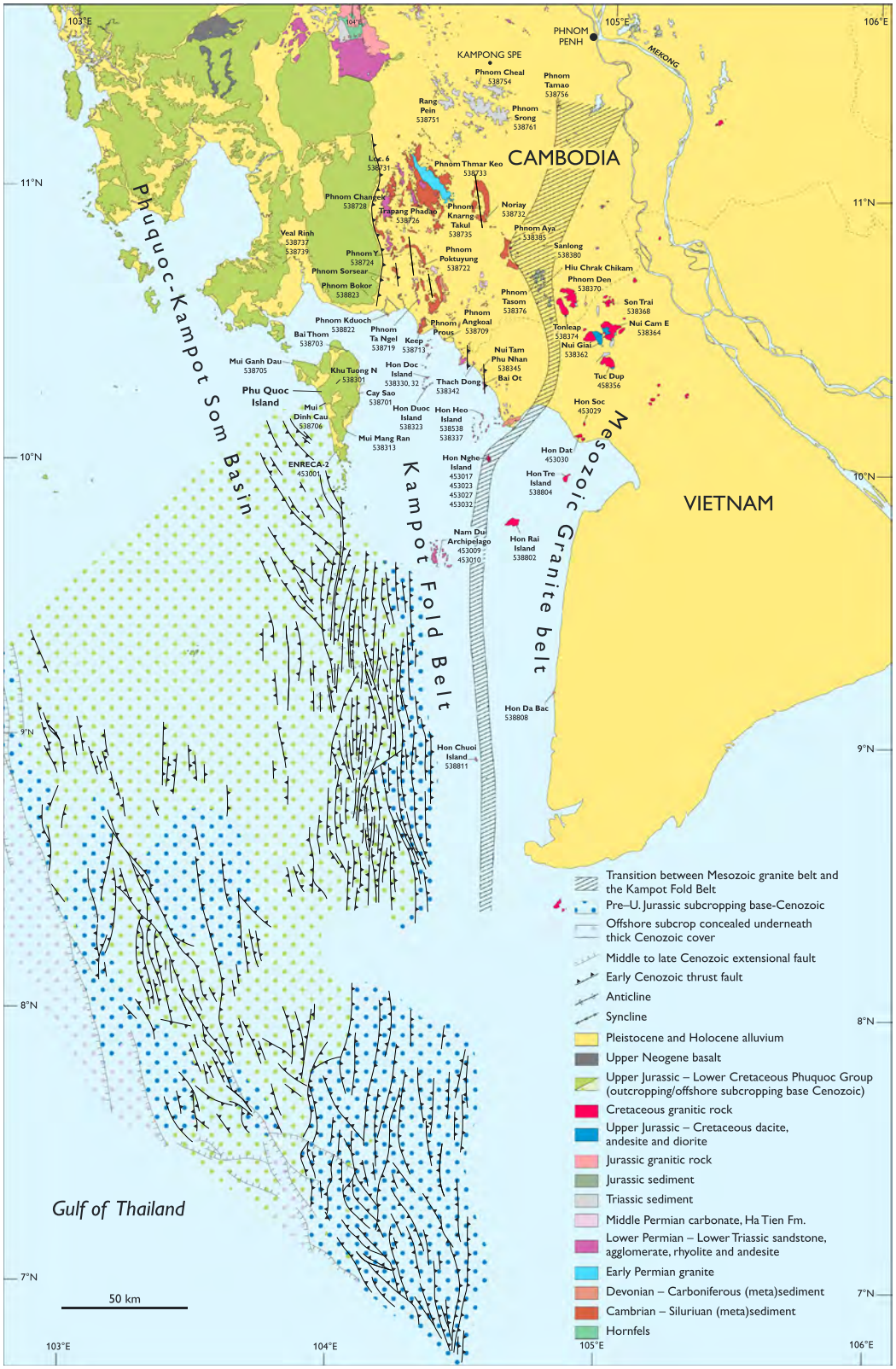


Figure 2. Geology of study area illustrating onshore and offshore Cretaceous basins, the Kampot Fold Belt, and the Mesozoic granite belt. Shaded area outlines the gradual transition between the Kampot Fold Belt and the Mesozoic granite belt characterized by pre-Cretaceous stratigraphy in outcrop sporadically intruded by Cretaceous and Jurassic plutons. Sampling sites with sample numbers are indicated. Geology compiled and modified to fit observations after *BRGM* [1971a, 1971b, 1973a, 1973b, 1973c], *Tien* [1991], *Hoa* [1996a, 1996b, 1996c, 1996d, 1996e], and *Fyhn et al.* [2010a].

Barber et al., 2011; *Searle and Morley*, 2011; *Ridd and Morley*, 2011; *Ridd*, 2012, 2015]. This study complements information from these better known fold belts located in more easily accessible parts of SE Asia and emphasizes the importance of early Cenozoic tectonism in the shaping of SE Asia.

2. Regional Setting

Cambodia and most of Vietnam form part of the Indochina Block of Sundaland (Figure 1). The Indochina block occupies most of the Indochina Peninsula, as well as parts of adjacent China and Thailand. The basement of the Indochina Block comprises a Precambrian metamorphic core (Kontum Massif) of amphibolite to granulite facies rocks exposed in Vietnam and a variety of Paleozoic and younger components. The onshore boundaries of the Indochina block are fairly well constrained, whereas the offshore boundaries are poorly defined. The offshore eastern boundary broadly corresponds to the eastern margin of Sundaland in the South China Sea region and to a possible cryptic Cretaceous suture toward Luconia offshore NW Borneo [*Hall*, 2012]. In the Gulf of Thailand the western edge of Indochina is mostly considered to roughly delineate the western margin of the Phuquoc-Kampot Som Basin [e.g., *Metcalfe*, 2013a], but *Sone et al*. [2012] speculated it to straddle the Cambodian coast toward SW Vietnam, and thus to form part of the area in focus in this paper. The core of Sundaland was subsequently welded together as Gondwana-derived continental fragments including Indochina and arc terrains collided during the mostly Triassic Indosinian orogeny (see *Metcalfe* [2013a] for additional details). The Indochina Block rifted from Gondwana during the Early Devonian and drifted northward, leading to the opening of the Paleotethys Ocean (Figure 3). During the Permian northward drift, the Sukhothai Arc rifted from western Indochina through back-arc extension and eventual back-arc seafloor spreading. By latest Permian and Early Triassic time, the back-arc basin closed causing the Indochina block and the Sukhothai Arc to collide. After the Indochina-Sukhothai Arc accretion, Indochina and Sukhothai collided with the Sibumatsu Block comprising western Sundaland during the Triassic.

Following the Indosinian Orogeny, oceanic convergence took place along the western margin of Sundaland during the Middle and Late Mesozoic (see *Sone and Metcalfe* [2008], *Morley* [2012], *Hall* [2012], and *Metcalfe* [2013a] for reviews). This resulted in accretion of arc terranes during the Cretaceous and possibly also collision with Gondwana-derived continental fragments. Eventually, the convergence led to the Indian-Eurasian collision commencing during the early Cenozoic.

Along the southeastern margin of Sundaland subduction of the paleo-Pacific plate took place during much of the Mesozoic [*Honza and Fujioka*, 2004; *Hall*, 2012; *Metcalfe*, 2013a]. A Jurassic-Cretaceous arc-related igneous belt stretches from Japan to southern Borneo and bears witness of this subduction [*Zhou and Li*, 2000; *Li et al*., 2004; *Nguyen et al*., 2004; *Thuy et al*., 2004; *Jiang et al*., 2009; *Wang et al*., 2011].

Between the subduction-related arcs and fringing the western and eastern Sundaland margins, a large basin developed in response to lithospheric flexure during latest Jurassic and Early Cretaceous time. The Khorat and Phuquoc-Kampot Som Basins make up erosional remnants of this once larger basin that may have extended from south China to Peninsular Malaysia [*Mouret et al*., 1993; *Heggemann et al*., 1994; *Carter and Bristow*, 2003; *Racey*, 2009; *Fyhn et al*., 2010a]. Upper Jurassic to Aptian deposits are dominantly nonmarine clastic sequences with only a few marine intercalations. Following a moderate Aptian to Albian structural inversion, Late Cretaceous (and possibly earliest Cenozoic) deposition resumed with sediments comprising evaporites and continental siliciclastics of the Maha Sakham and Phu Tok formations [*Lovatt-Smith et al*., 1996; *Meesook*, 2011; *Booth and Sattayarak*, 2011].

Paleo-Pacific subduction along the east Sundaland margin terminated between mid-Cretaceous and earliest Cenozoic time. The halt of paleo-Pacific subduction beneath Sundaland and southeastern China has been attributed to collision of Indochina and Luconia, now flooring part of the South China Sea (Figure 1) [*Hall et al*., 2009; *Fyhn et al*., 2010a; *Hall*, 2012] or retreat of the subduction zone away from Sundaland and southeastern China due to slab rollback causing back-arc extension and eventually seafloor spreading during Late Cretaceous time [*Honza and Fujioka*, 2004; *Morley*, 2012; *Metcalfe*, 2013a].

Cretaceous to early Cenozoic deformation affected much of Sundaland [*Morley et al*., 2011; *Searle and Morley*, 2011; *Watkinson et al*., 2011; *Morley*, 2012; *Cottam et al*., 2013]. In northern and western Sundaland this was mainly attributed to the convergence along the western Sundaland margin [*Lovatt-Smith et al*., 1996; *Watkinson et al*., 2008, 2011; *Morley et al*., 2011; *Booth and Sattayarak*, 2011; *Morley*, 2012]. *Fyhn et al*.

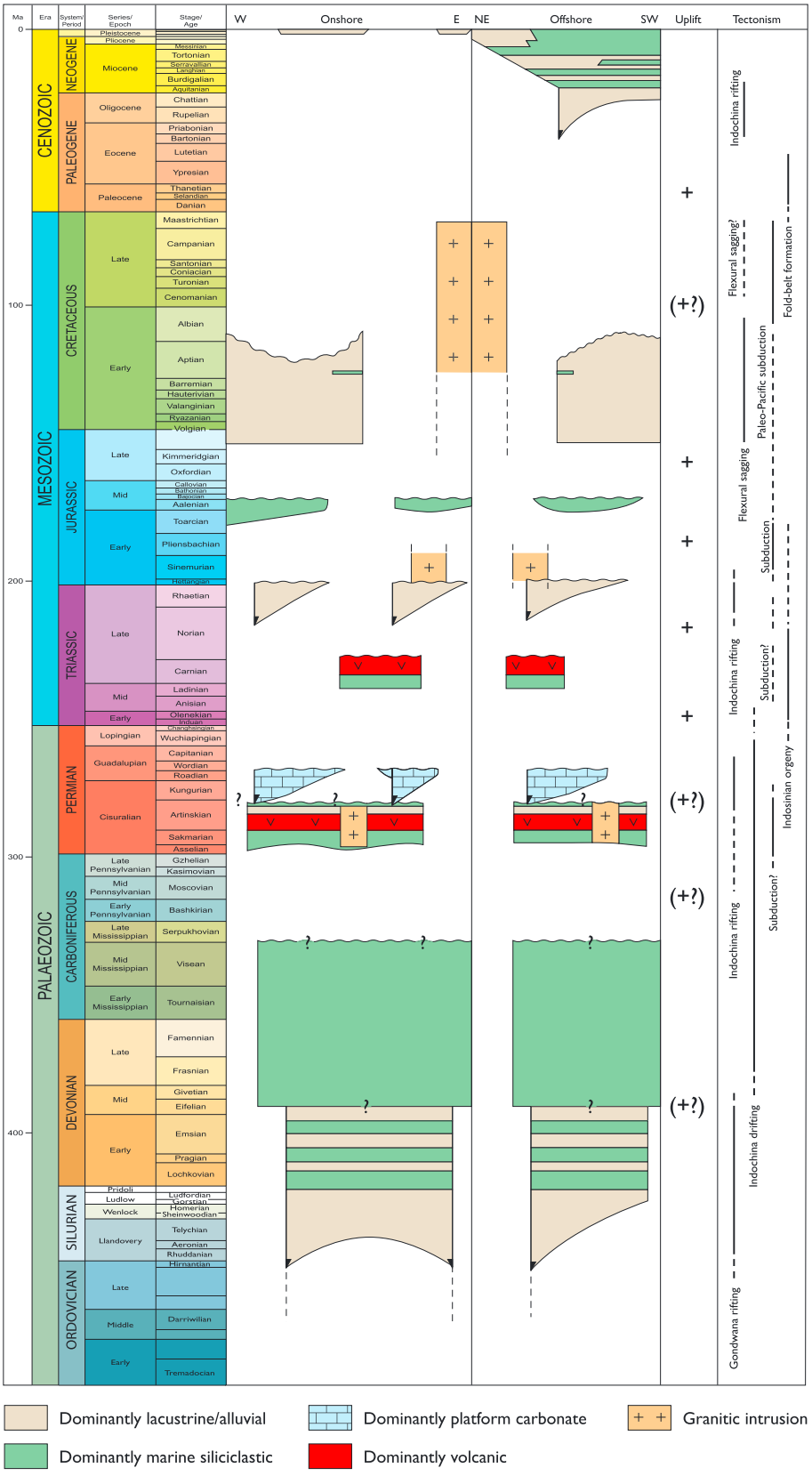


Figure 3. Stratigraphic chart covering SE Cambodia, SW Vietnam, and the adjacent offshore region in the Gulf of Thailand with regional tectonic events indicated.

[2010a] documented deformation along the eastern margin of Sundaland but related this to the termination of paleo-Pacific subduction beneath eastern Sundaland.

Since the early Cenozoic, rifting, transtensional tectonics, and postrift subsidence have shaped Sundaland. Cenozoic basins such as the Malay and Cuu Long occur in the Gulf of Thailand, offshore East Vietnam, and also exist in onshore Indochina. Many of these rift basins juxtapose older fold belts and suture zones and probably reactivated these zones of crustal weakness [*Ngah et al*., 1996; *Morley et al*., 2004; *Fyhn et al*., 2010b].

3. Data

Initial fieldwork was conducted in 2008 in SW Vietnam and followed up by more extensive fieldwork in SW Vietnam and Cambodia in 2012. During the fieldwork, samples were collected for AFTA and ZFTA (methods for constraining thermal histories based on analyses of fission track age and track length distributions in apatite and zircon [*Green and Duddy*, 2012]); biostratigraphic dating, sandstone provenance analysis, source rock screening, vitrinite reflectance measurements, and igneous rocks were sampled for radiometric dating. In addition, a Lower Cretaceous succession was cored in the 500 m deep Enhanced Research Capacity building (ENRECA)-2 stratigraphic well on Phu Quoc Island in order to investigate the stratigraphy and petroleum potential of the Phuquoc-Kampot Som Basin. AFTA was performed on an Aptian sandstone sample from the lowermost part of this well.

For the purpose of this study, fission track analyses on apatite from 38 samples and zircons from 25 samples have been conducted. In addition, we have also included analysis of apatite fission tracks of seven samples from SW Vietnam originally analyzed by *Fyhn et al*. [2010a]. Fission track data are summarized in Table 1, and full details of all fission track analyses are available in the supporting information (Texts S1–S4). Background on these techniques and methods employed to extract thermal history constraints from the data are provided by *Green and Duddy* [2012].

The strategy used in the sampling and ensuing analytical program was to investigate the broadest possible spectrum of rock lithologies and ages in the area and to obtain representative and regional sample coverage of the study area (Figure 2). Fresh rock samples were pursued during fieldwork; however, outcrops are limited in some areas and strong tropical weathering is ubiquitous so some samples are weathered.

Fission track age data quality depends on a range of factors including numbers of crystals analyzed and fission tracks counted and measured in the individual sample as well as the chemical composition of the individual grain analyzed. This information is provided in Table 1 and more elaborately in the supporting information. There is no simple formula to represent data quality for fission track data, and the ultimate guide is the 95% confidence limits on the resulting paleotemperature or timing constraint.

The quality of the AFTA data widely varies in this study, mainly as a result of variable apatite yields. Of 45 samples which yielded apatite, excellent apatite yields were only obtained from 20 samples, whereas lower numbers of single-grain ages ($<$15–20) were measured in the other 25 samples. A further 25 samples failed to yield any apatite. The chlorine content of individual apatite grains were measured and incorporated in the age analyses [*Green and Duddy*, 2012]. More than 100 confined track lengths were achieved in 10 samples, with between 12 and 83 lengths measured in another 14 samples and fewer lengths measured for the remaining samples. In most cases, the scarcity of measured track lengths reflects a combination of low uranium content and relatively young fission track ages. Thus, whereas high-quality data are available in some samples, in many other samples the data quality is less than normally considered desirable and some samples only provide broad constraints on cooling ages. However, the uniform consistency shown by AFTA data in a majority of samples (e.g., Figures 4 and 5) suggests that interpretations derived from higher-quality data can be applied with confidence to lesser quality data from adjacent samples. On this basis, the thermal history constraints derived from the AFTA data in the majority of samples are regarded as highly reliable, within the stated ranges of uncertainty (95% confidence limits), and the AFTA data set as a whole displays a high degree of consistency, providing a very clear regional interpretation which is regarded as highly reliable.

The quality of the ZFTA data discussed in this paper is very high, due largely to the excellent yields of zircon obtained from the samples analyzed, and results are regarded as highly reliable, although interpretation of the resulting age data is complex reflecting the geothermal history of the samples. In detail, zircon fission track ages were determined on at least 20 grains of zircon in 17 of the ZFTA samples, whereas between 10

Table 1. Apatite and Zircon Fission Track Data

Sample	Sample Location	Latitude (N)	Longitude (E)	Apatite FT Age[a]	Number of Apatite Grains	$\rho_D \times 10^6$ (N_D)[b]	ρ_s (N_s)[b]	ρ_i (N_i)[b]
453001	ENRECA-1 (497 mbKB)	10°02′04.2″	103°59′59.0″	*51,3 ± 3,6*	*20*	*1,481 (2318)*	*0,498 (340)*	*2,826 (1928)*
453009	Nam Du Island	09°40′33.0″	104°23′58.0″	*62,7 ± 5,9*	*12*	*1,476 (2318)*	*1,028 (152)*	*4,727 (699)*
453010	Nam Du Island	09°41′23.0″	104°21′05.0″					
453017	Hon Nghe	10°01′20.0″	104°32′50.0″	*58,8 ± 3,8*	*20*	*1,471 (2318)*	*0,858 (345)*	*4,200 (1689)*
453023	Hon Nghe	10°02′09.0″	104°32′53.0″	*89,6 ± 12,6*	*20*	*1,466 (2318)*	*0,233 (128)*	*0,804 (441)*
453027	Hon Nghe	10°02′04.0″	104°32′49.0″	*59,1 ± 7,9*	*11*	*1,461 (2318)*	*0,305 (71)*	*1,473 (343)*
453029	Hon Soc	10°09′05.0″	104°54′12.0″	*52,8 ± 3,3*	*20*	*1,456 (2318)*	*0,584 (379)*	*3,149 (2043)*
453030	Hon Dat	10°06′16.0″	104°53′50.0″	*53,4 ± 2,7*	*20*	*1,451 (2318)*	*1,488 (677)*	*7,908 (3598)*
453032	Hon Nghe Island	10°01′41.0″	104°33′37.0″					
538301	Khu Tuong N	10°17′37.7″	103°58′53.4″	*48,9 ± 6,0*	*8*	*1,239 (2040)*	*0,845 (84)*	*4,194 (417)*
538313	Mui Mang Ran	10°07′51.8″	104°01′53.0″	*57,3 ± 6,8*	*20*	*1,293 (2027)*	*0,691 (223)*	*3,153 (1018)*
538323	Hon Douc Isl.	10°17′08.5″	104°19′19.3″	*41,4 ± 15*	*5*	*1,246 (2040)*	*0,115 (9)*	*0,679 (53)*
538330	Hon Doc Isl. SE	10°19′17.4″	104°20′13.8″	*117,7 ± 19,5*	*9*	*1,253 (2040)*	*0,466 (56)*	*0,965 (116)*
538332	Hon Doc Isl. NE	10°19′15.9″	104°19′57.2″	*95,8 ± 40,9*	*5*	*1,292 (2027)*	*0,431 (42)*	*1,394 (136)*
538337	Hon Heo Isl. N	10°11′14.2″	104°31′45.7″	*31,2 ± 5,4*	*8*	*1,260 (2040)*	*0,431 (38)*	*3,416 (301)*
538338	Hon Heo Isl. SE	10°11′02.5″	104°31′57.2″	*44,9 ± 5,8*	*6*	*1,290 (2027)*	*0,991 (73)*	*5,406 (398)*
538342	Thach Dong	10°24′59.5″	104°28′31.7″	*49,5 ± 8,4*	*20*	*1,288 (2027)*	*0,179 (78)*	*1,053 (458)*
538345	Nui Tam Phu Nhan	10°22′57.1″	104°27′59.4″	*24,8 ± 12*	*1*	*1,266 (2040)*	*0,132 (1)*	*1,324 (10)*
538356	Tuc Dup	10°22′36.1″	104°57′33.3″	*35,3 ± 3,8*	*20*	*1,287 (2027)*	*0,210 (105)*	*1,451 (725)*
538362	Nui Giai	10°26′33.8″	104°57′41.9″	*49,3 ± 13,7*	*11*	*1,279 (2027)*	*0,385 (95)*	*1,447 (357)*
538364	Nui Cam E	10°30′14.7″	105°00′56.2″	*49,2 ± 6,1*	*20*	*1,285 (2070)*	*0,185 (83)*	*0,945 (424)*
538368	Son Trai	10°37′05.2″	104°59′39.7″	*21,3 ± 4,0*	*20*	*1,273 (2040)*	*0,085 (31)*	*0,999 (364)*
538370	Phnom Den	10°35′59.2″	104°54′18.6″	*54,8 ± 27,1*	*12*	*1,289 (2070)*	*0,013 (5)*	*0,061 (23)*
538374	Tonleap	10°33′24.6″	104°50′34.0″					
538376	Phnom Tasom	10°39′46.5″	104°43′03.2″	*30,2 ± 4,0*	*15*	*1,292 (2070)*	*0,253 (68)*	*2,125 (571)*
538380	Sanlong	10°47′56.8″	104°45′12.14″	*33,5 ± 4,2*	*15*	*1,295 (2070)*	*0,337 (76)*	*2,552 (575)*
538385	Phnom Aya	10°46′35.9″	104°37′48.4″	*41,3 ± 4,2*	*20*	*1,280 (2040)*	*0,287 (123)*	*1,741 (746)*
538701	Cay Sao	10°14′17.8″	104°04′37.7″	*53,8 ± 10*	*10*	*1,299 (2027)*	*0,854 (136)*	*3,517 (560)*
538703	Bai Thom	10°21′27.0″	104°04′52.8″	*68,2 ± 5*	*15*	*1,298 (2027)*	*1,108 (279)*	*3,991 (1005)*
538705	Mui Ganh Dau	10°22′29.6″	103°50′13.9″	*58,8 ± 17,5*	*6*	*1,287 (2040)*	*0,967 (59)*	*4,063 (248)*
538706	Mui Dinh Cau	10°13′02.7″	103°57′24.7″	*49,3 ± 9,1*	*8*	*1,295 (2027)*	*0,367 (36)*	*1,824 (179)*
538709	Phnom Angkoal	10°27′38.6″	104°23′31.8″					
538713	Keep	10°28′41.8″	104°17′42.2″					
538719	Phnom Ta Ngel	10°36′17.3″	104°17′15.9″					
538722	Phnom Poktuyung	10°43′01.3″	104°17′54.2″					
538724	Phnom Y	10°46′23.1″	104°13′22.4″	*42 ± 5,0*	*16*	*1,302 (2070)*	*0,340 (89)*	*2,063 (540)*
538726	Trapadang Phdao	10°51′20.5″	104°14′14.6″	*65 ± 11,3*	*22*	*1,306 (2070)*	*0569 (240)*	*1,741 (734)*
538728	Phnom Changek	10°58′49.7″	104°14′09.9″					
538731	Loc. 6	11°06′00′3″	104°13′50.3″	*43,8 ± 5,7*	*18*	*1,312 (2070)*	*0,406 (72)*	*2,378 (422)*
538732	Noriay	10°53′21.8″	104°32′27.2″	*51 ± 8,8*	*13*	*1,316 (2070)*	*0,315 (41)*	*1,589 (207)*
538733	Phnom Thmar Keo	11°05′37.2″	104°17′10.5″					
538735	Phnom Knarng Takul	11°01′16.2″	104°24′12.2″	*57,6 ± 9,1*	*12*	*1,323 (2070)*	*0,222 (51)*	*0,997 (229)*
538737	Vealrinh	10°43′33.5″	103°47′50.2″	*51,8 ± 5,1*	*15*	*1,326 (2070)*	*1,112 (135)*	*5,568 (676)*
538739	Vealrinh	10°43′33.5″	103°47′50.2″	*54,9 ± 8,0*	*8*	*1,293 (2040)*	*0,457 (59)*	*2,108 (272)*
538751	Rang Pein	11°22′41.1″	104°21′05.6″	*43,5 ± 4,6*	*20*	*1,333 (2070)*	*0,281 (114)*	*1,685 (684)*
538754	Phnom Cheal	11°23′21.8″	104°30′10.7″	*56,3 ± 4,1*	*20*	*1,336 (2070)*	*0,833 (274)*	*3,862 (1271)*
538756	Phnom Tamao	11°18′07.4″	104°47′59.1″	*37 ± 6,3*	*21*	*1,339 (2070)*	*0,147 (40)*	*1,042 (284)*
538761	Phnom Srong	11°15′55.5″	104°36′34.9″	*45,1 ± 5,5*	*20*	*1,343 (2070)*	*0,423 (82)*	*2,466 (478)*
538802	Hon Rai	09°48′01.5″	104° 39′10.2″	*53 ± 6,2*	*20*	*1,285 (2027)*	*0,165 (157)*	*0,751 (715)*
538804	Hon Tre S	09°57′28.0″	104°50′30.1″	*38,7 ± 4,2*	*20*	*1,284 (2027)*	*0,402 (254)*	*2,524 (1596)*
538808	Hon Da Bac	09°10′36.6″	104°48′16.9″	*53 ± 5,0*	*20*	*1,283 (2027)*	*0,534 (282)*	*2,487 (1313)*
538811	Hon Chuoi W	08°57′00.1″	104°31′24.2″	*26,9 ± 3,7*	*26*	*1,281 (2027)*	*0,148 (62)*	*1,335 (560)*
538822	Phnom Kduoch	10°33′15.2″	104°10′25.0″	*83,9 ± 14,6*	*5*	*1,299 (2070)*	*1,007 (45)*	*3,044 (136)*
538823	Phnom Bokor	10°36′04.2″	104°06′27.0″	*44,1 ± 5,0*	*17*	*1,319 (2070)*	*0,378 (97)*	*2,212 (568)*

[a]Sample location, fission track ages and analytical data fundamental to the age calculation and AFTA/ZFTA. Central age [*Galbraith*, 2005], used for samples containing a significant spread in single-grain ages (P(c2) < 5%), otherwise the pooled age is quoted. All ages were calculated using the zeta calibration approach of *Hurford and Green* [1983], using zeta values for CN5 glass of 380.4 ± 5.7. All errors quoted at ±1 s. All analytical details are as described by *Green* [1986], with the exception that thermal neutron irradiations for this study showed a significant flux gradient, and the appropriate value of r_D for each sample was determined by linear interpolation through the stack of grain mounts. Cl contents were determined as described in the text.

[b]ρ_D = glass dosimeter track density; ρ_s = spontaneous track density; ρ_i = induced track density. Numbers in parentheses show the number of tracks counted in determining all track densities.

[c]Probability that all single-grain ages belong to a single population [*Galbraith*, 2005].

[d]Standard deviation of the track length distribution.

Table 1. (continued)

Uranium Content (ppm)	$P(\chi^2)$[c]	Mean Track Length (μm)	(SD)[d]	Zircon FT Age[a]	Number of Zircon Grains	$\rho_D \times 10^6$ (N_D)[b]	ρs (N_s)[b]	ρ_i (N_i)[b]	Uranium Content (ppm)	$P(\chi^2)$
22	*3*	*13,23 ± 0,19*								
37	*53*	*12,94 ± 0,49*	*1.83*							
				64,7 ± 2,6	20	1,241 (1960)	8,780 (2061)	7,345 (1724)	316	6
33	*12*	*13,93 ± 0,18*	*1.94*							
6	*1*	*13,3 ± 0,55*	*2.95*							
11	*8*	*13,26 ± 0,32*	*2.07*							
25	*91*	*13,71 ± 0,18*	*1.87*	77,9 ± 3,0	20	1,242 (1960)	13,770 (2669)	9,571 (1855)	411	79
62	*27*	*13,61 ± 0,17*	*1.79*							
				62,8 ± 4,7	20	1,241 (1960)	6,072 (1861)	5,048 (1547)	217	<1
39	*66*	*13,06 ± 0,48*	*1.7*							
28	*>1*	*13,15 ± 0,89*	*2.2*							
6	*11*									
9	*42*	*13,03 ± 0,52*	*1.3*							
12	*<1*	*9,24 ± 1,0*		172,4 ± 7,7	20	1,348 (2121)	12,156 (2708)	4,112 (916)	163	43
31	*10*	*13,16 ± 0,46*	*1.3*							
48	*23*	*13,24 ± 1,2*	*2.9*							
9	*1*	*13,06 ± 0,46*	*1.3*	140,8 ± 11,5	16	1,348 (2121)	10,098 (1900)	4,103 (772)	162	<1
12	*100*	*13,59 ± 1,24*	*2.1*							
13	*8*	*13,13 ± 0,19*	*1.5*	80,9 ± 4,3	20	1,348 (2121)	6,956 (2145)	5,056 (1559)	200	<1
12	*>1*	*12,77 ± 0,61*	*1.2*							
8	*46*	*12,52 ± 0,55*	*2.6*	83,3 ± 4,0	20	1,348 (2121)	6,549 (2716)	4,632 (1921)	183	<1
9	*76*	*13,09 ± 0,22*	*1.9*							
1	*45*	*11,63 ± 2,49*	*4.3*							
				64,5 ± 3,3	20	1,348 (2121)	3,925 (1944)	3,560 (1763)	141	1
19	*90*	*12,51 ± 0,41*	*1.7*							
22	*9*	*13,17 ± 0,21*	*1.9*							
16	*25*	*13,79 ± 0,17*	*1.8*							
31	*1*	*13,73 ± 1,06*	*2.1*							
35	*14*	*12,49 ± 0,67*	*1.8*							
36	*>1*	*12,45 ± 0,68*	*1.9*							
16	*25*	*11,95 ± 1,0*								
				115,9 ± 8,0	12	1,348 (2121)	10,163 (1407)	5,172 (716)	204	1
				166,1 ± 8,3	20	1,348 (2121)	10,863 (3377)	3,780 (1175)	149	2
				136,3 ± 5,8	20	1,348 (2121)	10,806 (2693)	4,639 (1156)	183	24
				148,5 ± 8,8	20	1,348 (2121)	7,862 (2345)	3,101 (925)	123	1
18	*27*	*11,43 ± 0,77*	*1.9*							
15	*>1*	*12,88 ± 0,54*	*1.9*	117,5 ± 14,6	13	1,348 (2121)	8,737 (1765)	4,460 (901)	176	<1
				133,3 ± 5,3	20	1,348 (2121)	9,906 (3173)	4,349 (1393)	172	10
21	*84*	*13,57 ± 0,56*	*1.0*							
14	*29*	*13,82 ± 1,01*	*2.0*	135,8 ± 9,7	20	1,348 (2121)	7,392 (2754)	3,175 (1183)	126	<1
				125 ± 10	12	1,348 (2121)	11,234 (1216)	5,479 (593)	217	<1
9	*55*	*13,18 ± 0,79*	*1.8*	84,9 ± 6,9	10	1,348 (2121)	9,412 (1078)	6,339 (726)	251	<1
48	*21*	*12,68 ± 0,29*	*1.7*							
19	*93*	*13,21 ± 0,46*	*1.9*							
14	*66*	*12,96 ± 0,19*	*2.0*	75,1 ± 4,1	20	1,348 (2121)	5,770 (1732)	4,467 (1341)	177	2
33	*8*	*13,11 ± 0,15*	*1.5*	121 ± 5,2	20	1,348 (2121)	7,828 (2325)	3,791 (1126)	150	10
9	*96*	*13,45 ± 0,59*	*2.1*	132,6 ± 10,8	9	1,348 (2121)	11,123 (539)	4,912 (238)	194	13
21	*19*	*13,09 ± 0,17*	*1.8*	125,3 ± 5,9	20	1,348 (2121)	7,256 (1927)	3,393 (901)	134	17
7	*3*	*13,53 ± 0,25*	*1.7*	78,9 ± 4,4	20	1,348 (2121)	7,339 (2226)	5.,380 (1632)	213	<1
22	*>1*	*13,63 ± 0,15*	*1.5*	95 ± 3,4	20	1,348 (2121)	9,615 (3564)	5,941 (2202)	235	41
22	*4*	*12,85 ± 0,2*	*2.0*	130,6 ± 5,9	19	1,348 (2121)	12,119 (2143)	5,429 (960)	215	68
12	*8*	*12,17 ± 1,3*	*3.2*							
27	*21*	*13,81 ± 0,38*	*0.7*	113,4 ± 6,9	16	1,348 (2121)	12,930 (1668)	6,597 (851)	261	2
19	*71*	*12,67 ± 1,35*	*2.7*							

and 19 single-grain ages were measured in another 7 samples and 9 single-grain ages were measured in one sample. In most of the zircon grains analyzed, etched surfaces were of high quality, although large numbers of surface pits, fractures, inclusions, and refractive shadows from crystal faces caused problems for counting in some of the samples. In general, the grains that were eventually analyzed are relatively free of such features, but it is possible that these problems may contribute to the excessive scatter in single-grain

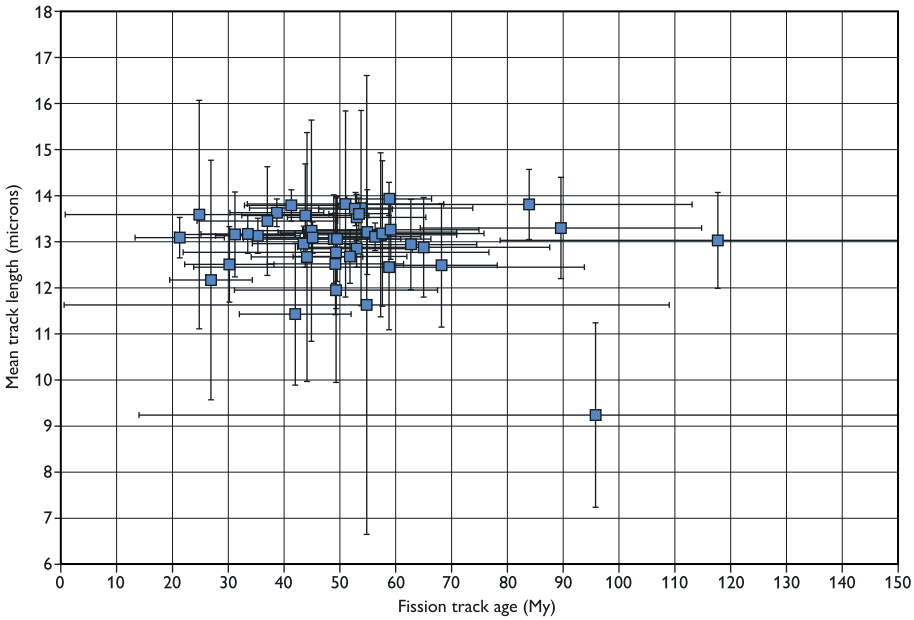


Figure 4. Mean track length plotted against apatite fission track age. The dense data grouping favors comparable cooling stories for most samples.

ages in some of the samples. Despite these problems, the resulting ZFTA data, and the resulting interpretations, are regarded as generally reliable within the stated ranges of uncertainty (95% confidence limits).

To obtain maturity measures independent from the thermochronology, vitrinite reflectance analyses were carried out. These measurements (Random, oil immersion) were carried out at 546 nm using monochromatic light. The readings were taken on polished blocks using a Leica DM 4000 M system. The average reflectance of each sample was calculated from a population selected from the total vitrinite reflectance histogram to omit reworked vitrinite and low-reflectance inertinite. Measurements were performed on mudrocks, limestones, and siltstones from a number of outcrops from some of which sandstones were collected for AFTA and ZFTA. In addition, vitrinite reflectance was measured throughout the ENRECA-2 core well in order to obtain information on the thermal maturity and paleogeothermal gradient in the area.

Approximately 300 km of 2-D seismic data covers the onshore Cambodian part of the study area crossing the Kampot Fold Belt and the eastern part of the Phuquoc-Kampot Som Basin. These data were collected in 2008 using state of the art acquisition and processing techniques and were made available to Geological Survey of Denmark and Greenland (GEUS) by PGS [*Hendrick et al*., 2009]. Much of these data were collected across the structurally complex Kampot Fold Belt, which challenges the regional stratigraphic and structural interpretation. Offshore Vietnam, ~18,000 line kilometers 2-D seismic data were included in the study. These data were made available to the Danish-Vietnamese research group by Vietnam Petroleum Institute and cover part of the Kampot Fold Belt and the Phuquoc-Kampot Som Basin. The offshore data were acquired in between the mid-1970s and 2004, and their quality therefore varies.

4. Geological Outline of Study Area

4.1. Stratigraphy

Cretaceous and older rocks crop out onshore in SE Cambodia and SW Vietnam and on islands immediately to the south (Figures 2 and 6). Offshore, similar age strata subcrop toward the base of middle or late Eocene to Neogene deposits [*Fyhn et al*., 2010a]. The observed offshore subcrop pattern of pre-Cenozoic strata was related to structural shortening and uplift caused by compression across the Kampot Fold Belt [*Fyhn et al*., 2010a].

The onshore is composed of Paleozoic and Mesozoic deposits and igneous rocks dated radiometrically in this study (zircon U/Pb ages, Figure 5). Paleogene rocks (cutting samples from exploration wells dated (middle? or) late Eocene to Oligocene) are known only from offshore rifts buried underneath Neogene strata [*Fyhn et al*., 2010b].

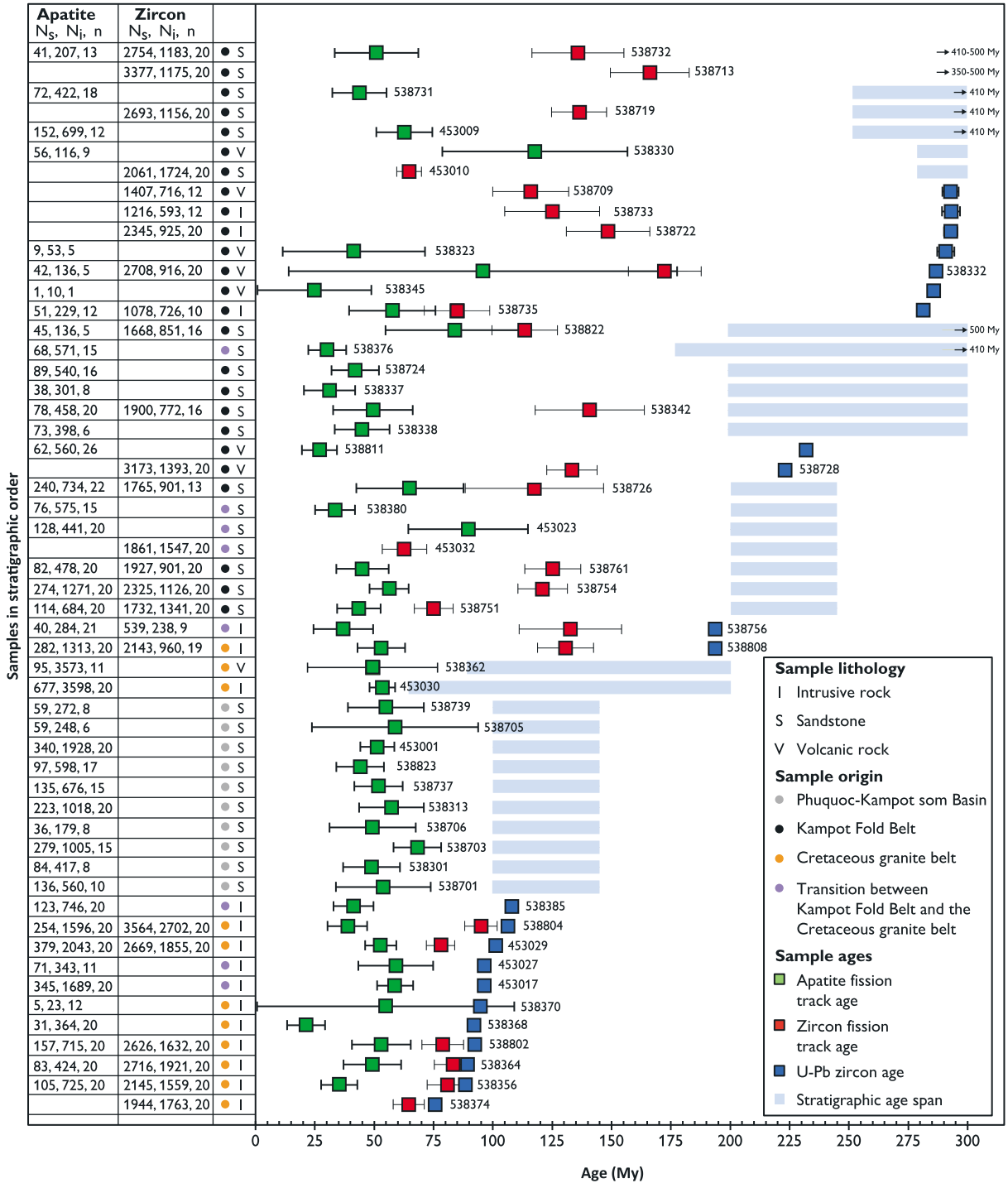


Figure 5. Apatite and zircon fission track ages plotted with stratigraphic (depositional) or radiometric (igneous) ages of samples. Error bars (quoted at 1σ) provide the best guide to the data quality of individual samples. In addition, numbers of counted spontaneous fission tracks (N_s), numbers of counted induced fission tracks (N_i), and numbers of analyzed grains (n) indicate the data material behind individual fission track ages. The statistical data quality generally improves with increasing N_s, N_i, and n. A full report of metadata is provided in Table 1.

Neogene strata onlap the pre-Cenozoic toward the northeast away from the center of the Gulf of Thailand (Figure 3). This causes the Neogene to wedge out and only comprise a fairly thin succession onshore.

The changing pre-Quaternary stratigraphy onshore documents considerable variation in Paleozoic and Mesozoic deposition that primarily reflects the alternating tectonic regimes governing the geologic

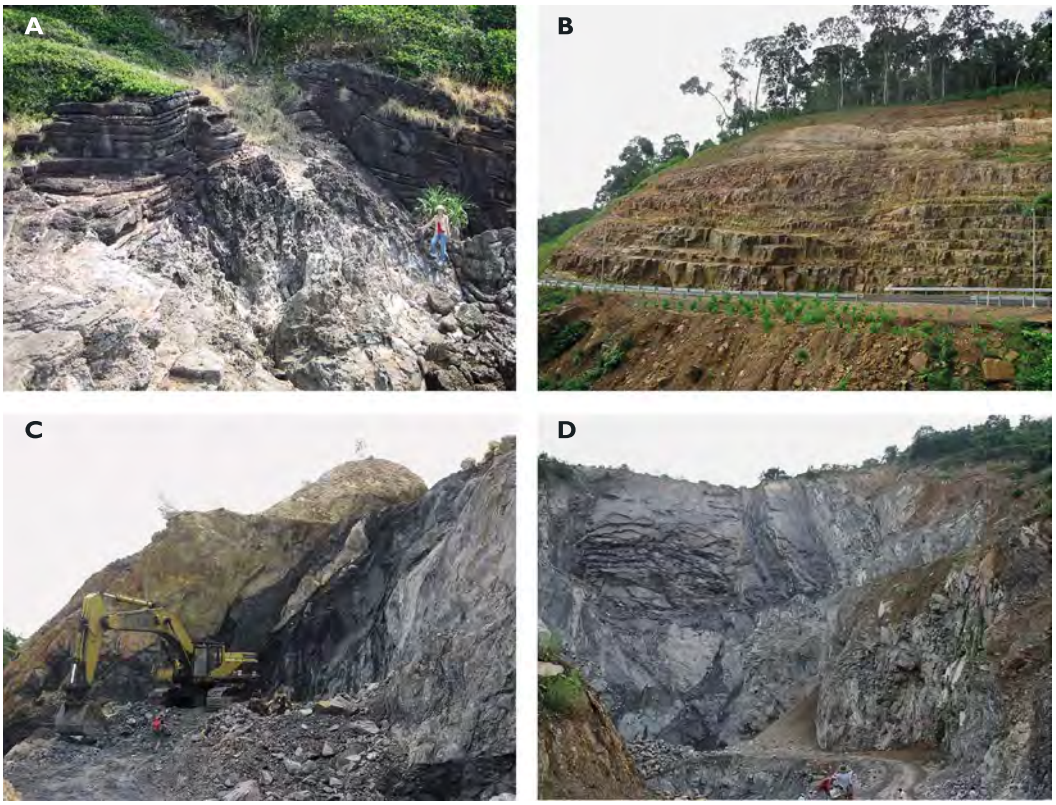


Figure 6. (a) Normal faulted, Lower Permian rhyolites with drag folding along fault plane, Hon Mau Island, Nam Du Archipelago. (b) Approximately 40 m high outcrop exposing very gently dipping, large-scale cross-bedded fluvial sandstones exposed in the Bokor Mountains (Phnom Bokor). (c) Part of thrust zone exposed in quarry at Bai Ot at the SW Vietnamese coast. The fault zone is a few hundred meters wide and thrust Devonian over Triassic deposits. Mélanges of Permian carbonates are contained within dark mylonitic kataclastites in the fault zone. (d) Approximately 80 m high exposure at Phnom Cheal from which steeply dipping Triassic sandstones is quarried.

development of Indochina (Figure 3). Stratigraphic investigations within the area, especially in Cambodia, are in their infancy. The ages inferred here are from existing geological maps, in certain cases revised to fit with igneous and unpublished detrital zircon ages measured recently by the GEUS (Figure 5).

Upper Jurassic to middle Cretaceous. The Phuquoc Group of Upper Jurassic(?) to mid-Cretaceous age is widespread west of the Kampot Fold Belt both onshore and offshore. The group is composed of alluvial and fluvial sandstones and floodplain mudstones although a few shallow marine intervals have been noted on Phu Quoc Island. The entire succession exceeds 4 km thickness and is considered equivalent to the Khorat Group in Thailand, Laos, central Vietnam, and northern Cambodia [*Fyhn et al*., 2010a; *Meesook*, 2011]. In this study, Barremian and Aptian mudstones and sandstones were investigated and sampled.

A mid-Cretaceous U/Pb zircon age (107.7 ± 0.2 Myr) was obtained for dacitic rocks at the gradual transition from the eastern rim of the Kampot Fold Belt toward the Mesozoic granite belt farther east (Figure 2) (sample 538385, Phnom Aya, Figure 5). This age is compatible with other radiometric ages of granitic rocks measured within this transition zone or east of the fold belt in the Mesozoic granite belt with modern U/Pb zircon ages ranging in between roughly 108 and 75 Myr (Figure 5). Elsewhere in Vietnam, U/Pb zircon ages up to 121 Myr have been measured from comparable granitic rocks [*Nguyen et al*., 2004]. Vintage ages (mostly K/Ar ages) may indicate a broader age spectrum spanning the entire Cretaceous and into the earliest Paleocene, although the accuracy of these K/Ar ages is unknown [*Fyhn et al*., 2010a].

Jurassic. Interbedded mudstones and sandstones are attributed to the Jurassic on Hon Nghe Island (Figure 2), and elsewhere in southwesternmost Vietnam based on lithologic comparisons with the Jurassic elsewhere in Vietnam [*Hoa*, 1996e]. In addition, Early Jurassic detrital zircon ages were measured and constitute the youngest zircons in sediments sampled at outcrops near Hiu Chrak Chikam mapped as Devonian to

Carboniferous in age (Figure 2). The deposits lack Late Jurassic and Early Cretaceous detrital zircons otherwise ubiquitous in the Upper Jurassic to mid-Cretaceous Phuquoc Group and the granite belt immediately east of the Kampot Fold Belt. A Lower to Middle Jurassic age is therefore tentatively suggested for these mixed sandstones and mudstones sampled near Hiu Chrak Chikam. Granites with Early Jurassic U/Pb zircon ages also exist in the region, documenting Jurassic magmatism.

Upper Triassic. Medium to coarse-grained quartzites occasionally conglomeratic interdigitating with up to few meters thick slaty beds crop out near Kampong Spe in the up to nearly 600 m high mountains. The succession has been mapped as Triassic in age [*BRGM*, 1973b]. This fits with abundant Early and Middle Triassic detrital zircons within the succession and the absence of younger zircons. Post-Triassic granites are mapped to intrude the quartzites. A comparable granite cropping out few tens of kilometers farther east yielded an Early Jurassic U/Pb zircon age (193.7 ± 1.3 Myr) (sample 538756, Phnom Tamao, Figure 5), which may support a Late Triassic age of the quartzites.

Middle Triassic. U/Pb dating of zircons from tuffs belonging to the Hon Ngang Formation on Hon Chuoi Island document an age around 230 Myr (Figure 2). Onshore SW Vietnam, sandstones similarly attributed to the Hon Ngang Formation are in places overthrusted by middle Permian and older strata and also considered as Middle Triassic in age [*Hoa*, 1996a, 1996b]. However, U/Pb zircon ages of rhyolites and andesites show that the formation in fact should be treated as two distinct formations of Triassic and Early Permian age, respectively (see Lower Permian section below).

Lower to Middle Permian. Permian limestone exposures are scattered across the region making up distinct isolated karst pinnacles. They include intervals of calcareous shales intercalated with thin-bedded marls [*Hoa*, 1996e]. A rich foraminifera fauna date the Ha Tien Formation in Vietnam to the Lower to Middle Permian (Artinskian to Guadalupian) [*Tri and Khuc*, 2011]. A similar age was assigned to carbonates in western Cambodia [*Ishii et al*., 1969; *Sone et al*., 2001].

Lower Permian. As mentioned above**,** the Hon Ngang Formation was considered as Middle Triassic in age [*Hoa*, 1996e]. However, the lower part of the formation cropping out amongst others on Hon Doc and Hon Duoc Islands (Figure 2) is Early Permian in age with measured radiometric ages ranging between roughly 293 and 283 Myr (Figure 4). Rhyolites, rhyolitic tuffs, andesites, and sandstones are the Lower Permian in the region.

Devonian to Carboniferous. Stratified fine to medium grained sandstones interbedded with siltstones and slates have been dated as Devonian to Carboniferous based on poorly preserved leaf imprints [*Hoa*, 1996e]. In addition, calcareous siltstones are believed to occur in the upper part of the unit. The upper part has been assigned to the Early Carboniferous based on its foraminifera assemblage [*Hoa*, 1996e].

Cambrian to Silurian. Quartzites, metaconglomerates, slates, and low-grade schists have been mapped as Cambrian to Silurian in age [*BRGM*, 1973b]. Reconnaissance detrital zircon dating supports Middle to Late Silurian ages; a single sample contained Carboniferous zircons suggesting that part of this succession is not easily distinguishable from the Devonian to Carboniferous units.

4.2. Structural Relations

Various kinds of structural features occur in the study area. The most severe deformation is observed within the Kampot Fold Belt. Both extensional and compressional structures have formed in the region, which most likely reflect its complex deformation history.

At the Nam Du Archipelago, Lower Permian strata are offset by normal faults with N-S and NE-SW strikes observed (Figure 6a). Moreover, Triassic strata are cut by normal faults and juxtaposed against limestones of presumed Early to Middle Permian age on Hon Nghe Island.

Within the Kampot Fold Belt, the pre-Cenozoic section has been severely shortened and all Paleozoic and Mesozoic deposits are often steeply inclined, sometimes vertical. Bedding strikes vary between NE, N, and NW. West of the Kampot Fold Belt on Phu Quouc Island and in the Bokor Mountains, the inclination of Cretaceous strata rarely exceeds 15° (Figure 6b), whereas bedding strikes are comparable to those observed in the Kampot Fold Belt. Seismic data confirm that deformation mostly affected the Kampot Fold Belt with deformation fading toward the west in the Phuquoc-Kampot Som Basin (Figure 7). The eastern part of the basin comprising the equivalent to Phu Quoc Island and the Bokor Mountains is more deformed than the central part of the basin.

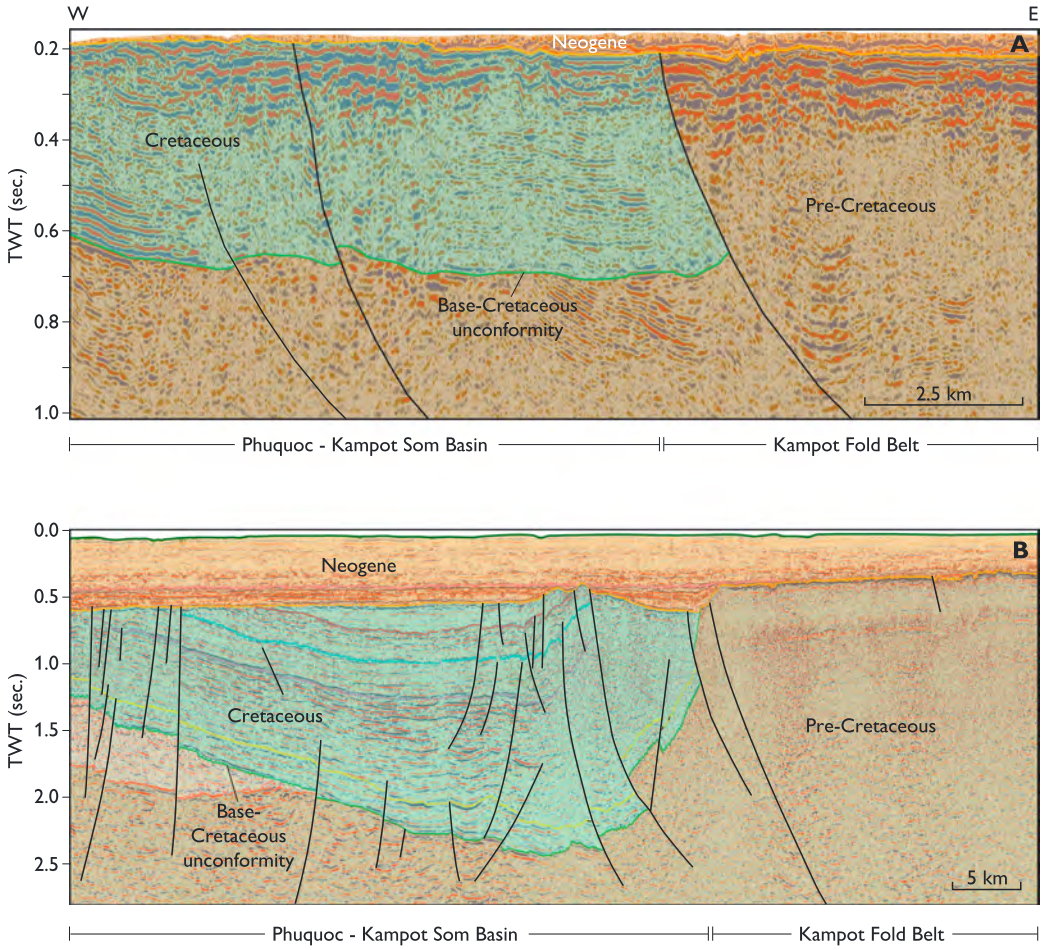


Figure 7. Seismic sections crossing from the Phuquoc-Kampot Som Basin and into the Kampot Fold Belt. (a) Onshore transect, courtesy of PGS. (b) Offshore transect, courtesy of PetroVietnam. Position of lines indicated in Figure 8. The degree of deformation increases toward the fold belt in the east and truncation of Cretaceous and older deposits occur toward the base of the Neogene. Erosion-resistant intervals delineate large buried hills. This erosion related relief recurs in the topography of the Bokor Mountains and on Phu Quoc Island. Truncation surfaces also exist at the base of the Cretaceous and at deeper levels suggesting pre-Cretaceous periods of uplift and denudation.

A plateau-like topography characterizes the tallest parts of the Bokor Mountains situated at roughly 800–1000 m above sea level (Figure 8). The plateau may represent a raised planation surface deeply incised by active rivers.

The Kampot Fold Belt contains abundant thrust faults with observed faults striking NE, N, and NW compatible with the most common attitude of bedding. However, detailed mapping of faults was not possible due to the restricted extent of outcrops.

In Bai Ot, southwesternmost Vietnam, Devonian to Carboniferous strata have been thrust into contact with Hon Ngang Formation sandstones allegedly of Triassic age [*Hoa*, 1996e]. The north-northwest striking fault zone is hundreds of meters wide and contains meters thick bands of dark mylonitic cataclasites and mélanges of carbonate rocks attributed to the Lower to Middle Permian Ha Tien Formation (Figure 6c). A few kilometers away in Nui Tam Phu Nhan, Permian Ha Tien carbonate rocks have also been thrusted over Triassic sandstones, which may represent the northern extension of the Bai Ot thrust zone (Figure 2).

Cataclasites are also observed near Thach Dong within Lower Permian rhyolites, and Ha Tien Formation carbonate rocks have been thrust over quartzites of presumed Early Paleozoic age near Phnom Sorsear.

Roughly N-S trending lineaments dominate the topography and geology of the study area (Figures 2 and 8). Offshore where Cretaceous strata subcrops below the Cenozoic, there is a prominent buried relief with similar lineaments (Figure 7). These lineaments are associated with N-S striking faults and folds within the offshore

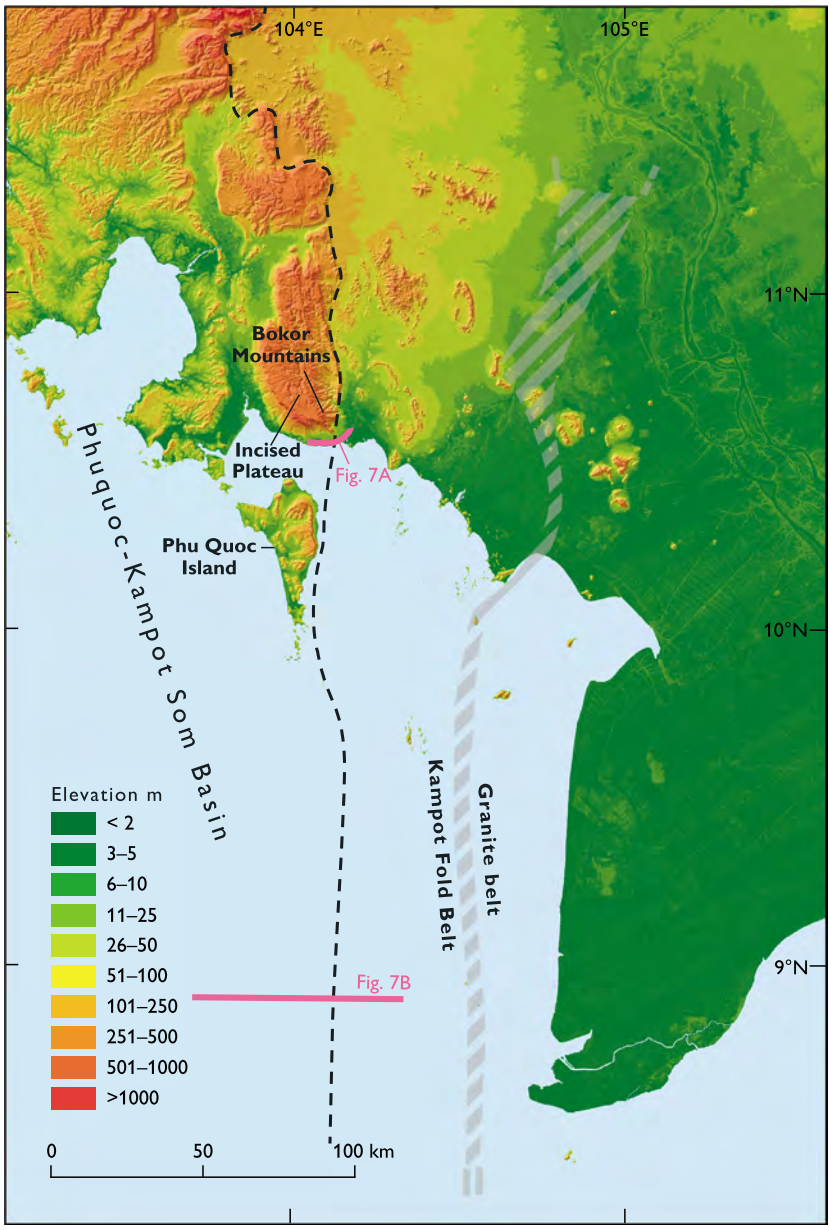


Figure 8. Onshore digital elevation model illustrating the mountainous Phuquoc-Kampot Som Basin and the flat to hilly Kampot Fold Belt and Mesozoic granite belt in the east. The incised plateau relief capping the Bokor Mountains may have formed during pulsed Cenozoic denudation.

part of the Phuquoc-Kampot Som Basin (Figure 2) [*Fyhn et al*., 2010a]. Onshore, the eastern flank of the Phuquoc-Kampot Som Basin outlines the most prominent sublongitudinal lineament. Apart from the apparent structural control, the relief along the eastern edge of the Phuquoc-Kampot Som Basin reflects Cretaceous erosion-resistant sandstone intervals. Similar erosion-resistant intervals were documented offshore by *Fyhn et al*. [2010a]. Terraces and escarpments along the flank of the Bokor Mountains, on the Phu Quoc Island and buried underneath Cenozoic strata offshore suggest a few intervals particularly resistant to erosion within the Phuquoc Group. These intervals may be stratigraphically equivalent to the Phu Phan Formation, but possibly also the Phra Wihan Formation in the Khorat Basin that crops out along the edge of the Khorat Plateau and give rise to a comparable relief [*Racey et al*., 1997; *Booth and Sattayarak*, 2011; *Meesook*, 2011].

Within the Kampot Fold Belt, NNW-SSE and N-S trending ridges and valleys probably reflect the structural grain. Furthermore, arch-shaped ridges delineate the erosional remnants of large folds with fold axes roughly striking N-S. Bedding within the fold limbs is steep suggesting they form part of large closed or even tight folds.

NW trending lineaments have been recognized especially within part of the Bokor Mountains and in the mountainous area south of Kampong Spe dominated by severely deformed Upper Triassic sandstones. Various orientations of topographic lineaments occur in this area, and apart from NW striking steep to subvertical bedding, we observed NE striking thrust faults and NE striking beddings in Rang Pein. Extensional faults also offset these Triassic deposits, and the Triassic succession in this area is speculated to form part of a local Triassic graben system compatible to roughly NW-SE to E-W trending Late Triassic grabens in northeastern Thailand. The NW-SE trend of lineaments in this area may thus reflect inherited structural fabric reactivated during deformation of the Kampot Fold Belt.

5. Thermal Maturity and Thermochronological Analysis

5.1. Vitrinite Reflectance Analyses

Vitrinite reflectance analyses were carried out on mudrocks and siltstones from a number of outcrops where sandstones were collected for AFTA and ZFTA as well as on carbonate rocks. Vitrinite reflectances (Ro) measured on Cretaceous mudstones from the Phuquoc-Kampot Som Basin regionally show values of approximately Ro 0.6%–0.9%, which is considered to be representative for Cretaceous near-surface rocks onshore in the Phuquoc-Kampot Som Basin. This is compatible with maximum paleotemperatures between ~100°C and 133°C assuming roughly linear heating rates of approximately 1°C/Myr [*Burnham and Sweeney*, 1989], which is close to the apatite fission track total annealing window. This is consistent with the thermal history solutions derived from AFTA data on samples from the same outcrops and comparable stratigraphic levels in the ENRECA-2 well drilled on the Phu Quoc Island. The vitrinite reflectance profile measured in the ENRECA-2 well record a slight increase from ~ Ro 0.60% to ~ Ro 0.63% in maturity across 500 m depth. The measured maturity gradient of the well section suggests a low to intermediate paleogeothermal gradient (25°C to 35°C/km) at the time of maximum maturation and certainly precludes a high geothermal gradient. This suggests that a few kilometers of denudation have occurred since the time of maximum paleotemperature.

Samples of pre-Cretaceous strata from the Kampot Fold Belt contained little organic matter. Vitrinite reflectances were obtained from samples of Lower to Middle Permian and Upper Triassic strata. The values are in the range Ro 3.3 to 4.5%, and no systematic variation between thermal maturities and depositional age was detected in the pre-Cretaceous succession. The regionally obtained Ro values of 3.3 to 4.5% are considered to be representable for most Permo-Triassic near-surface strata onshore in the Kampot Fold Belt, which is comparable to maximum long retention paleotemperatures of around 230°C to 300°C close to the zircon fission track total annealing window or within the partial annealing window [*Burnham and Sweeney*, 1989]. Detailed vitrinite reflectance assessment of thermally high maturity rocks with low organic content is difficult, and the measured pre-Cretaceous Ro values may only provide a general guideline of their maturity.

Although the fairly narrow thermal maturity range of different stratigraphic intervals does not suggest abnormally high paleogeothermal gradients within the Kampot Fold Belt, mid-Cretaceous to Late Cretaceous magmatism along the eastern transition of the belt and farther east may have been accompanied by a high geothermal gradient compared to that in the Phuquoc-Kampot Som Basin farther west. Nevertheless, vitrinite reflectance values between Ro 3.3% and 4.5% suggest kilometers of denudation since the time of deepest burial.

5.2. Description of Fission Track Data

5.2.1. Zircon Fission Track Data

All 25 measured zircon fission track ages are significantly younger than the rocks from which they were derived (Figure 5). This indicates substantial annealing of the fission tracks in the zircons, which is in accordance with the obtained vitrinite reflectance values between Ro 3.3% and 4.5% from the Kampot Fold Belt (section 4.1), suggesting maximum paleotemperatures in between ~230°C and 300°C. The zircon fission track age spectrum ranges from 172.4 ± 7.7 Myr to 62.8 ± 4.7 Myr. Zircon fission track age ages does not seem to vary systematically with neither rock age nor measured vitrinite reflectance (Figure 5). In the Kampot Fold Belt, the most common ages are Early Cretaceous ranging from 148.5 ± 8.8 Myr to 113.4 ± 6.9 Myr. In

addition, samples 538332 and 538713 from Hon Doc Island and Keep, respectively, (Figure 2) record slightly older ages of 172.4 ± 7.7 Myr and 166.1 ± 8.3 Myr. Four samples from the Kampot Fold Belt and the transition zone between the Kampot Fold Belt and the Mesozoic granite belt yield younger ages between 63 and 85 Myr (538735, 538751, 453010, and 453032 from Phnom Knarng Takul, Rang Pein, Nam Du archipelago, and Hon Nghe Island, respectively). One of these samples collected at Phnom Knarng Takul (sample 538735) is from a Permian granite pluton also sampled and analyzed at Phnom Thmar Keo (sample 538733) that yielded a zircon fission track age of 125 ± 10 Myr (Figure 2). The granite body forms a local mountain range and is continuous in between the two sample sites. Hence, no direct indications from outcrop or lithology suggest that the two neighboring areas have experienced a substantially different thermal evolution as potentially indicated by their zircon fission track ages. Similarly, a detrital zircon fission track age of sample 538751 of Triassic sandstones from Rang Pein (sample 538751) on 75.1 ± 4.1 Myr contrasts with Aptian ages of 121.0 ± 5.2 Myr and 125.3 ± 5.9 Myr (sample 538761 and 538754, respectively) of similar Triassic sandstones recorded by samples from Phnom Cheal and Phnom Srong nearby (Figure 2). The lithologies exposed at these three Triassic sample sites are virtually identical, and although the three exposures are not directly connected, no indications for a substantially different thermal evolution of the three samples have been observed. In contrast, the vitrinite reflectance values of these three sampled exposures are compatible in the range between 3.29% and 4.48% probably corresponding to maximum paleotemperatures between ~230 and 300°C.

In the south, a zircon fission track age of 64.7 ± 2.6 Myr is recorded by Lower Permian deposits from Nam Du Island (453010) (Figure 2). On Hon Nghe Island farther north, Triassic sandstones yield a central zircon fission track age of 62.8 ± 4.7 Myr (453032). The sandstones were sampled a few hundred meters away from a granitic intrusion with a zircon U/Pb age of 95 Myr that forms the core of the island with smaller dykes of similar age radiating from the central pluton and intruding the sediments along the coast of the island [*Fyhn et al.*, 2010a].

Rocks from the Mesozoic granite belt and Cretaceous granitic rocks from the transition zone separating the Kampot Fold Belt from the Mesozoic granite belt define a younger zircon fission track age trend (Figure 5). All analyzed ZFTA samples in this area are of Cretaceous and Jurassic granitic rocks. The zircon fission track ages vary with the U/Pb zircon ages recording the emplacement of the granites (Figure 5). Zircon fission track ages for all Cretaceous intrusions are 6–22 Myr younger than the emplacement ages of the respective granite samples. The two oldest granites sampled in or near the Mesozoic granite belt (538756, Phnom Tamao, and 538808, Hon Da Bac), recording coeval Early Jurassic emplacement ages yield comparable zircon fission track pooled ages of 132.6 ± 10.8 Myr and 130.6 ± 5.9 Myr, respectively, equivalent to the dominating zircon fission track age trend recorded in the Kampot Fold Belt (Figure 5).

5.2.2. Apatite Fission Track Data

All apatite fission track ages are significantly younger than the respective depositional or igneous ages of the samples (Figure 5). Furthermore, in all 16 samples with both apatite and zircon fission track ages the apatite ages are significantly younger than the zircon fission track ages. Measured apatite fission track ages range from ~20 to ~120 Myr but with the majority around 40 to 60 Myr (Figure 4). Cretaceous sandstones collected both from the Cambodian and Vietnamese part of the Phuquoc-Kampot Som Basin yield Paleocene to middle Eocene apatite fission track ages.

Pre-Cretaceous rocks sampled in the Kampot Fold Belt record a larger range in apatite fission track age. Like farther west, middle Eocene and Paleocene ages prevail. In addition, four samples yield ages between 83.9 and 117.7 Myr. A Triassic sandstone (sample 453023) from Hon Nghe Island record a central age of 89.6 ± 12.6 Myr. However, two other samples (453017 and 453027) collected nearby also on Ho Nghe Island obtained apatite fission track ages of 58.8 ± 3.8 Myr and 59.1 ± 7.9 Myr, respectively. Also, as mentioned above, a zircon fission track age of 62.8 ± 4.7 Myr (453032) was recorded from the island. Since there are no indications for strong temporal variations in the cooling history on this small island, we consider the oldest apatite fission track age as a statistical outlier.

Apatite fission track ages of 117.7 ± 19.5 Myr and 95.8 ± 40.9 Myr are recorded in two samples from Hon Doc Island (583330 and 58332, respectively). Although these two ages overlap within their 95% confidence limit, they differ substantially from the general age trend (Figure 4) and are associated with large uncertainties reflecting a substantial spread within a small number of individual apatite grain ages in each sample. For this reason, the departure of these ages from the main grouping in Figure 4, together with the value of 83.9 ± 14.6 Myr age recorded in sample 538822 from Phnom Kdouch, is thought to largely reflect the poorer data quality in these samples rather than any underlying difference in thermal history.

Of the 12 mid-Cretaceous granitic rock samples analyzed, 8 yield apatite fission track ages in between approximately 50 and 60 Myr comparable to most other samples. In addition, four samples from Son Trai, Tuc Dup, Hon Tre, and Phnom Aya yield ages of 21.3 ± 4.0, 35.3 ± 3.8, 38.7 ± 4.2, and 41.3 ± 4.2 My, respectively (538368, 538356, 538804, and 538585, respectively) (Figures 2 and 5). Nearby Phnom Aya, samples (538376 and 538380) from Phnom Tasom and Sanlong yield ages of 30.2 ± 4 and 33.5 ± 4.2 Myr. A Triassic tuff from Hon Chuoi (538811) records a comparable apatite fission track age of 26.9 ± 3.7 Myr. Comparable ages were derived from Nui Tam Phu Nhan (538345) (24.8 ± 26.1 Myr) and from Hon Heo (538337) (31.2 ± 5.4 Myr). However, only a single apatite crystal was derived from the Nui Tam Phu Nhan sample, and the resultant age is considered highly uncertain. On Hon Heo Island the age of 31.2 ± 5.4 Myr (528337) may also represent a statistical outlier since another sample from the island (538338) yielded an apatite fission track age of 44.9 ± 5.8 Myr within the dominant age range of the region. These two samples were collected roughly 500 m apart at similar altitude and represent rocks of similar age and related lithologies, and there is nothing to indicate that they should have experienced highly different cooling histories.

In all of the apatite samples, the mean apatite track lengths are shorter and/or the fission track ages younger than would be expected if the rocks had remained at present-day surface temperatures since their formation. However, no clear relation between mean track length and apatite fission track age can be discerned from the data (Figure 4). Most ages cluster around 40–60 Myr and most mean track lengths are around 13 μm. This suggests that most samples have undergone comparable overall thermal histories, with any differences between samples being fairly subtle. Nonetheless, both the 45 apatite fission track ages and AFTA data suggest some variations in the cooling pattern across the study area.

Thermal history constraints have been extracted from the AFTA data using methods explained in detail by *Green and Duddy* [2012]. By comparing the fission track age and track length distribution predicted from a range of candidate paleothermal scenarios with measured parameters, the range of conditions can be defined (maximum/peak paleotemperature and onset of cooling) giving predictions that are consistent with the measured data within 95% confidence limits. In practice, the variation of fission track age and track length distribution with wt % Cl is used to define the range of viable solutions. AFTA data cannot constrain the entire thermal history of a sample, so we focus on defining the main paleothermal episodes that exert a dominant control on the data in individual samples using assumed heating and cooling rates of 1°C/Myr and 10° C/Myr, respectively.

Timing constraints on the onset of cooling (95% confidence intervals) derived from AFTA data from samples in this study are compared in Figure 9, with samples grouped into the geographical regions illustrated in Figure 5. The size of these confidence regions in individual samples is affected by a combination of factors, including not only numbers of tracks counted for age determination and numbers of track lengths measured but also the Cl content for the apatite and the paleotemperature from which a sampled cooled in each episode. The timing of cooling below approximately 110°C can usually be determined with greatest precision, whereas uncertainty in the timing of cooling from paleotemperatures around 60–70°C is usually more dependent on the quality of the track length data. In general, the most important measure of the quality of an interpretation is the size of the 95% confidence interval on the timing of cooling and the maximum or peak paleotemperature in each episode [*Green and Duddy*, 2012].

The individually modeled temperature/time relation of all samples fits a regional thermal history characterized by three regionally synchronous cooling episodes (Figure 9). These three cooling episodes commenced during (1) the Paleocene (~60–55 Ma), (2) the middle to late Eocene (~40–35 Ma) and the early Miocene (~20–15 Ma). It should be noted that because AFTA data in nine of the samples were totally overprinted prior to the second cooling episode lasting from the middle to late Eocene, the earlier history is not defined, and Paleocene cooling may also have affected these samples.

The majority of the samples from across most of the study area cooled below ~110°C around the earliest (Paleocene) episode (modeled in 34 samples) (Figure 10b). In contrast, the second episode initiating during the middle to late Eocene seems focused particularly in samples from the north and east of the region within or near the Mesozoic granite belt (Figure 10c). AFTA data suggest that 11 samples in this area cooled below ~110°C in this middle to late Eocene episode, while farther west, a few samples cooled from lower temperatures around 90 to 100°C in this episode, following the initial (Paleocene) cooling below ~110°C also recorded in these samples.

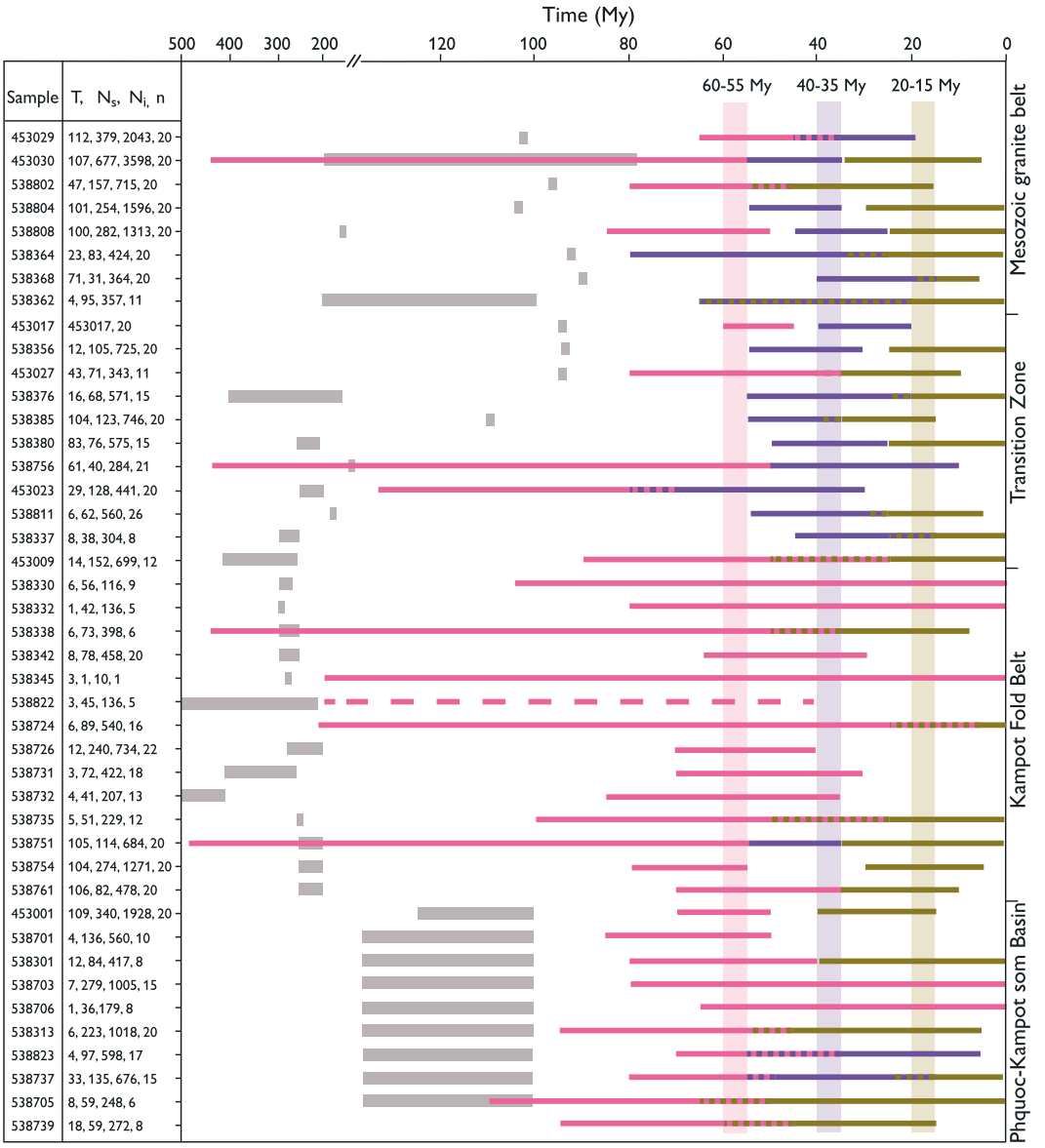


Figure 9. Timing constraints on the onset of cooling phases of individual samples suggested by AFTA data derived from modeling of track length distributions and apatite fission track ages. Depositional and igneous ages noted in grey. The colored bars represent the 95% confidence interval on the onset of cooling from each episode of elevated paleotemperatures. The width of each bar therefore provides an indication of how well each episode is constrained in individual samples and depends on the number of spontaneous fission tracks counted (N_s), the number of induced fission tracks counted (N_i), the number of track lengths measured (T), and also on the apatite Cl content and the magnitude of the paleotemperature from which the sample cooled in each event [*Green and Duddy*, 2012]. Data quality generally improves with increasing N_s, N_i, and T. Up to three cooling events can be discerned from individual samples distinguished from each other based on their timing and modeled temperature range. Overlap in the possible onset age of cooling events is indicated with dashed lines. The combined results of the samples are consistent with three major episodes of cooling which began in the intervals 65 to 60 Ma (pink), 40 to 35 Ma (purple), and 20 to 15 Ma (green). The cooling event that commenced around 40–35 Ma (purple) is best developed in samples from the Mesozoic granite belt and is interpreted to mainly reflect postintrusion cooling.

In contrast, Miocene cooling is identified in 27 samples (Figure 10d); those samples where it is not identified probably reflects data quality (particularly absence of track length data) rather than any real differences in thermal history. No discernible pattern of variation is evident in the Miocene paleotemperatures, with most samples consistent with cooling from a temperature of 60–70°C. Miocene cooling therefore seems to have been essentially uniform across the region.

6. Regional Interpretation and Synthesis of Thermochronological Data

Fission track analyses suggest that certain cooling episodes affected the entire study area, whereas one episode most dominantly appears from samples in the east. To investigate the mechanism behind these cooling events, below we discuss the fission track data in relation to the three distinct settings from which samples were derived: the Mesozoic granitebelt in the east, the (central) Kampot Fold Belt, and the Phuquoc-Kampot Som Basin in the west.

6.1. Mesozoic Granite Belt

The consistent relationship between zircon fission track ages of the Cretaceous granites and their emplacement ages (Figure 5) suggests that cooling into the partial annealing window of zircon occurred following the initial postmagmatic cooling of the individual plutons. The 6–22 Myr difference in U/Pb zircon age and zircon fission track age resembles the granite cooling pattern recorded farther north along the Cretaceous paleo-Pacific subduction arc in Korea [*Lim and Lee*, 2005], and comparable granite cooling rates have been reported from elsewhere [*Chesley et al*., 1993]. Similar to the Korean example, we interpret the zircon fission track ages of these granites to record primarily passive postemplacement cooling. The 6–22 Myr difference between U/Pb zircon ages and zircon fission track ages records the period in which the plutons cooled from roughly 900 to 300°C [*Lee et al*., 1997; *Green and Duddy*, 2012].

The young zircon fission track age of the Triassic sandstones (62.8 ± 4.7 Myr) from Hon Nghe Island (453032) is speculated to also record cooling in an area with local, high geothermal gradient associated with the neighboring mid-Cretaceous granite intrusion the contact with which is located between tens and a few hundred meters away from the sampling site. This is consistent with modeled heating of country rocks around middle-sized granite plutons emplaced within the upper crust [*Nabelek et al*., 2012]. Together with the early Paleocene zircon fission track age, the two apatite fission track ages of 58.8 ± 3.8 Myr and 59.1 ± 7.9 Myr (samples 453017 and 453027, respectively) also from Hon Nghe Island suggest very rapid Paleocene cooling around the island, which most likely record tectonic induced exhumation.

Because the Cretaceous granites all record Late Cretaceous to earliest Paleocene zircon fission track ages and thus gradually cooled through the complete zircon annealing window, it seems reasonable to assume that these rocks were emplaced within the upper crust with a background temperature below the complete zircon fission track annealing window. Emplacement in the upper crust is also suggested by the nature of the sedimentary and volcanic rocks in which they were intruded that are attributed to lithostratigraphic formations recurring in the Kampot Fold Belt. Apart from the contact metamorphism, these sedimentary and volcanic rocks resemble those of the Kampot Fold Belt where vitrinite reflectance values of comparable stratigraphic intervals suggest maximum paleotemperatures in the range 230–300°C, compatible with upper crustal conditions. Hence, no indications exist for a substantially different pre-Cenozoic burial history of outcropping rocks in the Mesozoic granite belt and rocks from the adjacent Kampot Fold Belt. Total uplift and denudation in the Mesozoic granite belt since Late Cretaceous time, therefore, most likely was comparable to that affecting the Kampot Fold Belt.

In comparison with the Cretaceous granites, zircon fission track ages in the Early Jurassic granites from Hon Da Bac (538808) and Phnom Tamao (538756) of 130.6 ± 5.9 Myr and 132.6 ± 10.8 Myr, respectively, sampled within the Mesozoic granite belt and at the transition between the granite belt and the Kampot Fold Belt, are substantially younger than their emplacement U/Pb ages. It is not clear whether these Early Cretaceous zircon fission track ages reflect postemplacement cooling or are analogous to the other Early Cretaceous zircon fission track ages obtained in the Kampot Fold Belt.

Regional late Paleocene cooling recorded by AFTA data affected the Mesozoic granite belt. Recognition of late Paleocene cooling in all settings across the entire study area suggest that cooling was dominantly an effect of exhumation. On Hon Nghe Island, the prevailing late Paleocene apatite fission track age (~59 Myr; 453017 and 453027) and the early Paleocene zircon fission track age (~62.8 ± 2.6 Myr; 453032) suggest approximately 200°C cooling during Paleocene time. This is in accordance with several kilometer-scale denudation, even with a high paleogeothermal gradient expected to have existed around the granite-cored island. Sedimentary rocks on Hon Nghe Island have only experienced contact metamorphism in upper crustal conditions. Hence, based on the rapid cooling rate suggested by ZFTA and AFTA ages (~200°C/4 Myr), fast exhumation most likely started during the earliest Paleocene or latest Cretaceous.

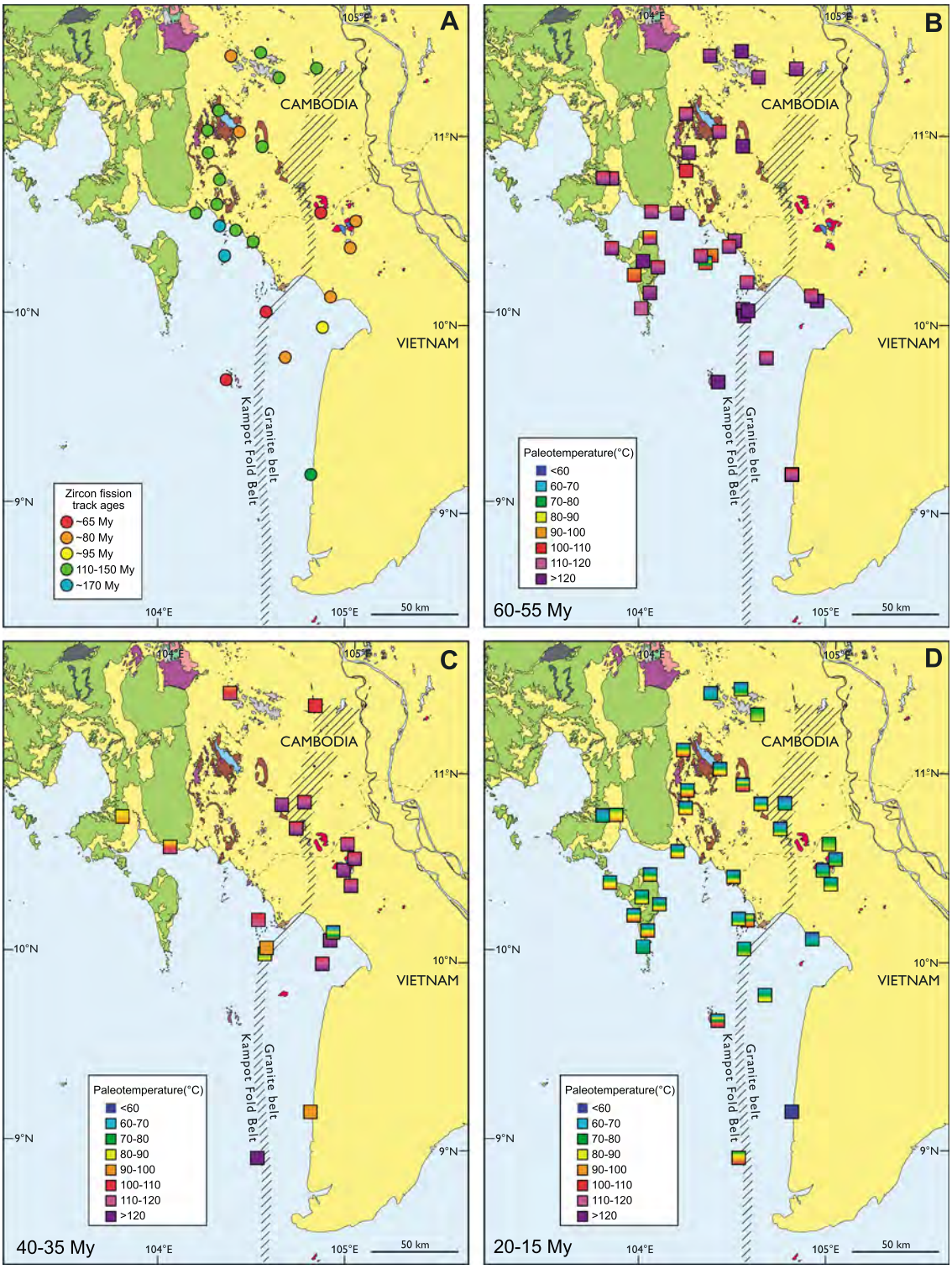


Figure 10. (a) Zircon fission track ages plotted on geological map. The youngest ages are predominantly confined to the Mesozoic granite belt in the east, reflecting cooling after Cretaceous magmatism, with older ages measured in samples from locations in the west. (b) Paleocene paleotemperatures derived from AFTA data marking the oldest regional cooling event that brought most samples into the partial annealing window. (c) Eocene paleotemperatures derived from AFTA data. By middle Eocene time all samples had cooled into the partial apatite fission track annealing window. Samples from the Mesozoic granite belt generally experienced the highest Eocene temperatures due to a locally high geothermal gradient after Cretaceous magmatism. (d) Miocene paleotemperatures derived from AFTA data. Regional Miocene cooling is suggested. The samples record comparable Miocene temperature regimes indicating that by Miocene time the granite belt in the east had cooled sufficiently for a uniform geothermal gradient to establish across the study area.

Despite regional Paleocene denudation, much of the Mesozoic granite belt did not cool sufficiently to retain apatite fission tracks during the Paleocene. Instead, these samples record middle to late Eocene cooling below ~110°C, whereas a number of key samples reveal the effect of both events. Although middle to late Eocene cooling is recorded in samples from the Phuquoc-Kampot Som Basin, the Kampot Fold Belt and the Mesozoic granite belt, by far the largest effect of this cooling episode is recorded within Cretaceous granites and near Cretaceous intrusions (Figure 10). This indicates that middle to late Eocene cooling was primarily a result of locally elevated basal heat flow and postemplacement granite cooling in the region of most recent magmatism. Uplift and erosion histories of the Mesozoic granite belt are therefore likely to resemble that of the areas farther west (Figure 11).

All samples show evidence of Miocene (20–15 Myr) cooling from paleotemperatures in the range 60–70°C or thereabouts similar to farther west, which is interpreted as representing the effects of regional exhumation across the region. Miocene paleotemperatures in samples that did not cool below ~110°C until the middle or late Eocene show no perceptible difference to those across the rest of the region, which presumably indicates that by Miocene time any residual effects of elevated heat flow due to Cretaceous igneous activity had dissipated.

6.2. Kampot Fold Belt

Rocks of the Kampot Fold Belt comprise a mixture of lithologies and ages, including Paleozoic to Mesozoic phyllites, quartzites, limestones, sandstones, and slates. Whereas zircon fission track ages from this region mostly fall between circa 110 and 150 Myr, two ages (538332 and 538713) from samples collected at Hon Doc Island and Keep, respectively, of around 170 Myr and four younger ages between circa 63 and 85 Myr (453010, 453032, 538735, and 538751 from the Nam Du archipelago, Hon Nghe, Knarng Takul, and Rang Pein, respectively) have been determined from Kampot Fold Belt samples. Of these four younger ages, at least one Paleocene zircon fission track age (453032 from Hon Nghe Island) is interpreted to reflect Paleocene denudation as elaborated above.

All of the measured zircon fission track ages are younger than the youngest (approximately Lower to Middle Jurassic) stratigraphic unit in the fold belt. However, interpretation of the Jurassic to mid-Cretaceous ZFTA ages is not straightforward. These ages could potentially reflect Early Cretaceous cooling associated with uplift, mountain building, and exhumation within the Kampot Fold Belt. The Phuquoc-Kampot Som Basin was suggested to have formed in response to flexural subsidence induced by mountain building east of the basin [*Fyhn et al*., 2010a], which may relate to the prevailing ZFTA ages.

However, the deformation of the Cretaceous sedimentary rocks in the eastern Phuquoc-Kampot Som Basin postdates the Early to mid-Cretaceous deposition, and most thickness variations within the Cretaceous succession were induced by postdepositional erosion along the base-Cenozoic unconformity [*Fyhn et al*., 2010a]. In contrast, the primary depositional thicknesses of the Cretaceous basin fill are remarkably uniform even toward the Kampot Fold Belt. Furthermore, Cretaceous depositional styles in the east of the basin resemble those in the west dominated by flood plain mudstones and braided to meandering river sandstones, and the substantial lateral facies variations that often occur in the proximal parts of foreland basins has not been resolved in either outcrop nor from seismic data. In fact, the lithostratigraphic units in the Phuquoc-Kampot Som Basin are virtually identical to those in the Khorat Basin farther north, which probably reflects that these two basins formed part of a single larger basin subsequently parted in two or more basins by erosion [e.g., *Fyhn et al*., 2010a; *Meesook*, 2011]. Detrital zircon analyses of sandstones from the Khorat Basin further suggest a Cathaysian sediment provenance [*Carter and Bristow*, 2003], and unpublished detrital zircon ages from Cretaceous sandstones collected by GEUS indicate a compatible provenance in the Phuquoc-Kampot Som Basin. All of the above are unexpected if mountain building and deformation in the Kampot Fold Belt took place at the same time as deposition in the Phuquoc-Kampot Som Basin but rather supports the model of *Carter and Bristow* [2003] suggesting the basin as a large foreland basin related to the rise of the Qinling Orogen in China.

Total zircon fission track resetting is expected at paleotemperatures slightly above 300°C, which corresponds to vitrinite reflectances in excess of approximately Ro 5% [*Green and Duddy*, 2012]. Partial annealing takes place at temperature between approximately 200 and 300°C corresponding to vitrinite reflectances around Ro 3–4% [*Green and Duddy*, 2012]. Vitrinite reflectances between Ro 3.3 and 4.5% of Permian and Triassic

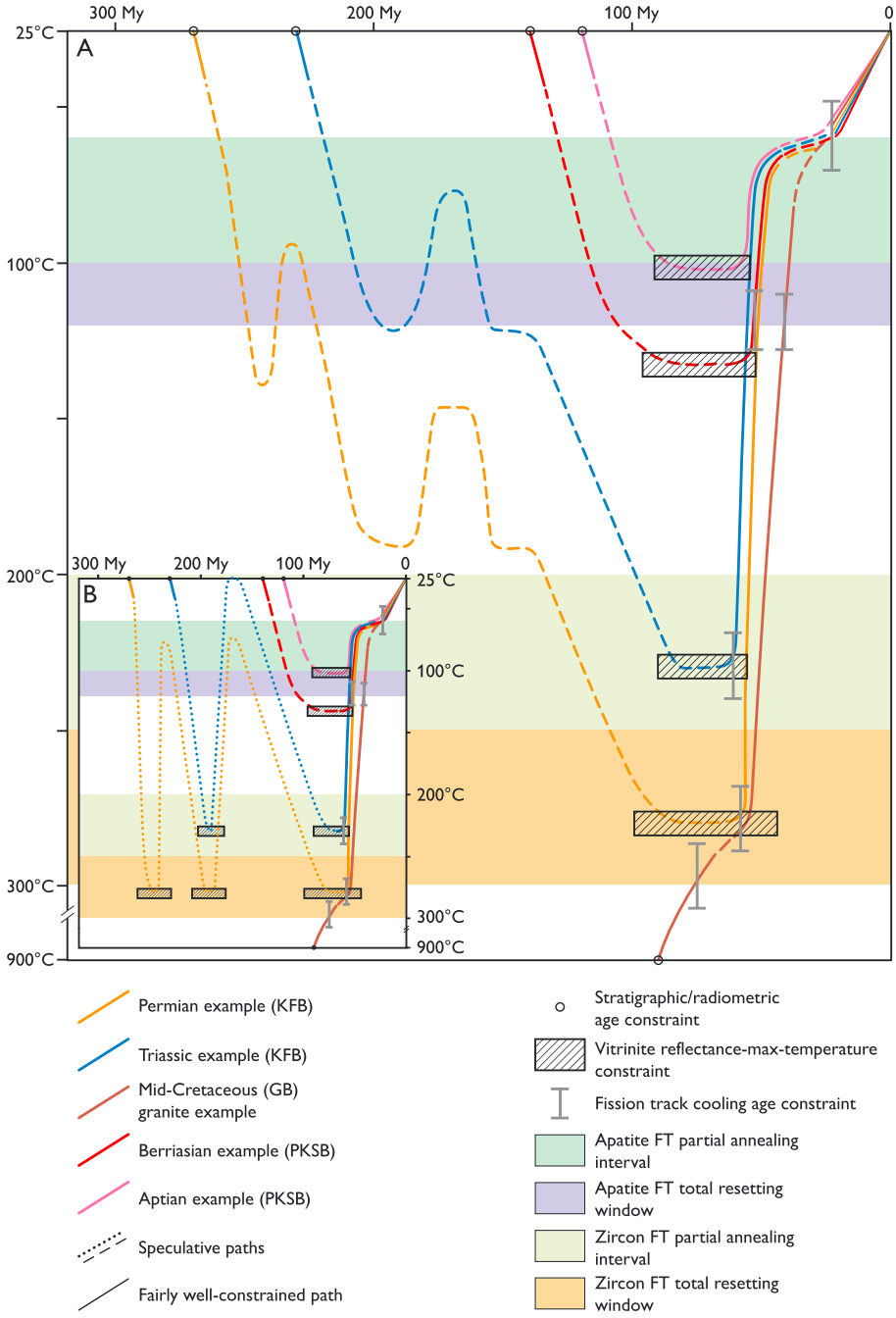


Figure 11. Schematic illustration of the preferred thermal history interpretation controlled primarily by the denudation/burial history and influenced by magmatism. Thermal histories are constrained by both zircon and apatite fission track analyses and vitrinite reflectance data that jointly constrain maximum paleotemperature and timing of cooling through total and partial annealing temperatures. (a and b) Different thermal histories that both fit the data constraints. The Cenozoic thermal history is fairly well constrained by fission track and vitrinite reflectance data and is therefore virtually identical in Figures 11a and 11b. In contrast, Paleozoic and Mesozoic thermal histories are only guided by depositional age and vitrinite reflectance that constrain the possible maximum paleotemperature. Substantial degrees of freedom thus exist as to the pre-Cenozoic thermal history. Figure 11a depicts a scenario where maximum paleotemperature (maximum burial by inference) took place immediately prior to Cenozoic cooling. Subordinate Paleozoic and Mesozoic cooling events are included to account for the pre-Cretaceous denudation events indicated by seismically observed pre-Cretaceous unconformities and supported by regional stratigraphy. Figure 11b depicts an end-member scenario where maximum paleotemperature (maximum burial by inference) was approached also in the Paleozoic and the Mesozoic followed by substantial cooling (denudation) and finally Cretaceous reburial prior to Cenozoic cooling. Figure 11b emphasize that pre-Cretaceous cooling (uplift) events affecting the analyzed samples are unlikely to have exceeded the total Cenozoic cooling.

strata from the Kampot Fold Belt suggest maximum temperatures of approximately 230°C to 300°C. This paleotemperature range is suggestive for roughly 6–11 km of burial and exhumation (assuming paleogeothermal gradients between 25°C and 35°C/km) and is only expected to have induced partial annealing in zircons and not a regionally full resetting of zircon ages. This is substantiated by the large ZFTA age spread of samples of same geological units collected close to each other (section 5.2.1). These are unlikely to have experienced substantially different thermal histories. The ZFTA age difference most likely represent slightly variable partial annealing further complicated by only partially annealed, detrital zircon fission tracks in sediments inherited from their hinterland areas. Since the Early Cretaceous zircon fission track ages are also difficult to reconcile with the above observations from the Phuquoc-Kampot Som basin, we interpret the scattered Early Cretaceous zircon ages to reflect varying degrees of partial resetting of the zircon fission track system. For this reason the timing of cooling from the paleothermal maximum is not fully constrained in all samples but seems to postdate the mid-Cretaceous.

The best indication is likely to be provided by the youngest measured zircon fission track ages, since these values reflect cooling from conditions that totally or nearly annealed all preexisting tracks. The youngest age from the Kampot Fold Belt of 64.7 ± 2.6 Myr was measured from a Permian sandstone from Nam Du Island (453010). This age is consistent with the 62.8 ± 4.7 Myr ZFTA cooling age recorded at the nearby Hon Nghe Island (453032) located in the transition zone between the fold belt and the Mesozoic granite belt.

Even so, earliest Paleocene rapid cooling from peak temperatures is consistent with the subsequent Paleocene cooling recorded by AFTA data. As in the Phuquoc-Kampot Som Basin and partly in the Mesozoic granite belt, AFTA suggest that the primary cooling episode below roughly 110°C initiated in the Paleocene. Very rapid cooling from more than roughly 250°C to less than approximately 110°C may thus have taken place across the Kampot Fold Belt between latest Cretaceous and Paleocene time. This is compatible with fast regional uplift and denudation and indicative of strong tectonism.

Middle to late Eocene cooling is also recorded by AFTA in three samples from the Kampot Fold Belt. Cooling from the highest temperatures is modeled in the two samples collected near the Mesozoic granite belt. It is not clear whether these samples record the late stages of post emplacement cooling of nearby Late Cretaceous granites similar to those in the Mesozoic granite belt or if they represent denudation-induced cooling. The fact that middle to late Eocene cooling is only recorded in a single sample farther from the granite belt favors postemplacement cooling as the dominant driving factor for this local cooling.

Similar to elsewhere in the area, Miocene cooling is pervasively recorded in the Kampot Fold Belt.

There are no clear systematic variations of zircon fission track ages with rock ages in the Kampot Fold Belt. Any potential cooling event older than the latest Cretaceous/Paleocene event (or Middle Jurassic if all zircon ages are treated as full reset ages) must thus have been overprinted by later heating. Vitrinite reflectance values of mid-Permian and Triassic rocks are comparable, which suggests that peak thermal maturities and deepest burial by inference was attained after the Middle Triassic. ZFTA data are interpreted to indicate exhumation from this deepest burial depth around latest Cretaceous to Paleocene time and certainly constrains exhumation to after the Jurassic. This also confines the magnitude of any previous cooling/denudation event within the Kampot Fold Belt after the mid-Permian (Figure 11). Any possible earlier events preceding the mid-Permian are unconstrained.

6.3. Phuquoc-Kampot Som Basin

AFTA data in all samples from the Phuquoc-Kampot Som Basin (453001, 538301, 5378313, 538701, 538703, 538705, 538706, 538737, 538739, and 538823) show a high degree of similarity all implying cooling from a paleotemperature >100°C in the middle Paleocene to earliest Eocene (60–55 Ma) interval defined above. Vitrinite reflectance data suggest maximum paleotemperatures experienced by these Cretaceous rocks between 100 and 133°C, which are highly consistent with Paleocene paleotemperatures constrained by AFTA data. The middle Paleocene to earliest Eocene AFTA age thus most likely falls close to the absolute onset age of the cooling (Figure 11).

Whether the middle to late Eocene cooling episode recorded in samples 538737 and 538823 (Veal Rinh and Phnom Bokor, respectively) represents a real event requires further analyses to clarify. However, any distinct cooling pulse commencing during the middle to late Eocene is likely to be much lower than the middle to late Eocene cooling experienced within the Mesozoic granite belt farther east.

Similar to farther east, Miocene (20–15 Ma) cooling from a temperature around 60 to 70°C are suggested ubiquitously by all AFTA samples from the Phuquoc-Kampot Som Basin. Miocene cooling thus appears uniform across the entire region and suggests Miocene denudation throughout the entire study area (Figure 10d).

Given the low to intermediate paleogeothermal gradient in the basin suggested by the vitrinite data profile in the ENRECA-2 well, exhumation is the likely primary cause of the observed cooling in this region. This is in line with distinct top Cretaceous erosion documented by seismic data (Figure 7). Provided a paleogeothermal gradient of roughly 25–35°C, cooling from between approximately 133–100°C to current average surface temperatures of around 25°C suggests between approximately 2.5 and 4.5 km exhumation. Available fission track data from the Phuquoc-Kampot Som Basin provide no conclusive insight as to whether cooling was more or less continuous, staged, or even interspersed with periods of reburial.

7. Discussion of Tectonism, Uplift and Denudation

7.1. Tectonic Mechanism Controlling Latest Cretaceous to Earliest Paleogene Denudation

The up to 1000 m high Bokor Mountains and the mountainous Phu Quoc Island rise high above the Kampot Fold Belt outlining a delta plain to the east with scattered tens to few hundred meters high hills (Figure 8). Thermochronologic data presented here document coeval denudation across the entire region, but together with maturity measures only 2.5–4.5 km post-Cretaceous denudation is suggested in the Bokor Mountains and on Phu Quoc Island both forming part of the Phuquoc-Kampot Som Basin. The stripped succession must have been deposited during the middle to Late Cretaceous and like the very closely resembling Phuquoc and Khorat Groups may have been analogous to the youngest part of the Khorat Group and the Maha Sakham and Phu Tok Formations from farther north in Sundaland.

Substantially deeper denudation is suggested by vitrinite reflectances, stratigraphy, and seismic imaging in the Kampot Fold Belt. Even though, middle to Late Cretaceous lateral variations in thermal gradient are likely to have affected the study area from the Mesozoic granite belt in the east to the Phuquoc-Kampot Som Basin in the west, this cannot explain the observed thermal maturity contrast between the Kampot Fold Belt and the Phuquoc-Kampot Som Basin. Thermal maturities of outcropping sediments in the Phuquoc-Kampot Som Basin range in between ~ Ro 0.6% and Ro 0.9%, whereas maturities of ~ Ro 4.5% occur only 10 km outside the basin. Such maturity difference corresponds to a 150°C–200°C higher paleoburial temperature in the Kampot Fold Belt, which cannot be explained by lateral regional variations in thermal gradient given their proximity, but must be a result of variation in maximum burial depth. Roughly 6–11 km of exhumation is thus suggested in the Kampot Fold Belt, which probably also took place in the Cretaceous igneous province farther east. Together with the regional structural outline, this suggests the present relief to be an artifact mainly of differential erosion between the Cretaceous Phuquoc Group in the Bokor Mountains and the generally more erodible pre-Cretaceous lithologies in the Kampot Fold Belt. A comparable relief exists at the rim of the Khorat Basin in NE Thailand and Laos and is also corroborated by offshore seismic data that depict a comparable relief of the base-Cenozoic unconformity. The unconformity delineates large buried hills in areas where Cretaceous strata subcrop the Cenozoic (Figure 7) [*Fyhn et al*., 2010a, 2010b]. Farther east, within the offshore extension of the Kampot Fold Belt, the relief fades as the surface here separates pre-Cretaceous from Cenozoic strata (Figure 7).

Fission track data suggest the present relief to be rooted in the strong uplift and denudation that most likely initiated during the latest Cretaceous to Paleocene. This is in accordance with the age of the regional base-Cenozoic unconformity that separates presumably pre–Upper Cretaceous rocks from post middle Eocene sediments offshore. The base-Cenozoic unconformity formed in response to uplift associated with massive thrusting and folding and caps dominantly north-south striking compressional structures (Figure 2). Comparable structures are documented in outcrop in this study, which were probably also active in between latest Cretaceous to Eocene time.

The observations made during this study, and the fact that all Cretaceous deposits in greater Indochina are deformed to variable degrees and subject to uplift and erosion [*Lovatt-Smith et al*., 1996; *Lovatt-Smith and Stokes*, 1997; *Stokes et al*., 1996; *Morley et al*., 2011; *Booth and Sattayarak*, 2011], document that latest Cretaceous and Paleocene deformation probably affected most of Indochina and eastern Thailand. This is corroborated by low-temperature geochronology that suggests regional uplift and denudation during the

earliest Cenozoic across much of eastern Sundaland [*Heggemann et al*., 1994; *Lovatt-Smith et al*., 1996; *Stokes et al*., 1996; *Racey et al*., 1997; *Upton*, 1999; *Carter et al*., 2000; *Meesook*, 2011]. This suggests a major tectonic reconfiguration taking place near the Cretaceous-Cenozoic boundary and affecting much if not all of central eastern Sundaland.

Clements et al. [2011] argued that Cretaceous intra-Sundaland subsidence and formation of the Khorat and the Phuquoc-Kampot Som basins occurred in response to subduction taking place along both sides of Sundaland. Consequently, when subduction ceased, flexural uplift took over causing denudation across much of Sundaland. Such a model may explain the protracted Cenozoic uplift history revealed by apatite fission track data in this study. However, sufficient evidence exists for a pronounced earliest Cenozoic deformation event linked with the onset of regional uplift [*Upton*, 1999; *Racey et al*., 1997; *Carter et al*., 2000; *Fyhn et al*., 2010a; *Morley*, 2012; this study]. Flexural uplift in the Indochina-Thailand region can hardly explain this earliest Cenozoic deformation and denudation phase alone.

Fyhn et al. [2010a] suggested the deformation to be a result of latest Cretaceous or Paleocene collision of Luconia and Sundaland, bringing paleo-Pacific subduction east of the Mesozoic granite belt to a rest. *Hall* [2012] envisaged Luconia as a large Cathaysia-derived continental fragment presently occupying much of the southern South China Sea but suggested a middle to Late Cretaceous collision confined by the youngest reliable ages of arc-related granites reported from the region. Although dating of SW Vietnamese granites is in its infancy, granites as young as late Campanian sampled east of the Kampot Fold Belt and dated in this study document magmatism younger than previously recognized with confidence (Figure 5), which could reflect continued subduction until latest Cretaceous time. Moreover, Late Cretaceous continental collision-induced uplift is not interpreted from the fission track confined rock cooling history in the study area.

In contrast, *Morley* [2012] considered the intensity of Late Cretaceous to Paleocene deformation in Indochina to increase toward the west and interpreted it to be rooted in oceanic convergence along the western margin of Sundaland and the brushing of the leading edge of India on the western margin of SE Asia already around the Paleocene.

Right-lateral shearing across the Ranong and Klong Marui faults initiated during or before Late Cretaceous time forced by plate convergence along western Sundaland (Figure 1) [*Watkinson et al*., 2008, 2011]. This shearing continued into the Cenozoic and peaked during middle Eocene time (48–40 Ma) as India brushed the margin and coupled with West Burma [*Watkinson et al*., 2011], thus slightly later than proposed by *Morley* [2012].

This documents an overlap with the latest Cretaceous to Paleocene onset of uplift across much of eastern Sundaland, but the peak right-lateral shearing in western Sundaland seems to slightly postdate the onset of uplift farther east. This questions if convergence along western Sundaland solely can explain the massive thrusting and uplift in eastern Sundaland or other mechanism rooted at the eastern Sundaland margin needs to be invoked for this event.

Morley [2012] further downplayed the importance of the Luconian collision and suggested collision-related deformation to be restricted to the earliest Cenozoic in NW Borneo. Instead, *Morley* [2012] and *Metcalfe* [2013a] suggested that the paleo-Pacific subduction zone retreated from SE Asia during the Cretaceous in response to slab rollback, back-arc rifting, and eventual seafloor spreading in the proto-South China Sea. Their models predict regional Cretaceous back-arc extension in Southeast China and southern Indochina including rifting and fault-controlled basin subsidence in the Phuquoc-Kampot Som Basin [*Morley*, 2012]. Even though Cretaceous rifting in SE China is well established, it remains speculative in Indochina, and the Phuquoc-Kampot Som Basin seems unaffected by Cretaceous rifting similar to the Khorat basin farther north [*Lovatt-Smith et al*., 1996; *Racey*, 2009; *Booth and Sattayarak*, 2011; *Fyhn et al*., 2010a].

Fission track data presented in this study are interpreted to reflect the latest Cretaceous or Paleocene onset of strong denudation induced by regional compression and uplift. The deformation and uplift recorded within the Kampot Fold Belt could therefore very well represent deformation rooted in the Luconian collision and termination of paleo-Pacific subduction east of the study area, although further analyses of the Cretaceous magmatic belt and the structural outline of the region especially to the east of our study area are needed before final conclusions can be made.

The middle to late Eocene cooling event recorded by AFTA data seems mainly to relate to postemplacement cooling within the Cretaceous igneous belt. Coeval cooling recorded by samples farther from the Cretaceous

igneous province could be associated with a denudation pulse possibly caused by renewed isostatic uplift due to crustal deformation. Any middle to late Eocene cooling seems more restricted compared with Paleocene to earliest Eocene cooling recorded by AFTA data. Enhanced uplift and denudation could be linked with the onset of regional rifting during the later Eocene. The nearest Eocene to Oligocene rift systems are located in the central Gulf of Thailand, offshore SE Vietnam, and have also been speculated to floor the Tonle Sap Basin to the north [*Fyhn et al*., 2010b; *Morley*, 2012].

Apart from the very strong erosional unconformity separating Eocene and younger deposits from pre-Cenozoic sediments offshore, no major regional intra-Cenozoic unconformities have been documented regionally to suggest repeated episodes of substantial exhumation in the region interspersed by episodes of reburial. This most likely also applies for the onshore area, as suggested by the highly similar structural and stratigraphic outline of the onshore and offshore regions.

The southwest to northeast Cenozoic onlap progression offshore along the base-Cenozoic unconformity recorded by seismic data indicates a gradual regional resumption of subsidence taking place since the Neogene in particular. Neogene cooling recorded by AFTA may reflect continuous and stable denudation of relief established during the early Cenozoic. However, the persistent record of this cooling event in most samples suggests a true discrete Neogene cooling pulse. A regression in the Vietnamese part of the Gulf of Thailand is recorded by a middle Miocene shift toward more terrestrial deposition [*Fyhn et al*., 2010b; *Petersen et al*., 2011]. Although speculative, the regression could be associated with increased sedimentation rates in the Gulf of Thailand forced by enhanced onshore denudation, although this potential link requires further investigation.

The presence of the plateau-like topography possibly representing a planation surface situated in roughly 800–1000 m altitude in the Bokor Mountains and deeply incised by active rivers could reflect two distinct phases of uplift and denudation (Figure 8). Although direct dating of the potential planation surface remains (e.g., through laterite dating), it might have formed following initial early Paleogene uplift. Incision of the planation surface may have taken place as uplift/denudation recommenced during the early Miocene as recorded by the Miocene AFTA cooling ages.

The very regional distribution of the Miocene event may suggest cooling resulting from enhanced denudation. Whether this enhanced denudation resulted from tectonically driven uplift or climatically controlled denudation remains uncertain. Basin inversion near the Oligocene to Miocene transition occurred in a number of basins along especially the eastern Vietnamese margin [*Fyhn et al*., 2009b, 2013; *Fyhn and Phach*, 2015] and also affected part of the Gulf of Thailand [*Ginger et al*., 1993; *Burton and Wood*, 2010] but did not strongly affect the Vietnamese part of the Gulf of Thailand [*Fyhn et al*., 2010b]. Published cross sections from the Ton Le Sap Basin have been used to suggest a base-Neogene erosive unconformity, but stratigraphic control of these cross sections are lacking, and the age of the observed unconformity requires confirmation from field work or drilling [*Morley*, 2012].

Alternatively, climate reorganization may have promoted faster erosion and denudation across the region. Early to middle Miocene enhanced SE Asian denudation forced by monsoon strengthening has been suggested [*Clift et al*., 2002, 2004, 2008; *Clift and Sun*, 2006; *Wei et al*., 2006], which could account for the observed cooling event in the study area. A similar interpretation for middle Miocene cooling recorded by fission track analysis along the South Chinese margin was proposed by *Yi et al*. [2009].

7.2. Pre-Cretaceous Denudation and Tectonism

The sedimentary and igneous shifts occurring throughout the Phanerozoic succession most likely largely reflect interchanging tectonic phases since the Silurian. For instance, the Permo-Triassic shift from carbonate to siliciclastic deposition may denote Indosinian tectonism within or near the Kampot Fold Belt similar to elsewhere in Indochina [*Booth and Sattayarak*, 2011]. Moreover, the extensional faulting offsetting Permo-Triassic strata most likely stems from late Paleozoic(?) and Late Triassic rifting affecting much of Indochina. Even so, these events are likely to have induced local or regional uplift in the study area, which was overprinted by subsequent reburial (Figure 11).

The absence of exposed key stratigraphic surfaces in the region hampers analyses of pre-Cretaceous tectonism. It is not clear if the major depositional shifts were gradual or if they were abrupt and is outlined by erosional unconformities. Offshore, sub-Cretaceous unconformities imaged by seismic data suggest pre-

Cretaceous tectonism and periods of denudation (Figure 7). The unconformities are difficult to correlate regionally, and it is not possible to assess their pre-Cretaceous tectonic significance. Furthermore, the lack of stratigraphic constraints from wells makes the ages of these unconformities and the associated pre-Cretaceous tectonism uncertain.

Permian, Triassic, and Jurassic magmatism within the study area similarly record tectonic active periods. The magmatism has not yet been classified, and the associated tectonic setting remains uncertain, and it is unknown if the magmatism was accompanied by uplift and erosion.

Direct evidence for Indosinian deformation within the Kampot Fold Belt remains to be documented, and the comparable thermal maturities of mid-Permian and Triassic strata favor the strongest exhumation of these samples to have occurred after the Triassic. The strong post-Triassic tectonism indicated by highly deformed Upper Triassic and Jurassic strata and Paleozoic strata thrusted over Triassic deposits probably primarily record the latest Cretaceous and Paleocene structuring constrained by thermochronology in this study.

8. Conclusion

1. The geology along the Kampot Fold Belt contains a remarkable record of Phanerozoic deposition, magmatism, and tectonics in southern Indochina. In particular, the deformation and denudation history of the fold belt and the flanking Cretaceous Phuquoc-Kampot Som Basin and the Mesozoic granite belt in south Indochina document an important, but scarcely known orogenic event in Indochina that bridges the subduction-dominated Mesozoic era with the rift-dominated mid-Cenozoic era.
2. The changing depositional style and the variable igneous activity around the Kampot Fold Belt testify to a highly variable tectonic setting through Paleozoic and Mesozoic time in southern Cambodia and Vietnam. Pre-Cretaceous unconformities imaged seismically suggest that deposition was interspersed with periods of uplift and that the main depositional units likely are separated by hiatuses, although these unconformities are rarely exposed in the Kampot Fold Belt.
3. AFTA data define three major cooling periods affecting the study area during Paleocene, middle to late Eocene, and Miocene times.
4. Major latest Cretaceous to Paleocene thrusting, folding, and exhumation took place within the Kampot Fold Belt, and deformation and denudation similarly affected the adjacent Phuquoc-Kampot Som Basin and the Mesozoic granite belt in the east. The latest Cretaceous to Cenozoic exhumation of examined samples from the Kampot Fold Belt equals or more likely exceeded any other late Paleozoic to Mesozoic exhumation event as documented by fission track and thermal maturity data.
5. The Kampot Fold Belt probably experienced between roughly 6 and 11 km of Cenozoic denudation, whereas the Cretaceous Phuquoc-Kampot Som Basin only suffered roughly 2.5–4.5 km of exhumation. The present relief between the Bokor Mountains situated high above the Kampot Fold Belt is thus an artifact of differential erosion.
6. A few kilometers thick middle to Upper Cretaceous succession was deposited on top of the Phuquoc Group in the Phuquoc-Kampot Som Basin but was stripped off during latest Cretaceous to Cenozoic uplift and denudation. The succession may have been analogous to the youngest part of the Khorat Group and the Maha Sakham and Phu Tok Formations in central Indochina.
7. Latest Cretaceous to Paleocene orogenesis affected most of greater Indochina and probably reflects a significant plate reorganization along Sundaland's margin. The orogeny is difficult to reconcile solely with a collision event along western Sundaland but may have formed in response to collisional processes taking place along eastern Sundaland or a combination of collisions along both east and west Sundaland. Further analyses and dating of especially Late Mesozoic Vietnamese arc granites are required in the evaluation of the latest Cretaceous local plate tectonic configuration.
8. AFTA data document delayed cooling of the Mesozoic granite belt relative to the area farther west due to local elevated thermal gradients persisting up to a few tens of million years after Cretaceous granite emplacement. The thermal gradient had returned to present-day levels by early Miocene time.
9. Miocene cooling from low temperatures is recorded across the entire study area, which may reflect a renewed denudation pulse driven by either regional tectonism or climate-enhanced erosion. An incised plateau situated in up to 800–1000 m altitude in the Bokor Mountain may record this late-stage uplift and denudation.

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Supporting Information for

Cenozoic deformation and exhumation of the Kampot Fold Belt and implications for south Indochina tectonics

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Introduction

This supporting information provides details on the fission track analyses data including raw-data, sample information, fission track measurements details, methodology and interpretation procedures. S4 provides information on vitrinite reflectance measurements.

S1: Sample Details, Geological Data and Apatite Compositions

S1.1 Sample details

A total of 81 samples from outcrops in Vietnam and Cambodia were provided by Michael Fyhn, GEUS, Copenhagen. Of these, 63 samples were processed for AFTA, with 38 samples yielding apatite grains suitable for analysis. A total of 25 samples were also processed for ZFTA, with zircons suitable for analysis obtained from all these samples. Details of all samples submitted for AFTA and ZFTA, including sample locations, and stratigraphic ages (as provided by the client) are summarized in Table S1.1, together with the yields of apatite and zircon from each sample after mineral separation.

S1.2 Stratigraphic details

Details of stratigraphic ages of each sample were provided by the client. Age assignments for each sample are summarized in Table S1.1. The chronostratigraphic assignment of each sample was converted to a chronometric (numerical) scale using Gradstein et al. (2004), and the corresponding values are also listed for each sample in Table S1.1.

Any slight errors in the estimated chronometric ages of each sample are not expected to affect the thermal history interpretation of the AFTA, ZFTA or VR data to any significant degree in this study.

For some samples of igneous rock, additional information on emplacement ages from U-Pb dating was provided during the course of the project. This information is summarized in Table S3.1. All of these ages are highly consistent with ZFTA data in corresponding samples. Any revision of individual sample ages resulting from this information has no effect on the thermal history interpretation of AFTA data, for reasons discussed in S2.4.

S1.3 Present temperatures

For the outcrop samples analyses in this study a value of 25°C has been assumed. In practice this value has no significant influence on the interpretation of the results of this study.

S1.4 Grain morphologies

The zircon and apatite grains in samples analyzed for this study showed a variety of morphologies. Detrital apatite grains from sedimentary rocks showed euhedral, subeuhedral to sub-rounded, rounded and angular forms in each sample. No clear trends were evident which could be interpreted in terms of systematic variation in sedimentary provenance between different samples. Accessory apatites and zircons obtained from igneous samples were dominated by euhedral forms.

S1.5 Apatite compositions

The annealing kinetics of fission tracks in apatite is affected by chemical composition, specifically the Cl (chlorine) content, as explained in detail in Supporting information chapter S3. In all samples collected for this study, Cl contents were measured in all apatite grains analyzed (i.e. for both fission track age determination and track length measurement).

The measured compositions in individual grains have been used in interpreting the fission track data, using methods outlined in Supporting information chapter S3.

Chlorine contents were measured using a fully automated Jeol JXA-5A electron microprobe equipped with a computer controlled X-Y-Z stage and three computer controlled wavelength dispersive crystal spectrometers, with an accelerating voltage of 15kV and beam current of 25nA. The beam was defocused to 20 μm diameter to avoid problems associated with apatite decomposition, which occurs under a fully focused 1μm - 5μm beam. The X-Y co-ordinates of all analyzed grains were transferred from the Autoscan Fission Track Stage to a file suitable for direct input into the electron microprobe. A digital image of every analyzed grain is collected, stored and used to verify grain identification prior to probe analysis. Cl count rates from the analyzed grains were converted to wt% Cl by reference to those from a Durango apatite standard (Melbourne University Standard APT151), analyzed at regular intervals. This approach implicitly takes into account atomic number absorption and fluorescence matrix effects, which are normally calculated explicitly when analyzing for all elements. A value of 0.43 wt% Cl was used for the Durango standard, based on repeated measurements on the same single fragment using pure rock salt (NaCl) as a standard for Cl. This approach gives essentially identical results to Cl contents determined from full compositional measurements, but has the advantage of reducing analytical time by a factor of ten or more.

Lower limits of detection for Cl content have been calculated for typical analytical conditions (beam current, counting time, etc.) and are listed in Table S1.4. Errors in wt% composition are given as a percentage and quoted at ±1σ for Cl determinations. A generalized summary of error values for various values of wt% Cl is presented in Table S1.5.

Results of analyses from this study

Apatites obtained from the samples analyzed for this study show a wide variety in the distributions of Cl content. Most samples show a familiar pattern, typical of the distribution of Cl contents found in detrital apatites from common quartzose sandstone samples. The majority of grains have Cl contents between 0 and 0.1 wt%, while a smaller number of grains give values up to ~0.5 wt% (close to the value found in the Durango apatite on which our original kinetic model of fission track behavior was based, see Supporting information chapter S3), and a very few grains contain higher amounts up to ~1 wt% Cl and above. In contrast, apatite grains obtained from samples GC1017-2, -3 and -7 are dominated by grains containing between 1.0 and 1.5 wt% Cl, with a very small number of apatites in samples GC1011-2 and -3 containing less than 0.1 wt% Cl. In sample GC1017-5, Cl contents between 0.5 and 0.7 wt% are dominant while in samples GC1017-6, GC1129-21 and GC1129-24 most grains contain between 0.1 and 0.3 wt% Cl. In sample GC1129-43 most grains contain between 0.2 and 0.3 wt% Cl. These differing patterns presumably reflect different apatite provenances relating to the nature of the magmas from which they were ultimately derived.

In all samples analyzed for this report, the measured distribution of compositions has been employed in interpreting the AFTA data, using methods outlined in Supporting information chapter S3.

S1: References

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Table S1.1 Details of fission track samples and apatite yields - samples from Vietnam and Cambodia

Sample number	Depth TVDrKB (m)	Sample type	Stratigraphic Subdivision	Stratigraphic age (Ma)	Present temperature *1 (°C)	Raw weight (g)	Washed weight (g)	Apatite yield *2
ENRECA								
GC1017-1	496-497	core	Phuquoc Fm (E. Cretaceous)	125-100	30	1264	200	excellent

*1 See Appendix A for discussion of present temperature data.

*2 Yield based on quantity of mineral suitable for age determination. Excellent: >20 grains; Good: 15-19 grains; Fair: 10-14 grains; Poor: 5-9 grains; Very Poor: <5 grains.

Table S1.2 Summary of the ENRECA-2 well stratigraphy, sample 453001

	KB elevation (mAMSL)	Ground level (m)	Stratigraphic Interval	Depth of Top TVD rKB (m)	Age of Top (Ma)
ENRECA					
	0	0	*Unconformity*	0	0
			E. Cretaceous	0	100
			TD	500	125

All depths quoted are with respect to KB, except where otherwise stated.

Table S1.3 Details of fission track samples and apatite and zircon yields - outcrop samples from Vietnam and Cambodia

Sample number	Source #	Location Digital Lat/Long Elevation (m)	Stratigraphic Subdivision	Stratigraphic age (Ma)	Raw weight (g)	Washed weight (g)	Apatite yield *1	Zircon yield *2
outcrop samples								
GC1017-2	453009	Hon Nganh Island 9.675967 104.399623	Devonian / Triassic?	416-359	1658	700	fair	-
GC1017-3	453017	Hon Nghe Island 10.022213 104.547355	Jurassic / Cretaceous?	200-65	1317	240	excellent	-
GC1017-4	453023	Hon Nghe Island 10.035926 104.548066	Triassic?	251-200	1937	400	excellent	-
GC1017-5	453027	Hon Nghe Island 10.034567 104.547028	Jurassic / Cretaceous?	200-65	2357	680	fair	-
GC1017-6	453029	N. Hon Sic quarry 10.151492 104.903601	L. Creataceous	100-65	1738	180	excellent	excellent
GC1017-7	453030	Hon Dat 10.104621 104.897344	Jurassic / E. Cretaceous?	200-100	2120	380	excellent	-
GC1129-1	538701	Phuquoc (Pq.) Cay Sao 10.238278 104.077139	Cretaceous	145-65	950	200	fair	-
GC1129-2	538702	Pq. Khu Tuong S 10.282194 103.978222	Cretaceous	145-65	1210	140	none	-
GC1129-3	538301	Pq. Khu Tuong N 10.293806 103.981500	Cretaceous	145-65	1110	200	poor	-
GC1129-4	538310	Pq. Khu Tuong N 10.293806 103.981500	Cretaceous	145-65	1220	250	none	-
GC1129-5	538703	Pq. Bai Thom 10.357500 104.081333	Cretaceous	145-65	1040	140	good	-
GC1129-6	538704	Pq. Mui Ganh Dau 10.373944 103.837889	Cretaceous	145-65	880	220	none	-
GC1129-7	538706	Pq. Mui Dinh Cau 10.217417 103.956861	Cretaceous	145-65	1050	210	poor	-
GC1129-8	538313	Pq. Mui Mang Ran 10.131056 104.031389	Cretaceous	145-65	1310	420	excellent	-
GC1129-9	538317	Pq. Duong To 10.142056 104.003806	Cretaceous	145-65		Not	Processed	-
GC1129-10	538318	Pq. Trung Hung Dao 10.055667 104.011472	Cretaceous	145-65		Not	Processed	-

147 **Table S1.3** Continued

Sample number	Source #	Location Digital Lat/Long Elevation (m)	Stratigraphic Subdivision	Stratigraphic age (Ma)	Raw weight (g)	Washed weight (g)	Apatite yield *1	Zircon yield *2
GC1129-11	538323	Hon Douc Isl. 10.285694 104.322028	Triassic/Permian?	299-199	2220	470	poor	-
GC1129-12	538330	Hon Doc Isl. SE 10.321500 104.337167	Cret./Trias./Perm?	299-65	1860	370	poor	-
GC1129-13	538332	Hon Doc Isl. NE 10.321083 104.332556	Cret./Trias./Perm?	299-65	1330	230	poor	excellent
GC1129-14	538337	Hon Heo Isl. N 10.187278 104.529361	Triassic?	251-199	1880	480	poor	-
GC1129-15	538338	Hon Heo Isl. SE 10.184028 104.532556	Triassic?	251-199	2140	500	poor	-
GC1129-16	538342	Thach Ding 10.416528 104.475472	Triassic	251-199	1560	490	excellent	excellent
GC1129-17	538343	Mui Nai Nui Den 10.374944 104.445944	Triassic	251-199	1110	170	none	-
GC1129-18	538345	Nui Tam Phu Nhan 10.382528 104.466500	Triassic	251-199	1270	310	very poor	-
GC1129-19	538349	Bai Ot 10.302806 104.538111	Triassic	251-199	1720	140	none	-
GC1129-20	538353	Hon Chong 10.152861 104.606472	Devonian	416-359	1800	210	none	-
GC1129-21	538356	Tuc Dup 10.376694 104.959250	Cretaceous	145-65	1310	100	excellent	excellent
GC1129-22	538357	Hon Trerm 10.142333 104.627667	Devonian	416-359	1740	300	none	-
GC1129-23	538802	Hon Rai 9.800417 104.652833	Cretaceous	145-65	3110	280	excellent	excellent
GC1129-24	538804	Hon Tre S 9.957778 104.841694	Cretaceous	145-65	4270	520	excellent	excellent
GC1129-25	538806	Hon Tre W 9.966972 104.828694	Cretaceous	145-65		Not	Processed	-
GC1129-26	538808	Hon Da Bac 9.176833 104.804694	Cretaceous	145-65	3050	290	excellent	excellent

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151 **Table S1.3** Continued

Sample number	Source #	Location Digital Lat/Long Elevation (m)	Stratigraphic Subdivision	Stratigraphic age (Ma)	Raw weight (g)	Washed weight (g)	Apatite yield *1	Zircon yield *2
GC1129-27	538811	Hon Chuoi W 8.950028 104.523389	Permian?	299-251	3280	600	excellent	-
GC1129-28	538359	Tapa Pagoda 10.414056 104.992500	Triassic	251-199		Not	Processed	-
GC1129-29	538360	Nam Qui 10.439250 104.994500	Triassic	251-199		Not	Processed	-
GC1129-30	538362	Nui Giai 10.442722 104.961639	Jurassic?	199-145	1960	340	fair	-
GC1129-31	538364	Nui Cam E 10.504083 105.015611	Cretaceous	145-65	1580	100	excellent	excellent
GC1129-32	538366	Nui Cam W 10.511972 104.962917	Cretaceous	145-65		Not	Processed	-
GC1129-33	538368	Son Trai 10.618111 104.994361	Cretaceous	145-65	1340	150	excellent	-
GC1129-34	538370	Phnom Den 10.599778 104.905167	Cretaceous	145-65	1630	170	fair	-
GC1129-35	538372	Tonl Porgrar 10.582250 104.876861	Cretaceous	145-65		Not	Processed	-
GC1129-36	538374	Tonleap 10.556833 104.842778	Cretaceous	145-65	3320	410	none	excellent
GC1129-37	538376	Phnom Tasom 10.662917 104.717556	Devonian	416-359	2190	450	good	-
GC1129-38	538377	Bophal 10.598167 104.777667	Devonian	416-359	1470	340	none	-
GC1129-39	538378	Hiu Chrak Chikam 10.625833 104.729222	Devonian	416-359		Not	Processed	-
GC1129-40	538379	Phnom Pontamao 10.705556 104.754889	Devonian	416-359		Not	Processed	-
GC1129-41	538380	Sanlong 10.799111 104.753372	Triassic?	251-199	2360	180	good	-
GC1129-42	538384	Phnom Tany 10.772167 104.658083	Permian?	299-251	2670	350	none	-

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Table S1.3 Continued

Sample number	Source #	Location Digital Lat/Long Elevation (m)	Stratigraphic Subdivision	Stratigraphic age (Ma)	Raw weight (g)	Washed weight (g)	Apatite yield *1	Zircon yield *2
GC1129-43	538385	Phnom Aya 10.776639 104.630111	Permian?	299-251	1510	520	excellent	-
GC1129-44	538394	Phnom Neang Tou 10.585083 104.524528	Devonian	416-359	1640	250	none	-
GC1129-45	538398	Phnom Worr East 10.595278 104.391694	Devonian	416-359	1570	160	none	-
GC1129-46	538707	Phnom Lok 10.448972 104.443556	Devonian?	416-359	2440	550	none	-
GC1129-47	538709	Phnom Angkoal 10.460722 104.392167	Permian?	299-251	2440	650	none	excellent
GC1129-48	538711	Phnom Prous 10.533889 104.316639	Permian?	299-251		Not	Processed	-
GC1129-49	538713	Keep 10.478278 104.295056	Cambrian	542-488	1280	160	none	excellent
GC1129-50	538719	Phnom Ta Ngel 10.604806 104.287750	Cambrian	542-488	1920	600	none	excellent
GC1129-51	538814	Phnom Worr South 10.552889 104.330500	Cret./Perm./Trias?	299-65	2110	400	none	-
GC1129-52	538815	Phnom Worr South 10.552889 104.330500	Cambro-Silur / Devonian	542-359	2870	540	none	-
GC1129-53	538820	Phnom Sorsear 10.671556 104.182889	Devonian (?)	416-359	1310	320	none	-
GC1129-54	538822	Phnom Kduoch 10.554222 104.173611	Cam-Sil(?) Map LowCret	542-99	1530	260	poor	excellent
GC1129-55	538720	Phnom Poktuyung 10.717028 104.298389	Cambrian	542-488		Not	Processed	-
GC1129-56	538722	Phnom Poktuyung 10.717028 104.298389	Permian / Triassic?	299-199	1450	280	none	excellent
GC1129-57	538724	Phnom Y 10.773083 104.222889	Permian / Triassic?	299-199	1310	360	good	-
GC1129-58	538725	Phnom Mean Boon 10.820250 104.233556	Cambrian	542-488		Not	Processed	-

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160 **Table S1.3** Continued

Sample number	Source #	Location Digital Lat/Long Elevation (m)	Stratigraphic Subdivision	Stratigraphic age (Ma)	Raw weight (g)	Washed weight (g)	Apatite yield *1	Zircon yield *2
GC1129-59	538726	Trapadang Phdao 10.855694 104.237389	Permian / Triassic	299-199	2100	280	excellent	excellent
GC1129-60	538728	Phnom Changek 10.980472 104.236083	Permian / Triassic?	299-199	1400	480	none	excellent
GC1129-61	538730	Loc. 5 11.012944 104.240639	Cambrian	542-488		Not	Processed	-
GC1129-62	538731	Loc. 6 11.100083 104.230639	Cambrian	542-488	820	100	good	-
GC1129-63	538732	Noriay 10.889389 104.540889	Triassic?	251-199	1740	330	fair	excellent
GC1129-64	538823	Phnom Bokor 11.417833 104.107500	Lower Cretaceous	145-99	2400	640	good	-
GC1129-65	538834	Phnom Bokor 10.608028 104.104667	Lower Cretaceous	145-99	1690	300	none	-
GC1129-66	538846	Phnom Bokor 10.622083 104.082028	Lower Cretaceous	145-99	2450	180	none	-
GC1129-67	538733	Phnom Thmar Keo 11.093667 104.286250	Permian/Triassic?	299-199	2110	380	none	excellent
GC1129-68	538735	Phnom Knarng Takul 11.021167 104.403389	Permian/Triassic?	299-199	1860	290	fair	excellent
GC1129-69	538737	Vealrinh 10.725972 103.797278	Cretaceous	145-65	1420	200	good	-
GC1129-70	538751	Rang Pein 10.251119 103.797278	Triassic	251-199	1560	200	excellent	excellent
GC1129-71	538753	Prak Sambo 10.251119 103.797278	Triassic	251-199		Not	Processed	-
GC1129-72	538754	Phnom Cheal 11.389389 104.502972	Triassic	251-199	2530	280	excellent	excellent
GC1129-73	538756	Phnom Tamao 11.302056 104.799750	Cretaceous	145-65	1440	80	excellent	good
GC1129-74	538758	Phnom Totear 11.281278 104.759861	Triassic	251-199		Not	Processed	-

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Table S1.3 Continued

Sample number	Source #	Location Digital Lat/Long Elevation (m)	Stratigraphic Subdivision	Stratigraphic age (Ma)	Raw weight (g)	Washed weight (g)	Apatite yield *1	Zircon yield *2
GC1129-76	538761	Phnom Srong 11.265417 104.609694	Triassic	251-199	1390	100	excellent	excellent
GC1129-77	538763	Phnom Krang 11.207917 104.486361	Triassic	251-199		Not	Processed	-
GC1129-78	453010	Nam Du Island 9.685904 104.346763	Permian	299-251	860	140	none	excellent
GC1129-79	453032	Hon Nge Island 10.026098 104.561174	Triassic	251-199	1480	220	none	excellent
GC1129-80	538705	Mui Ganh Dau 10.374889 103.837194	Lower Cretaceous	145-100	1190	410	poor	-
GC1129-81	538739	Vealrinh 10.725972 103.797278	Lower Cretaceous	145-100	600	180	poor	-
GC1129-82	538747	Vealrinh 10.725972 103.797278	Lower Cretaceous	145-100	830	200	none	-

*1 Yield based on quantity of mineral suitable for age determination. Excellent: >20 grains; Good: 15-19 grains; Fair: 10-14 grains; Poor: 5-9 grains; Very Poor: <5 grains.

*2 Yield based on quantity of mineral suitable for age determination. Excellent: >10 grains; Good: 7-9 grains; Fair: 3-6 grains; Poor: <3 grains.

Table S1.4 Lower Limits of Detection for Apatite Analyses

Element	LLD (95% c.l.)		LLD (99% c.l.)	
	(wt%)	(ppm)	(wt%)	(ppm)
Cl	0.01	126	0.02	182

Table S1.5 Per cent errors in chlorine content

Chlorine content (wt%)	Error (%)
0.01	9.3
0.02	8.7
0.05	7.3
0.10	6.1
0.20	4.7
0.50	3.2
1.00	2.3
1.50	1.9
2.00	1.7
2.50	1.5
3.00	1.4

Errors quoted are at 1σ. See Appendix A for more details.

S2: Sample Preparation, Analytical Details and Data Presentation

S2.1 Sample Preparation

Core and outcrop samples are crushed in a jaw crusher and then ground to sand grade in a rotary disc mill. Cuttings samples are washed and dried before grinding to sand grade. The ground material is then washed to remove dust, dried and processed by conventional heavy liquid and magnetic separation techniques to recover heavy minerals. Apatite grains are mounted in epoxy resin on glass slides, polished and etched for 20 sec in 5M HNO3 at 20°C to reveal the fossil fission tracks. Zircon grains were embedded in FEP Teflon between heated microscope slides on a hot plate, then polished and etched in a molten KOH:NaOH eutectic mixture at ~220°C (Gleadow et al., 1976), for an appropriate time (typically between 10 and 100 hours) to reveal properly etched fission tracks in a high proportion of the grains present.

After etching, all mounts are cut down to 1.5 X 1 cm, and cleaned in detergent, alcohol and distilled water. The mounts are then sealed in intimate contact with low-uranium muscovite detectors within heat-shrink plastic film. Each batch of mounts is stacked between two pieces of uranium standard glass, which has been prepared in similar fashion. The stack is then inserted into an aluminum can for irradiation.

After irradiation, the mica detectors are removed from the grain mounts and standard glasses and etched in hydrofluoric acid to reveal the fission tracks produced by induced fission of 235U in the apatite and standard glass.

S2.2 Analytical Details

Fission track ages

Fission track ages are determined by the external detector method or EDM (Gleadow, 1981). The EDM has the advantage of allowing fission track ages to be determined on single grains. In apatite, tracks are counted in 20 grains from each mount wherever possible. In zircon, 10 grains are normally analyzed. In those samples where the desired number is not present, all available grains are counted, the actual number depending on the availability of suitably etched and oriented grains. Only grains oriented with surfaces parallel to the crystallographic c-axis are analyzed. Such grains can be identified on the basis of the etching characteristics, as well as from morphological evidence in euhedral grains. The grain mount is scanned sequentially, and for apatite the first 20 suitably oriented grains identified are analyzed, while for zircon the first 10 suitable grains are used.

Fission track ages are calculated using the standard fission track age equation using the zeta calibration method (equation five of Hurford and Green, 1983), viz:

$$\text{F.T. AGE} = \frac{1}{\lambda_D} \ln\left[1 + \left(\frac{\zeta \lambda_D \rho_s g \rho_D}{\rho_i}\right)\right] \qquad \text{B.1}$$

where: λD = Total decay constant of 238U (= 1.55125 x 10-10)
ζ = Zeta calibration factor
ρs = Spontaneous track density
ρi = Induced track density
ρD = Track density from uranium standard glass
g = A geometry factor (= 0.5)

Tracks are counted within an eyepiece graticule divided into 100 grid squares. In each grain, the number of spontaneous tracks (Ns) within a certain number of grid squares (Na) is recorded. The number of induced tracks (Ni) in the corresponding location within the mica external detector is then counted. Spontaneous and induced track densities (ρs and ρi, respectively) are calculated by dividing the track counts by the total area counted, given by the product of Na and the area or each grid square (determined by calibration against a ruled stage graticule or diffraction grating). Fission track ages may be calculated by substituting track counts (Ns and Ni) for track densities (ρs and ρi) in equation B.1, since the areas cancel in the ratio.

Translation between apatite grains in the grain mount and external detector locations corresponding to each grain is carried out using Autoscan™ microcomputer-controlled automatic stages (Smith and Leigh Jones, 1985). This system allows repeated movement between grain and detector, and all grain locations are stored for later reference if required.

Neutron irradiations are carried out in a well-thermalised flux (X-7 facility; Cd ratio for Au ~98) in the Australian Atomic Energy Commission's HIFAR research reactor. Total neutron fluence is monitored by counting tracks in mica external detectors attached to two pieces of Corning Glass Works standard glass (CN5 containing ~11 ppm uranium for apatite, and U3 containing ~55 ppm uranium for zircon) included in the irradiation canister at each end of the sample stack. In determining track densities in external detectors irradiated adjacent to uranium standard glasses, 25 fields are normally counted in each detector. The total track count (ND) is divided by the total area counted to obtain the track density (ρD). The positions of the counted fields are arranged in a 5 X 5 grid covering the whole area of the detector. For typical track densities of between ~5 X 105 and 5 X 106, this is a convenient arrangement to sample across the detector while gathering sufficient counts to achieve a precision of ~±2% in a reasonable time.

A small flux gradient is often present in the irradiation facility over the length of the sample package. If a detectable gradient is present, the track count in the external detector adjacent to each standard glass is converted to a track density ρD and a value for each mount in the stack is calculated by linear interpolation. When no detectable gradient is present, the track counts in the two external detectors are pooled to give a single value of ρD which is used to calculate fission track ages for each sample. A Zeta calibration factor (ζ) has been determined empirically for each observer by analyzing a set of carefully chosen age standards with independently known K-Ar ages, following the methods outlined by Hurford and Green (1983) and Green (1985).

All track counting is carried out using Zeiss(R) Axioplan microscopes, with an overall linear magnification of 1068 x using dry objectives. For further details and background information on practical aspects of fission track age determination, see e.g. Fleischer, Price and Walker (1975), Naeser (1979) and Hurford (1986).

Track length measurements

For track length studies in apatite, the full lengths of "confined" fission tracks are measured. Confined tracks are those which do not intersect the polished surface but have been etched from other tracks or fractures, so that the whole length of the track is etched. Confined track lengths are measured using a digitizing tablet connected to a microcomputer, superimposed on the microscope field of view via a projection tube.

With this system, calibrated against a stage graticule ruled in 2 μm divisions, individual tracks can be measured to a precision of ± 0.2 μm. Tracks are measured only in prismatic grains, characterized by sharp polishing scratches with well-etched tracks of narrow cone angle in all orientations, because of the anisotropy of annealing of fission tracks in apatite (as discussed by Green et al. 1986). Tracks are also measured following the recommendations of Laslett et al. (1982), the most important of which is that only horizontal tracks should be measured. One hundred tracks are measured whenever possible. In apatite samples with low track density, or in those samples in which only a small number of apatite grains are obtained, fewer confined tracks may be available. In such cases, the whole mount is scanned to measure as many confined tracks as possible.

Integrated fission track age and length measurement

Fission track age determination and length measurement are now made in a single pass of the grain mount, in an integrated approach. The location of each grain in which tracks are either counted or measured is recorded for future reference. Thus, track length measurements can be tied to age determination in individual grains. As a routine procedure we do not measure the age of every grain in which lengths are determined, as this would be much too time-consuming. Likewise we do not only measure ages in grain in which lengths are measured, as this would bias the age data against low track density grains. Nevertheless, the ability to determine the fission track age of certain grains from which length data originate can be a particularly useful aid to interpretation in some cases. Grain location data are not provided in this report, but are available on request.

S2.3 Data Presentation

Fission track age data

Data sheets summarizing the fission track age data for both apatite and zircon, including full details of fission track age data for individual grains in each sample, together with the primary counting results and statistical data, are given in the following pages.

Individual grain fission track ages are calculated from the ratio of spontaneous to induced fission track counts for each grain using equation B.1, and errors in the single grain ages are calculated using Poissonian statistics, as explained in more detail by Galbraith (1981) and Green (1981). All errors are quoted as ±1σ throughout this report, unless otherwise stated. The variability of fission track ages between individual apatite grains within each sample can be assessed using a chi-squared (χ^2) statistic (Galbraith, 1981), the results of which are summarized for each sample in the data sheets. If all the grains counted belong to a single age population, the probability of obtaining the observed χ^2 value, for ν degrees of freedom (where ν = number of crystals -1), is listed in the data sheets as $P(\chi^2)$ or P(chi squared).

A $P(\chi^2)$ value greater than 5% can be taken as evidence that all grains are consistent with a single population of fission track age. In this case, the best estimate of the fission track age of the sample is given by the "pooled age", calculated from the ratio of the total spontaneous and induced track counts in all grains analyzed. Errors for the pooled age are calculated using the "conventional" technique outlined by Green (1981), based on the total number of tracks counted for each track density measurement (see also Galbraith, 1981).

A $P(\chi^2)$ value of less than 5% denotes a significant spread of single grain ages, suggesting real differences exist between the fission track ages of individual apatite grains. A significant spread in grain ages can result either from inheritance of detrital grains from mixed source areas (in sedimentary rocks), or from differential annealing in apatite grains of different composition, within a narrow range of temperature. Calculation of the pooled age inherently

assumes that only a single population of ages is present, and is thus not appropriate to samples containing a significant spread of fission track ages. In such cases, Galbraith has recently devised a means of estimating the modal age of a distribution of single grain fission track ages which is referred to as the "central age". Calculation of the central age assumes that all single grain ages belong to a Normal distribution of ages, with a standard deviation (σ) known as the "age dispersion". An iterative algorithm (Galbraith and Laslett, 1993) is used to provide estimates of the central age with its associated error, and the age dispersion, which are all quoted in the data sheets. Note that this treatment replaces use of the "mean age", which has used been in the past for those samples in which P(χ2)<5%. For samples in which P(χ2)>5%, the central age and the pooled age should be equal, and the age dispersion should be less than ~10%.
Table S2.1 summarizes the fission track age data in apatite from each sample analyzed, while Table S2.4 summarizes similar data for zircon in each sample analyzed.

Construction of radial plots of single grain age data

Single grain age data are best represented in the form of radial plot diagrams (Galbraith, 1988, 1990). As illustrated in Figure B.1, these plots display the variation of individual grain ages in a plot of y against x, where:

$$y = (zj - zo) / \sigma i \qquad x = 1/\sigma j \qquad \text{B.2}$$

and; zj = Fission track age of grain j
zo = A reference age
σj = Error in age for grain j

In this plot, all points on a straight line from the origin define a single value of fission track age, and, at any point, the value of x is a measure of the precision of each individual grain age. Therefore, precise individual grain ages fall to the right of the plot (small error, high x), which is useful, for example, in enabling precise, young grains to be identified. The age scale is shown radially around the perimeter of the plot (in Ma). If all grains belong to a single age population, all data should scatter between y = +2 and y = -2, equivalent to scatter within ±2σ. Scatter outside these boundaries shows a significant spread of individual grain ages, as also reflected in the values of P(χ2) and age dispersion.
In detail, rather than using the fission track age for each grain as in equation B.2, we use:

$$zj = \frac{Nsj}{Nij} \qquad sj = \{1/Nsj + 1/Nij\} \qquad \text{B.3}$$

as we are interested in displaying the scatter within the data from each sample in comparison with that allowed by the Poissonian uncertainty in track counts, without the additional terms which are involved in determination of the fission track age (ρ_D, ζ, etc).
Zero ages cannot be displayed in such a plot. This can be achieved using a modified plot, (Galbraith, 1990) with:

$$z_j = \arcsin\sqrt{\left\{\frac{N_{sj}+3/8}{N_{sj}+N_{ij}+3/4}\right\}} \qquad \sigma_j = \frac{1}{2}\sqrt{\left\{\frac{1}{N_{sj}+N_{ij}}\right\}} \qquad \text{B.4}$$

Note that the numerical terms in the equation for z_j are standard terms, introduced for statistical reasons. Using this arc-sin transformation, zero ages plot on a diagonal line which slopes from upper left to lower right. Note that this line does not go through the origin. Figure B.2 illustrates this difference between conventional and arc-sin radial plots, and also provides a simple guide to the structure of radial plots.

Use of arc-sin radial plots is particularly useful in assessing the relative importance of zero ages. For instance, grains with $N_s = 0$, $N_i = 1$ are compatible with ages up to ~900 Ma (at the 95% confidence level), whereas grains with $N_s = 0$, $N_i = 50$ are only compatible with ages up to ~14 Ma. The two data would readily be distinguishable on the radial plot as the 0,50 datum would plot well to the right (high x) compared to the 0,1 datum.

In this study the value of z corresponding to the stratigraphic age of each sample (or the midpoint of the range where appropriate) is adopted as the reference value, z_o. This allows rapid assessment of the fission track age of individual grains in relation to the stratigraphic age, which is a key component in the interpretation of AFTA data, as explained in more detail in S3.

Note that the x axis of the radial plot is normally not labelled, as this would obscure the age scale around the plot. In general labelling is not considered necessary, as we are concerned only with relative variation within the data, rather than absolute values of precision.

Radial plots of the single grain age data in apatite and zircon from each sample analyzed in this report are shown on the fission track age data summary sheets at the end of Supporting information 2. Use of radial plots to provide thermal history information is explained in Supporting information 3 and Figure S3.7.

Track length data

Distributions of confined track lengths in apatite from each sample are shown as simple histograms on the fission track age data summary sheets at the end of this Appendix.

For every track length measurement, the length is recorded to the nearest 0.1 µm, but the measurements have been grouped into 1 µm intervals for construction of these histograms. Each distribution has been normalized to 100 tracks for each sample to facilitate comparison.

A summary of the length distribution in each sample is presented in Table S2.2, which also shows the mean track length in each sample and its associated error, the standard deviation of each distribution and the number of tracks (N) measured in each sample. The angle which each confined track makes with the crystallographic c-axis is also routinely recorded, as is the width of each fracture within which tracks are revealed. These data are not provided in this report, but can be supplied on request.

Track length data is not routinely collected in zircon.

Breakdown of data into compositional groups

In Table S2.3, AFTA data are grouped into compositional intervals of 0.1 wt% Cl width. Parameters for each interval represent the data from all grains with Cl contents within each interval. Also shown are the parameters for each compositional interval predicted from the Default Thermal History (see Section 2.1). These data form the basis of interpretation of the AFTA data, which takes full account of the influence of Cl content on annealing kinetics, as described in Supporting information chapter S3. Distributions of Cl contents in all apatites analyzed from each sample (i.e. for both age and length determinations) are shown on the fission track age data summary sheets at the end of this chapter.

Plots of fission track age against Cl content for individual apatite grains

Fission track ages of single apatite grains within individual samples are plotted against the Cl content of each grain on the fission track age data summary sheets at the end of this Appendix. These plots are useful in assessing the degree of annealing, as expressed by the fission track age data. For example, if grains with a range of Cl contents from zero to some upper limit all give similar fission track ages which are significantly less than the stratigraphic age, then grains with these compositions must have been totally annealed. Alternatively, if fission track age falls rapidly with decreasing Cl content, the sample displays a high degree of partial annealing.

S2.4 A note on terminology

Note that throughout this report, the term "fission track age" is understood to denote the parameter calculated from the fission track age equation, using the observed spontaneous and induced track counts (either pooled for all grains or for individual grains). The resulting number (with units of Ma) should not be taken as possessing any significance in terms of events taking place at the time indicated by the measured fission track age, but should rather be regarded as a measure of the integrated thermal history of the sample, and should be interpreted in that light using the principles outlined in Supporting information chapter S3. Use of the term "apparent age" is not considered to be useful in this regard, as almost every fission track age should be regarded as an apparent age, in the classic sense, and repeated use becomes cumbersome.

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Table S2.1: Apatite fission track analytical results - samples from Vietnam and Cambodia

Sample number	Number of grains	ρ_D (N_D) x10^6/cm	ρ_s (N_s) x10^6/cm	ρ_i (N_i) x10^6/cm	Uranium content (ppm)	$P(\chi^2)$ (%)	Age dispersion (%)	Fission track age (Ma)
outcrop samples								
GC1017-2	12	1.476 (2318)	1.028 (152)	4.727 (699)	37	53	2	62.7 ± 5.9
GC1017-3	20	1.471 (2318)	0.858 (345)	4.200 (1689)	33	12	15	58.8 ± 3.8
GC1017-4	20	1.466 (2318)	0.233 (128)	0.804 (441)	6	1	40	83.0 ± 8.7 *89.6 ± 12.6**
GC1017-5	11	1.461 (2318)	0.305 (71)	1.473 (343)	11	8	28	59.1 ± 7.9
GC1017-6	20	1.456 (2318)	0.584 (379)	3.149 (2043)	25	91	<1	52.8 ± 3.3
GC1017-7	20	1.451 (2318)	1.488 (677)	7.908 (3598)	62	27	5	53.4 ± 2.7
GC1129-1	10	1.299 (2027)	0.854 (136)	3.517 (560)	31	1	40	59.7 ± 5.9 *53.8 ± 10.0**
GC1129-2			No apatite					
GC1129-3	8	1.239 (2040)	0.845 (84)	4.194 (417)	39	66	<1	48.9 ± 6.0
GC1129-4			No apatite					
GC1129-5	15	1.298 (2027)	1.108 (279)	3.991 (1005)	35	14	17	68.2 ± 5.0
GC1129-6			No apatite					
GC1129-7	8	1.295 (2027)	0.367 (36)	1.824 (179)	16	25	29	49.3 ± 9.1
GC1129-8	20	1.293 (2027)	0.691 (223)	3.153 (1018)	28	<1	35	53.6 ± 4.2 *57.3 ± 6.8**
GC1129-11	5	1.246 (2040)	0.115 (9)	0.679 (53)	6	11	46	41.4 ± 15.0
GC1129-12	9	1.253 (2040)	0.466 (56)	0.965 (116)	9	42	10	117.7 ± 19.5
GC1129-13	5	1.292 (2027)	0.431 (42)	1.394 (136)	12	<1	85	75.4 ± 13.5 *95.8 ± 40.9**
GC1129-14	8	1.260 (2040)	0.431 (38)	3.416 (301)	31	10	44	31.2 ± 5.4
GC1129-15	6	1.290 (2027)	0.991 (73)	5.406 (398)	48	23	3	44.9 ± 5.8
GC1129-16	20	1.288 (2027)	0.179 (78)	1.053 (458)	9	1	45	41.6 ± 5.2 *49.5 ± 8.4**
GC1129-17			No apatite					

520 **Table S2.1:** Continued

Sample number	Number of grains	ρ_D (N_D) x10^6/cm	ρ_s (N_s) x10^6/cm	ρ_i (N_i) x10^6/cm	Uranium content (ppm)	$P(\chi^2)$ (%)	Age dispersion (%)	Fission track age (Ma)
GC1129-18	1	1.266 (2040)	0.132 (1)	1.324 (10)	12	100	<1	24.8 ± 26.1
GC1129-19			No apatite					
GC1129-20			No apatite					
GC1129-21	20	1.287 (2027)	0.210 (105)	1.451 (725)	13	8	29	35.3 ± 3.8
GC1129-22			No apatite					
GC1129-23	20	1.285 (2027)	0.165 (157)	0.751 (715)	7	3	31	53.5 ± 4.9
								*53.0 ± 6.2**
GC1129-24	20	1.284 (2027)	0.402 (254)	2.524 (1596)	22	<1	34	38.7 ± 2.8
								*38.7 ± 4.2**
GC1129-26	20	1.283 (2027)	0.534 (282)	2.487 (1313)	22	4	25	52.2 ± 3.7
								*53.0 ± 5.0**
GC1129-27	26	1.281 (2027)	0.148 (62)	1.335 (560)	12	8	38	26.9 ± 3.7
GC1129-30	11	1.279 (2027)	0.385 (95)	1.447 (357)	13	<1	77	64.4 ± 7.6
								*49.3 ± 13.7**
GC1129-31	20	1.285 (2070)	0.185 (83)	0.945 (424)	8	46	5	49.2 ± 6.1
GC1129-33	20	1.273 (2040)	0.085 (31)	0.999 (364)	9	76	1	21.3 ± 4.0
GC1129-34	12	1.289 (2070)	0.013 (5)	0.061 (23)	1	45	302	54.8 ± 27.1
GC1129-36			No apatite					
GC1129-37	15	1.292 (2070)	0.253 (68)	2.125 (571)	19	90	<1	30.2 ± 4.0
GC1129-38			No apatite					
GC1129-41	15	1.295 (2070)	0.337 (76)	2.552 (575)	22	9	45	33.5 ± 4.2
GC1129-42			No apatite					
GC1129-43	20	1.280 (2040)	0.287 (123)	1.741 (746)	16	25	22	41.3 ± 4.2
GC1129-44			No apatite					
GC1129-45			No apatite					
GC1129-46			No apatite					

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523 **Table S2.1:** Continued

Sample number	Number of grains	ρ_D (N_D) x10^6/cm	ρ_s (N_s) x10^6/cm	ρ_i (N_i) x10^6/cm	Uranium content (ppm)	$P(\chi^2)$ (%)	Age dispersion (%)	Fission track age (Ma)
GC1129-47			No apatite					
GC1129-49			No apatite					
GC1129-50			No apatite					
GC1129-51			No apatite					
GC1129-52			No apatite					
GC1129-53			No apatite					
GC1129-54	5	1.299 (2070)	1.007 (45)	3.044 (136)	27	21	5	83.9 ± 14.6
GC1129-56			No apatite					
GC1129-57	16	1.302 (2070)	0.340 (89)	2.063 (540)	18	27	24	42.0 ± 5.0
GC1129-59	22	1.306 (2070)	0.569 (240)	1.741 (734)	15	<1	67	83.3 ± 6.6 *65.0 ± 11.3**
GC1129-60			No apatite					
GC1129-62	18	1.312 (2070)	0.406 (72)	2.378 (422)	21	84	<1	43.8 ± 5.7
GC1129-63	13	1.316 (2070)	0.315 (41)	1.589 (207)	14	29	17	51.0 ± 8.8
GC1129-64	17	1.319 (2070)	0.378 (97)	2.212 (568)	19	71	2	44.1 ± 5.0
GC1129-65			No apatite					
GC1129-66			No apatite					
GC1129-67			No apatite					
GC1129-68	12	1.323 (2070)	0.222 (51)	0.997 (229)	9	55	1	57.6 ± 9.1
GC1129-69	15	1.326 (2070)	1.112 (135)	5.568 (676)	48	21	13	51.8 ± 5.1
GC1129-70	20	1.333 (2070)	0.281 (114)	1.685 (684)	14	66	8	43.5 ± 4.6
GC1129-72	20	1.336 (2070)	0.833 (274)	3.862 (1271)	33	8	19	56.3 ± 4.1
GC1129-73	21	1.339 (2070)	0.147 (40)	1.042 (284)	9	96	<1	37.0 ± 6.3

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526 **Table S2.1:** Continued

Sample number	Number of grains	ρ_D (N_D) x10^6/cm	ρ_s (N_s) x10^6/cm	ρ_i (N_i) x10^6/cm	Uranium content (ppm)	P(χ^2) (%)	Age dispersion (%)	Fission track age (Ma)
GC1129-76	20	1.343 (2070)	0.423 (82)	2.466 (478)	21	19	34	45.1 ± 5.5
GC1129-78			No apatite					
GC1129-79			No apatite					
GC1129-80	6	1.287 (2040)	0.967 (59)	4.063 (248)	36	<1	58	59.9 ± 8.8 *58.8 ± 17.5**
GC1129-81	8	1.293 (2040)	0.457 (59)	2.108 (272)	19	93	<1	54.9 ± 8.0
GC1129-82			No apatite					
ENRECA								
GC1017-1	20	1.481 (2318)	0.498 (340)	2.826 (1928)	22	3	8	51.1 ± 3.3 *51.3 ± 3.6**

ρ_s = spontaneous track density; ρ_i = induced track density; ρ_D = track density in glass standard external detector. Brackets show number of tracks counted. ρ D and ρi measured in mica external detectors; ρ_s measured in internal surfaces.
*Central age, used where sample contains a significant spread of single grain ages (P(χ^2)<5%). Errors quoted at 1 σ
Ages calculated using dosimeter glass CN5, with a zeta of 380.4 ± 5.7 (Analyst: C. O'Brien) for samples; GC1129-1, GC1129-5 - 8, GC1129-13, GC1129-15 - 16, GC1129-21 - 30
CN5, with a zeta of 392.9 ± 7.4 (Analyst: M. Moore) for samples; GC1017-1 - 7, GC1129-3, GC1129-11 - 12, GC1129-14, GC1129-18, GC1129-31 - 81

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Table S2.2: Length distribution summary data - samples from Vietnam and Cambodia

Sample number	Mean track length (µm)	Standard deviation (µm)	Number of tracks (N)	Number of tracks in Length Intervals (µm)																			
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
outcrop samples																							
GC1017-2	12.94 ± 0.49	1.83	14	-	-	-	-	-	-	-	-	-	1	1	2	2	2	6	-	-	-	-	-
GC1017-3	13.93 ± 0.18	1.94	111	-	-	-	-	-	-	-	-	2	2	5	8	16	18	27	19	10	2	2	-
GC1017-4	13.30 ± 0.55	2.95	29	-	-	-	-	1	1	-	-	-	1	2	2	1	4	10	6	1	-	-	-
GC1017-5	13.26 ± 0.32	2.07	43	-	-	-	-	-	-	-	-	2	2	2	1	12	8	7	4	5	-	-	-
GC1017-6	13.71 ± 0.18	1.87	112	-	-	-	-	-	1	-	-	-	4	5	10	9	23	30	23	6	1	-	-
GC1017-7	13.61 ± 0.17	1.79	107	-	-	-	-	-	-	-	-	2	5	4	9	12	22	33	13	6	1	-	-
GC1129-1	13.73 ± 1.06	2.12	4	-	-	-	-	-	-	-	-	-	-	1	-	-	-	1	2	-	-	-	-
GC1129-2	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-3	13.06 ± 0.48	1.66	12	-	-	-	-	-	-	-	-	-	-	1	3	3	1	2	2	-	-	-	-
GC1129-4	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-5	12.49 ± 0.67	1.77	7	-	-	-	-	-	-	-	-	-	1	1	-	-	5	-	-	-	-	-	-
GC1129-6	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-7	11.95	-	1	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-
GC1129-8	13.15 ± 0.89	2.18	6	-	-	-	-	-	-	-	-	-	1	-	1	1	-	2	1	-	-	-	-
GC1129-11	No confined	tracks	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-12	13.03 ± 0.52	1.28	6	-	-	-	-	-	-	-	-	-	-	1	-	2	1	2	-	-	-	-	-
GC1129-13	9.24	-	1	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-
GC1129-14	13.16 ± 0.46	1.30	8	-	-	-	-	-	-	-	-	-	-	-	2	2	2	2	-	-	-	-	-
GC1129-15	13.24 ± 1.20	2.95	6	-	-	-	-	-	-	-	-	1	-	1	-	-	1	2	-	-	1	-	-
GC1129-16	13.06 ± 0.46	1.29	8	-	-	-	-	-	-	-	-	-	-	1	-	2	3	2	-	-	-	-	-
GC1129-17	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-18	13.59 ± 1.24	2.14	3	-	-	-	-	-	-	-	-	-	-	-	1	-	-	2	-	-	-	-	-
GC1129-19	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-20	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-21	13.13 ± 0.19	1.47	61	-	-	-	-	-	-	-	-	-	2	3	10	11	16	16	1	2	-	-	-
GC1129-22	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-23	13.53 ± 0.25	1.72	47	-	-	-	-	-	-	-	1	-	-	2	6	5	12	13	7	1	-	-	-
GC1129-24	13.63 ± 0.15	1.54	101	-	-	-	-	-	-	-	1	1	1	4	3	17	30	28	14	2	-	-	-
GC1129-26	12.85 ± 0.20	2.00	100	-	-	-	-	-	-	-	2	2	8	8	11	10	22	31	5	1	-	-	-
GC1129-27	12.17 ± 1.30	3.19	6	-	-	-	-	-	-	-	1	-	-	1	1	-	-	2	1	-	-	-	-
GC1129-30	12.77 ± 0.61	1.22	4	-	-	-	-	-	-	-	-	-	-	-	1	2	-	1	-	-	-	-	-
GC1129-31	12.52 ± 0.55	2.63	23	-	-	-	-	-	-	2	-	-	1	1	3	7	4	3	1	-	-	1	-
GC1129-33	13.09 ± 0.22	1.88	71	-	-	-	-	-	-	1	-	-	1	4	15	11	12	18	7	-	1	1	-
GC1129-34	11.63 ± 2.49	4.31	3	-	-	-	-	-	-	1	-	-	-	-	-	-	1	1	-	-	-	-	-
GC1129-36	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-37	12.51 ± 0.41	1.66	16	-	-	-	-	-	-	-	-	-	2	1	1	6	3	1	2	-	-	-	-
GC1129-38	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-41	13.17 ± 0.21	1.88	83	-	-	-	-	1	-	-	1	1	2	2	10	14	25	19	7	-	1	-	-
GC1129-42	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-43	13.79 ± 0.17	1.77	104	-	-	-	-	-	-	-	2	-	2	3	6	14	20	34	17	6	-	-	-
GC1129-44	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-45	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-46	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

538 **Table S2.2:** Continued

Sample number	Mean track length (µm)	Standard deviation (µm)	Number of tracks (N)	Number of tracks in Length Intervals (µm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
GC1129-47	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-49	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-50	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-51	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-52	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-53	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-54	13.81 ± 0.38	0.66	3	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	-	-	-	-	-
GC1129-56	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-57	11.43 ± 0.77	1.88	6	-	-	-	-	-	-	-	-	-	1	2	-	2	1	-	-	-	-	-	-
GC1129-59	12.88 ± 0.54	1.86	12	-	-	-	-	-	-	-	-	1	1	-	1	-	5	4	-	-	-	-	-
GC1129-60	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-62	13.57 ± 0.56	0.97	3	-	-	-	-	-	-	-	-	-	-	-	-	1	1	1	-	-	-	-	-
GC1129-63	13.82 ± 1.01	2.02	4	-	-	-	-	-	-	-	-	-	-	-	1	-	1	1	-	1	-	-	-
GC1129-64	12.67 ± 1.35	2.70	4	-	-	-	-	-	-	-	-	-	1	-	-	1	-	1	1	-	-	-	-
GC1129-65	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-66	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-67	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-68	13.18 ± 0.79	1.76	5	-	-	-	-	-	-	-	-	-	-	-	1	2	-	1	1	-	-	-	-
GC1129-69	12.68 ± 0.29	1.66	33	-	-	-	-	-	-	-	-	-	2	4	5	5	11	4	2	-	-	-	-
GC1129-70	12.96 ± 0.19	1.98	105	-	-	-	1	-	-	-	2	1	4	7	13	17	25	24	10	1	-	-	-
GC1129-72	13.11 ± 0.15	1.51	104	-	-	-	-	-	-	-	-	1	2	6	11	26	30	19	6	3	-	-	-
GC1129-73	13.45 ± 0.59	2.05	12	-	-	-	-	-	-	-	-	1	-	-	2	-	4	2	2	1	-	-	-
GC1129-76	13.09 ± 0.17	1.79	106	-	-	-	-	-	-	-	1	2	2	9	14	22	23	16	14	3	-	-	-
GC1129-78	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-79	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GC1129-80	12.45 ± 0.68	1.93	8	-	-	-	-	-	-	-	-	-	1	1	1	1	2	2	-	-	-	-	-
GC1129-81	13.21 ± 0.46	1.94	18	-	-	-	-	-	-	-	-	1	-	1	3	2	-	11	-	-	-	-	-
GC1129-82	No apatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ENRECA																							
GC1017-1	13.23 ± 0.19	2.02	109	-	-	-	-	-	2	-	-	-	2	6	13	25	24	19	12	3	2	-	1

Track length measurements by: C. O'Brien for samples; GC1129-1
GC1129-5 - 8
GC1129-16
GC1129-21 - 30
M. Moore for samples; GC1017-1 - 7
GC1129-3
GC1129-12 - 14
GC1129-18
GC1129-31 - 81

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543 **Table S2.3:** AFTA Data in Compositional Groups

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (μm)	Mean Track length (μm)	Error in length (μm)	Std deviation (μm)	Number of lengths	Number of grains	Number of tracks in length interval (μm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
outcrop samples																															
1017-2†	**379**	**62.7**	**5.9**	**53.0**	**12**	**14.5**	**12.9**	**0.5**	**1.8**	**14**	**6**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**2**	**2**	**2**	**6**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	369	72.1	57.0	30.1	2	14.1	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.3 - 0.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.7 - 0.8	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.8 - 0.9	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.9 - 1.0	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.0 - 1.1	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.1 - 1.2	381	57.4	10.3	35.5	3	14.5	14.9	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
1.2 - 1.3	381	45.8	9.8	73.5	2	14.5	12.4	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1.3 - 1.4	381	73.0	13.2	83.8	3	14.5	13.1	0.8	1.9	6	2	0	0	0	0	0	0	0	0	0	0	1	1	0	1	3	0	0	0	0	0
1.4 - 1.5	381	74.4	12.4	17.7	2	14.5	12.5	0.8	2.0	6	2	0	0	0	0	0	0	0	0	0	1	0	1	1	1	2	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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545 **Table S2.3:** Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (μm)	Mean Track length (μm)	Error in length (μm)	Std deviation (μm)	Number of lengths	Number of grains	Number of tracks in length interval (μm)																				
												1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1017-3†	**131**	**58.8**	**3.8**	**12.1**	**20**	**14.6**	**13.9**	**0.2**	**1.9**	**111**	**27**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**2**	**5**	**8**	**16**	**18**	**27**	**19**	**10**	**2**	**2**	**0**
0.0 - 0.1	128	45.2	17.2	93.3	4	14.3	12.2	0.5	1.8	12	4	0	0	0	0	0	0	0	0	1	0	2	2	2	3	2	0	0	0	0	0
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	129	24.3	9.6	100.0	1	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.3 - 0.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	131	0.0	0.0	0.0	0	14.5	14.9	0.3	0.4	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
0.7 - 0.8	131	51.4	25.0	100.0	1	14.6	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.8 - 0.9	131	61.8	12.8	100.0	1	14.6	14.5	0.4	1.5	11	2	0	0	0	0	0	0	0	0	0	0	0	1	1	2	3	2	2	0	0	0
0.9 - 1.0	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.0 - 1.1	131	68.2	11.3	51.1	2	14.6	13.9	0.7	2.6	12	4	0	0	0	0	0	0	0	0	0	1	0	1	4	0	3	1	0	1	1	0
1.1 - 1.2	132	54.5	6.6	13.8	4	14.6	13.6	0.3	1.7	32	6	0	0	0	0	0	0	0	0	0	1	3	0	7	5	9	5	2	0	0	0
1.2 - 1.3	132	70.9	17.4	2.2	3	14.7	14.2	0.4	1.8	24	5	0	0	0	0	0	0	0	0	1	0	0	3	1	2	7	8	2	0	0	0
1.3 - 1.4	132	59.4	8.8	13.0	2	14.7	14.0	0.5	1.6	9	3	0	0	0	0	0	0	0	0	0	0	0	1	1	3	2	1	1	0	0	0
1.4 - 1.5	132	77.3	19.1	100.0	1	14.7	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1.5 - 1.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.6 - 1.7	132	56.5	11.0	100.0	1	14.7	15.5	0.6	1.9	9	1	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	1	3	1	1	0
1017-4†	**219**	**89.6**	**12.6**	**0.9**	**20**	**14.2**	**13.3**	**0.5**	**3.0**	**29**	**16**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**1**	**2**	**2**	**1**	**4**	**10**	**6**	**1**	**0**	**0**	**0**
0.0 - 0.1	217	93.5	18.4	56.4	6	14.2	13.2	0.7	3.2	24	12	0	0	0	0	1	1	0	0	0	1	2	2	0	3	7	6	1	0	0	0
0.1 - 0.2	217	33.1	14.3	87.7	2	14.2	14.4	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0.2 - 0.3	218	91.7	25.7	41.8	3	14.3	14.2	0.6	0.9	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0
0.3 - 0.4	219	138.6	40.0	17.0	3	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.4 - 0.5	220	71.6	16.5	34.0	3	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.5 - 0.6	220	52.1	40.1	100.0	1	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.6 - 0.7	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.7 - 0.8	222	43.5	14.8	100.0	1	14.5	13.3	1.0	1.4	2	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0
0.8 - 0.9	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.9 - 1.0	223	238.0	81.0	100.0	1	14.6	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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547 **Table S2.3:** Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (μm)	Mean Track length (μm)	Error in length (μm)	Std deviation (μm)	Number of lengths	Number of grains	Number of tracks in length interval (μm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1017-5†	**130**	**59.1**	**7.9**	**8.1**	**11**	**14.5**	**13.3**	**0.3**	**2.1**	**43**	**22**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**2**	**2**	**1**	**12**	**8**	**7**	**4**	**5**	**0**	**0**	**0**
0.0 - 0.1	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.1 - 0.2	129	47.2	12.8	13.7	2	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.2 - 0.3	129	22.0	16.2	100.0	1	14.4	12.1	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
0.3 - 0.4	130	0.0	0.0	0.0	0	14.4	14.3	0.9	1.6	3	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0
0.4 - 0.5	130	0.0	0.0	0.0	0	14.5	12.1	0.6	1.9	9	2	0	0	0	0	0	0	0	0	1	0	2	0	3	2	1	0	0	0	0	0
0.5 - 0.6	130	60.7	16.3	10.9	3	14.5	13.4	1.0	2.6	7	6	0	0	0	0	0	0	0	0	1	0	0	0	2	1	1	0	2	0	0	0
0.6 - 0.7	131	73.4	13.9	24.4	5	14.5	13.6	0.4	2.0	23	11	0	0	0	0	0	0	0	0	0	2	0	1	5	5	4	3	3	0	0	0
1017-6†	**81**	**52.8**	**3.3**	**90.9**	**20**	**14.4**	**13.7**	**0.2**	**1.9**	**112**	**38**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**4**	**5**	**10**	**9**	**23**	**30**	**23**	**6**	**1**	**0**	**0**
0.0 - 0.1	80	0.0	0.0	0.0	0	14.4	13.9	0.3	1.1	12	4	0	0	0	0	0	0	0	0	0	0	0	1	1	4	4	2	0	0	0	0
0.1 - 0.2	81	51.4	6.0	75.4	6	14.4	13.6	0.3	1.8	40	14	0	0	0	0	0	1	0	0	0	0	0	5	4	9	14	6	1	0	0	0
0.2 - 0.3	81	51.9	4.1	77.1	12	14.5	13.8	0.3	2.2	53	17	0	0	0	0	0	0	0	0	0	4	5	4	2	8	11	13	5	1	0	0
0.3 - 0.4	81	58.7	8.1	38.1	2	14.5	13.9	0.4	1.2	7	3	0	0	0	0	0	0	0	0	0	0	0	0	2	2	1	2	0	0	0	0
1017-7†	**149**	**53.4**	**2.7**	**26.9**	**20**	**14.6**	**13.6**	**0.2**	**1.8**	**107**	**11**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**5**	**4**	**9**	**12**	**22**	**33**	**13**	**6**	**1**	**0**	**0**
0.0 - 0.1	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.3 - 0.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.7 - 0.8	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.8 - 0.9	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.9 - 1.0	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.0 - 1.1	149	50.0	6.4	32.5	3	14.6	13.3	1.0	2.7	8	1	0	0	0	0	0	0	0	0	0	2	0	1	0	0	3	0	2	0	0	0
1.1 - 1.2	149	54.3	6.0	2.3	6	14.6	13.4	0.3	1.9	28	3	0	0	0	0	0	0	0	0	1	1	2	2	5	4	9	3	1	0	0	0
1.2 - 1.3	149	55.8	4.1	61.0	8	14.6	13.7	0.2	1.7	50	5	0	0	0	0	0	0	0	0	1	1	2	4	4	16	12	7	2	1	0	0
1.3 - 1.4	149	0.0	0.0	0.0	0	14.6	13.7	0.5	1.6	11	1	0	0	0	0	0	0	0	0	0	1	0	0	2	2	5	1	0	0	0	0
1.4 - 1.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.5 - 1.6	149	56.4	7.8	51.6	2	14.7	14.1	0.5	1.6	10	1	0	0	0	0	0	0	0	0	0	0	0	2	1	0	4	2	1	0	0	0
1.6 - 1.7	149	49.3	7.3	100.0	1	14.7	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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Table S2.3: Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-1†	**102**	**53.8**	**10.0**	**0.5**	**10**	**14.4**	**13.7**	**1.1**	**2.1**	**4**	**4**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**1**	**2**	**0**	**0**	**0**	**0**
0.0 - 0.1	102	60.3	11.0	29.1	6	14.3	10.6	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	103	49.5	17.0	0.0	3	14.4	14.6	0.5	0.7	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
0.3 - 0.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	103	45.1	13.7	100.0	1	14.5	15.1	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1129-3†	**102**	**48.9**	**6.0**	**66.0**	**8**	**14.4**	**13.1**	**0.5**	**1.7**	**12**	**6**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**3**	**3**	**1**	**2**	**2**	**0**	**0**	**0**	**0**
0.0 - 0.1	102	58.2	11.9	58.2	4	14.3	12.7	0.8	2.0	7	3	0	0	0	0	0	0	0	0	0	0	1	3	1	0	0	2	0	0	0	0
0.1 - 0.2	102	41.5	10.1	41.5	3	14.4	13.3	1.1	1.5	2	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0
0.2 - 0.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.3 - 0.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	104	47.1	8.9	100.0	1	14.6	13.7	0.6	1.0	3	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0
1129-5†	**102**	**68.2**	**5.0**	**13.8**	**15**	**14.4**	**12.5**	**0.7**	**1.8**	**7**	**6**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**5**	**0**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	102	64.8	5.6	12.1	12	14.3	11.7	1.1	2.1	4	4	0	0	0	0	0	0	0	0	0	1	1	0	0	2	0	0	0	0	0	0
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.3 - 0.4	103	65.8	14.6	21.4	2	14.5	13.3	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	104	82.2	12.1	100.0	1	14.6	13.6	0.4	0.5	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0
1129-7†	**102**	**49.3**	**9.1**	**24.9**	**8**	**14.4**	**11.9**	**0.0**	**0.0**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	102	38.7	9.4	33.5	6	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	103	75.4	21.6	39.4	2	14.4	11.9	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

550

551 **Table S2.3:** Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm)																				
												1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-16†	**218**	**49.5**	**8.4**	**0.9**	**20**	**14.2**	**13.1**	**0.5**	**1.3**	**8**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**2**	**3**	**2**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	216	64.3	21.6	1.9	5	14.2	13.4	0.4	0.9	6	2	0	0	0	0	0	0	0	0	0	0	0	0	2	2	2	0	0	0	0	0
0.1 - 0.2	217	91.2	61.8	100.0	1	14.2	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.2 - 0.3	218	39.8	16.2	26.1	2	14.3	12.0	1.5	2.1	2	1	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0
0.3 - 0.4	219	39.6	14.3	0.1	5	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.4 - 0.5	219	43.6	21.2	58.6	2	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.5 - 0.6	220	60.9	34.1	64.8	2	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.6 - 0.7	221	54.2	30.0	94.9	3	14.5	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1129-18†	**219**	**24.8**	**26.1**	**100.0**	**1**	**14.4**	**13.6**	**1.2**	**2.1**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**2**	**0**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.3 - 0.4	219	24.8	26.1	100.0	1	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.4 - 0.5	219	0.0	0.0	0.0	0	14.4	13.6	1.2	2.1	3	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	2	0	0	0	0	0
1129-21†	**102**	**35.3**	**3.8**	**8.2**	**20**	**14.4**	**13.1**	**0.2**	**1.5**	**61**	**37**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**3**	**10**	**11**	**16**	**16**	**1**	**2**	**0**	**0**	**0**
0.0 - 0.1	102	11.1	11.4	100.0	1	14.3	12.8	0.5	1.0	4	1	0	0	0	0	0	0	0	0	0	0	0	0	3	0	1	0	0	0	0	0
0.1 - 0.2	102	37.7	4.2	10.9	16	14.4	13.1	0.2	1.5	52	32	0	0	0	0	0	0	0	0	0	2	3	10	7	15	13	1	1	0	0	0
0.2 - 0.3	103	22.8	9.0	23.9	3	14.4	14.2	0.6	1.4	5	4	0	0	0	0	0	0	0	0	0	0	0	0	1	1	2	0	1	0	0	0
1129-23†	**102**	**53.0**	**6.2**	**3.4**	**20**	**14.3**	**13.5**	**0.3**	**1.7**	**47**	**25**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**2**	**6**	**5**	**12**	**13**	**7**	**1**	**0**	**0**	**0**
0.0 - 0.1	102	61.3	6.5	12.6	13	14.3	13.2	0.5	2.2	18	10	0	0	0	0	0	0	0	1	0	0	0	5	2	1	6	2	1	0	0	0
0.1 - 0.2	102	38.1	6.7	20.0	7	14.4	13.7	0.3	1.4	29	15	0	0	0	0	0	0	0	0	0	0	2	1	3	11	7	5	0	0	0	0
1129-24†	**102**	**38.7**	**4.2**	**0.1**	**20**	**14.4**	**13.6**	**0.2**	**1.5**	**101**	**46**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**4**	**3**	**17**	**30**	**28**	**14**	**2**	**0**	**0**	**0**
0.0 - 0.1	102	43.7	18.0	100.0	1	14.3	14.4	0.8	1.8	5	3	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	3	0	0	0	0
0.1 - 0.2	102	39.4	4.5	0.0	18	14.4	13.6	0.2	1.5	95	42	0	0	0	0	0	0	0	1	1	1	4	2	17	29	28	10	2	0	0	0
0.2 - 0.3	103	0.0	0.0	0.0	0	14.4	15.2	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0.3 - 0.4	103	22.5	9.6	100.0	1	14.5	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1129-26†	**102**	**53.0**	**5.0**	**3.8**	**20**	**14.3**	**12.8**	**0.2**	**2.0**	**99**	**47**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**2**	**8**	**8**	**11**	**10**	**22**	**31**	**5**	**1**	**0**	**0**	**0**
0.0 - 0.1	102	56.2	6.6	1.3	14	14.3	12.5	0.3	2.3	58	31	0	0	0	0	0	0	0	2	2	6	6	7	4	12	14	4	1	0	0	0
0.1 - 0.2	102	42.8	6.8	79.0	5	14.4	13.2	0.3	1.6	33	13	0	0	0	0	0	0	0	0	0	2	2	3	4	8	13	1	0	0	0	0
0.2 - 0.3	103	58.3	18.8	100.0	1	14.4	13.4	0.3	1.0	8	3	0	0	0	0	0	0	0	0	0	0	0	1	2	2	3	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

552

553 **Table S2.3:** Continued

Cl	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm)																				
Wt %												1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-27†	**266**	**26.9**	**3.7**	**8.3**	**26**	**14.3**	**12.2**	**1.3**	**3.2**	**6**	**5**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**1**	**0**	**0**	**2**	**1**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	263	29.1	7.5	12.0	7	14.1	10.9	3.6	5.0	2	2	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0
0.1 - 0.2	265	30.6	8.4	7.5	6	14.2	12.6	1.5	2.1	2	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0
0.2 - 0.3	266	72.7	33.9	100.0	1	14.2	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.3 - 0.4	267	27.6	9.3	92.0	8	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.4 - 0.5	267	16.6	9.9	38.3	2	14.3	10.2	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.7 - 0.8	270	18.2	5.7	5.3	2	14.5	15.9	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1129-30†	**167**	**49.3**	**13.7**	**0.0**	**11**	**14.3**	**12.8**	**0.6**	**1.2**	**4**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**2**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	166	59.9	28.8	3.7	3	14.2	12.6	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0.1 - 0.2	166	67.2	30.8	0.0	3	14.3	12.8	0.9	1.5	3	1	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0
0.2 - 0.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.3 - 0.4	168	15.8	7.3	96.3	3	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.4 - 0.5	168	57.9	19.5	10.7	2	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1129-31†	**102**	**49.2**	**6.1**	**45.6**	**20**	**14.3**	**12.5**	**0.5**	**2.6**	**23**	**15**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**0**	**0**	**1**	**1**	**3**	**7**	**4**	**3**	**1**	**0**	**0**	**1**	**0**
0.0 - 0.1	102	49.2	6.1	45.6	20	14.3	12.5	0.5	2.6	23	15	0	0	0	0	0	0	2	0	0	1	1	3	7	4	3	1	0	0	1	0
1129-33†	**102**	**21.3**	**4.0**	**75.6**	**20**	**14.3**	**13.1**	**0.2**	**1.9**	**71**	**53**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**4**	**15**	**11**	**12**	**18**	**7**	**0**	**1**	**1**	**0**
0.0 - 0.1	102	21.3	4.0	75.6	20	14.3	13.1	0.2	1.9	71	53	0	0	0	0	0	0	1	0	0	1	4	15	11	12	18	7	0	1	1	0
1129-34†	**102**	**54.8**	**27.1**	**44.7**	**12**	**14.3**	**11.6**	**2.5**	**4.3**	**3**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	102	54.8	27.1	44.7	12	14.3	11.6	2.5	4.3	3	3	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	0	0	0
1129-37†	**369**	**30.2**	**4.0**	**90.2**	**15**	**14.1**	**12.5**	**0.4**	**1.7**	**16**	**5**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**1**	**1**	**6**	**3**	**1**	**2**	**0**	**0**	**0**	**0**
0.0 - 0.1	369	30.2	4.0	90.2	15	14.1	12.5	0.4	1.7	16	5	0	0	0	0	0	0	0	0	0	2	1	1	6	3	1	2	0	0	0	0
1129-41†	**216**	**33.5**	**4.2**	**8.9**	**15**	**14.2**	**13.2**	**0.2**	**1.9**	**82**	**41**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**1**	**2**	**2**	**10**	**14**	**25**	**19**	**7**	**0**	**1**	**0**	**0**
0.0 - 0.1	216	34.2	4.4	6.9	13	14.2	13.3	0.2	1.7	59	26	0	0	0	0	0	0	0	1	1	1	2	5	12	16	15	6	0	0	0	0
0.1 - 0.2	217	16.9	12.4	100.0	1	14.2	12.8	0.6	2.6	18	11	0	0	0	0	1	0	0	0	0	1	0	4	2	6	2	1	0	1	0	0
0.2 - 0.3	218	0.0	0.0	0.0	0	14.3	12.9	0.7	1.3	3	3	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0
0.3 - 0.4	219	0.0	0.0	0.0	0	14.3	14.3	0.6	0.8	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0
0.4 - 0.5	219	63.3	70.8	100.0	1	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

554

555 **Table S2.3:** Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-43†	**266**	**41.3**	**4.2**	**25.0**	**20**	**14.3**	**13.8**	**0.2**	**1.8**	**104**	**70**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**0**	**2**	**3**	**6**	**14**	**20**	**34**	**17**	**6**	**0**	**0**	**0**
0.0 - 0.1	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.1 - 0.2	265	15.4	9.2	100.0	1	14.2	14.3	0.1	0.3	3	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0
0.2 - 0.3	266	44.4	4.8	29.4	16	14.2	13.8	0.2	1.8	82	56	0	0	0	0	0	0	0	1	0	2	3	6	13	12	25	15	5	0	0	0
0.3 - 0.4	267	33.6	10.8	75.3	3	14.3	13.6	0.4	1.8	19	12	0	0	0	0	0	0	0	1	0	0	0	0	1	8	6	2	1	0	0	0
1129-54†	**307**	**83.9**	**14.6**	**20.9**	**5**	**14.2**	**13.8**	**0.4**	**0.7**	**3**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**1**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	306	80.6	16.0	12.2	4	14.1	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.1 - 0.2	308	96.1	34.1	100.0	1	14.2	13.8	0.4	0.7	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0
1129-57†	**239**	**42.0**	**5.0**	**27.4**	**16**	**14.2**	**11.4**	**0.8**	**1.9**	**6**	**5**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**2**	**0**	**2**	**1**	**0**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	239	34.7	5.0	63.7	14	14.2	11.7	0.9	2.0	5	4	0	0	0	0	0	0	0	0	0	1	1	0	2	1	0	0	0	0	0	0
0.1 - 0.2	240	69.8	14.3	72.9	2	14.2	10.1	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1129-59†	**240**	**65.0**	**11.3**	**0.0**	**22**	**14.2**	**12.9**	**0.5**	**1.9**	**12**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**5**	**4**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	239	72.7	19.7	0.0	10	14.2	12.7	0.6	1.9	11	2	0	0	0	0	0	0	0	0	1	1	0	1	0	5	3	0	0	0	0	0
0.1 - 0.2	240	47.1	10.1	39.8	5	14.2	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.2 - 0.3	241	28.8	10.8	70.7	2	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.3 - 0.4	242	104.1	41.3	100.0	1	14.3	14.6	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0.4 - 0.5	242	42.6	46.0	100.0	1	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	244	100.7	27.5	9.4	3	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1129-62†	**490**	**43.8**	**5.7**	**84.3**	**18**	**14.0**	**13.6**	**0.6**	**1.0**	**3**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	489	40.8	6.3	68.5	13	14.0	13.2	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0.1 - 0.2	491	52.8	12.5	83.6	5	14.1	13.8	0.9	1.3	2	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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557 **Table S2.3:** Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-63†	**217**	**51.0**	**8.8**	**28.9**	**13**	**14.2**	**13.8**	**1.0**	**2.0**	**4**	**2**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**1**	**0**	**0**	**0**
0.0 - 0.1	216	49.0	9.7	39.6	10	14.2	14.6	0.8	1.4	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0
0.1 - 0.2	217	113.9	68.5	100.0	1	14.2	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.2 - 0.3	218	128.0	90.6	100.0	1	14.3	11.4	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0.3 - 0.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.7 - 0.8	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.8 - 0.9	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.9 - 1.0	222	26.7	16.2	100.0	1	14.6	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1129-64†	**118**	**44.1**	**5.0**	**71.0**	**17**	**14.3**	**12.7**	**1.3**	**2.7**	**4**	**4**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**0**	**0**	**0**
0.0 - 0.1	118	36.5	8.2	54.2	6	14.3	14.3	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0.1 - 0.2	119	47.2	6.1	67.1	11	14.3	12.1	1.7	3.0	3	3	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	0	0	0
1129-68†	**239**	**57.6**	**9.1**	**55.2**	**12**	**14.2**	**13.2**	**0.8**	**1.8**	**5**	**4**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**2**	**0**	**1**	**1**	**0**	**0**	**0**	**0**
0.0 - 0.1	239	48.9	10.4	35.2	8	14.2	12.1	0.5	0.9	3	3	0	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0	0	0
0.1 - 0.2	240	72.1	16.8	79.0	4	14.2	14.9	0.8	1.2	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
1129-69†	**102**	**51.8**	**5.1**	**21.1**	**15**	**14.4**	**12.7**	**0.3**	**1.7**	**33**	**9**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2**	**4**	**5**	**5**	**11**	**4**	**2**	**0**	**0**	**0**	**0**
0.0 - 0.1	102	29.6	7.2	63.8	9	14.3	12.1	0.7	1.7	6	4	0	0	0	0	0	0	0	0	0	1	1	0	2	2	0	0	0	0	0	0
0.1 - 0.2	102	65.9	8.6	40.5	5	14.4	12.4	0.6	2.0	12	3	0	0	0	0	0	0	0	0	0	1	2	3	0	3	2	1	0	0	0	0
0.2 - 0.3	103	48.9	8.7	100.0	1	14.4	13.4	0.4	1.2	12	1	0	0	0	0	0	0	0	0	0	0	0	2	2	5	2	1	0	0	0	0
0.3 - 0.4	103	0.0	0.0	0.0	0	14.5	12.1	0.8	1.4	3	1	0	0	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0	0	0	0
1129-70†	**217**	**43.5**	**4.6**	**66.2**	**20**	**14.2**	**13.0**	**0.2**	**2.0**	**105**	**61**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**2**	**1**	**4**	**7**	**13**	**17**	**25**	**24**	**10**	**1**	**0**	**0**	**0**
0.0 - 0.1	216	36.7	5.3	31.9	11	14.2	12.7	0.3	1.9	45	30	0	0	0	0	0	0	0	2	1	0	4	7	10	11	8	1	1	0	0	0
0.1 - 0.2	217	54.4	9.1	91.9	7	14.2	13.0	0.3	1.5	30	15	0	0	0	0	0	0	0	0	0	1	2	4	6	10	3	4	0	0	0	0
0.2 - 0.3	218	48.7	14.2	90.8	2	14.3	13.3	0.7	2.9	18	9	0	0	0	1	0	0	0	0	0	1	1	1	0	2	9	3	0	0	0	0
0.3 - 0.4	219	0.0	0.0	0.0	0	14.3	13.1	0.9	2.5	7	3	0	0	0	0	0	0	0	0	0	2	0	0	1	1	1	2	0	0	0	0
0.4 - 0.5	219	0.0	0.0	0.0	0	14.4	13.7	0.5	1.2	5	4	0	0	0	0	0	0	0	0	0	0	0	1	0	1	3	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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Table S2.3: Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-72†	**218**	**56.3**	**4.1**	**8.0**	**20**	**14.3**	**13.1**	**0.1**	**1.5**	**104**	**48**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**2**	**6**	**11**	**26**	**30**	**19**	**6**	**3**	**0**	**0**	**0**
0.0 - 0.1	216	40.0	5.6	81.3	6	14.2	13.4	0.4	1.5	15	9	0	0	0	0	0	0	0	0	0	0	1	3	1	5	4	0	1	0	0	0
0.1 - 0.2	217	67.7	11.1	25.5	4	14.2	12.5	0.7	2.0	8	7	0	0	0	0	0	0	0	0	0	2	0	0	2	2	2	0	0	0	0	0
0.2 - 0.3	218	45.5	7.9	80.9	4	14.3	13.4	0.3	1.4	21	10	0	0	0	0	0	0	0	0	0	0	2	0	8	3	6	1	1	0	0	0
0.3 - 0.4	219	99.2	41.3	62.4	2	14.3	12.4	0.3	1.2	18	8	0	0	0	0	0	0	0	0	0	0	2	4	6	5	0	1	0	0	0	0
0.4 - 0.5	219	0.0	0.0	0.0	0	14.4	12.5	0.7	2.1	10	3	0	0	0	0	0	0	0	0	1	0	1	1	3	2	1	1	0	0	0	0
0.5 - 0.6	220	70.1	7.6	21.3	4	14.4	13.3	0.2	1.0	21	6	0	0	0	0	0	0	0	0	0	0	0	2	4	9	5	1	0	0	0	0
0.6 - 0.7	221	0.0	0.0	0.0	0	14.5	14.8	0.6	1.4	5	2	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0
0.7 - 0.8	221	0.0	0.0	0.0	0	14.5	12.5	0.5	0.9	3	2	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
0.8 - 0.9	222	0.0	0.0	0.0	0	14.5	13.9	0.6	1.1	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	0	0	0	0
1129-73†	**102**	**37.0**	**6.3**	**96.0**	**21**	**14.3**	**13.5**	**0.6**	**2.1**	**12**	**8**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**2**	**0**	**4**	**2**	**2**	**1**	**0**	**0**	**0**
0.0 - 0.1	102	37.0	6.3	96.0	21	14.3	13.5	0.6	2.1	12	8	0	0	0	0	0	0	0	0	1	0	0	2	0	4	2	2	1	0	0	0
1129-76†	**217**	**45.1**	**5.5**	**18.8**	**20**	**14.2**	**13.1**	**0.2**	**1.8**	**106**	**56**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**2**	**2**	**9**	**14**	**22**	**23**	**16**	**14**	**3**	**0**	**0**	**0**
0.0 - 0.1	216	40.3	5.8	14.5	14	14.2	12.8	0.3	1.6	40	22	0	0	0	0	0	0	0	0	0	1	7	5	8	10	5	3	1	0	0	0
0.1 - 0.2	217	39.5	17.3	26.0	3	14.2	12.9	0.7	2.2	9	8	0	0	0	0	0	0	0	1	0	0	0	1	3	0	3	1	0	0	0	0
0.2 - 0.3	218	65.6	73.4	100.0	1	14.3	13.2	0.3	2.0	32	13	0	0	0	0	0	0	0	0	1	1	2	6	3	7	6	5	1	0	0	0
0.3 - 0.4	219	77.7	22.2	100.0	1	14.3	13.4	0.4	1.8	21	9	0	0	0	0	0	0	0	0	1	0	0	2	6	5	2	4	1	0	0	0
0.4 - 0.5	219	0.0	0.0	0.0	0	14.4	13.1	0.0	0.0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0.5 - 0.6	220	65.6	51.9	100.0	1	14.4	13.3	1.0	1.7	3	3	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	0	0	0	0
1129-80†	**119**	**58.8**	**17.5**	**0.0**	**6**	**14.3**	**12.5**	**0.7**	**1.9**	**8**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**2**	**2**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	119	69.6	23.0	0.0	4	14.3	12.5	0.7	1.9	8	3	0	0	0	0	0	0	0	0	0	1	1	1	1	2	2	0	0	0	0	0
0.1 - 0.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.2 - 0.3	119	29.7	12.8	100.0	1	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.3 - 0.4	120	42.0	45.4	100.0	1	14.4	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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Table S2.3: Continued

Cl Wt %	Default fission track age* (Ma)	Measured fission track age (Ma)	Error in age (Ma)	P (χ^2)	Number of grains	Default fission track length* (µm)	Mean Track length (µm)	Error in length (µm)	Std deviation (µm)	Number of lengths	Number of grains	Number of tracks in length interval (µm) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1129-81†	**119**	**54.9**	**8.0**	**93.3**	**8**	**14.4**	**13.2**	**0.5**	**1.9**	**18**	**8**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**3**	**2**	**0**	**11**	**0**	**0**	**0**	**0**	**0**
0.0 - 0.1	119	57.5	10.9	81.9	4	14.3	12.5	1.0	2.7	7	3	0	0	0	0	0	0	0	0	1	0	1	1	0	0	4	0	0	0	0	0
0.1 - 0.2	119	37.3	17.9	100.0	1	14.3	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.2 - 0.3	119	54.0	14.9	57.8	2	14.4	14.0	0.4	1.1	9	4	0	0	0	0	0	0	0	0	0	0	0	1	1	0	7	0	0	0	0	0
0.3 - 0.4	120	0.0	0.0	0.0	0	14.4	12.4	0.4	0.6	2	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
0.4 - 0.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.5 - 0.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.6 - 0.7	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.7 - 0.8	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.8 - 0.9	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
0.9 - 1.0	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.0 - 1.1	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.1 - 1.2	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.2 - 1.3	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.3 - 1.4	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.4 - 1.5	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.5 - 1.6	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.6 - 1.7	122	84.1	56.1	100.0	1	14.7	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENRECA																															
1017-1†	**111**	**51.3**	**3.6**	**2.6**	**20**	**14.1**	**13.2**	**0.2**	**2.0**	**109**	**28**	**0**	**0**	**0**	**0**	**0**	**2**	**0**	**0**	**0**	**2**	**6**	**13**	**25**	**24**	**19**	**12**	**3**	**2**	**0**	**1**
0.0 - 0.1	110	51.0	3.4	5.5	17	14.1	13.2	0.2	1.8	75	20	0	0	0	0	0	1	0	0	0	1	5	8	14	21	15	7	2	1	0	0
0.1 - 0.2	111	43.8	10.3	79.5	2	14.2	13.0	0.5	2.2	22	5	0	0	0	0	0	0	0	0	0	1	1	5	8	1	1	4	0	0	0	1
0.2 - 0.3	111	0.0	0.0	0.0	0	14.2	13.0	0.4	0.5	2	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
0.3 - 0.4	112	0.0	0.0	0.0	0	14.3	13.7	1.3	3.6	8	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	3	1	1	1	0	0
0.4 - 0.5	112	172.2	89.1	100.0	1	14.3	13.2	0.5	0.7	2	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0

*Fission Track Age and Mean Track Length predicted from the Default Thermal History (i.e. if the sample has not been hotter in the past)
†Combined data for all compositional groups

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563 **Table S2.4:** Zircon fission track analytical results - samples from Vietnam and Cambodia

Sample number	Number of grains	ρ_D (N_D) x10^6/cm	ρ_s (Ns) x10^6/cm	ρ_i (Ni) x10^6/cm	Uranium content (ppm)	P(χ^2) (%)	Age dispersion (%)	Fission track age (Ma)
outcrop samples								
GC1017-6	20	1.242 (1960)	13.770 (2669)	9.571 (1855)	411	79	<1	77.9 ± 3.0
GC1129-13	20	1.348 (2121)	12.156 (2708)	4.112 (916)	163	43	3	172.4 ± 7.7
GC1129-16	16	1.348 (2121)	10.098 (1900)	4.103 (772)	162	<1	26	143.8 ± 7.0 *140.8 ± 11.5**
GC1129-21	20	1.348 (2121)	6.956 (2145)	5.056 (1559)	200	<1	15	80.8 ± 3.3 *80.9 ± 4.3**
GC1129-23	20	1.348 (2121)	7.339 (2226)	5.380 (1632)	213	<1	17	80.1 ± 3.2 *78.9 ± 4.4**
GC1129-24	20	1.348 (2121)	9.615 (3564)	5.941 (2202)	235	41	2	95.0 ± 3.4
GC1129-26	19	1.348 (2121)	12.119 (2143)	5.429 (960)	215	68	<1	130.6 ± 5.9
GC1129-31	20	1.348 (2121)	6.549 (2716)	4.632 (1921)	183	<1	13	83.0 ± 3.2 *83.3 ± 4.0**
GC1129-36	20	1.348 (2121)	3.925 (1944)	3.560 (1763)	141	1	14	64.8 ± 2.6 *64.5 ± 3.3**
GC1129-47	12	1.348 (2121)	10.163 (1407)	5.172 (716)	204	1	16	115.1 ± 5.9 *115.9 ± 8.0**
GC1129-49	20	1.348 (2121)	10.863 (3377)	3.780 (1175)	149	2	12	167.7 ± 6.9 *166.1 ± 8.3**
GC1129-50	20	1.348 (2121)	10.806 (2693)	4.639 (1156)	183	24	4	136.3 ± 5.8
GC1129-54	16	1.348 (2121)	12.930 (1668)	6.597 (851)	261	2	15	114.8 ± 5.5 *113.4 ± 6.9**
GC1129-56	20	1.348 (2121)	7.862 (2345)	3.101 (925)	123	1	17	148.2 ± 6.7 *148.5 ± 8.8**
GC1129-59	13	1.348 (2121)	8.737 (1765)	4.460 (901)	176	<1	41	114.8 ± 5.4 *117.5 ± 14.6**
GC1129-60	20	1.348 (2121)	9.906 (3173)	4.349 (1393)	172	10	7	133.3 ± 5.3
GC1129-63	20	1.348 (2121)	7.392 (2754)	3.175 (1183)	126	<1	25	136.2 ± 5.7 *135.8 ± 9.7**
GC1129-67	12	1.348 (2121)	11.234 (1216)	5.479 (593)	217	<1	19	120.1 ± 6.6 *125.0 ± 10.0**
GC1129-68	10	1.348 (2121)	9.412 (1078)	6.339 (726)	251	<1	19	87.2 ± 4.7 *84.9 ± 6.9**
GC1129-70	20	1.349 (2121)	5.770 (1732)	4.467 (1341)	177	2	14	75.9 ± 3.3 *75.1 ± 4.1**
GC1129-72	20	1.349 (2121)	7.828 (2325)	3.791 (1126)	150	10	12	121.0 ± 5.2

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565 **Table S2.4:** Continued

Sample number	Number of grains	ρ_D (N_D) x10^6/cm	ρ_s (Ns) x10^6/cm	ρ_i (Ni) x10^6/cm	Uranium content (ppm)	$P(\chi^2)$ (%)	Age dispersion (%)	Fission track age (Ma)
GC1129-73	9	1.349 (2121)	11.123 (539)	4.912 (238)	194	13	16	132.6 ± 10.8
GC1129-76	20	1.349 (2121)	7.256 (1927)	3.393 (901)	134	17	10	125.3 ± 5.9
GC1129-78	20	1.241 (1960)	8.780 (2061)	7.345 (1724)	316	6	9	64.7 ± 2.6
GC1129-79	20	1.241 (1960)	6.072 (1861)	5.048 (1547)	217	<1	27	65.2 ± 2.7 *62.8 ± 4.7**

ρ_s = spontaneous track density; ρ_i = induced track density; ρ_D = track density in glass standard external detector. Brackets show number of tracks counted. ρ D and ρ_i measured in mica external detectors; ρ_s measured in internal surfaces.
*Central age, used where sample contains a significant spread of single grain ages ($P(\chi^2)$<5%). Errors quoted at 1 σ
Ages calculated using dosimeter glass U3, with a zeta of 87.7 ± 0.8 (Analyst: P. F. Green) for samples; GC1017-6
GC1129-13 - 79

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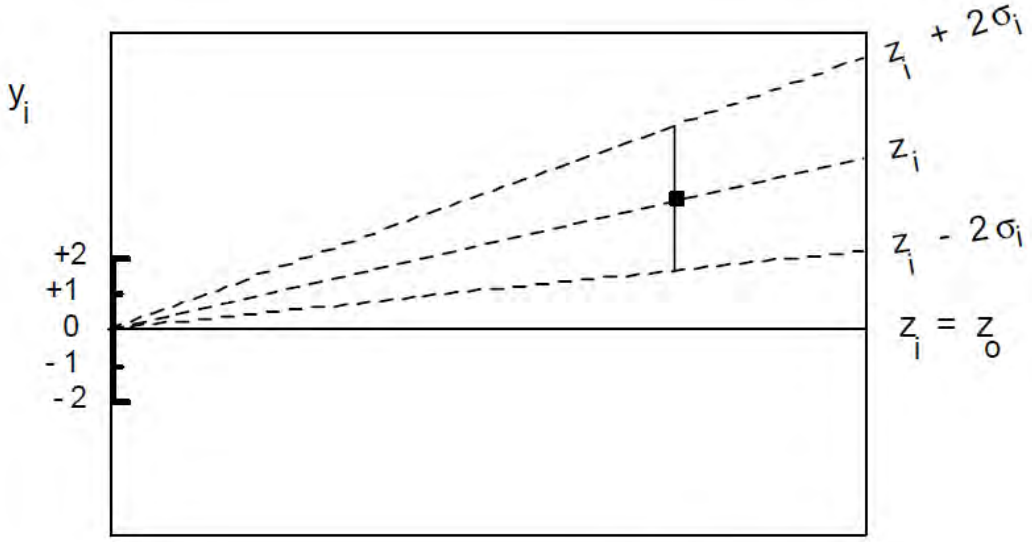
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Estimates	z_i
Standard errors	σ_i
Reference value	z_o
Standardised estimates	$y_i = (z_i - z_o) / \sigma_i$
Precision	$x_i = 1 / \sigma_i$

PLOT y_i against x_i



Slope of line from origin through data point

$= y_i / x_i$

$= \{(z_i - z_o)/\sigma_i\} / \{1/\sigma_i\}$

$= z_i - z_o$

Key Points:

1. Radial lines emanating from the origin correspond to fixed values of z
2. Data points with higher values of xi have greater precision.
3. Error bars on all points are the same size in this plot.

Figure S2.1 Basic construction of a radial plot. In AFTA, the estimates zi correspond to the fission track age values for individual apatite grains. Any convenient value of age can be chosen as the reference value corresponding to the horizontal in the radial plot. Radial lines emanating from the origin with positive slopes correspond to fission track ages greater than the reference value. Lines with negative slopes correspond to fission track ages less than the reference value.

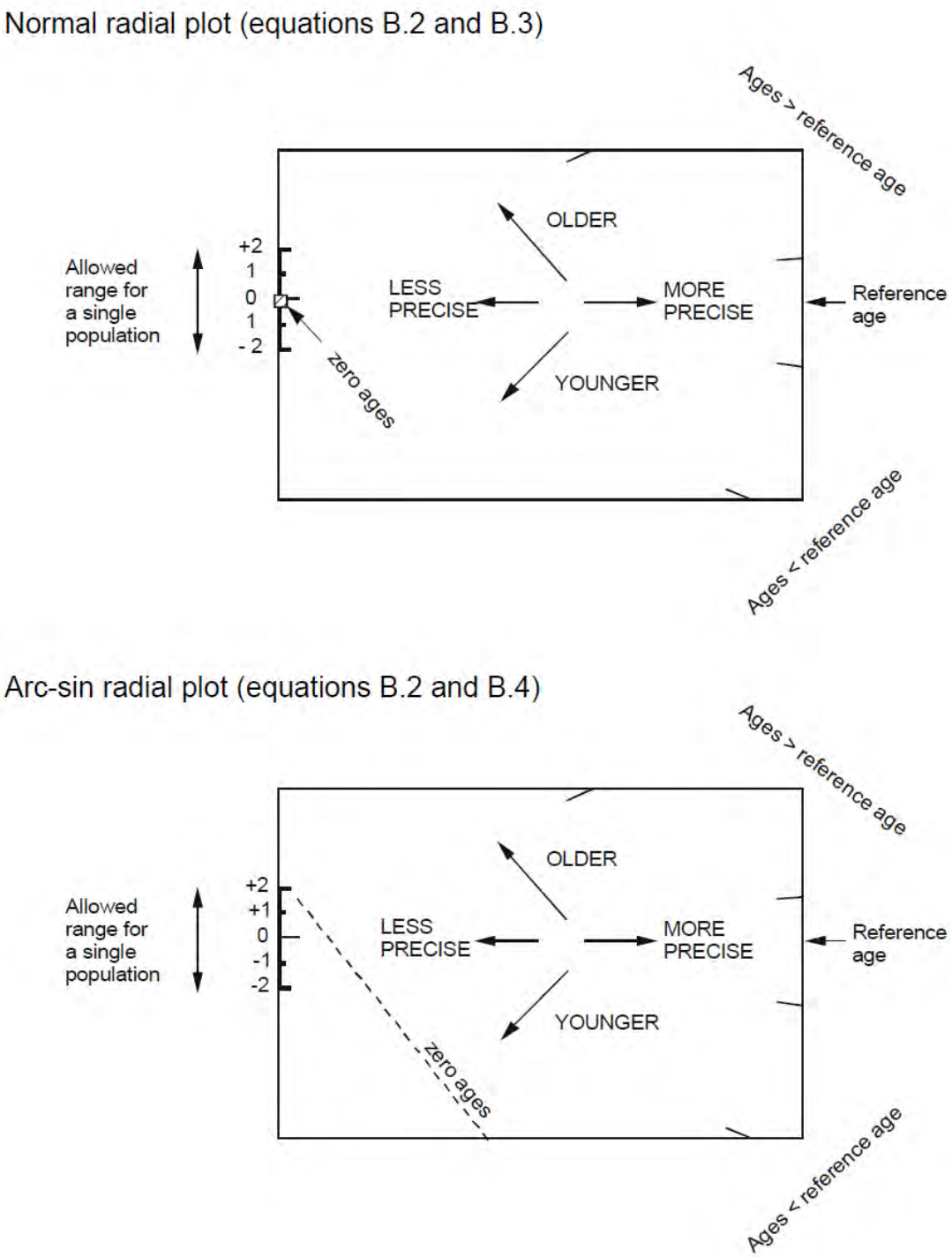


Figure S2.2 Simplified structure of Normal and Arc-sin radial plots.

Fission Track Age Data Sheets - Glossary

N_s	= Number of spontaneous tracks in N_a grid squares
N_i	= Number of induced tracks in N_a grid squares
N_a	= Number of grid squares counted in each grain
RATIO	= N_S/N_i
U (ppm)	= Uranium content of each grain (= U content of standard glass * $\rangle_i/\rho_D$)
Cl (wt%)	= Weight percent chlorine content of each grain
ρ_s	= Spontaneous track density (ρ_s) = N_s/ (N_a*area of basic unit)
ρ_i	= Induced track density (ρ_i) = $N_i/(N_a$*area of basic unit)
F.T. AGE	= Fission track age, calculated using equation B.1
Area of basic unit	= Area of one grid square
Chi squared	= χ^2 parameter, used to assess variation of single grain ages within the sample
P(chi squared)	= Probability of obtaining observed χ^2 value for the relevant number of degrees of freedom, if all grains belong to a single population
Age Dispersion	= % variation in single grain ages - see discussion in text re "Central age"
N_s/N_i	= Pooled ratio, total spontaneous tracks divided by total induced tracks for all grains
Mean ratio	= Mean of (N_s/N_i) for individual grains
Zeta	= Calibration constant, determined empirically for each observer
ρ_D	= Track density (ρ_D) from uranium standard glass (interpolated from values at each end of stack)
ND	= Total number of tracks counted for determining ρ_D
POOLED AGE	= Fission track age calculated from pooled ratio N_s/N_i. Valid only when $P(\chi^2) > 5\%$

CENTRAL AGE = Alternative to pooled age when $P(\chi^2) < 5\%$

Key to Figures:

A: Radial plot of single grain ages *(See Figures B.1 and B.2 for details of radial plot construction)*	B: Distribution of Cl contents in apatite grains
C: Single grain age vs weight % Cl for individual apatite grains.	D: Distribution of confined track lengths

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GC1017-1 Apatite
Counted by: MEM

ENRECA 496.46-496.73m 453001

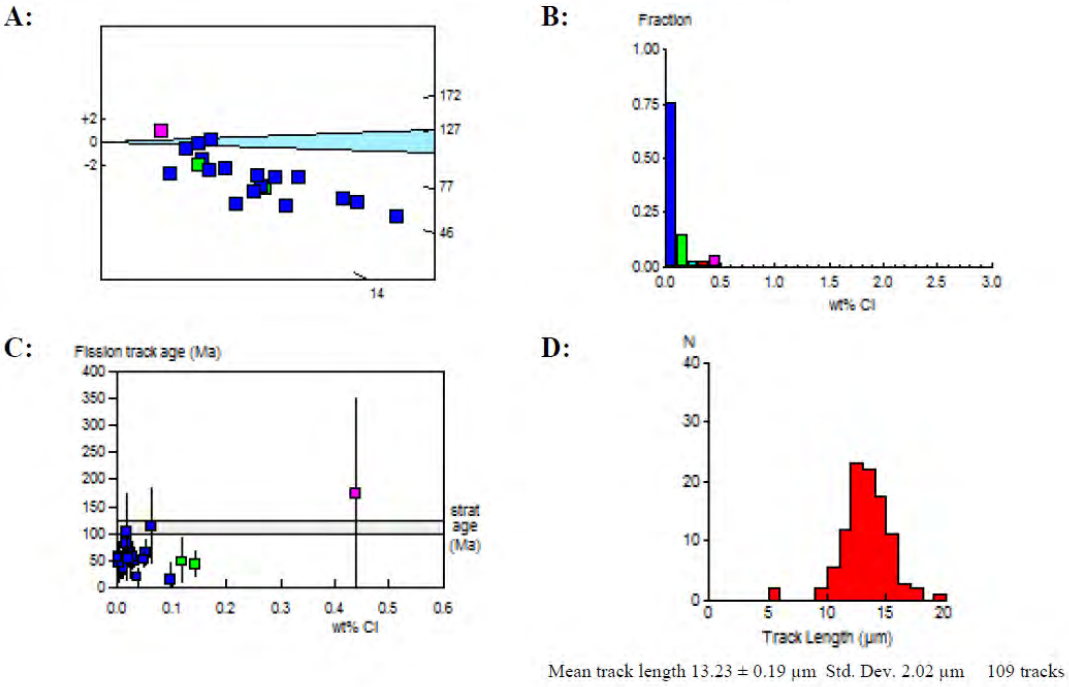
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-5	3	5	76	100	7.945E+04	1.208E+06	0.066	9.3	0.04	19.1 ± 8.8
G1087-5	4	15	103	50	4.767E+05	3.273E+06	0.146	25.2	0.14	42.2 ± 11.7
G1087-5	5	8	37	40	3.178E+05	1.470E+06	0.216	11.3	0.02	62.6 ± 24.5
G1087-5	6	15	98	36	6.621E+05	4.326E+06	0.153	33.3	0.01	44.4 ± 12.4
G1087-5	7	6	36	100	9.534E+04	5.721E+05	0.167	4.4	0.12	48.3 ± 21.3
G1087-5	8	55	326	24	3.642E+06	2.158E+07	0.169	166.1	0.03	48.9 ± 7.3
G1087-5	9	7	25	30	3.708E+05	1.324E+06	0.280	10.2	0.02	81.0 ± 34.7
G1087-5	10	6	10	30	3.178E+05	5.297E+05	0.600	4.1	0.44	172.2 ± 89.1
G1087-5	13	15	134	49	4.864E+05	4.346E+06	0.112	33.5	0.01	32.5 ± 8.9
G1087-5	16	11	57	60	2.913E+05	1.510E+06	0.193	11.6	0.00	55.9 ± 18.5
G1087-5	18	11	91	60	2.913E+05	2.410E+06	0.121	18.6	0.01	35.1 ± 11.2
G1087-5	19	7	44	80	1.390E+05	8.740E+05	0.159	6.7	0.00	46.1 ± 18.8
G1087-5	20	11	31	56	3.121E+05	8.797E+05	0.355	6.8	0.02	102.4 ± 36.1
G1087-5	23	31	141	36	1.368E+06	6.224E+06	0.220	47.9	0.05	63.7 ± 12.8
G1087-5	24	1	20	35	4.540E+04	9.080E+05	0.050	7.0	0.10	14.5 ± 14.9
G1087-5	26	43	243	48	1.424E+06	8.045E+06	0.177	61.9	0.05	51.3 ± 8.6
G1087-5	27	15	38	60	3.973E+05	1.006E+06	0.395	7.7	0.06	113.8 ± 34.9
G1087-5	28	22	112	70	4.994E+05	2.543E+06	0.196	19.6	0.03	56.9 ± 13.4
G1087-5	30	39	216	80	7.747E+05	4.290E+06	0.181	33.0	0.02	52.3 ± 9.2
G1087-5	32	17	90	40	6.754E+05	3.575E+06	0.189	27.5	0.00	54.7 ± 14.6
		340	1928		4.984E+05	2.826E+06		21.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 32.777 with 19 degrees of freedom
P(χ^2) = 2.6%
Age Dispersion = 8.422% (did not converge)

Ns / Ni = 0.176 ± 0.010
Mean Ratio = 0.207 ± 0.028

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.481E+06cm^{-2} ND = 2318
ρ_D interpolated between top of can; ρ = 1.501E+06cm^{-2} ND = 1181
bottom of can; ρ = 1.445E+06cm^{-2} ND = 1137

POOLED AGE = 51.1 ± 3.3 Ma
CENTRAL AGE = 51.3 ± 3.6 Ma



Mean track length 13.23 ± 0.19 µm Std. Dev. 2.02 µm 109 tracks

639

640 **Figure S2.3:** Apatite Fission Track Age Data of sample 453001

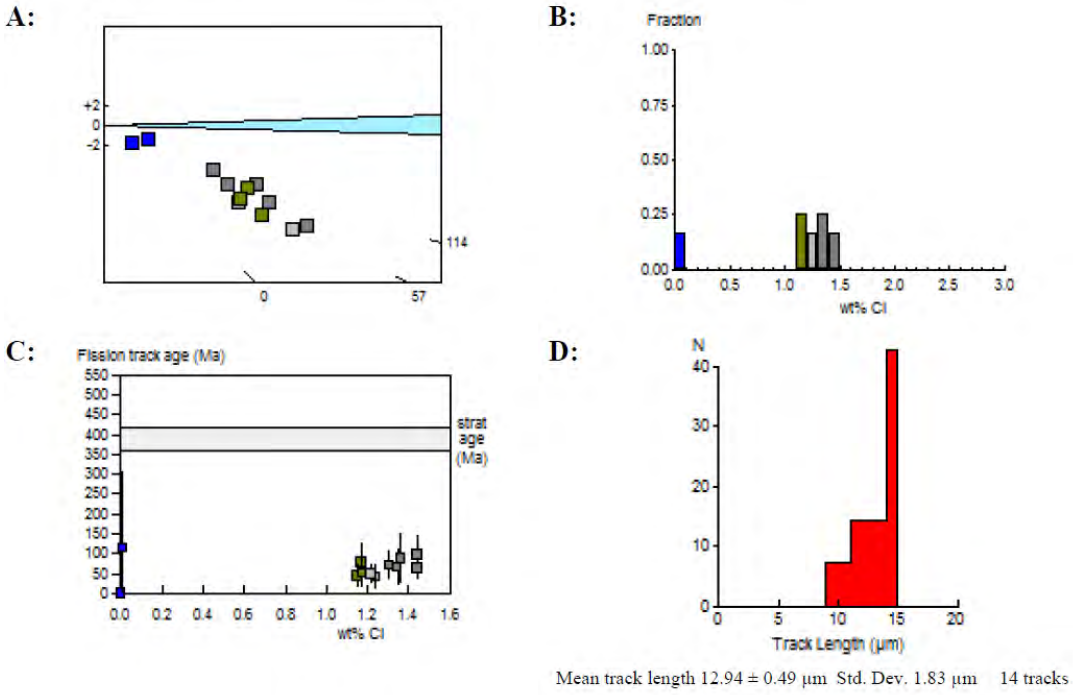
GC1017-2 Apatite 453009
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-6	4	21	62	12	2.781E+06	8.210E+06	0.339	63.4	1.45	97.5 ± 24.8
G1087-6	5	8	56	9	1.413E+06	9.888E+06	0.143	76.4	1.24	41.3 ± 15.6
G1087-6	8	26	120	25	1.653E+06	7.628E+06	0.217	58.9	1.44	62.5 ± 13.6
G1087-6	9	18	108	20	1.430E+06	8.581E+06	0.167	66.3	1.22	48.1 ± 12.3
G1087-6	11	19	77	18	1.677E+06	6.798E+06	0.247	52.5	1.31	71.2 ± 18.3
G1087-6	12	0	3	64	0.000E+00	7.449E+04	0.000	0.6	0.00	0.0 ± 239.4
G1087-6	13	10	44	9	1.766E+06	7.769E+06	0.227	60.0	1.35	65.6 ± 23.0
G1087-6	14	16	58	18	1.413E+06	5.120E+06	0.276	39.5	1.17	79.5 ± 22.6
G1087-6	16	2	5	18	1.766E+05	4.414E+05	0.400	3.4	0.01	114.9 ± 96.2
G1087-6	17	10	33	10	1.589E+06	5.244E+06	0.303	40.5	1.36	87.3 ± 31.6
G1087-6	18	12	77	20	9.534E+05	6.118E+06	0.156	47.3	1.15	45.0 ± 14.0
G1087-6	19	10	56	12	1.324E+06	7.416E+06	0.179	57.3	1.17	51.6 ± 17.8
		152	699		1.028E+06	4.727E+06		36.5		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 10.002 with 11 degrees of freedom
P(χ^2) = 53.0%
Age Dispersion = 2.495% (did not converge)
Ns / Ni = 0.217 ± 0.019
Mean Ratio = 0.221 ± 0.030

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.476E+06cm^{-2} ND = 2318
ρ_D interpolated between top of can; ρ = 1.501E+06cm^{-2} ND = 1181
bottom of can; ρ = 1.445E+06cm^{-2} ND = 1137

POOLED AGE = 62.7 ± 5.9 Ma
CENTRAL AGE = 62.8 ± 5.9 Ma



Mean track length 12.94 ± 0.49 µm Std. Dev. 1.83 µm 14 tracks

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642 **Figure S2.4:** Apatite Fission Track Age Data of sample 453009

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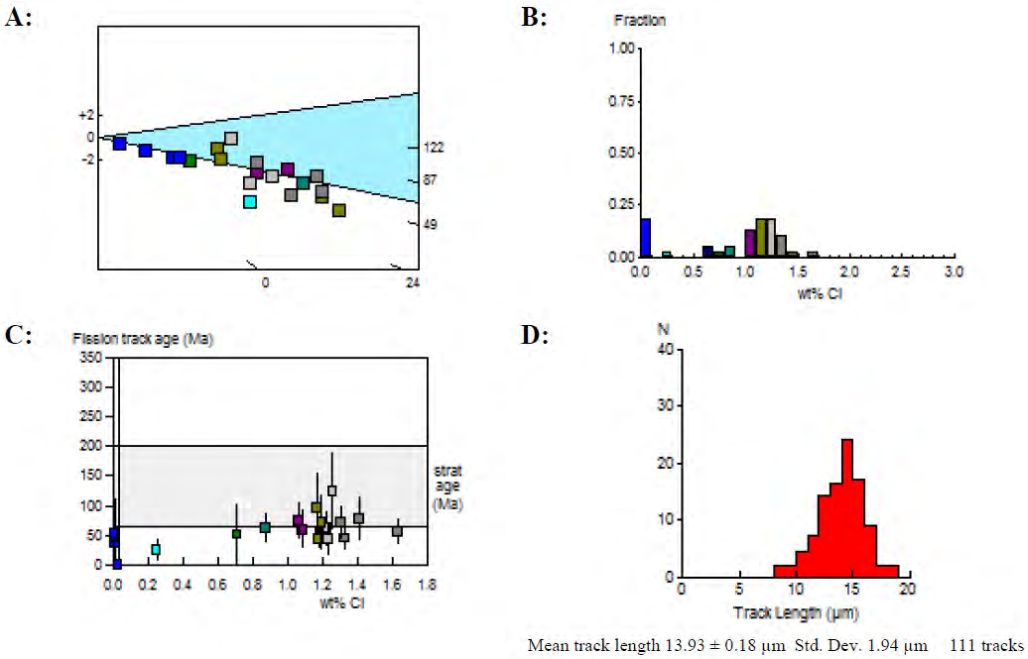
GC1017-3 Apatite 453017
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-7	7	5	28	30	2.648E+05	1.483E+06	0.179	11.5	0.71	51.4 ± 25.0
G1087-7	8	30	167	40	1.192E+06	6.634E+06	0.180	51.4	1.19	51.7 ± 10.4
G1087-7	9	7	83	24	4.635E+05	5.496E+06	0.084	42.6	0.25	24.3 ± 9.6
G1087-7	11	1	8	24	6.621E+04	5.297E+05	0.125	4.1	0.01	36.0 ± 38.2
G1087-7	12	0	2	36	0.000E+00	8.828E+04	0.000	0.7	0.02	0.0 ± 466.2
G1087-7	14	20	127	30	1.059E+06	6.727E+06	0.157	52.1	1.33	45.3 ± 11.0
G1087-7	15	3	19	70	6.810E+04	4.313E+05	0.158	3.3	0.00	45.5 ± 28.3
G1087-7	17	14	42	12	1.854E+06	5.562E+06	0.333	43.1	1.17	95.6 ± 29.6
G1087-7	18	4	22	100	6.356E+04	3.496E+05	0.182	2.7	0.00	52.3 ± 28.5
G1087-7	19	30	197	35	1.362E+06	8.944E+06	0.152	69.3	1.18	43.9 ± 8.7
G1087-7	22	21	99	24	1.390E+06	6.555E+06	0.212	50.8	1.23	61.0 ± 14.8
G1087-7	23	17	82	18	1.501E+06	7.239E+06	0.207	56.1	1.08	59.6 ± 16.0
G1087-7	27	12	48	12	1.589E+06	6.356E+06	0.250	49.3	1.20	71.8 ± 23.3
G1087-7	28	29	112	30	1.536E+06	5.933E+06	0.259	46.0	1.06	74.4 ± 15.6
G1087-7	29	12	79	16	1.192E+06	7.846E+06	0.152	60.8	1.23	43.7 ± 13.6
G1087-7	30	32	163	42	1.211E+06	6.167E+06	0.196	47.8	1.63	56.5 ± 11.0
G1087-7	32	21	49	20	1.669E+06	3.893E+06	0.429	30.2	1.26	122.7 ± 32.2
G1087-7	33	21	78	16	2.086E+06	7.747E+06	0.269	60.0	1.42	77.3 ± 19.1
G1087-7	40	37	149	30	1.960E+06	7.892E+06	0.248	61.2	1.30	71.4 ± 13.3
G1087-7	42	29	135	30	1.536E+06	7.151E+06	0.215	55.4	0.87	61.8 ± 12.8
		345	1689		8.579E+05	4.200E+06		32.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 26.337 with 19 degrees of freedom
P(χ^2) = 12.1%
Age Dispersion = 14.878% (did not converge)
Ns / Ni = 0.204 ± 0.012
Mean Ratio = 0.199 ± 0.020

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.471E+06cm^{-2} ND = 2318
ρ_D interpolated between top of can: ρ = 1.501E+06cm^{-2} ND = 1181
bottom of can: ρ = 1.445E+06cm^{-2} ND = 1137

POOLED AGE = 58.8 ± 3.8 Ma
CENTRAL AGE = 59.3 ± 4.5 Ma



Mean track length 13.93 ± 0.18 µm Std. Dev. 1.94 µm 111 tracks

Figure S2.5: Apatite Fission Track Age Data of sample 453017

GC1017-4 Apatite 453023
Counted by: MEM

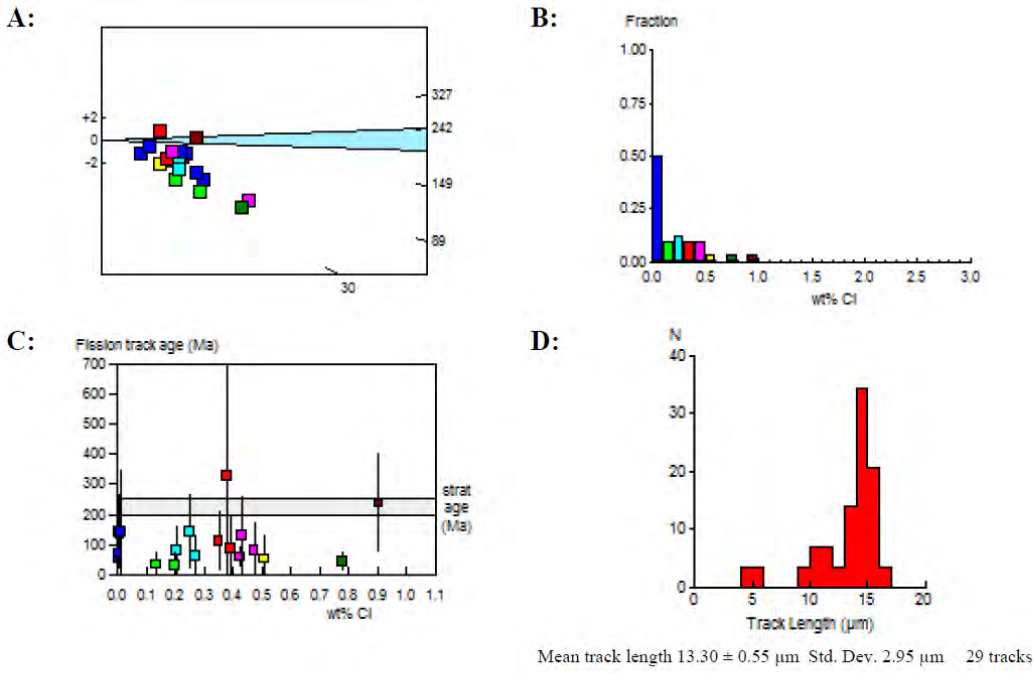
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-8	3	4	33	50	1.271E+05	1.049E+06	0.121	8.2	0.13	34.8 ± 18.5
G1087-8	5	14	69	25	8.899E+05	4.386E+06	0.203	34.1	0.42	58.2 ± 17.1
G1087-8	6	7	6	15	7.416E+05	6.356E+05	1.167	4.9	0.38	327.5 ± 182.4
G1087-8	7	7	33	100	1.112E+05	5.244E+05	0.212	4.1	0.00	60.8 ± 25.4
G1087-8	8	7	18	28	3.973E+05	1.022E+06	0.389	7.9	0.35	111.0 ± 49.5
G1087-8	11	4	14	24	2.648E+05	9.270E+05	0.286	7.2	0.47	81.8 ± 46.4
G1087-8	12	9	19	60	2.384E+05	5.032E+05	0.474	3.9	0.01	135.0 ± 54.7
G1087-8	13	8	16	16	7.945E+05	1.589E+06	0.500	12.4	0.25	142.4 ± 61.8
G1087-8	15	5	18	35	2.270E+05	8.172E+05	0.278	6.4	0.20	79.5 ± 40.2
G1087-8	16	2	11	60	5.297E+04	2.913E+05	0.182	2.3	0.51	52.1 ± 40.1
G1087-8	19	1	5	100	1.589E+04	7.945E+04	0.200	0.6	0.00	57.3 ± 62.8
G1087-8	20	8	16	45	2.825E+05	5.650E+05	0.500	4.4	0.00	142.4 ± 61.8
G1087-8	21	3	6	80	5.959E+04	1.192E+05	0.500	0.9	0.01	142.4 ± 100.8
G1087-8	26	16	19	15	1.695E+06	2.013E+06	0.842	15.7	0.91	238.0 ± 81.0
G1087-8	27	7	28	70	1.589E+05	6.356E+05	0.250	4.9	0.00	71.6 ± 30.3
G1087-8	28	2	19	35	9.080E+04	8.626E+05	0.105	6.7	0.20	30.2 ± 22.5
G1087-8	29	10	66	12	1.324E+06	8.740E+06	0.152	68.0	0.78	43.5 ± 14.8
G1087-8	30	4	13	24	2.648E+05	8.607E+05	0.308	6.7	0.39	88.0 ± 50.4
G1087-8	31	6	13	42	2.270E+05	4.919E+05	0.462	3.8	0.43	131.5 ± 65.0
G1087-8	32	4	19	36	1.766E+05	8.387E+05	0.211	6.5	0.27	60.3 ± 33.2
		128	441		2.333E+05	8.036E+05		6.3		

Area of basic unit = 6.293E-07 cm-2
χ^2 = 36.606 with 19 degrees of freedom
P(χ^2) = 0.9%
Age Dispersion = 40.145%

Ns / Ni = 0.290 ± 0.029
Mean Ratio = 0.367 ± 0.058

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.466E+06cm-2 ND = 2318
ρ_D interpolated between top of can: ρ = 1.501E+06cm-2 ND = 1181
bottom of can: ρ = 1.445E+06cm-2 ND = 1137

POOLED AGE = 83.0 ± 8.7 Ma
CENTRAL AGE = 89.6 ± 12.6 Ma



Mean track length 13.30 ± 0.55 µm Std. Dev. 2.95 µm 29 tracks

Figure S2.6: Apatite Fission Track Age Data of sample 453023

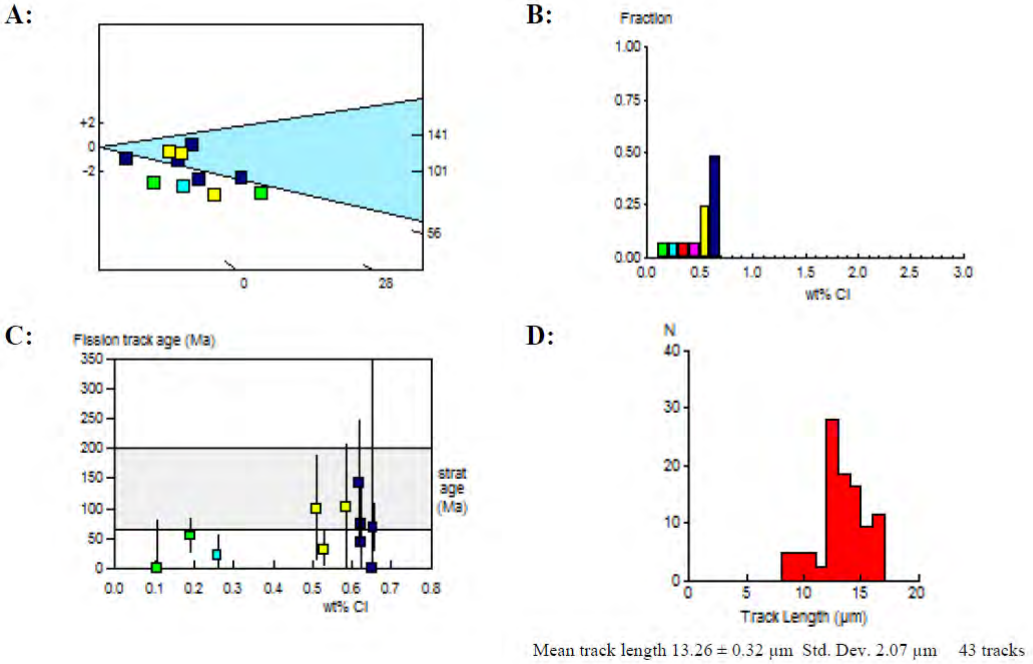
GC1017-5 Apatite 453027
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-9	3	0	3	10	0.000E+00	4.767E+05	0.000	3.7	0.65	0.0 ± 237.0
G1087-9	7	5	19	20	3.973E+05	1.510E+06	0.263	11.8	0.62	75.1 ± 37.8
G1087-9	9	0	12	36	0.000E+00	5.297E+05	0.000	4.1	0.11	0.0 ± 40.4
G1087-9	13	16	85	40	6.356E+05	3.377E+06	0.188	26.4	0.19	53.8 ± 14.7
G1087-9	18	5	33	30	2.648E+05	1.748E+06	0.152	13.6	0.62	43.3 ± 20.8
G1087-9	19	5	14	20	3.973E+05	1.112E+06	0.357	8.7	0.59	101.7 ± 53.0
G1087-9	21	5	46	49	1.621E+05	1.492E+06	0.109	11.6	0.53	31.1 ± 14.7
G1087-9	24	11	22	25	6.992E+05	1.398E+06	0.500	10.9	0.62	141.9 ± 52.6
G1087-9	25	15	63	30	7.945E+05	3.337E+06	0.238	26.0	0.65	68.0 ± 19.6
G1087-9	28	2	26	80	3.973E+04	5.164E+05	0.077	4.0	0.26	22.0 ± 16.2
G1087-9	31	7	20	30	3.708E+05	1.059E+06	0.350	8.3	0.51	99.7 ± 43.9
		71	343		3.049E+05	1.473E+06		11.5		

Area of basic unit = 6.293E-07 cm-2
χ^2 = 16.700 with 10 degrees of freedom
P(χ^2) = 8.1%
Age Dispersion = 28.388% (did not converge)
Ns / Ni = 0.207 ± 0.027
Mean Ratio = 0.203 ± 0.048

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.461E+06cm-2 ND = 2318
ρ_D interpolated between top of can: ρ = 1.501E+06cm-2 ND = 1181
bottom of can: ρ = 1.445E+06cm-2 ND = 1137

POOLED AGE = 59.1 ± 7.9 Ma
CENTRAL AGE = 59.5 ± 9.9 Ma



Mean track length 13.26 ± 0.32 µm Std. Dev. 2.07 µm 43 tracks

Figure S2.7: Apatite Fission Track Age Data of sample 453027

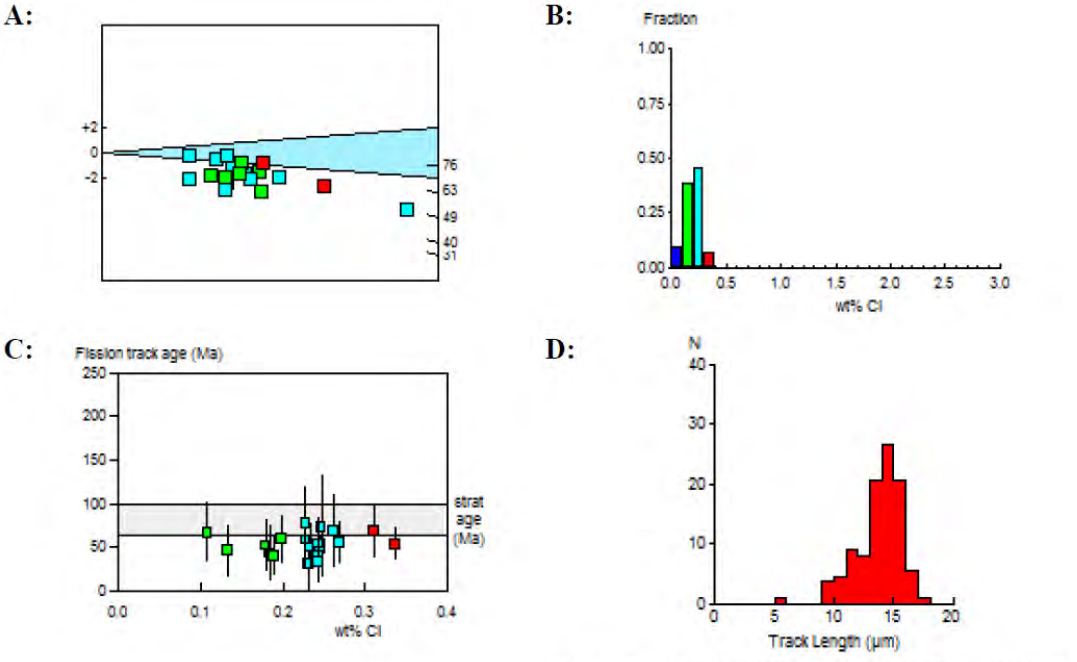
GC1017-6 Apatite 453029
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-10	4	18	97	60	4.767E+05	2.569E+06	0.186	20.1	0.25	52.8 ± 13.6
G1087-10	5	40	212	70	9.080E+05	4.813E+06	0.189	37.7	0.34	53.7 ± 9.4
G1087-10	6	13	54	40	5.164E+05	2.145E+06	0.241	16.8	0.26	68.5 ± 21.2
G1087-10	7	4	36	40	1.589E+05	1.430E+06	0.111	11.2	0.23	31.7 ± 16.7
G1087-10	8	8	31	70	1.816E+05	7.037E+05	0.258	5.5	0.25	73.4 ± 29.2
G1087-10	9	69	404	70	1.566E+06	9.171E+06	0.171	71.8	0.25	48.7 ± 6.5
G1087-10	14	17	89	50	5.403E+05	2.829E+06	0.191	22.2	0.24	54.4 ± 14.5
G1087-10	15	10	71	40	3.973E+05	2.821E+06	0.141	22.1	0.24	40.1 ± 13.6
G1087-10	16	22	106	49	7.135E+05	3.438E+06	0.208	26.9	0.20	59.1 ± 13.9
G1087-10	20	15	73	50	4.767E+05	2.320E+06	0.205	18.2	0.23	58.5 ± 16.7
G1087-10	22	17	97	50	5.403E+05	3.083E+06	0.175	24.1	0.23	49.9 ± 13.2
G1087-10	23	8	53	30	4.238E+05	2.807E+06	0.151	22.0	0.18	43.0 ± 16.4
G1087-10	24	17	63	50	5.403E+05	2.002E+06	0.270	15.7	0.23	76.7 ± 21.1
G1087-10	27	19	81	24	1.258E+06	5.363E+06	0.235	42.0	0.11	66.7 ± 17.1
G1087-10	30	26	134	70	5.902E+05	3.042E+06	0.194	23.8	0.27	55.2 ± 11.9
G1087-10	32	15	82	50	4.767E+05	2.606E+06	0.183	20.4	0.18	52.1 ± 14.7
G1087-10	38	11	68	40	4.370E+05	2.701E+06	0.162	21.2	0.13	46.1 ± 15.0
G1087-10	39	16	114	48	5.297E+05	3.774E+06	0.140	29.6	0.19	40.0 ± 10.7
G1087-10	40	8	70	60	2.119E+05	1.854E+06	0.114	14.5	0.24	32.6 ± 12.2
G1087-10	42	26	108	70	5.902E+05	2.452E+06	0.241	19.2	0.31	68.5 ± 15.1
		379	2043		5.841E+05	3.149E+06		24.7		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 11.405 with 19 degrees of freedom
P(χ^2) = 90.9%
Age Dispersion = 0.014% (did not converge)
Ns / Ni = 0.186 ± 0.010
Mean Ratio = 0.188 ± 0.010

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.456E+06cm^{-2} ND = 2318
ρ_D interpolated between top of can: ρ = 1.501E+06cm^{-2} ND = 1181
bottom of can: ρ = 1.445E+06cm^{-2} ND = 1137

POOLED AGE = 52.8 ± 3.3 Ma
CENTRAL AGE = 52.8 ± 3.3 Ma



Mean track length 13.71 ± 0.18 µm Std. Dev. 1.87 µm 112 tracks

Figure S2.8: Apatite Fission Track Age Data of sample 453029

GC1017-7 Apatite 453030
Counted by: MEM

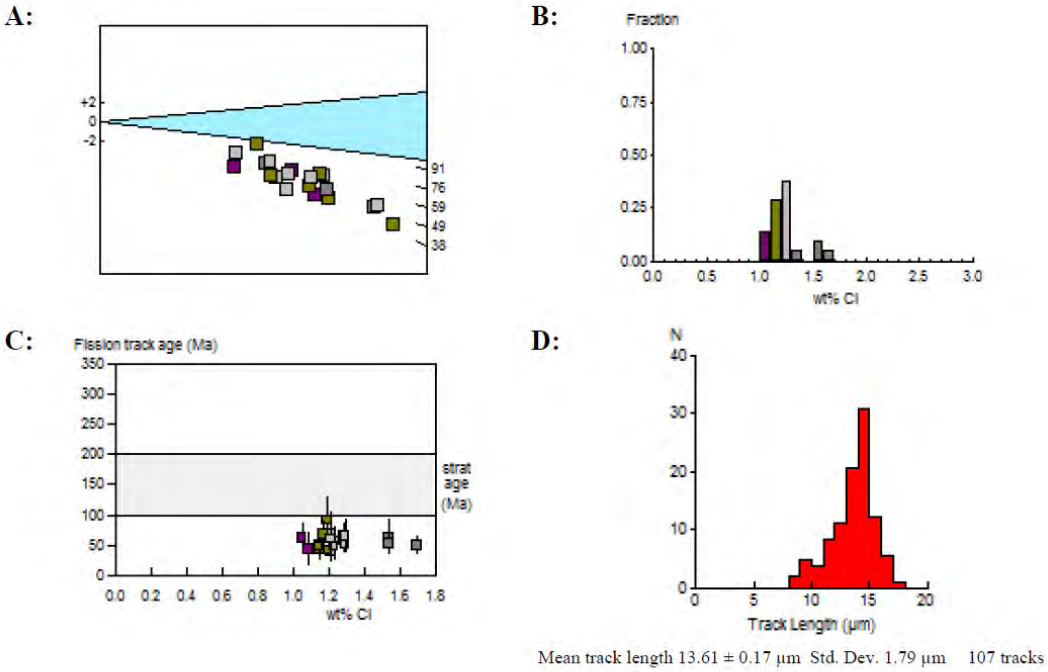
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1087-11	3	46	202	36	2.030E+06	8.916E+06	0.228	70.1	1.29	64.6 ± 10.7
G1087-11	4	30	93	21	2.270E+06	7.037E+06	0.323	55.3	1.19	91.3 ± 19.3
G1087-11	5	33	149	40	1.311E+06	5.919E+06	0.221	46.5	1.05	62.8 ± 12.2
G1087-11	6	21	153	35	9.534E+05	6.946E+06	0.137	54.6	1.21	39.0 ± 9.1
G1087-11	7	30	148	30	1.589E+06	7.839E+06	0.203	61.6	1.23	57.5 ± 11.6
G1087-11	8	18	74	21	1.362E+06	5.600E+06	0.243	44.0	1.22	68.9 ± 18.2
G1087-11	9	25	112	20	1.986E+06	8.899E+06	0.223	69.9	1.54	63.3 ± 14.1
G1087-11	10	55	372	60	1.457E+06	9.852E+06	0.148	77.4	1.20	42.0 ± 6.2
G1087-11	11	35	225	48	1.159E+06	7.449E+06	0.156	58.5	1.15	44.2 ± 8.1
G1087-11	12	30	198	40	1.192E+06	7.866E+06	0.152	61.8	1.09	43.0 ± 8.5
G1087-11	14	22	131	24	1.457E+06	8.674E+06	0.168	68.2	1.23	47.7 ± 11.1
G1087-11	15	12	79	12	1.589E+06	1.046E+07	0.152	82.2	1.08	43.1 ± 13.4
G1087-11	16	40	215	50	1.271E+06	6.833E+06	0.186	53.7	1.54	52.8 ± 9.2
G1087-11	17	55	317	50	1.748E+06	1.007E+07	0.174	79.2	1.70	49.3 ± 7.3
G1087-11	18	33	186	36	1.457E+06	8.210E+06	0.177	64.5	1.15	50.4 ± 9.6
G1087-11	19	21	125	24	1.390E+06	8.276E+06	0.168	65.0	1.14	47.7 ± 11.3
G1087-11	20	46	194	42	1.740E+06	7.340E+06	0.237	57.7	1.17	67.2 ± 11.2
G1087-11	21	59	324	60	1.563E+06	8.581E+06	0.182	67.4	1.29	51.7 ± 7.5
G1087-11	22	27	117	24	1.788E+06	7.747E+06	0.231	60.9	1.29	65.4 ± 14.1
G1087-11	23	39	184	50	1.239E+06	5.848E+06	0.212	46.0	1.21	60.1 ± 10.7
		677	3598		1.488E+06	7.908E+06		62.2		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 22.304 with 19 degrees of freedom
P(χ^2) = 26.9%
Age Dispersion = 5.483% (did not converge)

Ns / Ni = 0.188 ± 0.008
Mean Ratio = 0.196 ± 0.010

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.451E+06cm^{-2} ND = 2318
ρ_D interpolated between top of can: ρ = 1.501E+06cm^{-2} ND = 1181
bottom of can: ρ = 1.445E+06cm^{-2} ND = 1137

POOLED AGE = 53.4 ± 2.7 Ma
CENTRAL AGE = 53.6 ± 2.8 Ma



Mean track length 13.61 ± 0.17 µm Std. Dev. 1.79 µm 107 tracks

Figure S2.9: Apatite Fission Track Age Data of sample 453030

GC1129-1 Apatite 538701
Counted by: COB

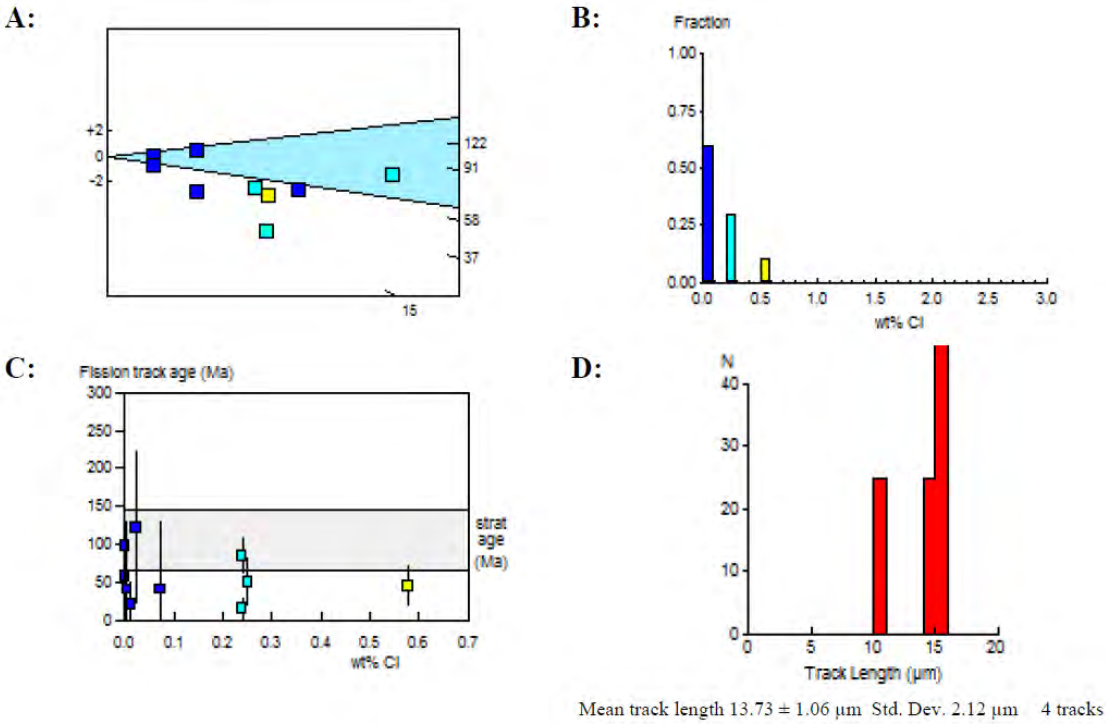
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-1	3	13	71	25	8.263E+05	4.513E+06	0.183	39.6	0.58	45.1 ± 13.7
G1219-1	4	23	96	32	1.142E+06	4.767E+06	0.240	41.8	0.00	58.9 ± 13.8
G1219-1	5	5	78	9	8.828E+05	1.377E+07	0.064	120.8	0.24	15.8 ± 7.3
G1219-1	6	2	5	21	1.513E+05	3.783E+05	0.400	3.3	0.00	98.1 ± 82.1
G1219-1	7	12	59	40	4.767E+05	2.344E+06	0.203	20.6	0.25	50.1 ± 15.9
G1219-1	8	2	24	18	1.766E+05	2.119E+06	0.083	18.6	0.01	20.6 ± 15.1
G1219-1	9	9	18	20	7.151E+05	1.430E+06	0.500	12.5	0.02	122.4 ± 50.1
G1219-1	10	1	6	10	1.589E+05	9.534E+05	0.167	8.4	0.07	41.1 ± 44.4
G1219-1	11	1	6	28	5.675E+04	3.405E+05	0.167	3.0	0.00	41.1 ± 44.4
G1219-1	12	68	197	50	2.161E+06	6.261E+06	0.345	54.9	0.24	84.7 ± 12.1
		136	560		8.542E+05	3.517E+06		30.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 23.356 with 9 degrees of freedom
P(χ^2) = 0.5%
Age Dispersion = 40.234%

Ns / Ni = 0.243 ± 0.023
Mean Ratio = 0.235 ± 0.044

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.299E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 59.7 ± 5.9 Ma
CENTRAL AGE = 53.8 ± 10.0 Ma



Mean track length 13.73 ± 1.06 µm Std. Dev. 2.12 µm 4 tracks

660

661 **Figure S2.10:** Apatite Fission Track Age Data of sample 538701

GC1129-3 Apatite 538301
Counted by: MEM

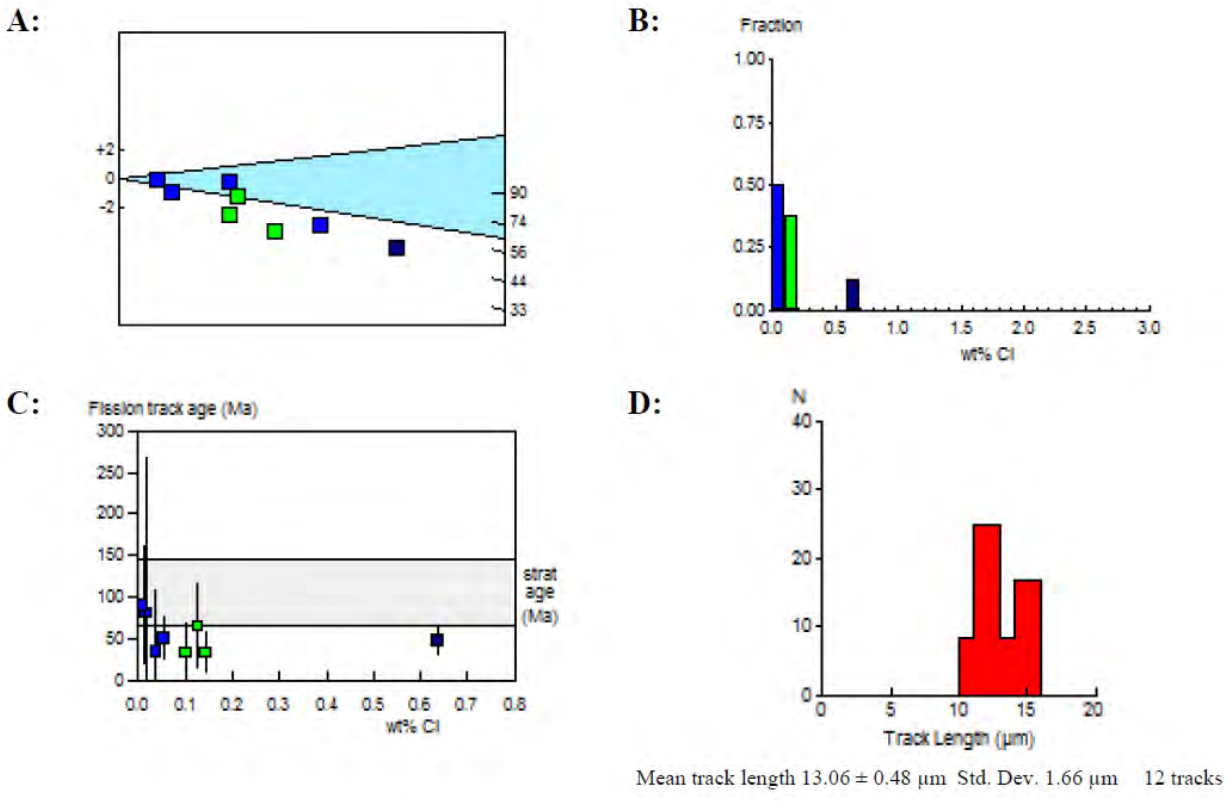
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-1	3	4	29	25	2.543E+05	1.843E+06	0.138	17.0	0.10	33.5 ± 17.9
G1224-1	4	1	7	12	1.324E+05	9.270E+05	0.143	8.5	0.04	34.7 ± 37.1
G1224-1	6	1	3	9	1.766E+05	5.297E+05	0.333	4.9	0.02	80.7 ± 93.2
G1224-1	7	19	91	36	8.387E+05	4.017E+06	0.209	36.9	0.05	50.6 ± 12.9
G1224-1	8	9	24	20	7.151E+05	1.907E+06	0.375	17.5	0.01	90.7 ± 35.5
G1224-1	9	8	30	6	2.119E+06	7.945E+06	0.267	73.1	0.13	64.6 ± 25.8
G1224-1	10	8	58	20	6.356E+05	4.608E+06	0.138	42.4	0.14	33.5 ± 12.7
G1224-1	13	34	175	30	1.801E+06	9.270E+06	0.194	85.3	0.64	47.1 ± 8.9
		84	417		8.448E+05	4.194E+06		38.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 4.997 with 7 degrees of freedom
P(χ^2) = 66.0%
Age Dispersion = 0.006% (did not converge)

Ns / Ni = 0.201 ± 0.024
Mean Ratio = 0.225 ± 0.032

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.239E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can; ρ = 1.239E+06cm^{-2} ND = 975
bottom of can; ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 48.9 ± 6.0 Ma
CENTRAL AGE = 48.9 ± 6.0 Ma



Mean track length 13.06 ± 0.48 µm Std. Dev. 1.66 µm 12 tracks

Figure S2.11: Apatite Fission Track Age Data of sample 538301

GC1129-5 Apatite 538703
Counted by: COB

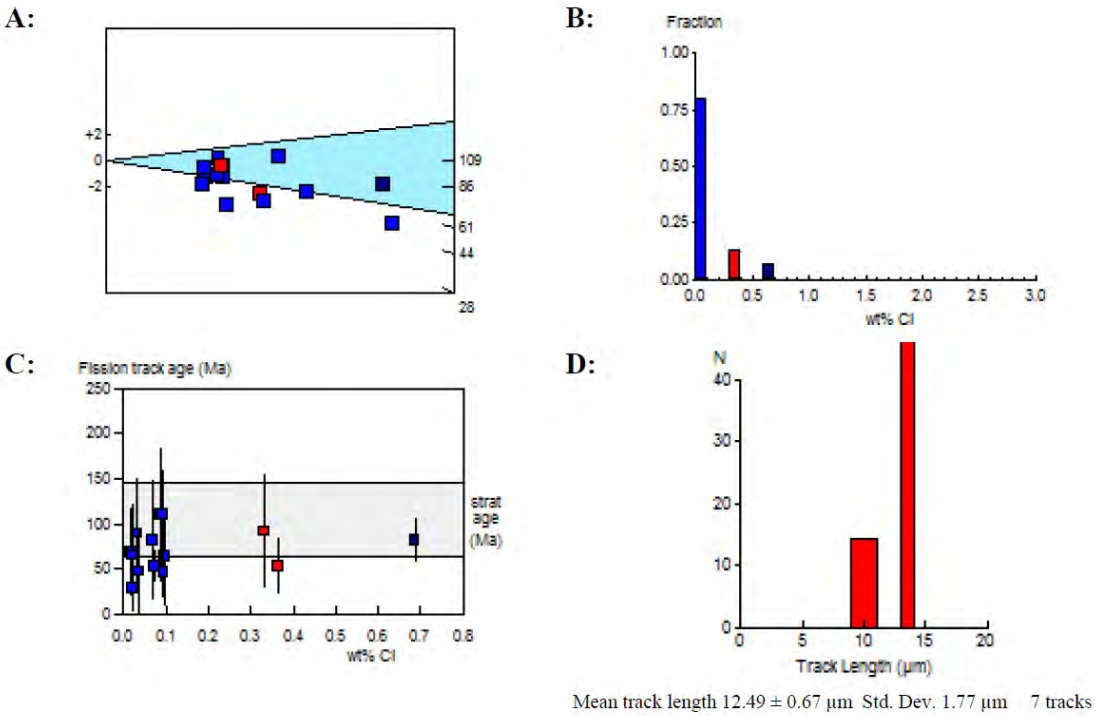
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-2	3	7	27	30	3.708E+05	1.430E+06	0.259	12.6	0.10	63.7 ± 27.1
G1219-2	4	14	65	54	4.120E+05	1.913E+06	0.215	16.8	0.37	53.0 ± 15.7
G1219-2	5	13	29	30	6.886E+05	1.536E+06	0.448	13.5	0.09	109.7 ± 36.7
G1219-2	6	10	36	18	8.828E+05	3.178E+06	0.278	27.9	0.02	68.2 ± 24.5
G1219-2	8	12	33	9	2.119E+06	5.827E+06	0.364	51.2	0.03	89.2 ± 30.1
G1219-2	9	31	69	18	2.737E+06	6.091E+06	0.449	53.5	0.09	110.0 ± 24.0
G1219-2	10	9	32	30	4.767E+05	1.695E+06	0.281	14.9	0.02	69.1 ± 26.1
G1219-2	11	28	105	60	7.416E+05	2.781E+06	0.267	24.4	0.02	65.5 ± 14.0
G1219-3	3	5	43	27	2.943E+05	2.531E+06	0.116	22.2	0.02	28.6 ± 13.5
G1219-3	4	8	24	12	1.059E+06	3.178E+06	0.333	27.9	0.07	81.7 ± 33.4
G1219-3	5	12	32	8	2.384E+06	6.356E+06	0.375	55.9	0.33	91.8 ± 31.2
G1219-3	6	64	191	40	2.543E+06	7.588E+06	0.335	66.7	0.69	82.1 ± 12.1
G1219-3	7	48	223	16	4.767E+06	2.215E+07	0.215	194.7	0.07	52.9 ± 8.5
G1219-3	8	5	26	18	4.414E+05	2.295E+06	0.192	20.2	0.04	47.3 ± 23.1
G1219-3	9	13	69	30	6.886E+05	3.655E+06	0.188	32.1	0.09	46.3 ± 14.1
		279	1005		1.108E+06	3.991E+06		35.1		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 19.742 with 14 degrees of freedom
P(χ^2) = 13.8%
Age Dispersion = 16.530%

Ns / Ni = 0.278 ± 0.019
Mean Ratio = 0.288 ± 0.025

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.298E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 68.2 ± 5.0 Ma
CENTRAL AGE = 68.6 ± 6.1 Ma



Mean track length 12.49 ± 0.67 µm Std. Dev. 1.77 µm 7 tracks

Figure S2.12: Apatite Fission Track Age Data of sample 538703

GC1129-7 Apatite 538706
Counted by: COB

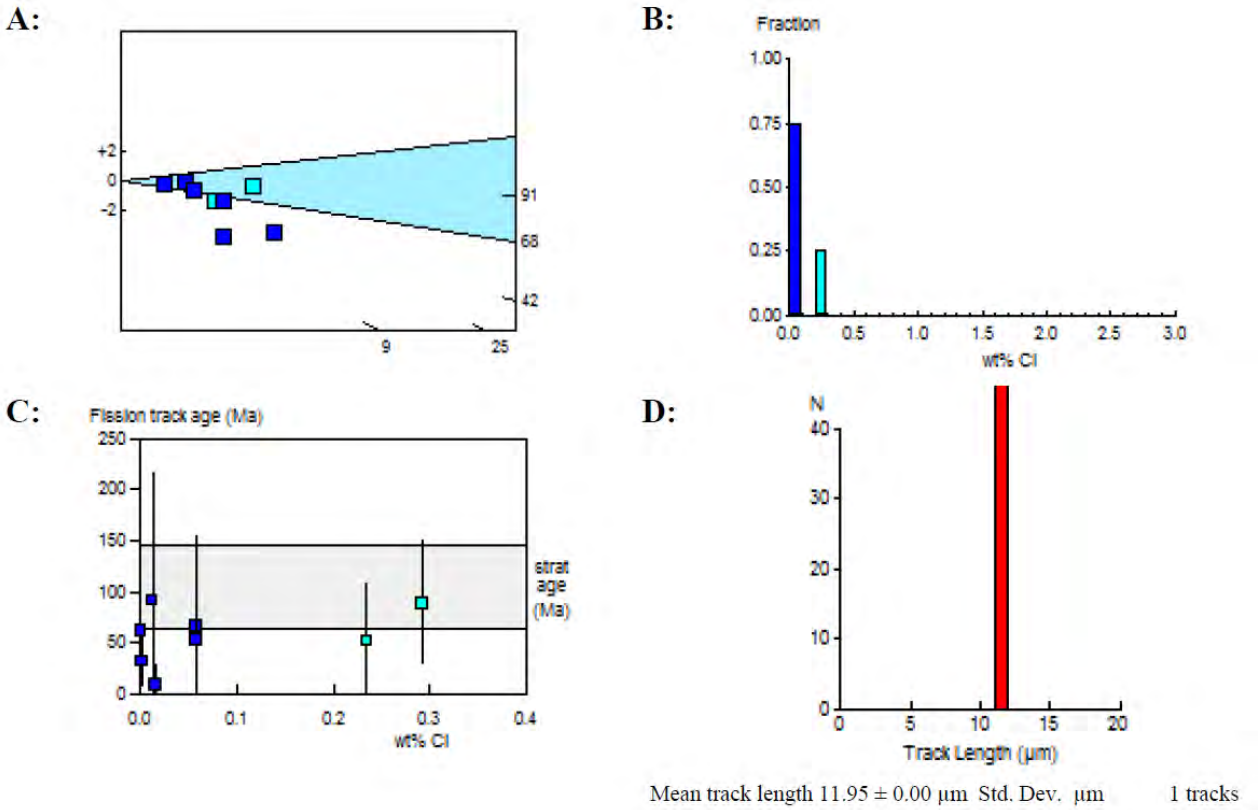
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-4	3	3	8	30	1.589E+05	4.238E+05	0.375	3.7	0.01	91.7 ± 62.2
G1219-4	5	1	4	15	1.059E+05	4.238E+05	0.250	3.7	0.00	61.3 ± 68.6
G1219-4	7	3	11	24	1.986E+05	7.283E+05	0.273	6.4	0.06	66.8 ± 43.6
G1219-4	8	12	33	21	9.080E+05	2.497E+06	0.364	22.0	0.29	89.0 ± 30.1
G1219-4	9	4	19	18	3.531E+05	1.677E+06	0.211	14.8	0.24	51.7 ± 28.5
G1219-5	3	1	27	16	9.932E+04	2.682E+06	0.037	23.6	0.02	9.1 ± 9.3
G1219-5	5	7	54	16	6.952E+05	5.363E+06	0.130	47.2	0.00	31.8 ± 12.8
G1219-5	6	5	23	16	4.966E+05	2.284E+06	0.217	20.1	0.06	53.3 ± 26.3
		36	179		3.667E+05	1.824E+06		16.1		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 9.053 with 7 degrees of freedom
P(χ^2) = 24.9%
Age Dispersion = 29.081%

Ns / Ni = 0.201 ± 0.037
Mean Ratio = 0.232 ± 0.040

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.295E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 49.3 ± 9.1 Ma
CENTRAL AGE = 50.2 ± 10.9 Ma



Mean track length 11.95 ± 0.00 µm Std. Dev. µm 1 tracks

Figure S2.13: Apatite Fission Track Age Data of sample 538706

GC1129-8 Apatite 538313
Counted by: COB

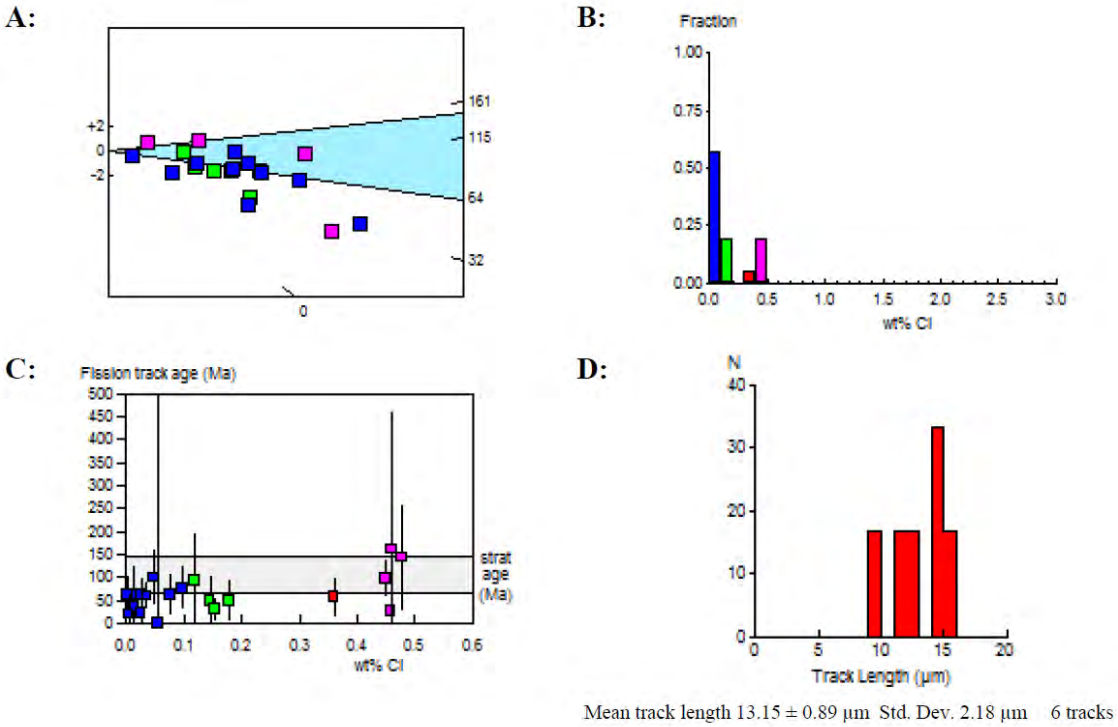
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-6	3	10	17	9	1.766E+06	3.002E+06	0.588	26.5	0.48	143.0 ± 57.1
G1219-6	4	9	39	14	1.022E+06	4.427E+06	0.231	39.0	0.36	56.5 ± 20.9
G1219-6	5	4	20	8	7.945E+05	3.973E+06	0.200	35.0	0.15	49.0 ± 26.9
G1219-6	6	15	37	18	1.324E+06	3.266E+06	0.405	28.8	0.05	98.9 ± 30.4
G1219-6	7	5	20	32	2.483E+05	9.932E+05	0.250	8.8	0.01	61.2 ± 30.6
G1219-6	9	6	30	30	3.178E+05	1.589E+06	0.200	14.0	0.18	49.0 ± 22.0
G1219-6	10	0	2	6	0.000E+00	5.297E+05	0.000	4.7	0.06	0.0 ± 400.9
G1219-6	11	15	145	50	4.767E+05	4.608E+06	0.103	40.6	0.46	25.4 ± 6.9
G1219-6	12	15	58	25	9.534E+05	3.687E+06	0.259	32.5	0.00	63.3 ± 18.4
G1219-6	13	15	48	70	3.405E+05	1.090E+06	0.313	9.6	0.10	76.4 ± 22.7
G1219-6	14	35	88	42	1.324E+06	3.329E+06	0.398	29.4	0.45	97.1 ± 19.6
G1219-6	15	7	58	42	2.648E+05	2.194E+06	0.121	19.3	0.15	29.6 ± 11.9
G1219-6	16	26	177	36	1.148E+06	7.813E+06	0.147	68.9	0.01	36.0 ± 7.6
G1219-6	18	15	60	20	1.192E+06	4.767E+06	0.250	42.0	0.03	61.2 ± 17.7
G1219-6	19	1	12	20	7.945E+04	9.534E+05	0.083	8.4	0.00	20.5 ± 21.3
G1219-6	20	5	13	10	7.945E+05	2.066E+06	0.385	18.2	0.12	93.9 ± 49.5
G1219-6	22	23	94	21	1.740E+06	7.113E+06	0.245	62.7	0.04	59.9 ± 14.0
G1219-6	23	2	3	8	3.973E+05	5.959E+05	0.667	5.3	0.46	161.9 ± 147.8
G1219-6	24	5	57	24	3.311E+05	3.774E+06	0.088	33.3	0.03	21.5 ± 10.1
G1219-6	25	10	40	28	5.675E+05	2.270E+06	0.250	20.0	0.08	61.2 ± 21.7
		223	1018		6.908E+05	3.153E+06		27.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 44.820 with 19 degrees of freedom
P(χ^2) = 0.1%
Age Dispersion = 34.680%

Ns / Ni = 0.219 ± 0.016
Mean Ratio = 0.259 ± 0.037

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.293E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can; ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can; ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 53.6 ± 4.2 Ma
CENTRAL AGE = 57.3 ± 6.8 Ma



Mean track length 13.15 ± 0.89 µm Std. Dev. 2.18 µm 6 tracks

Figure S2.14: Apatite Fission Track Age Data of sample 538313

GC1129-11 Apatite 538323
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-2	6	2	2	30	1.059E+05	1.059E+05	1.000	1.0	0.00	240.3 ± 240.4
G1224-2	7	0	2	24	0.000E+00	1.324E+05	0.000	1.2	0.00	0.0 ± 399.2
G1224-2	14	1	3	20	7.945E+04	2.384E+05	0.333	2.2	0.00	81.1 ± 93.7
G1224-2	19	5	45	30	2.648E+05	2.384E+06	0.111	21.8	0.13	27.1 ± 12.8
G1224-2	20	1	1	20	7.945E+04	7.945E+04	1.000	0.7	0.01	240.3 ± 339.9
		9	53		1.153E+05	6.792E+05		6.2		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 7.604 with 4 degrees of freedom
P(χ^2) = 10.7%
Age Dispersion = 45.624% (did not converge)
Ns / Ni = 0.170 ± 0.061
Mean Ratio = 0.489 ± 0.215

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.246E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975
bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 41.4 ± 15.0 Ma
CENTRAL AGE = 54.9 ± 25.5 Ma

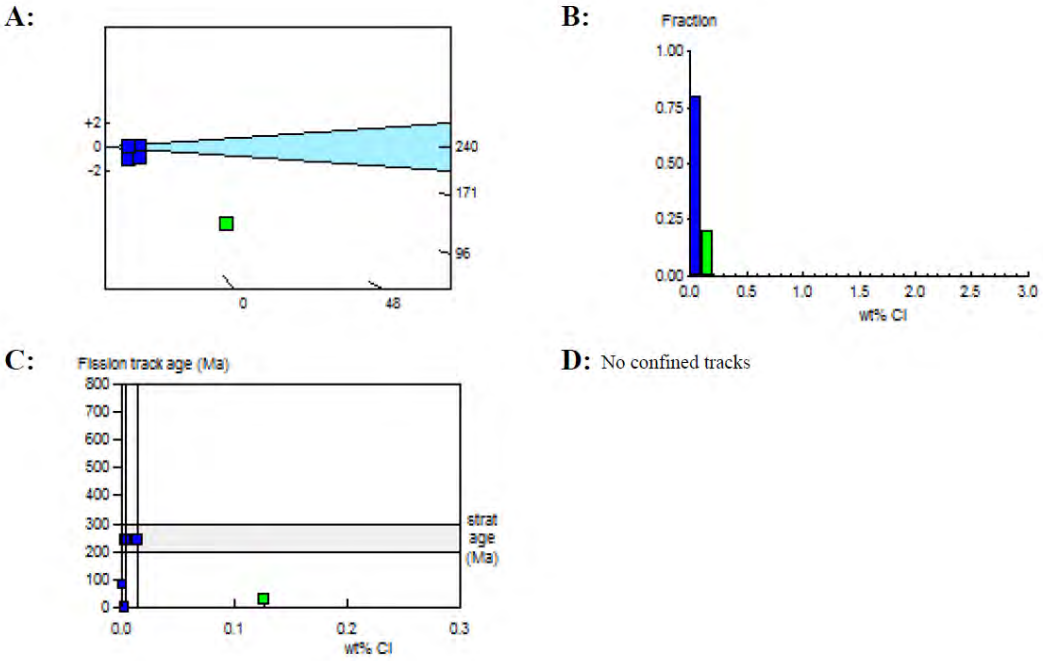


Figure S2.15: Apatite Fission Track Age Data of sample 538323

GC1129-12 Apatite 538330
Counted by: MEM

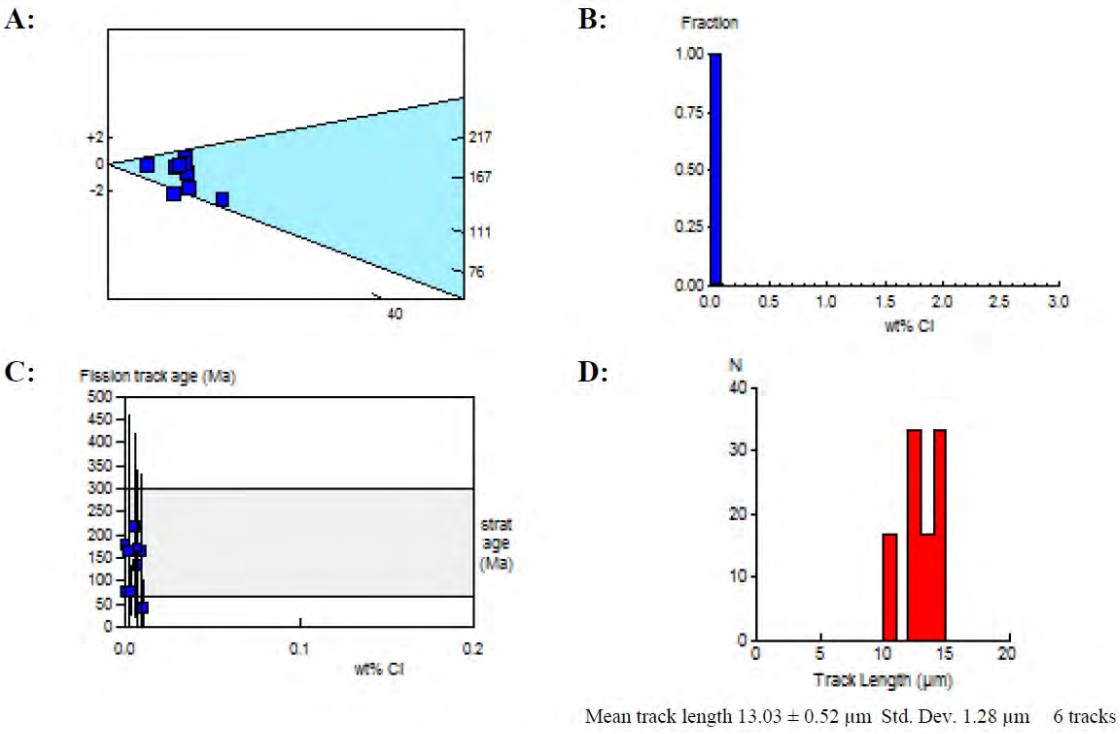
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-3	3	7	13	20	5.562E+05	1.033E+06	0.538	9.4	0.01	131.2 ± 61.6
G1224-3	4	9	10	12	1.192E+06	1.324E+06	0.900	12.0	0.01	217.8 ± 100.3
G1224-3	9	8	11	30	4.238E+05	5.827E+05	0.727	5.3	0.00	176.6 ± 82.2
G1224-3	10	5	16	12	6.621E+05	2.119E+06	0.313	19.3	0.00	76.5 ± 39.2
G1224-3	11	2	12	12	2.648E+05	1.589E+06	0.167	14.5	0.01	40.9 ± 31.3
G1224-3	13	6	9	30	3.178E+05	4.767E+05	0.667	4.3	0.01	162.0 ± 85.5
G1224-3	14	2	3	10	3.178E+05	4.767E+05	0.667	4.3	0.00	162.0 ± 148.0
G1224-3	15	10	32	40	3.973E+05	1.271E+06	0.313	11.6	0.00	76.5 ± 27.8
G1224-3	17	7	10	25	4.449E+05	6.356E+05	0.700	5.8	0.01	170.0 ± 83.9
		56	116		4.659E+05	9.651E+05		8.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 8.145 with 8 degrees of freedom
P(χ^2) = 41.9%
Age Dispersion = 10.339% (did not converge)

Ns / Ni = 0.483 ± 0.079
Mean Ratio = 0.555 ± 0.080

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.253E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975
bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 117.7 ± 19.5 Ma
CENTRAL AGE = 118.4 ± 20.1 Ma



Mean track length 13.03 ± 0.52 µm Std. Dev. 1.28 µm 6 tracks

Figure S2.16: Apatite Fission Track Age Data of sample 538330

GC1129-13 Apatite 538332
Counted by: COB

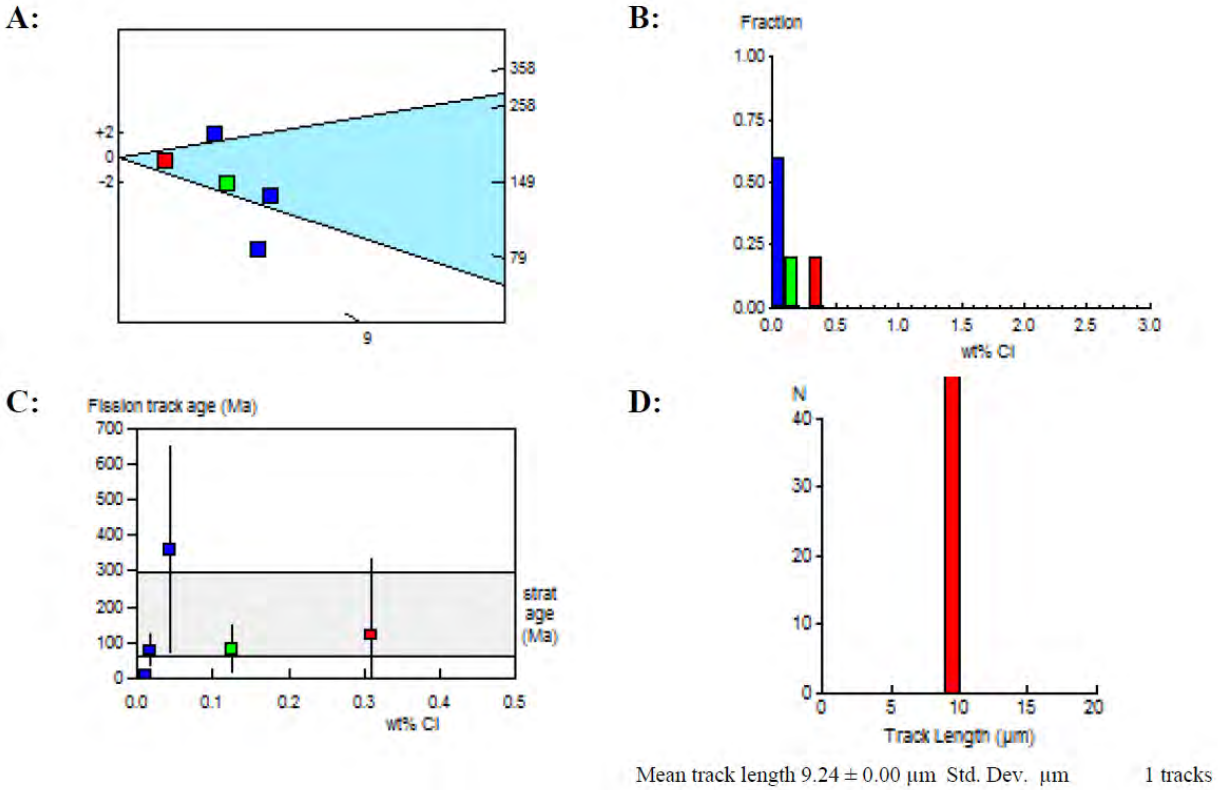
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-7	3	8	24	20	6.356E+05	1.907E+06	0.333	16.8	0.13	81.4 ± 33.3
G1219-7	4	15	47	63	3.783E+05	1.185E+06	0.319	10.5	0.02	77.9 ± 23.2
G1219-7	5	2	51	24	1.324E+05	3.377E+06	0.039	29.8	0.01	9.6 ± 6.9
G1219-7	6	15	10	28	8.513E+05	5.675E+05	1.500	5.0	0.04	358.3 ± 146.6
G1219-7	7	2	4	20	1.589E+05	3.178E+05	0.500	2.8	0.31	121.7 ± 105.4
		42	136		4.306E+05	1.394E+06		12.3		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 30.292 with 4 degrees of freedom
P(χ^2) = 0.0%
Age Dispersion = 84.918%

Ns / Ni = 0.309 ± 0.055
Mean Ratio = 0.538 ± 0.252

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.292E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can; ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can; ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 75.4 ± 13.5 Ma
CENTRAL AGE = 95.8 ± 40.9 Ma



Mean track length 9.24 ± 0.00 µm Std. Dev. µm 1 tracks

Figure S2.17: Apatite Fission Track Age Data of sample 538332

GC1129-14 Apatite 538337
Counted by: MEM

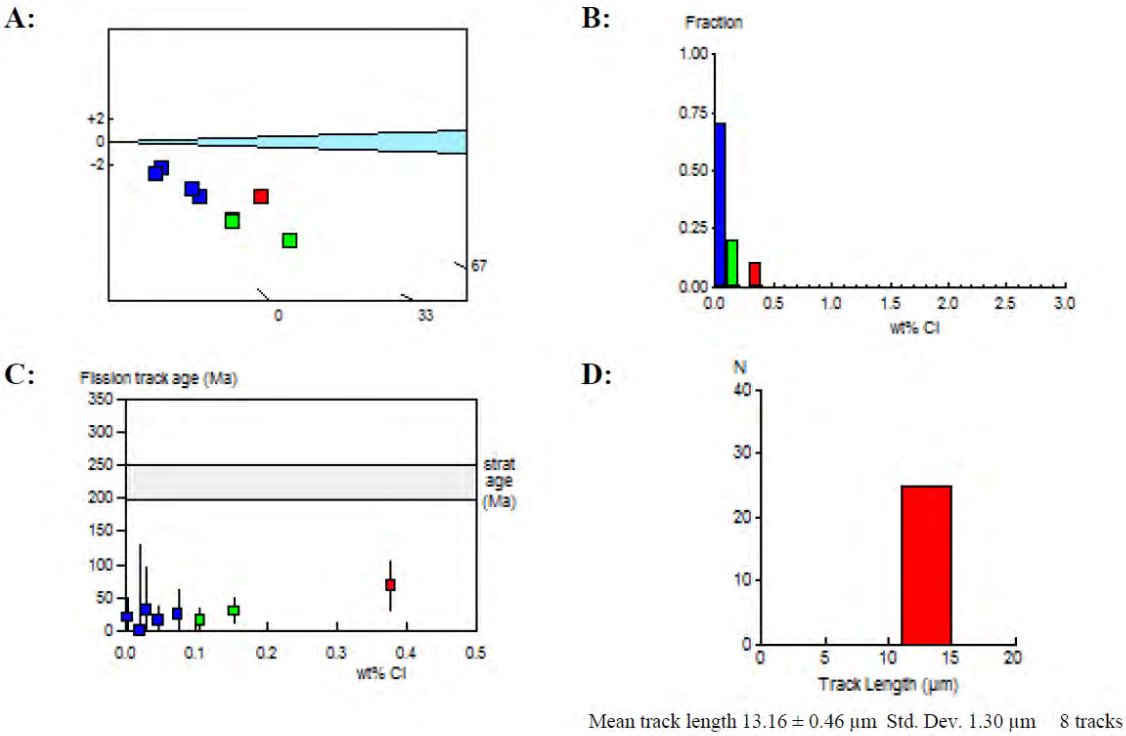
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-4	3	11	92	30	5.827E+05	4.873E+06	0.120	44.1	0.15	29.5 ± 9.5
G1224-4	4	1	8	18	8.828E+04	7.063E+05	0.125	6.4	0.03	30.9 ± 32.7
G1224-4	6	16	58	20	1.271E+06	4.608E+06	0.276	41.7	0.38	67.9 ± 19.3
G1224-4	7	2	25	15	2.119E+05	2.648E+06	0.080	24.0	0.00	19.8 ± 14.5
G1224-4	15	3	45	20	2.384E+05	3.575E+06	0.067	32.4	0.05	16.5 ± 9.8
G1224-4	16	3	46	16	2.980E+05	4.569E+06	0.065	41.3	0.11	16.1 ± 9.6
G1224-4	30	2	20	15	2.119E+05	2.119E+06	0.100	19.2	0.07	24.7 ± 18.3
G1224-4	31	0	7	6	0.000E+00	1.854E+06	0.000	16.8	0.02	0.0 ± 65.4
		38	301		4.313E+05	3.416E+06		30.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 11.925 with 7 degrees of freedom
P(χ^2) = 10.3%
Age Dispersion = 44.295%

Ns / Ni = 0.126 ± 0.022
Mean Ratio = 0.104 ± 0.028

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.260E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975
bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 31.2 ± 5.4 Ma
CENTRAL AGE = 28.9 ± 7.3 Ma



Mean track length 13.16 ± 0.46 µm Std. Dev. 1.30 µm 8 tracks

Figure S2.17: Apatite Fission Track Age Data of sample 538337

GC1129-15 Apatite 538338
Counted by: COB

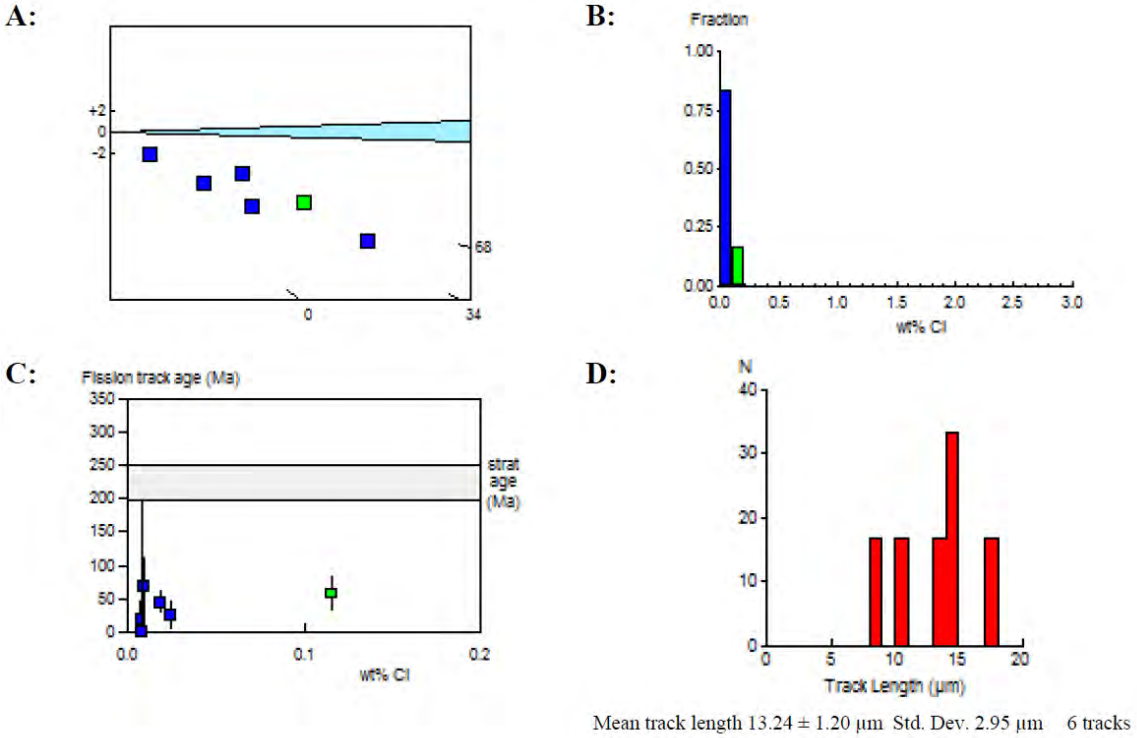
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-8	4	31	173	45	1.095E+06	6.109E+06	0.179	54.0	0.02	43.8 ± 8.6
G1219-8	6	22	94	20	1.748E+06	7.469E+06	0.234	66.0	0.12	57.2 ± 13.6
G1219-8	7	12	43	9	2.119E+06	7.592E+06	0.279	67.1	0.01	68.1 ± 22.3
G1219-8	8	6	57	25	3.814E+05	3.623E+06	0.105	32.0	0.02	25.8 ± 11.1
G1219-9	4	2	26	9	3.531E+05	4.591E+06	0.077	40.6	0.01	18.8 ± 13.8
G1219-9	6	0	5	9	0.000E+00	8.828E+05	0.000	7.8	0.01	0.0 ± 99.1
		73	398		9.915E+05	5.406E+06		47.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 6.888 with 5 degrees of freedom
P(χ^2) = 22.9%
Age Dispersion = 3.075% (did not converge)

Ns / Ni = 0.183 ± 0.023
Mean Ratio = 0.146 ± 0.043

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.290E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 44.9 ± 5.8 Ma
CENTRAL AGE = 44.8 ± 5.9 Ma



Mean track length 13.24 ± 1.20 µm Std. Dev. 2.95 µm 6 tracks

Figure S2.18: Apatite Fission Track Age Data of sample 538338

GC1129-16 Apatite 538342
Counted by: COB

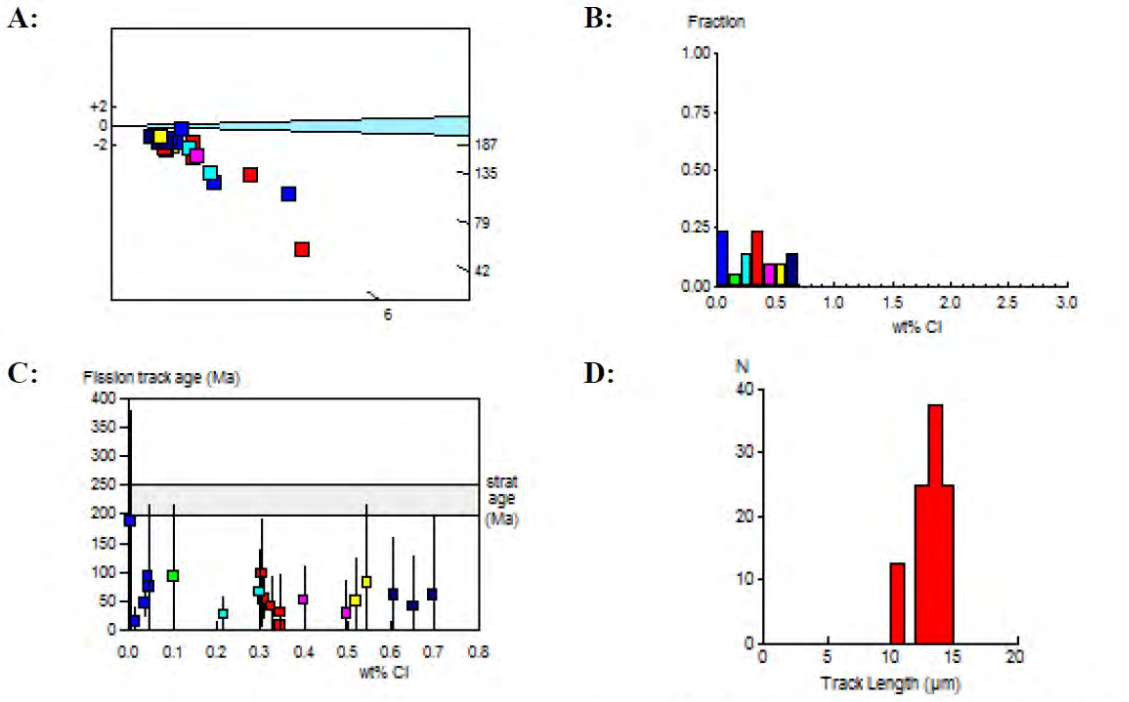
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-10	3	2	32	42	7.567E+04	1.211E+06	0.063	10.7	0.01	15.3 ± 11.2
G1219-10	4	1	4	25	6.356E+04	2.543E+05	0.250	2.3	0.70	60.9 ± 68.2
G1219-10	5	11	50	40	4.370E+05	1.986E+06	0.220	17.6	0.31	53.7 ± 17.9
G1219-10	6	2	10	28	1.135E+05	5.675E+05	0.200	5.0	0.52	48.8 ± 37.8
G1219-10	7	3	111	100	4.767E+04	1.764E+06	0.027	15.6	0.35	6.6 ± 3.9
G1219-10	8	1	9	9	1.766E+05	1.589E+06	0.111	14.1	0.50	27.2 ± 28.6
G1219-10	9	3	8	14	3.405E+05	9.080E+05	0.375	8.0	0.10	91.2 ± 61.8
G1219-10	10	3	28	40	1.192E+05	1.112E+06	0.107	9.8	0.22	26.2 ± 15.9
G1219-10	11	1	6	20	7.945E+04	4.767E+05	0.167	4.2	0.65	40.7 ± 44.0
G1219-10	12	16	84	60	4.238E+05	2.225E+06	0.190	19.7	0.03	46.5 ± 12.7
G1219-10	13	3	10	40	1.192E+05	3.973E+05	0.300	3.5	0.04	73.1 ± 48.1
G1219-10	14	3	8	24	1.986E+05	5.297E+05	0.375	4.7	0.04	91.2 ± 61.8
G1219-10	15	1	8	36	4.414E+04	3.531E+05	0.125	3.1	0.35	30.5 ± 32.4
G1219-10	16	7	9	24	4.635E+05	5.959E+05	0.778	5.3	0.00	187.7 ± 94.7
G1219-10	17	3	18	20	2.384E+05	1.430E+06	0.167	12.7	0.33	40.7 ± 25.4
G1219-10	18	6	15	40	2.384E+05	5.959E+05	0.400	5.3	0.30	97.2 ± 47.0
G1219-10	19	4	15	24	2.648E+05	9.932E+05	0.267	8.8	0.30	65.0 ± 36.6
G1219-10	20	4	19	40	1.589E+05	7.548E+05	0.211	6.7	0.40	51.4 ± 28.3
G1219-10	21	2	8	15	2.119E+05	8.475E+05	0.250	7.5	0.61	60.9 ± 48.2
G1219-10	22	2	6	50	6.356E+04	1.907E+05	0.333	1.7	0.55	81.1 ± 66.3
		78	458		1.794E+05	1.053E+06		9.3		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 36.502 with 19 degrees of freedom
P(χ^2) = 0.9%
Age Dispersion = 44.681%

Ns / Ni = 0.170 ± 0.021
Mean Ratio = 0.246 ± 0.036

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.288E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 41.6 ± 5.2 Ma
CENTRAL AGE = 49.5 ± 8.4 Ma



Mean track length 13.06 ± 0.46 µm Std. Dev. 1.29 µm 8 tracks

Figure S2.19: Apatite Fission Track Age Data of sample 538342

GC1129-18 Apatite 538345
Counted by: MEM

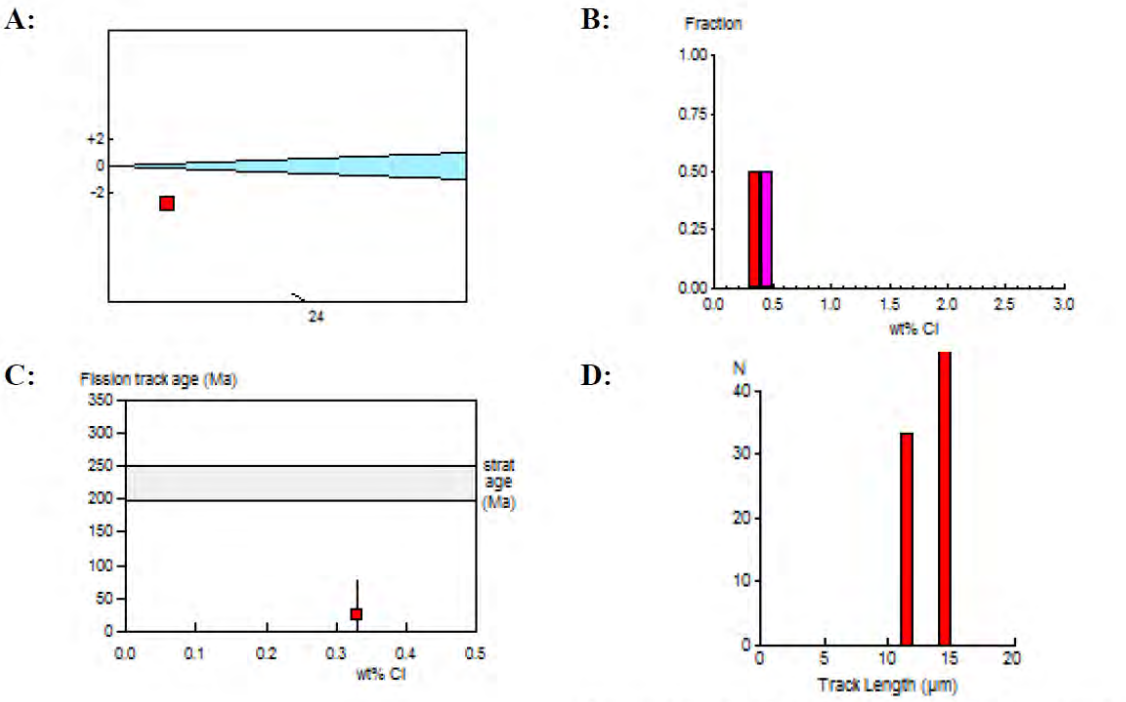
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-5	9	1	10	12	1.324E+05	1.324E+06	0.100	11.9	0.33	24.8 ± 26.1
		1	10		1.324E+05	1.324E+06		11.9		

Area of basic unit = 6.293E-07 cm^{-2}

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.266E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975
bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

Ns / Ni = 0.100 ± 0.105

POOLED AGE = **24.8 ± 26.1 Ma**



Mean track length 13.59 ± 1.24 µm Std. Dev. 2.14 µm 3 tracks

Figure S2.20: Apatite Fission Track Age Data of sample 538345

GC1129-21 Apatite 538356
Counted by: COB

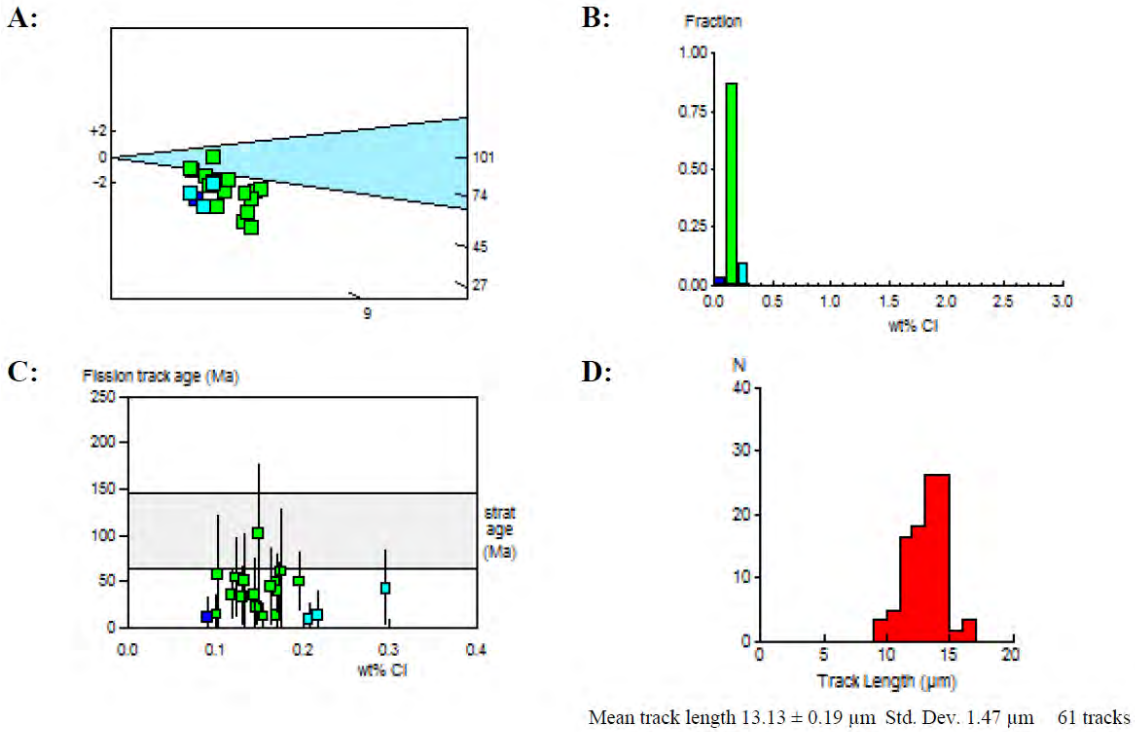
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-11	3	3	53	35	1.362E+05	2.406E+06	0.057	21.3	0.17	13.8 ± 8.2
G1219-11	4	11	56	80	2.185E+05	1.112E+06	0.196	9.9	0.17	47.9 ± 15.8
G1219-11	5	12	59	50	3.814E+05	1.875E+06	0.203	16.6	0.20	49.6 ± 15.8
G1219-11	6	8	54	42	3.027E+05	2.043E+06	0.148	18.1	0.12	36.1 ± 13.7
G1219-11	7	5	36	40	1.986E+05	1.430E+06	0.139	12.7	0.13	33.9 ± 16.2
G1219-11	8	8	36	32	3.973E+05	1.788E+06	0.222	15.8	0.12	54.1 ± 21.2
G1219-11	9	4	17	24	2.648E+05	1.126E+06	0.235	10.0	0.10	57.3 ± 31.9
G1219-11	12	1	22	28	5.675E+04	1.249E+06	0.045	11.1	0.09	11.1 ± 11.4
G1219-11	13	5	55	40	1.986E+05	2.185E+06	0.091	19.4	0.15	22.2 ± 10.4
G1219-11	14	5	24	40	1.986E+05	9.534E+05	0.208	8.4	0.13	50.8 ± 25.0
G1219-11	15	4	16	16	3.973E+05	1.589E+06	0.250	14.1	0.18	60.9 ± 34.1
G1219-11	16	10	24	20	7.945E+05	1.907E+06	0.417	16.9	0.15	101.2 ± 38.2
G1219-11	17	8	49	48	2.648E+05	1.622E+06	0.163	14.4	0.17	39.8 ± 15.2
G1219-11	18	5	28	45	1.766E+05	9.888E+05	0.179	8.8	0.16	43.5 ± 21.2
G1219-11	19	1	19	32	4.966E+04	9.435E+05	0.053	8.4	0.22	12.9 ± 13.2
G1219-11	20	3	60	56	8.513E+04	1.703E+06	0.050	15.1	0.16	12.2 ± 7.2
G1219-11	21	4	27	40	1.589E+05	1.073E+06	0.148	9.5	0.15	36.1 ± 19.4
G1219-11	22	2	34	27	1.177E+05	2.001E+06	0.059	17.7	0.10	14.4 ± 10.5
G1219-11	23	1	27	54	2.943E+04	7.945E+05	0.037	7.0	0.21	9.1 ± 9.2
G1219-11	27	5	29	45	1.766E+05	1.024E+06	0.172	9.1	0.30	42.1 ± 20.4
		105	725		2.101E+05	1.451E+06		12.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 28.100 with 19 degrees of freedom
P(χ^2) = 8.2%
Age Dispersion = 29.147%

Ns / Ni = 0.145 ± 0.015
Mean Ratio = 0.154 ± 0.021

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.287E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 35.3 ± 3.8 Ma
CENTRAL AGE = 35.5 ± 4.5 Ma



Mean track length 13.13 ± 0.19 µm Std. Dev. 1.47 µm 61 tracks

Figure S2.21: Apatite Fission Track Age Data of sample 538356

GC1129-23 Apatite 538802
Counted by: COB

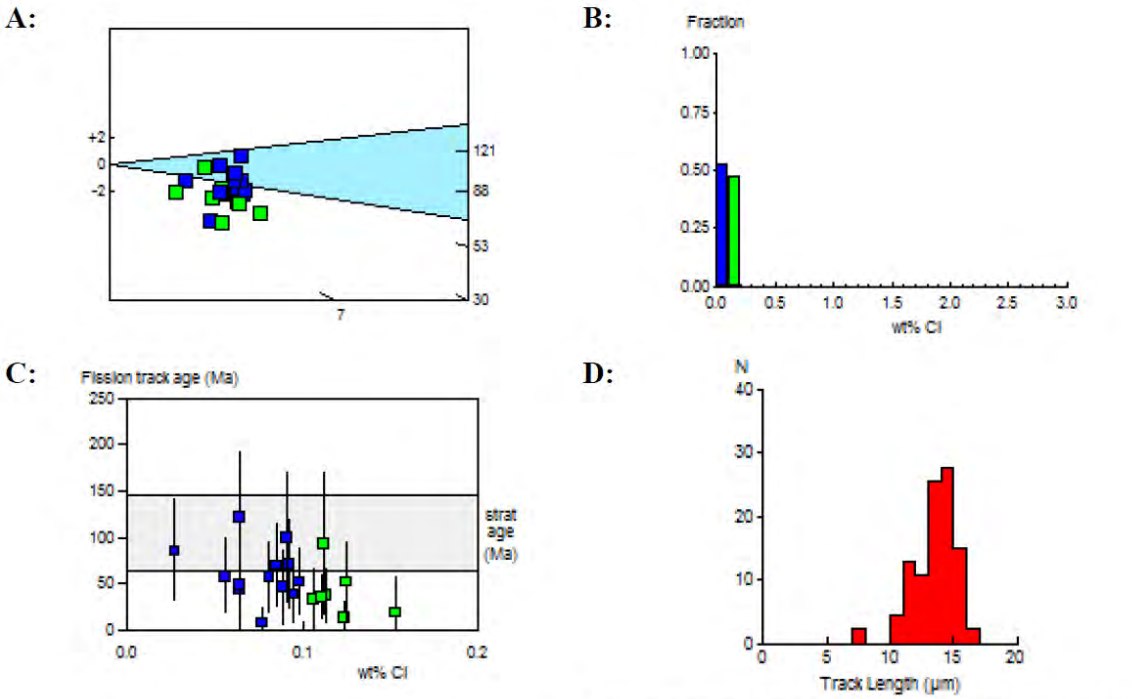
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-12	3	7	44	100	1.112E+05	6.992E+05	0.159	6.2	0.10	38.8 ± 15.8
G1219-12	4	10	42	90	1.766E+05	7.416E+05	0.238	6.6	0.06	57.9 ± 20.4
G1219-12	5	18	36	90	3.178E+05	6.356E+05	0.500	5.6	0.06	121.1 ± 35.1
G1219-12	6	10	47	70	2.270E+05	1.067E+06	0.213	9.5	0.10	51.8 ± 18.1
G1219-12	7	6	34	63	1.513E+05	8.576E+05	0.176	7.6	0.06	43.0 ± 19.1
G1219-12	8	8	21	60	2.119E+05	5.562E+05	0.381	4.9	0.11	92.5 ± 38.5
G1219-12	9	7	46	100	1.112E+05	7.310E+05	0.152	6.5	0.11	37.1 ± 15.1
G1219-12	10	11	47	100	1.748E+05	7.469E+05	0.234	6.6	0.08	57.0 ± 19.1
G1219-12	11	4	30	100	6.356E+04	4.767E+05	0.133	4.2	0.11	32.5 ± 17.3
G1219-12	12	3	15	35	1.362E+05	6.810E+05	0.200	6.0	0.06	48.7 ± 30.8
G1219-12	13	12	42	50	3.814E+05	1.335E+06	0.286	11.8	0.09	69.5 ± 22.8
G1219-12	14	11	38	80	2.185E+05	7.548E+05	0.289	6.7	0.09	70.4 ± 24.2
G1219-12	15	11	27	80	2.185E+05	5.363E+05	0.407	4.8	0.09	98.8 ± 35.5
G1219-12	16	1	31	60	2.648E+04	8.210E+05	0.032	7.3	0.08	7.9 ± 8.0
G1219-12	17	9	62	100	1.430E+05	9.852E+05	0.145	8.7	0.11	35.4 ± 12.7
G1219-12	18	2	38	70	4.540E+04	8.626E+05	0.053	7.7	0.12	12.9 ± 9.3
G1219-12	19	7	33	90	1.236E+05	5.827E+05	0.212	5.2	0.13	51.6 ± 21.5
G1219-12	20	13	37	56	3.689E+05	1.050E+06	0.351	9.3	0.03	85.3 ± 27.6
G1219-12	21	6	32	70	1.362E+05	7.264E+05	0.188	6.4	0.09	45.7 ± 20.4
G1219-12	22	1	13	48	3.311E+04	4.304E+05	0.077	3.8	0.15	18.8 ± 19.5
		157	715		1.650E+05	7.514E+05		6.7		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 31.670 with 19 degrees of freedom
P(χ^2) = 3.4%
Age Dispersion = 30.824%

Ns / Ni = 0.220 ± 0.019
Mean Ratio = 0.221 ± 0.027

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.285E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 53.5 ± 4.9 Ma
CENTRAL AGE = 53.0 ± 6.2 Ma



Mean track length 13.53 ± 0.25 µm Std. Dev. 1.72 µm 47 tracks

Figure S2.22: Apatite Fission Track Age Data of sample 538802

GC1129-24 Apatite 538804

Counted by: COB

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-13	3	3	45	40	1.192E+05	1.788E+06	0.067	15.9	0.12	16.3 ± 9.7
G1219-13	4	8	36	32	3.973E+05	1.788E+06	0.222	15.9	0.20	54.0 ± 21.2
G1219-13	5	15	72	40	5.959E+05	2.860E+06	0.208	25.4	0.16	50.7 ± 14.4
G1219-13	6	10	52	36	4.414E+05	2.295E+06	0.192	20.4	0.14	46.8 ± 16.2
G1219-13	7	6	65	50	1.907E+05	2.066E+06	0.092	18.3	0.31	22.5 ± 9.6
G1219-13	8	7	39	24	4.635E+05	2.582E+06	0.179	22.9	0.08	43.7 ± 18.0
G1219-13	9	37	122	100	5.880E+05	1.939E+06	0.303	17.2	0.12	73.6 ± 14.0
G1219-13	10	17	147	100	2.701E+05	2.336E+06	0.116	20.7	0.15	28.2 ± 7.3
G1219-13	11	3	38	30	1.589E+05	2.013E+06	0.079	17.9	0.12	19.3 ± 11.6
G1219-13	12	10	61	45	3.531E+05	2.154E+06	0.164	19.1	0.16	39.9 ± 13.7
G1219-13	13	22	165	72	4.855E+05	3.642E+06	0.133	32.3	0.11	32.5 ± 7.4
G1219-13	15	12	24	35	5.448E+05	1.090E+06	0.500	9.7	0.11	121.0 ± 42.9
G1219-13	16	3	48	35	1.362E+05	2.179E+06	0.063	19.3	0.11	15.2 ± 9.1
G1219-13	17	4	53	32	1.986E+05	2.632E+06	0.075	23.4	0.11	18.4 ± 9.6
G1219-13	18	21	179	40	8.343E+05	7.111E+06	0.117	63.1	0.12	28.6 ± 6.6
G1219-13	19	11	100	64	2.731E+05	2.483E+06	0.110	22.0	0.13	26.8 ± 8.5
G1219-13	20	12	87	60	3.178E+05	2.304E+06	0.138	20.5	0.18	33.6 ± 10.4
G1219-13	21	10	48	40	3.973E+05	1.907E+06	0.208	16.9	0.12	50.7 ± 17.7
G1219-13	22	13	86	80	2.582E+05	1.708E+06	0.151	15.2	0.14	36.8 ± 11.0
G1219-13	23	30	129	50	9.534E+05	4.100E+06	0.233	36.4	0.19	56.5 ± 11.6
		254	1596		4.016E+05	2.524E+06		22.4		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 45.816 with 19 degrees of freedom

P(χ^2) = 0.1%

Age Dispersion = 33.927%

Ns / Ni = 0.159 ± 0.011

Mean Ratio = 0.168 ± 0.023

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass

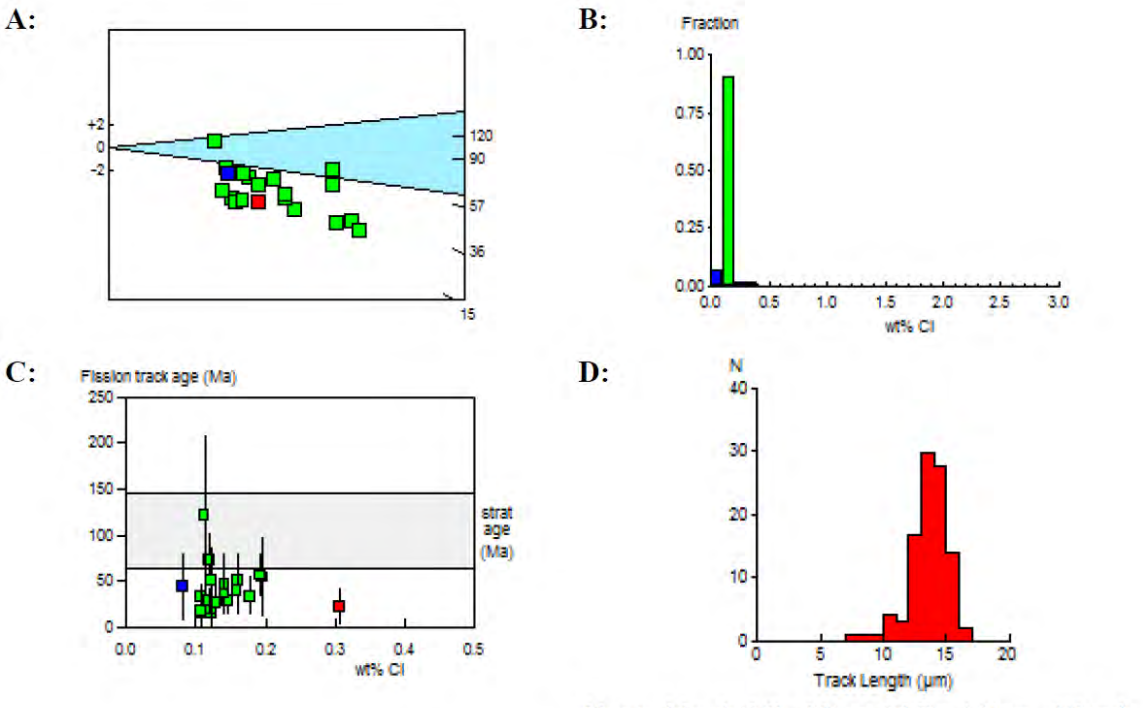
ρ = 1.284E+06cm^{-2} ND = 2027

ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022

bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 38.7 ± 2.8 Ma

CENTRAL AGE = 38.7 ± 4.2 Ma



Mean track length 13.63 ± 0.15 µm Std. Dev. 1.54 µm 101 tracks

721

722 **Figure S2.23:** Apatite Fission Track Age Data of sample 538804

GC1129-26 Apatite 538808
Counted by: COB

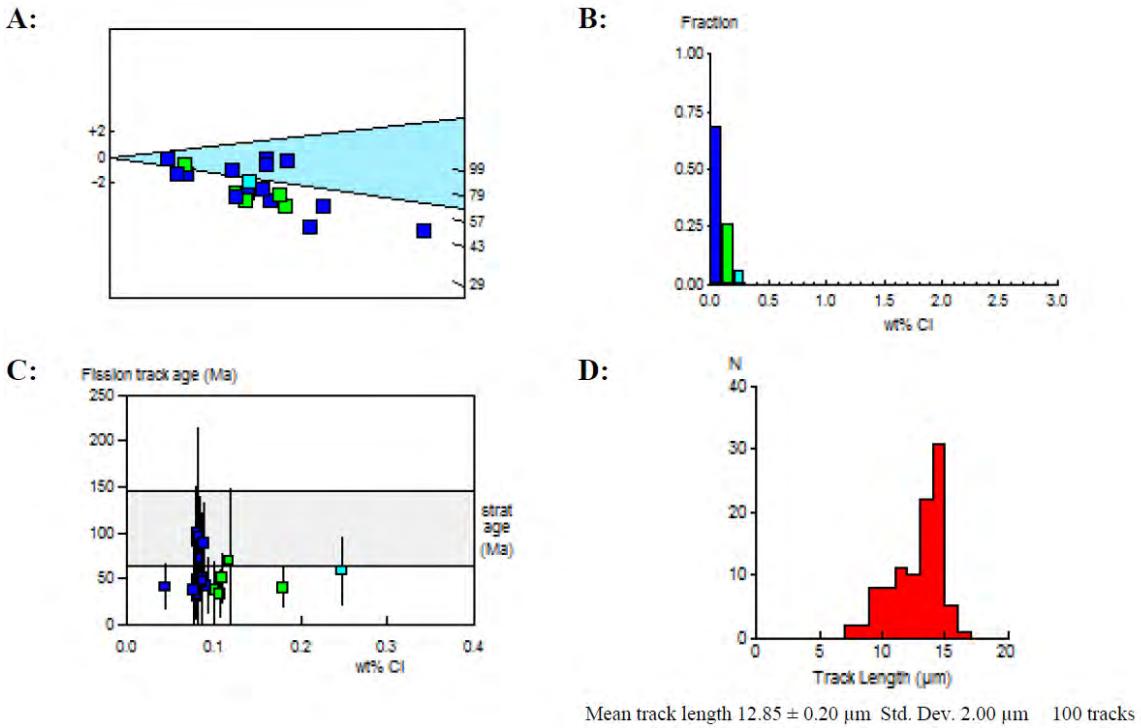
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-14	3	3	8	20	2.384E+05	6.356E+05	0.375	5.6	0.08	90.8 ± 61.6
G1219-14	4	7	45	50	2.225E+05	1.430E+06	0.156	12.7	0.10	37.8 ± 15.4
G1219-14	5	14	114	50	4.449E+05	3.623E+06	0.123	32.2	0.08	29.9 ± 8.5
G1219-14	6	52	263	63	1.312E+06	6.634E+06	0.198	59.0	0.08	48.1 ± 7.4
G1219-14	7	23	56	35	1.044E+06	2.543E+06	0.411	22.6	0.08	99.4 ± 24.8
G1219-14	8	9	52	40	3.575E+05	2.066E+06	0.173	18.4	0.09	42.1 ± 15.2
G1219-14	9	3	16	12	3.973E+05	2.119E+06	0.188	18.8	0.09	45.6 ± 28.7
G1219-14	10	29	73	80	5.760E+05	1.450E+06	0.397	12.9	0.08	96.2 ± 21.3
G1219-14	11	7	52	70	1.589E+05	1.180E+06	0.135	10.5	0.11	32.8 ± 13.2
G1219-14	12	12	71	36	5.297E+05	3.134E+06	0.169	27.9	0.04	41.1 ± 12.9
G1219-14	13	6	45	30	3.178E+05	2.384E+06	0.133	21.2	0.08	32.4 ± 14.1
G1219-14	15	14	85	32	6.952E+05	4.221E+06	0.165	37.5	0.18	40.1 ± 11.6
G1219-14	16	4	14	15	4.238E+05	1.483E+06	0.286	13.2	0.12	69.3 ± 39.4
G1219-14	17	2	13	20	1.589E+05	1.033E+06	0.154	9.2	0.08	37.4 ± 28.4
G1219-14	18	11	37	12	1.457E+06	4.900E+06	0.297	43.5	0.09	72.1 ± 24.8
G1219-14	19	21	58	50	6.674E+05	1.843E+06	0.362	16.4	0.09	87.7 ± 22.5
G1219-14	20	13	62	50	4.132E+05	1.970E+06	0.210	17.5	0.09	51.0 ± 15.6
G1219-14	21	24	122	90	4.238E+05	2.154E+06	0.197	19.1	0.09	47.8 ± 10.8
G1219-14	22	16	77	54	4.708E+05	2.266E+06	0.208	20.1	0.11	50.5 ± 13.9
G1219-14	23	12	50	30	6.356E+05	2.648E+06	0.240	23.5	0.25	58.3 ± 18.8
		282	1313		5.341E+05	2.487E+06		22.1		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 31.221 with 19 degrees of freedom
P(χ^2) = 3.8%
Age Dispersion = 24.540%

Ns / Ni = 0.215 ± 0.014
Mean Ratio = 0.229 ± 0.021

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.283E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 52.2 ± 3.7 Ma
CENTRAL AGE = 53.0 ± 5.0 Ma



Mean track length 12.85 ± 0.20 µm Std. Dev. 2.00 µm 100 tracks

723

724 **Figure S2.24**: Apatite Fission Track Age Data of sample 538808

GC1129-27 Apatite 538811
Counted by: COB

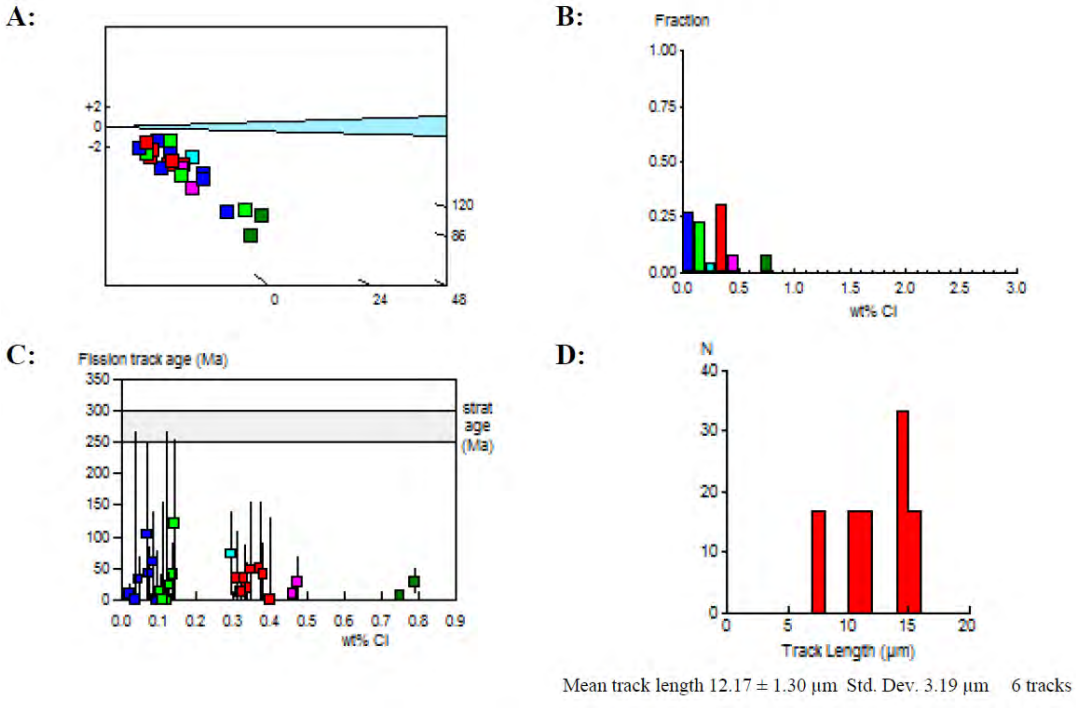
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-15	3	5	29	40	1.986E+05	1.152E+06	0.172	10.2	0.07	41.9 ± 20.3
G1219-15	5	1	13	32	4.966E+04	6.456E+05	0.077	5.7	0.34	18.7 ± 19.4
G1219-15	7	6	20	32	2.980E+05	9.932E+05	0.300	8.8	0.29	72.7 ± 33.9
G1219-15	8	1	26	36	4.414E+04	1.148E+06	0.038	10.2	0.46	9.4 ± 9.5
G1219-15	9	0	4	9	0.000E+00	7.063E+05	0.000	6.3	0.12	0.0 ± 133.1
G1219-15	10	1	5	16	9.932E+04	4.966E+05	0.200	4.4	0.37	48.6 ± 53.2
G1219-15	11	4	30	40	1.589E+05	1.192E+06	0.133	10.6	0.04	32.4 ± 17.3
G1219-16	3	3	18	24	1.986E+05	1.192E+06	0.167	10.6	0.14	40.5 ± 25.3
G1219-16	4	1	19	30	5.297E+04	1.006E+06	0.053	9.0	0.32	12.8 ± 13.1
G1219-16	5	3	18	50	9.534E+04	5.721E+05	0.167	5.1	0.38	40.5 ± 25.3
G1219-16	6	0	11	28	0.000E+00	6.243E+05	0.000	5.6	0.09	0.0 ± 37.9
G1219-16	7	2	18	24	1.324E+05	1.192E+06	0.111	10.6	0.47	27.0 ± 20.1
G1219-16	8	3	7	24	1.986E+05	4.635E+05	0.429	4.1	0.07	103.5 ± 71.5
G1219-16	9	0	7	30	0.000E+00	3.708E+05	0.000	3.3	0.40	0.0 ± 64.4
G1219-16	10	1	19	21	7.567E+04	1.438E+06	0.053	12.8	0.10	12.8 ± 13.1
G1219-16	11	9	76	32	4.469E+05	3.774E+06	0.118	33.6	0.79	28.8 ± 10.2
G1219-16	13	3	12	15	3.178E+05	1.271E+06	0.250	11.3	0.08	60.6 ± 39.1
G1219-16	14	1	7	32	4.966E+04	3.476E+05	0.143	3.1	0.31	34.7 ± 37.1
G1219-16	15	2	14	16	1.986E+05	1.390E+06	0.143	12.4	0.33	34.7 ± 26.2
G1219-16	16	6	62	24	3.973E+05	4.105E+06	0.097	36.6	0.13	23.5 ± 10.1
G1219-16	17	2	49	28	1.135E+05	2.781E+06	0.041	24.8	0.02	9.9 ± 7.2
G1219-16	18	0	6	15	0.000E+00	6.356E+05	0.000	5.7	0.11	0.0 ± 77.9
G1219-16	19	5	10	15	5.297E+05	1.059E+06	0.500	9.4	0.14	120.6 ± 66.1
G1219-16	20	2	71	24	1.324E+05	4.701E+06	0.028	41.9	0.75	6.9 ± 4.9
G1219-16	21	0	4	15	0.000E+00	4.238E+05	0.000	3.8	0.03	0.0 ± 132.9
G1219-16	22	1	5	15	1.059E+05	5.297E+05	0.200	4.7	0.35	48.5 ± 53.2
		62	560		1.477E+05	1.335E+06		11.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 35.276 with 25 degrees of freedom
P(χ^2) = 8.3%
Age Dispersion = 38.190% (did not converge)

Ns / Ni = 0.111 ± 0.015
Mean Ratio = 0.131 ± 0.025

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.281E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 26.9 ± 3.7 Ma
CENTRAL AGE = 28.4 ± 4.6 Ma



Mean track length 12.17 ± 1.30 µm Std. Dev. 3.19 µm 6 tracks

Figure S2.25: Apatite Fission Track Age Data of sample 538811

GC1129-30 Apatite 53836
Counted by: COB

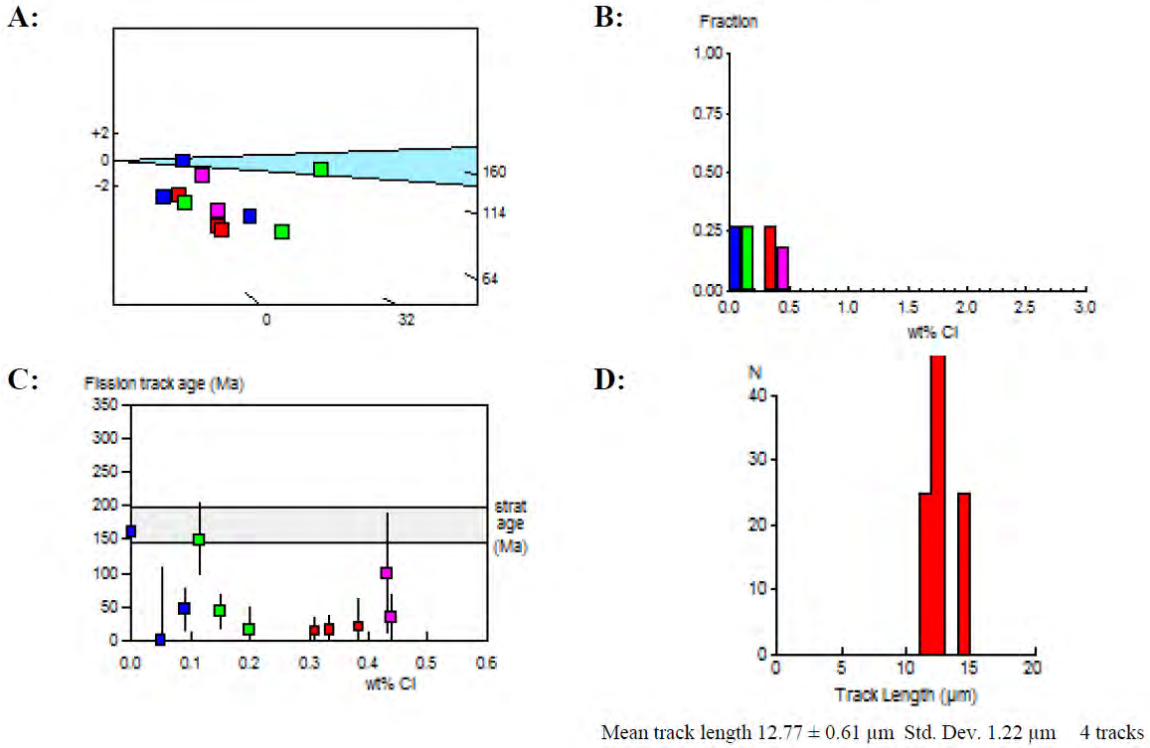
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1219-17	4	6	9	28	3.405E+05	5.108E+05	0.667	4.6	0.00	160.2 ± 84.5
G1219-17	5	2	31	49	6.486E+04	1.005E+06	0.065	9.0	0.34	15.7 ± 11.4
G1219-17	6	50	81	30	2.648E+06	4.290E+06	0.617	38.2	0.12	148.4 ± 27.0
G1219-17	7	7	17	48	2.317E+05	5.628E+05	0.412	5.0	0.43	99.4 ± 44.7
G1219-17	8	1	12	21	7.567E+04	9.080E+05	0.083	8.1	0.38	20.2 ± 21.1
G1219-17	9	1	15	32	4.966E+04	7.449E+05	0.067	6.6	0.20	16.2 ± 16.7
G1219-17	10	2	34	42	7.567E+04	1.286E+06	0.059	11.5	0.31	14.3 ± 10.4
G1219-17	11	9	48	21	6.810E+05	3.632E+06	0.188	32.4	0.09	45.4 ± 16.6
G1219-17	12	0	8	15	0.000E+00	8.475E+05	0.000	7.6	0.05	0.0 ± 54.8
G1219-17	13	13	73	70	2.951E+05	1.657E+06	0.178	14.8	0.15	43.2 ± 13.0
G1219-17	14	4	29	36	1.766E+05	1.280E+06	0.138	11.4	0.44	33.5 ± 17.9
		95	357		3.851E+05	1.447E+06		12.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 46.997 with 10 degrees of freedom
P(χ^2) = 0.0%
Age Dispersion = 77.133%

Ns / Ni = 0.266 ± 0.031
Mean Ratio = 0.225 ± 0.070

Ages calculated using a zeta of 380.4 ± 5.7 for CN5 glass
ρ = 1.279E+06cm^{-2} ND = 2027
ρ_D interpolated between top of can: ρ = 1.299E+06cm^{-2} ND = 1022
bottom of can: ρ = 1.278E+06cm^{-2} ND = 1005

POOLED AGE = 64.4 ± 7.6 Ma
CENTRAL AGE = 49.3 ± 13.7 Ma



Mean track length 12.77 ± 0.61 µm Std. Dev. 1.22 µm 4 tracks

Figure S2.26: Apatite Fission Track Age Data of sample 538836

729

GC1129-31 Apatite 538364
Counted by: MEM

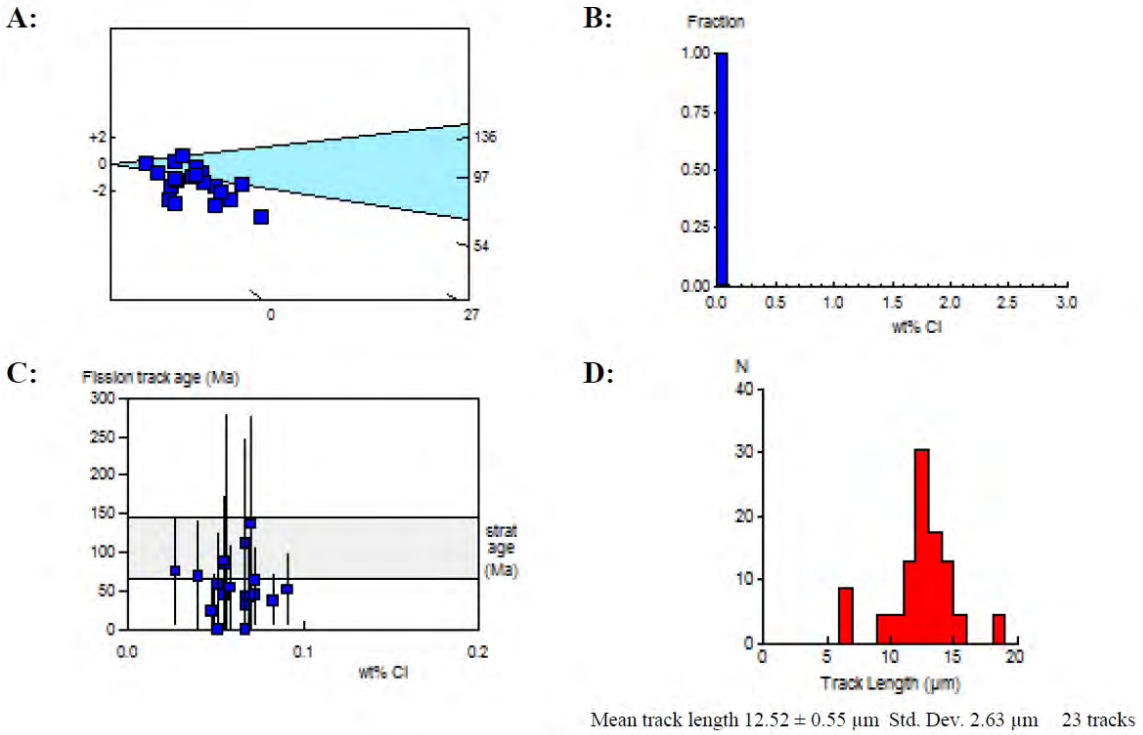
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-1	3	4	9	30	2.119E+05	4.767E+05	0.444	4.2	0.07	111.3 ± 66.9
G1220-1	4	6	29	28	3.405E+05	1.646E+06	0.207	14.6	0.09	52.0 ± 23.4
G1220-1	5	6	20	36	2.648E+05	8.828E+05	0.300	7.8	0.03	75.3 ± 35.1
G1220-1	6	1	11	20	7.945E+04	8.740E+05	0.091	7.8	0.05	22.9 ± 23.9
G1220-1	7	2	12	10	3.178E+05	1.907E+06	0.167	16.9	0.07	41.9 ± 32.1
G1220-1	8	2	11	30	1.059E+05	5.827E+05	0.182	5.2	0.06	45.7 ± 35.2
G1220-1	10	1	3	16	9.932E+04	2.980E+05	0.333	2.6	0.06	83.6 ± 96.6
G1220-1	11	5	23	40	1.986E+05	9.137E+05	0.217	8.1	0.06	54.7 ± 27.0
G1220-1	12	6	11	40	2.384E+05	4.370E+05	0.545	3.9	0.07	136.3 ± 69.3
G1220-1	13	6	40	60	1.589E+05	1.059E+06	0.150	9.4	0.08	37.8 ± 16.6
G1220-1	14	6	17	30	3.178E+05	9.005E+05	0.353	8.0	0.06	88.5 ± 42.1
G1220-1	15	1	6	20	7.945E+04	4.767E+05	0.167	4.2	0.07	41.9 ± 45.3
G1220-1	16	6	33	49	1.946E+05	1.070E+06	0.182	9.5	0.07	45.7 ± 20.3
G1220-1	18	3	32	50	9.534E+04	1.017E+06	0.094	9.0	0.05	23.6 ± 14.3
G1220-1	19	8	64	48	2.648E+05	2.119E+06	0.125	18.8	0.07	31.5 ± 11.8
G1220-1	21	4	17	30	2.119E+05	9.005E+05	0.235	8.0	0.05	59.1 ± 32.9
G1220-1	23	5	18	40	1.986E+05	7.151E+05	0.278	6.3	0.04	69.8 ± 35.3
G1220-1	24	0	11	21	0.000E+00	8.324E+05	0.000	7.4	0.05	0.0 ± 39.3
G1220-1	27	0	13	35	0.000E+00	5.902E+05	0.000	5.2	0.07	0.0 ± 32.6
G1220-1	28	11	44	80	2.185E+05	8.740E+05	0.250	7.8	0.07	62.8 ± 21.3
		83	424		1.850E+05	9.450E+05		8.4		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 19.006 with 19 degrees of freedom
P(χ^2) = 45.6%
Age Dispersion = 5.300% (did not converge)

Ns / Ni = 0.196 ± 0.023
Mean Ratio = 0.216 ± 0.030

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.285E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can: ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can: ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 49.2 ± 6.1 Ma
CENTRAL AGE = 49.3 ± 6.1 Ma



Mean track length 12.52 ± 0.55 µm Std. Dev. 2.63 µm 23 tracks

730
731 **Figure S2.27:** Apatite Fission Track Age Data of sample 538364

GC1129-33 Apatite 538368

Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-6	4	3	13	36	1.324E+05	5.738E+05	0.231	5.1	0.00	57.5 ± 36.8
G1224-6	5	5	42	48	1.655E+05	1.390E+06	0.119	12.5	0.01	29.7 ± 14.1
G1224-6	6	1	8	16	9.932E+04	7.945E+05	0.125	7.1	0.00	31.2 ± 33.1
G1224-6	7	0	8	16	0.000E+00	7.945E+05	0.000	7.1	0.02	0.0 ± 56.3
G1224-6	9	0	16	40	0.000E+00	6.356E+05	0.000	5.7	0.01	0.0 ± 25.6
G1224-6	10	1	10	18	8.828E+04	8.828E+05	0.100	7.9	0.01	25.0 ± 26.2
G1224-6	11	0	16	16	0.000E+00	1.589E+06	0.000	14.2	0.01	0.0 ± 25.6
G1224-6	12	3	22	30	1.589E+05	1.165E+06	0.136	10.4	0.01	34.0 ± 21.0
G1224-6	13	1	6	20	7.945E+04	4.767E+05	0.167	4.3	0.00	41.5 ± 44.9
G1224-6	14	0	5	9	0.000E+00	8.828E+05	0.000	7.9	0.02	0.0 ± 101.0
G1224-6	19	4	76	60	1.059E+05	2.013E+06	0.053	18.0	0.00	13.2 ± 6.8
G1224-6	20	2	14	24	1.324E+05	9.270E+05	0.143	8.3	0.01	35.6 ± 27.0
G1224-6	21	0	9	18	0.000E+00	7.945E+05	0.000	7.1	0.00	0.0 ± 49.0
G1224-6	23	3	33	30	1.589E+05	1.748E+06	0.091	15.7	0.01	22.7 ± 13.7
G1224-6	24	2	19	20	1.589E+05	1.510E+06	0.105	13.5	0.00	26.3 ± 19.5
G1224-6	25	1	20	36	4.414E+04	8.828E+05	0.050	7.9	0.01	12.5 ± 12.8
G1224-6	26	1	7	16	9.932E+04	6.952E+05	0.143	6.2	0.00	35.6 ± 38.1
G1224-6	27	2	6	60	5.297E+04	1.589E+05	0.333	1.4	0.01	82.8 ± 67.7
G1224-6	28	1	21	36	4.414E+04	9.270E+05	0.048	8.3	0.01	11.9 ± 12.2
G1224-6	29	1	13	30	5.297E+04	6.886E+05	0.077	6.2	0.01	19.2 ± 19.9
		31	364		8.508E+04	9.990E+05		8.9		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 14.468 with 19 degrees of freedom

P(χ^2) = 75.6%

Age Dispersion = 0.902% (did not converge)

Ns / Ni = 0.085 ± 0.016

Mean Ratio = 0.096 ± 0.019

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass

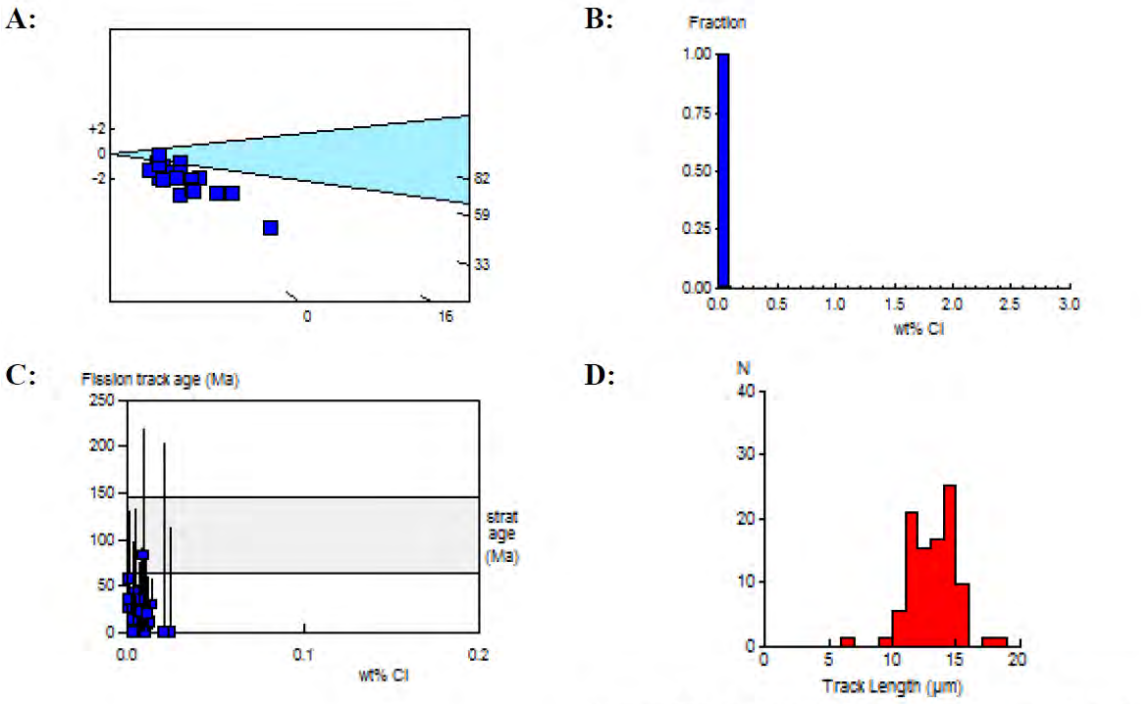
ρ = 1.273E+06cm^{-2} ND = 2040

ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975

bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 21.3 ± 4.0 Ma

CENTRAL AGE = 21.3 ± 4.0 Ma



Mean track length 13.09 ± 0.22 µm Std. Dev. 1.88 µm 71 tracks

Figure S2.28: Apatite Fission Track Age Data of sample 538368

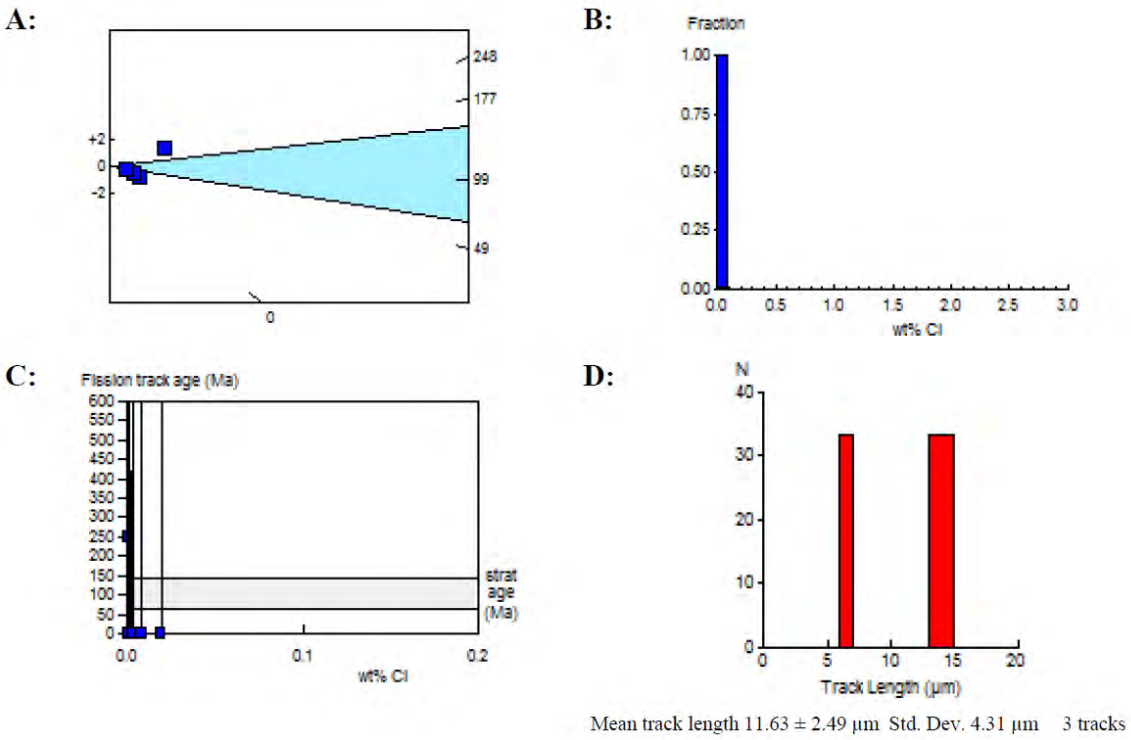
734

GC1129-34 Apatite 538370
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-2	3	0	1	42	0.000E+00	3.783E+04	0.000	0.3	0.00	0.0 ± 1796.6
G1220-2	7	0	1	40	0.000E+00	3.973E+04	0.000	0.4	0.02	0.0 ± 1796.6
G1220-2	11	0	1	70	0.000E+00	2.270E+04	0.000	0.2	0.00	0.0 ± 1796.6
G1220-2	16	5	5	49	1.621E+05	1.621E+05	1.000	1.4	0.00	248.3 ± 157.2
G1220-2	20	0	1	49	0.000E+00	3.243E+04	0.000	0.3	0.00	0.0 ± 1796.6
G1220-2	21	0	1	15	0.000E+00	1.059E+05	0.000	0.9	0.00	0.0 ± 1796.6
G1220-2	24	0	3	49	0.000E+00	9.729E+04	0.000	0.9	0.00	0.0 ± 210.0
G1220-2	32	0	2	36	0.000E+00	8.828E+04	0.000	0.8	0.00	0.0 ± 412.0
G1220-2	35	0	3	35	0.000E+00	1.362E+05	0.000	1.2	0.00	0.0 ± 210.0
G1220-2	44	0	2	50	0.000E+00	6.356E+04	0.000	0.6	0.00	0.0 ± 412.0
G1220-2	46	0	2	60	0.000E+00	5.297E+04	0.000	0.5	0.00	0.0 ± 412.0
G1220-2	47	0	1	100	0.000E+00	1.589E+04	0.000	0.1	0.01	0.0 ± 1796.6
		5	23		1.335E+04	6.143E+04		0.5		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 10.957 with 11 degrees of freedom
P(χ^2) = 44.7%
Age Dispersion = 302.123% (did not converge)
Ns / Ni = 0.217 ± 0.107
Mean Ratio = 0.083 ± 0.083

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.289E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059
POOLED AGE = 54.8 ± 27.1 Ma
CENTRAL AGE = 43.8 ± 13.5 Ma



Mean track length 11.63 ± 2.49 µm Std. Dev. 4.31 µm 3 tracks

735
736 **Figure S2.29:** Apatite Fission Track Age Data of sample 538370

GC1129-37 Apatite 538376

Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-3	5	1	35	50	3.178E+04	1.112E+06	0.029	9.8	0.05	7.2 ± 7.4
G1220-3	6	15	123	40	5.959E+05	4.886E+06	0.122	43.1	0.06	30.9 ± 8.5
G1220-3	7	2	26	40	7.945E+04	1.033E+06	0.077	9.1	0.05	19.5 ± 14.3
G1220-3	8	1	12	21	7.567E+04	9.080E+05	0.083	8.0	0.05	21.1 ± 22.0
G1220-3	9	24	196	30	1.271E+06	1.038E+07	0.122	91.6	0.06	31.0 ± 6.8
G1220-3	10	3	13	15	3.178E+05	1.377E+06	0.231	12.2	0.06	58.3 ± 37.4
G1220-3	11	1	17	20	7.945E+04	1.351E+06	0.059	11.9	0.04	14.9 ± 15.4
G1220-3	12	1	4	25	6.356E+04	2.543E+05	0.250	2.2	0.07	63.1 ± 70.6
G1220-3	13	1	14	30	5.297E+04	7.416E+05	0.071	6.5	0.06	18.1 ± 18.7
G1220-3	14	1	7	15	1.059E+05	7.416E+05	0.143	6.5	0.04	36.2 ± 38.7
G1220-3	15	2	10	16	1.986E+05	9.932E+05	0.200	8.8	0.03	50.6 ± 39.2
G1220-3	16	3	15	15	3.178E+05	1.589E+06	0.200	14.0	0.05	50.6 ± 32.0
G1220-3	18	6	32	30	3.178E+05	1.695E+06	0.188	15.0	0.05	47.4 ± 21.1
G1220-3	19	3	22	30	1.589E+05	1.165E+06	0.136	10.3	0.03	34.5 ± 21.3
G1220-3	21	4	45	50	1.271E+05	1.430E+06	0.089	12.6	0.07	22.5 ± 11.8
		68	571		2.531E+05	2.125E+06		18.7		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 7.749 with 14 degrees of freedom

P(χ^2) = 90.2%

Age Dispersion = 0.001% (did not converge)

Ns / Ni = 0.119 ± 0.015

Mean Ratio = 0.133 ± 0.017

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass

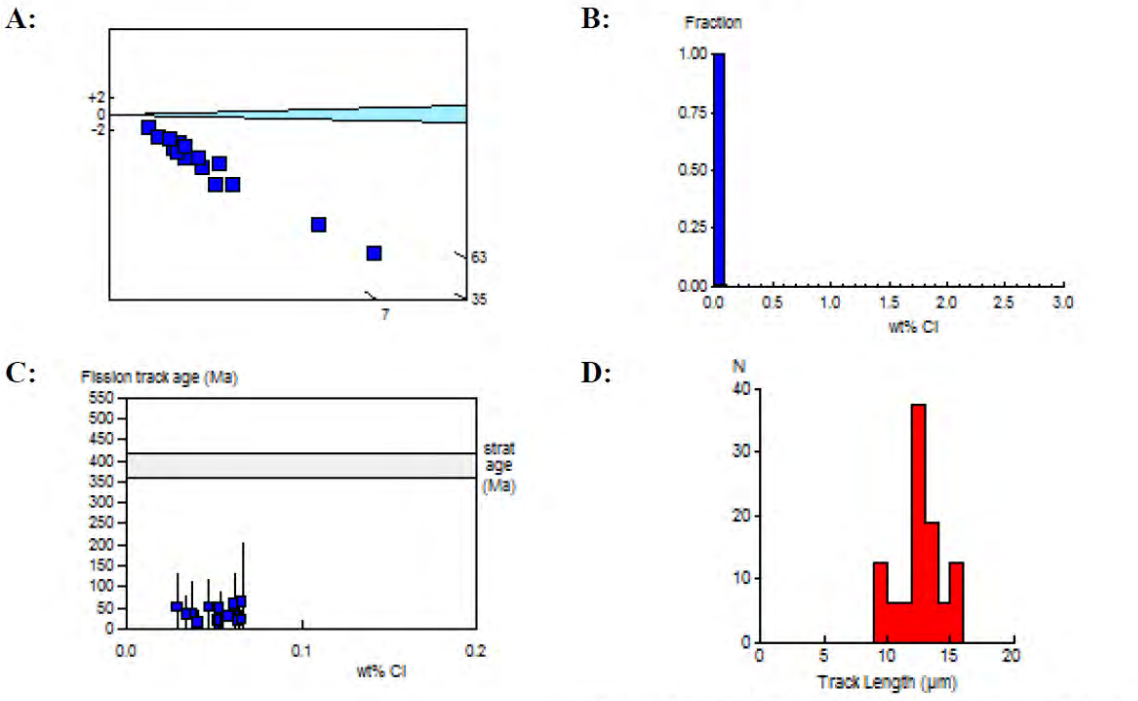
ρ = 1.292E+06cm^{-2} ND = 2070

ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011

bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 30.2 ± 4.0 Ma

CENTRAL AGE = 30.2 ± 4.0 Ma



Mean track length 12.51 ± 0.41 µm Std. Dev. 1.66 µm 16 tracks

Figure S2.30: Apatite Fission Track Age Data of sample 538376

GC1129-41 Apatite 538380

Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-4	8	0	4	16	0.000E+00	3.973E+05	0.000	3.5	0.03	0.0 ± 138.8
G1220-4	9	2	30	10	3.178E+05	4.767E+06	0.067	42.0	0.12	16.9 ± 12.4
G1220-4	14	1	9	12	1.324E+05	1.192E+06	0.111	10.5	0.07	28.2 ± 29.8
G1220-4	21	3	27	28	1.703E+05	1.532E+06	0.111	13.5	0.02	28.2 ± 17.2
G1220-4	26	1	4	12	1.324E+05	5.297E+05	0.250	4.7	0.46	63.3 ± 70.8
G1220-4	31	3	39	40	1.192E+05	1.549E+06	0.077	13.6	0.03	19.5 ± 11.7
G1220-4	32	0	3	9	0.000E+00	5.297E+05	0.000	4.7	0.05	0.0 ± 211.1
G1220-4	33	4	64	8	7.945E+05	1.271E+07	0.063	111.9	0.04	15.9 ± 8.2
G1220-4	34	6	31	24	3.973E+05	2.053E+06	0.194	18.1	0.05	49.1 ± 21.9
G1220-4	43	3	14	12	3.973E+05	1.854E+06	0.214	16.3	0.02	54.3 ± 34.6
G1220-4	45	0	9	16	0.000E+00	8.939E+05	0.000	7.9	0.05	0.0 ± 49.9
G1220-4	53	22	74	80	4.370E+05	1.470E+06	0.297	12.9	0.09	75.2 ± 18.4
G1220-4	57	2	34	50	6.356E+04	1.081E+06	0.059	9.5	0.05	15.0 ± 10.9
G1220-4	58	29	229	25	1.843E+06	1.456E+07	0.127	128.1	0.07	32.1 ± 6.4
G1220-4	59	0	4	16	0.000E+00	3.973E+05	0.000	3.5	0.07	0.0 ± 138.8
		76	575		3.373E+05	2.552E+06		22.5		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 21.500 with 14 degrees of freedom

P(χ^2) = 8.9%

Age Dispersion = 45.064%

Ns / Ni = 0.132 ± 0.016

Mean Ratio = 0.105 ± 0.025

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass

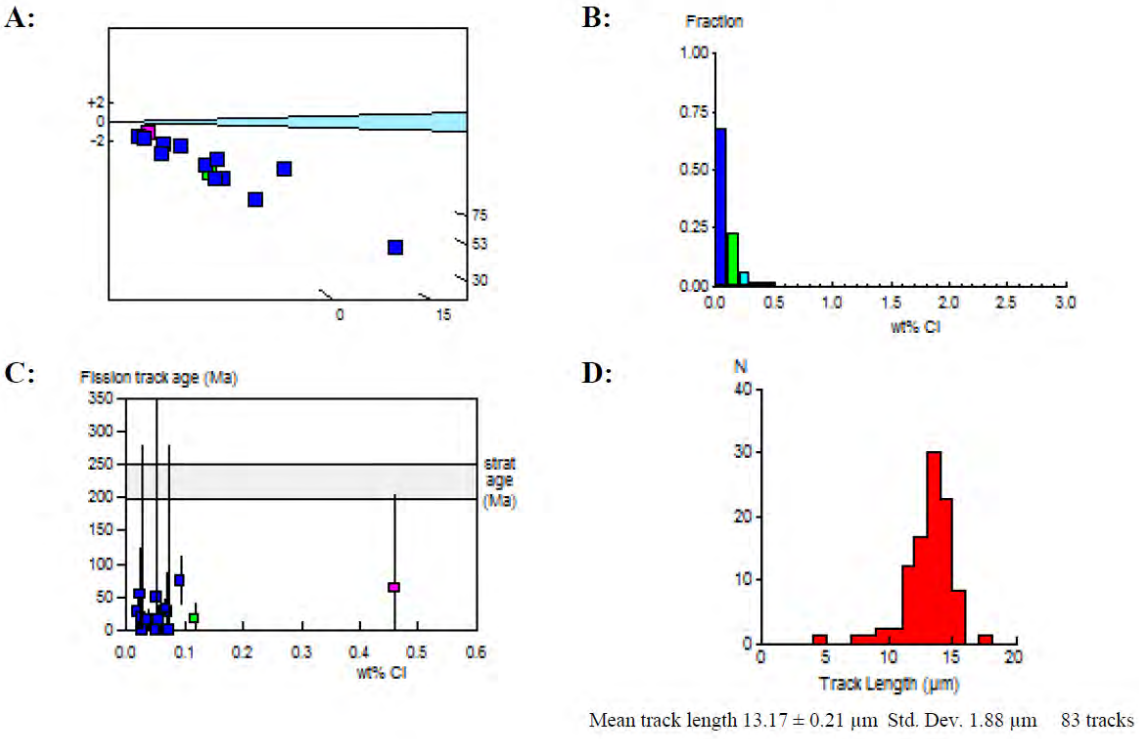
ρ = 1.295E+06cm^{-2} ND = 2070

ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011

bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 33.5 ± 4.2 Ma

CENTRAL AGE = 31.0 ± 6.3 Ma



Mean track length 13.17 ± 0.21 µm Std. Dev. 1.88 µm 83 tracks

Figure S2.31: Apatite Fission Track Age Data of sample 538380

GC1129-43 Apatite 538385

Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-7	19	7	25	30	3.708E+05	1.324E+06	0.280	11.8	0.23	70.0 ± 30.0
G1224-7	24	5	33	30	2.648E+05	1.748E+06	0.152	15.6	0.28	38.0 ± 18.3
G1224-7	28	3	22	20	2.384E+05	1.748E+06	0.136	15.6	0.27	34.2 ± 21.1
G1224-7	33	6	44	40	2.384E+05	1.748E+06	0.136	15.6	0.31	34.2 ± 14.9
G1224-7	34	5	39	30	2.648E+05	2.066E+06	0.128	18.4	0.28	32.2 ± 15.3
G1224-7	36	2	11	20	1.589E+05	8.740E+05	0.182	7.8	0.27	45.6 ± 35.0
G1224-7	41	20	71	60	5.297E+05	1.880E+06	0.282	16.7	0.28	70.4 ± 17.9
G1224-7	51	22	138	24	1.457E+06	9.137E+06	0.159	81.4	0.23	40.0 ± 9.2
G1224-7	54	5	18	15	5.297E+05	1.907E+06	0.278	17.0	0.25	69.5 ± 35.2
G1224-7	68	2	23	36	8.828E+04	1.015E+06	0.087	9.0	0.22	21.8 ± 16.1
G1224-7	71	3	55	48	9.932E+04	1.821E+06	0.055	16.2	0.28	13.7 ± 8.1
G1224-7	84	8	45	35	3.632E+05	2.043E+06	0.178	18.2	0.26	44.5 ± 17.1
G1224-7	85	3	16	20	2.384E+05	1.271E+06	0.188	11.3	0.33	47.0 ± 29.6
G1224-7	86	7	30	25	4.449E+05	1.907E+06	0.233	17.0	0.21	58.4 ± 24.6
G1224-7	87	5	23	24	3.311E+05	1.523E+06	0.217	13.6	0.20	54.4 ± 26.9
G1224-7	88	11	37	25	6.992E+05	2.352E+06	0.297	20.9	0.29	74.3 ± 25.6
G1224-7	89	0	13	20	0.000E+00	1.033E+06	0.000	9.2	0.23	0.0 ± 32.4
G1224-7	90	2	22	100	3.178E+04	3.496E+05	0.091	3.1	0.40	22.8 ± 16.9
G1224-7	91	4	32	30	2.119E+05	1.695E+06	0.125	15.1	0.25	31.4 ± 16.7
G1224-7	92	3	49	49	9.729E+04	1.589E+06	0.061	14.2	0.19	15.4 ± 9.2
		123	746		2.870E+05	1.741E+06		15.5		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 22.727 with 19 degrees of freedom

P(χ^2) = 25.0%

Age Dispersion = 22.152% (did not converge)

Ns / Ni = 0.165 ± 0.016

Mean Ratio = 0.163 ± 0.019

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass

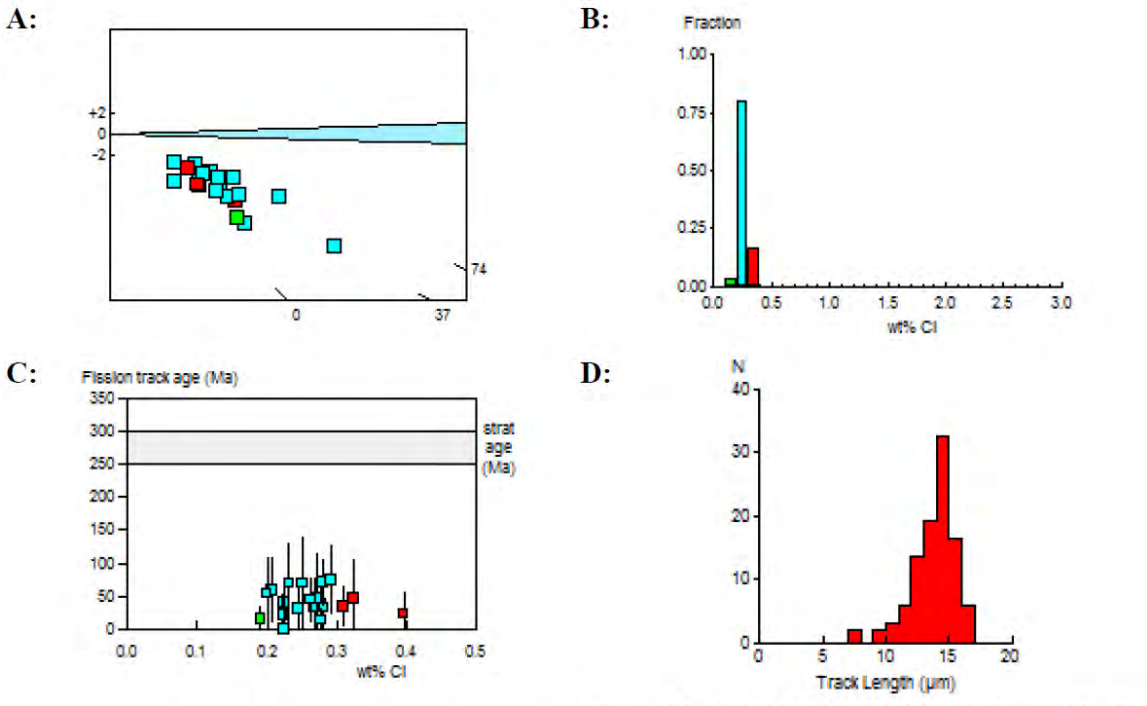
ρ = 1.280E+06cm^{-2} ND = 2040

ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975

bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 41.3 ± 4.2 Ma

CENTRAL AGE = 41.0 ± 4.8 Ma



Mean track length 13.79 ± 0.17 µm Std. Dev. 1.77 µm 104 tracks

Figure S2.32: Apatite Fission Track Age Data of sample 538385

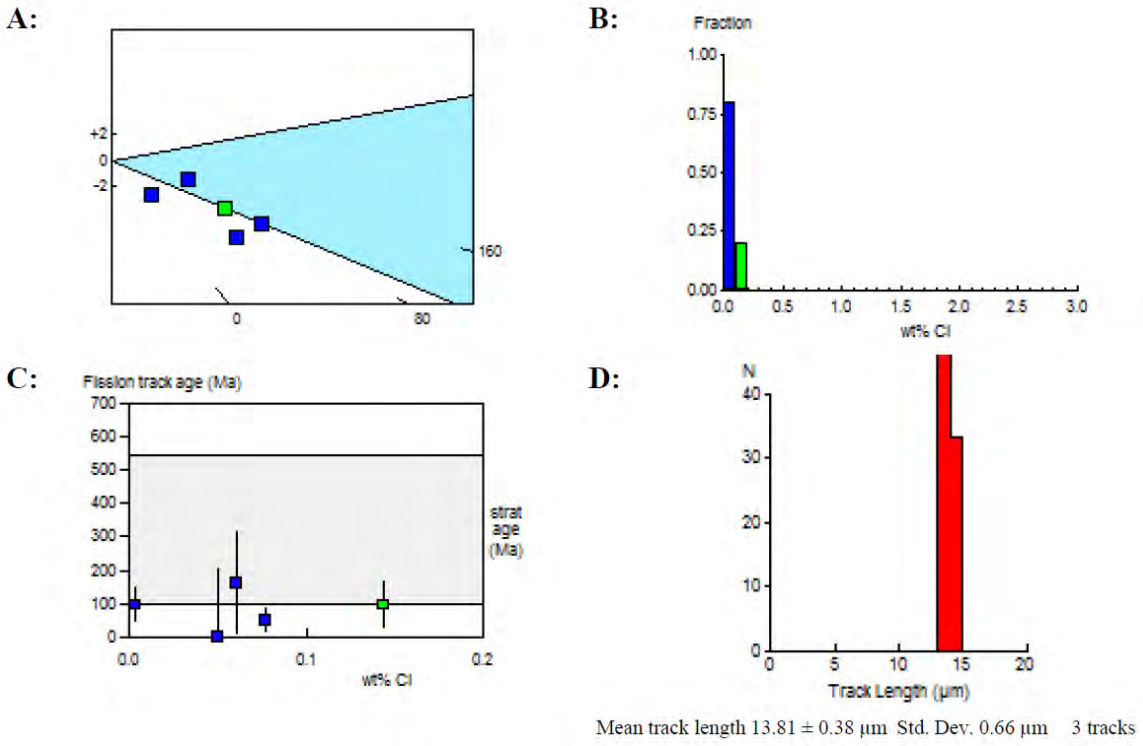
GC1129-54 Apatite 538822
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-5	4	0	5	12	0.000E+00	6.621E+05	0.000	5.8	0.05	0.0 ± 103.0
G1220-5	8	7	11	15	7.416E+05	1.165E+06	0.636	10.2	0.06	160.4 ± 77.7
G1220-5	10	8	41	14	9.080E+05	4.654E+06	0.195	40.8	0.08	49.6 ± 19.2
G1220-5	11	19	50	9	3.355E+06	8.828E+06	0.380	77.5	0.00	96.2 ± 26.1
G1220-5	17	11	29	21	8.324E+05	2.194E+06	0.379	19.3	0.14	96.1 ± 34.1
		45	136		1.007E+06	3.044E+06		26.7		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 5.874 with 4 degrees of freedom
P(χ^2) = 20.9%
Age Dispersion = 4.939% (did not converge)
Ns / Ni = 0.331 ± 0.057
Mean Ratio = 0.318 ± 0.106

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.299E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 83.9 ± 14.6 Ma
CENTRAL AGE = 83.9 ± 14.8 Ma



Mean track length 13.81 ± 0.38 µm Std. Dev. 0.66 µm 3 tracks

Figure S2.33: Apatite Fission Track Age Data of sample 538822

GC1129-57 Apatite 538724
Counted by: MEM

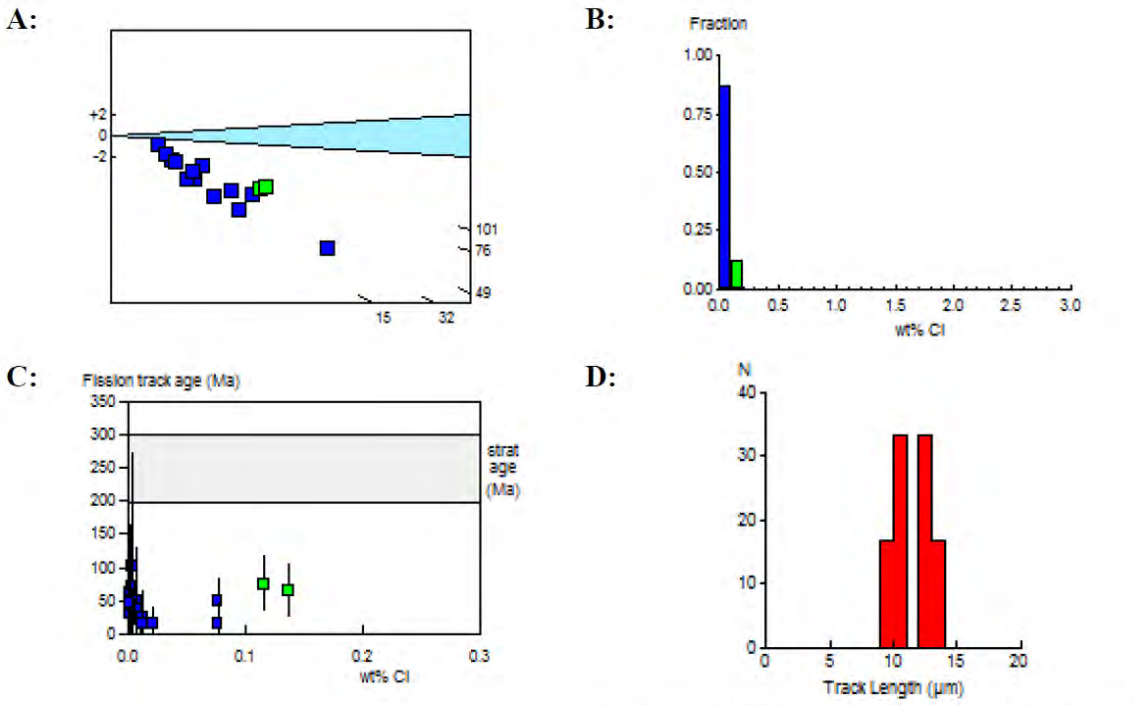
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-6	6	2	5	15	2.119E+05	5.297E+05	0.400	4.6	0.00	101.5 ± 85.0
G1220-6	7	2	10	20	1.589E+05	7.945E+05	0.200	7.0	0.01	51.0 ± 39.5
G1220-6	8	10	52	21	7.567E+05	3.935E+06	0.192	34.4	0.08	49.0 ± 17.0
G1220-6	14	4	48	25	2.543E+05	3.051E+06	0.083	26.7	0.01	21.3 ± 11.1
G1220-6	15	14	55	40	5.562E+05	2.185E+06	0.255	19.1	0.14	64.8 ± 19.5
G1220-6	16	2	20	16	1.986E+05	1.986E+06	0.100	17.4	0.01	25.5 ± 18.9
G1220-6	18	2	8	10	3.178E+05	1.271E+06	0.250	11.1	0.00	63.6 ± 50.3
G1220-6	19	1	17	15	1.059E+05	1.801E+06	0.059	15.8	0.01	15.0 ± 15.5
G1220-6	20	6	39	24	3.973E+05	2.582E+06	0.154	22.6	0.01	39.2 ± 17.2
G1220-6	22	17	58	40	6.754E+05	2.304E+06	0.293	20.2	0.12	74.5 ± 20.7
G1220-6	26	6	21	32	2.980E+05	1.043E+06	0.286	9.1	0.00	72.7 ± 33.7
G1220-6	27	15	130	80	2.980E+05	2.582E+06	0.115	22.6	0.00	29.4 ± 8.1
G1220-6	28	1	17	30	5.297E+04	9.005E+05	0.059	7.9	0.08	15.0 ± 15.5
G1220-6	30	2	31	20	1.589E+05	2.463E+06	0.065	21.6	0.02	16.5 ± 12.0
G1220-6	31	3	18	12	3.973E+05	2.384E+06	0.167	20.9	0.00	42.5 ± 26.5
G1220-6	33	2	11	16	1.986E+05	1.092E+06	0.182	9.6	0.00	46.3 ± 35.7
		89	540		3.400E+05	2.063E+06		18.1		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 17.796 with 15 degrees of freedom
P(χ^2) = 27.4%
Age Dispersion = 24.360%

Ns / Ni = 0.165 ± 0.019
Mean Ratio = 0.179 ± 0.025

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.302E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can: ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can: ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 42.0 ± 5.0 Ma
CENTRAL AGE = 42.3 ± 5.9 Ma



Mean track length 11.43 ± 0.77 µm Std. Dev. 1.88 µm 6 tracks

Figure S2.34: Apatite Fission Track Age Data of sample 538724

GC1129-59 Apatite 538726
Counted by: MEM

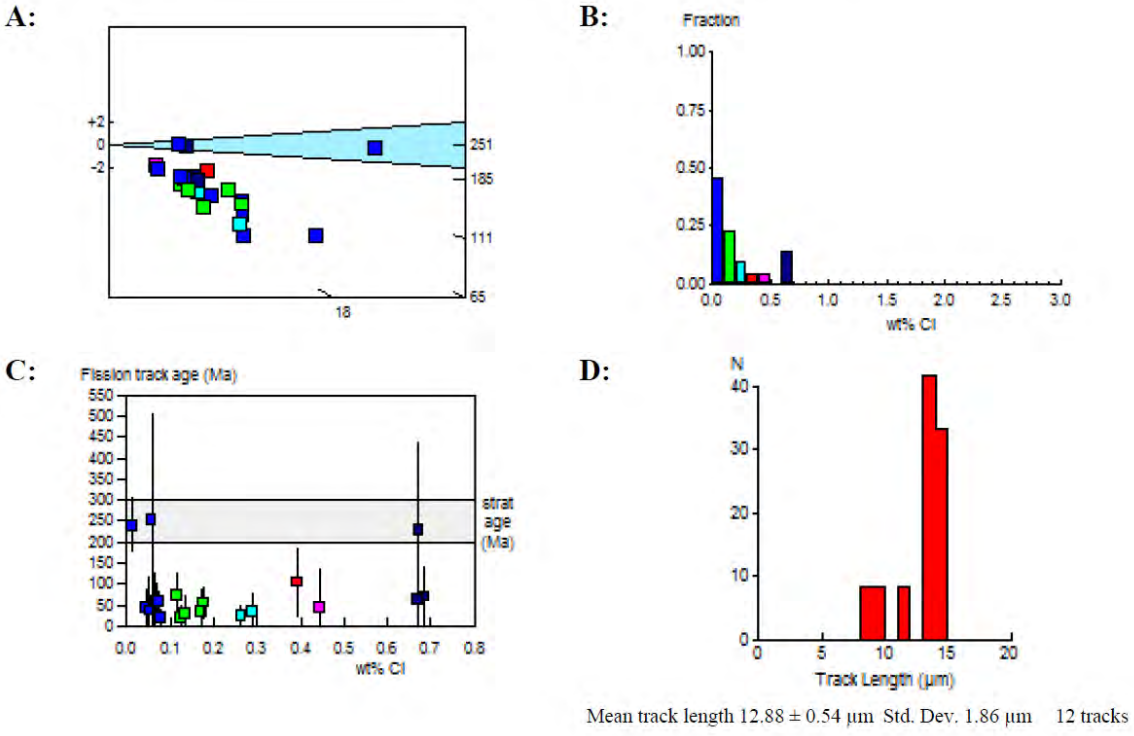
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-7	3	109	115	49	3.535E+06	3.729E+06	0.948	32.6	0.01	238.6 ± 32.6
G1220-7	4	2	15	15	2.119E+05	1.589E+06	0.133	13.9	0.17	34.1 ± 25.7
G1220-7	6	9	10	24	5.959E+05	6.621E+05	0.900	5.8	0.67	226.8 ± 104.4
G1220-7	7	9	22	50	2.860E+05	6.992E+05	0.409	6.1	0.39	104.1 ± 41.3
G1220-7	11	1	6	25	6.356E+04	3.814E+05	0.167	3.3	0.44	42.6 ± 46.0
G1220-7	14	8	49	40	3.178E+05	1.947E+06	0.163	17.0	0.05	41.7 ± 16.0
G1220-7	15	5	29	30	2.648E+05	1.536E+06	0.172	13.4	0.05	44.1 ± 21.4
G1220-7	18	5	18	35	2.270E+05	8.172E+05	0.278	7.1	0.69	70.9 ± 35.9
G1220-7	19	11	46	70	2.497E+05	1.044E+06	0.239	9.1	0.07	61.0 ± 20.6
G1220-7	20	3	15	30	1.589E+05	7.945E+05	0.200	6.9	0.06	51.1 ± 32.3
G1220-8	5	3	22	30	1.589E+05	1.165E+06	0.136	10.1	0.29	35.0 ± 21.5
G1220-8	6	10	35	15	1.059E+06	3.708E+06	0.286	32.3	0.11	73.1 ± 26.3
G1220-8	10	2	27	15	2.119E+05	2.860E+06	0.074	24.9	0.13	19.0 ± 13.9
G1220-8	12	10	46	30	5.297E+05	2.437E+06	0.217	21.2	0.18	55.7 ± 19.5
G1220-8	13	4	54	32	1.986E+05	2.682E+06	0.074	23.4	0.08	19.0 ± 9.9
G1220-8	14	3	14	16	2.980E+05	1.390E+06	0.214	12.1	0.07	54.9 ± 34.9
G1220-8	15	5	20	24	3.311E+05	1.324E+06	0.250	11.5	0.67	64.0 ± 32.0
G1220-8	16	2	18	28	1.135E+05	1.022E+06	0.111	8.9	0.13	28.5 ± 21.3
G1220-8	17	5	49	24	3.311E+05	3.244E+06	0.102	28.3	0.26	26.2 ± 12.3
G1220-8	18	8	8	40	3.178E+05	3.178E+05	1.000	2.8	0.06	252.2 ± 126.3
G1220-8	19	25	110	24	1.655E+06	7.283E+06	0.227	63.4	0.07	58.2 ± 13.0
G1220-8	20	1	7	24	6.621E+04	4.635E+05	0.143	4.0	0.05	36.6 ± 39.2
		240	734		5.692E+05	1.741E+06		15.2		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 121.061 with 21 degrees of freedom
P(χ^2) = 0.0%
Age Dispersion = 66.516%

Ns / Ni = 0.327 ± 0.024
Mean Ratio = 0.293 ± 0.059

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.306E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 83.3 ± 6.6 Ma
CENTRAL AGE = 65.0 ± 11.3 Ma



748
749 **Figure S2.35:** Apatite Fission Track Age Data of sample 538726

GC1129-62 Apatite 538731

Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-9	6	5	23	16	4.966E+05	2.284E+06	0.217	19.8	0.12	55.8 ± 27.6
G1220-9	7	5	29	12	6.621E+05	3.840E+06	0.172	33.4	0.02	44.3 ± 21.5
G1220-9	8	1	12	18	8.828E+04	1.059E+06	0.083	9.2	0.11	21.4 ± 22.3
G1220-9	9	2	18	16	1.986E+05	1.788E+06	0.111	15.5	0.05	28.6 ± 21.3
G1220-9	10	0	19	18	0.000E+00	1.677E+06	0.000	14.6	0.05	0.0 ± 21.9
G1220-9	18	7	43	12	9.270E+05	5.694E+06	0.163	49.5	0.08	41.8 ± 17.1
G1220-9	19	1	12	8	1.986E+05	2.384E+06	0.083	20.7	0.06	21.4 ± 22.3
G1220-9	21	11	40	16	1.092E+06	3.973E+06	0.275	34.5	0.09	70.5 ± 24.1
G1220-9	25	5	17	9	8.828E+05	3.002E+06	0.294	26.1	0.17	75.4 ± 38.4
G1220-9	28	7	44	20	5.562E+05	3.496E+06	0.159	30.4	0.07	40.9 ± 16.7
G1220-9	29	3	18	16	2.980E+05	1.788E+06	0.167	15.5	0.12	42.8 ± 26.7
G1220-9	30	1	7	9	1.766E+05	1.236E+06	0.143	10.7	0.05	36.7 ± 39.3
G1220-9	33	6	43	25	3.814E+05	2.733E+06	0.140	23.7	0.09	35.9 ± 15.7
G1220-9	35	1	13	9	1.766E+05	2.295E+06	0.077	19.9	0.04	19.8 ± 20.6
G1220-9	37	5	22	32	2.483E+05	1.092E+06	0.227	9.5	0.03	58.3 ± 28.9
G1220-9	40	8	37	10	1.271E+06	5.880E+06	0.216	51.1	0.10	55.5 ± 21.7
G1220-9	45	3	10	12	3.973E+05	1.324E+06	0.300	11.5	0.05	76.9 ± 50.7
G1220-9	48	1	15	24	6.621E+04	9.932E+05	0.067	8.6	0.00	17.2 ± 17.7
		72	422		4.057E+05	2.378E+06		20.7		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 11.248 with 17 degrees of freedom

P(χ^2) = 84.3%

Age Dispersion = 0.188% (did not converge)

Ns / Ni = 0.171 ± 0.022

Mean Ratio = 0.161 ± 0.020

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass

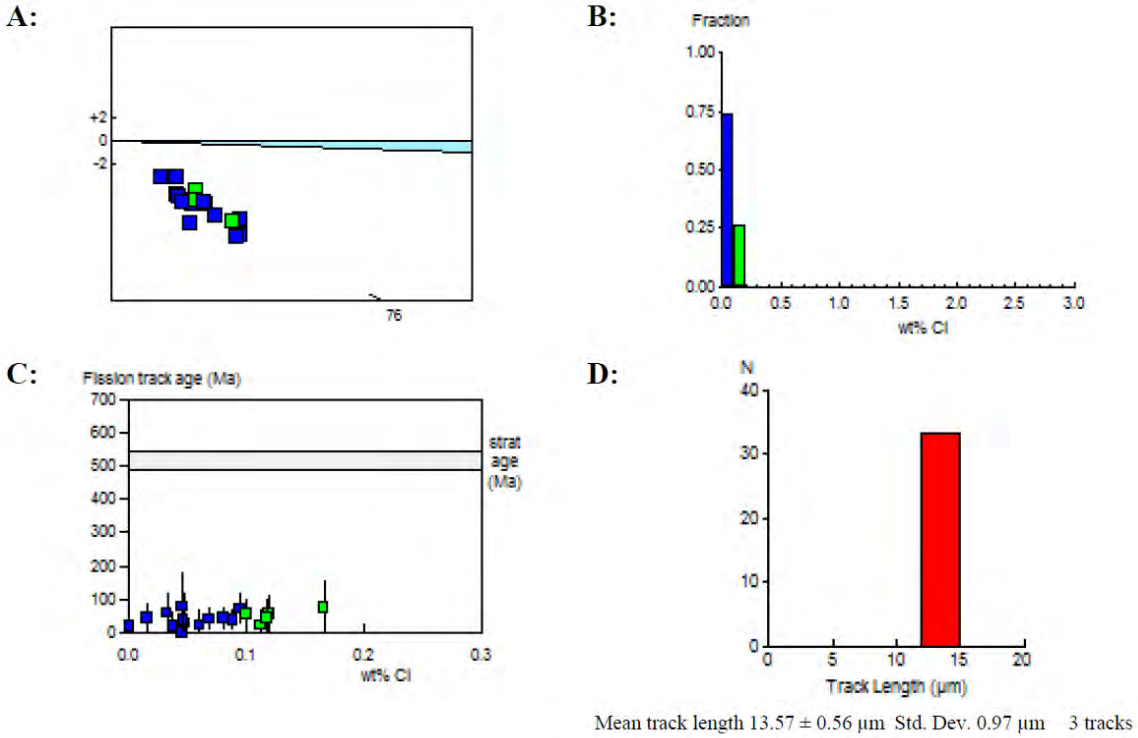
ρ = 1.312E+06cm^{-2} ND = 2070

ρ_D interpolated between top of can: ρ = 1.285E+06cm^{-2} ND = 1011

bottom of can: ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 43.8 ± 5.7 Ma

CENTRAL AGE = 43.8 ± 5.7 Ma



Mean track length 13.57 ± 0.56 µm Std. Dev. 0.97 µm 3 tracks

Figure S2.36: Apatite Fission Track Age Data of sample 538731

GC1129-63 Apatite 538732
Counted by: MEM

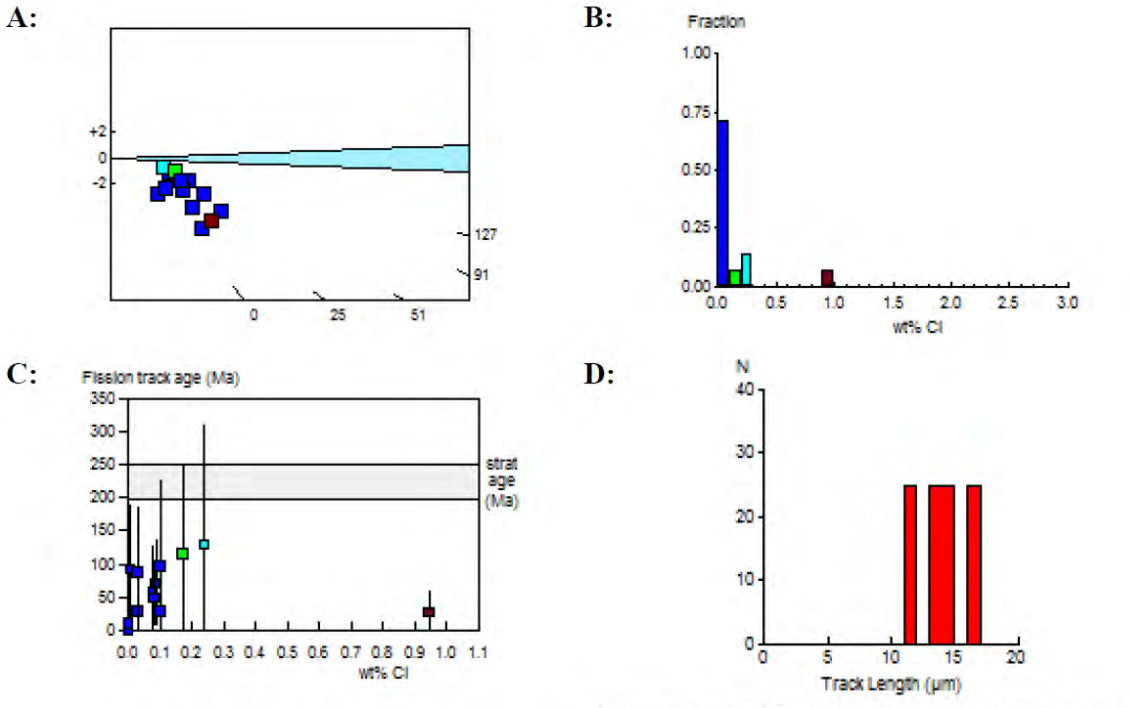
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-10	3	6	22	10	9.534E+05	3.496E+06	0.273	30.3	0.09	70.1 ± 32.4
G1220-10	6	3	8	10	4.767E+05	1.271E+06	0.375	11.0	0.10	96.2 ± 65.2
G1220-10	7	2	19	25	1.271E+05	1.208E+06	0.105	10.5	0.10	27.2 ± 20.2
G1220-10	9	3	6	20	2.384E+05	4.767E+05	0.500	4.1	0.24	128.0 ± 90.6
G1220-10	12	4	9	18	3.531E+05	7.945E+05	0.444	6.9	0.17	113.9 ± 68.5
G1220-10	16	0	7	10	0.000E+00	1.112E+06	0.000	9.6	0.00	0.0 ± 68.3
G1220-10	20	5	14	16	4.966E+05	1.390E+06	0.357	12.0	0.01	91.7 ± 47.8
G1220-10	23	3	14	18	2.648E+05	1.236E+06	0.214	10.7	0.08	55.2 ± 35.1
G1220-10	24	1	9	10	1.589E+05	1.430E+06	0.111	12.4	0.03	28.7 ± 30.2
G1220-10	26	4	12	10	6.356E+05	1.907E+06	0.333	16.5	0.03	85.6 ± 49.5
G1220-10	27	1	26	14	1.135E+05	2.951E+06	0.038	25.6	0.00	9.9 ± 10.1
G1220-10	29	6	32	25	3.814E+05	2.034E+06	0.188	17.6	0.08	48.3 ± 21.5
G1220-10	32	3	29	21	2.270E+05	2.194E+06	0.103	19.0	0.95	26.7 ± 16.2
		41	207		3.147E+05	1.589E+06		13.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 14.188 with 12 degrees of freedom
P(χ^2) = 28.9%
Age Dispersion = 16.564% (did not converge)

Ns / Ni = 0.198 ± 0.034
Mean Ratio = 0.234 ± 0.044

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.316E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can: ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can: ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 51.0 ± 8.8 Ma
CENTRAL AGE = 51.4 ± 9.3 Ma



Mean track length 13.82 ± 1.01 µm Std. Dev. 2.02 µm 4 tracks

Figure S2.37: Apatite Fission Track Age Data of sample 538732

GC1129-64 Apatite 538823
Counted by: MEM

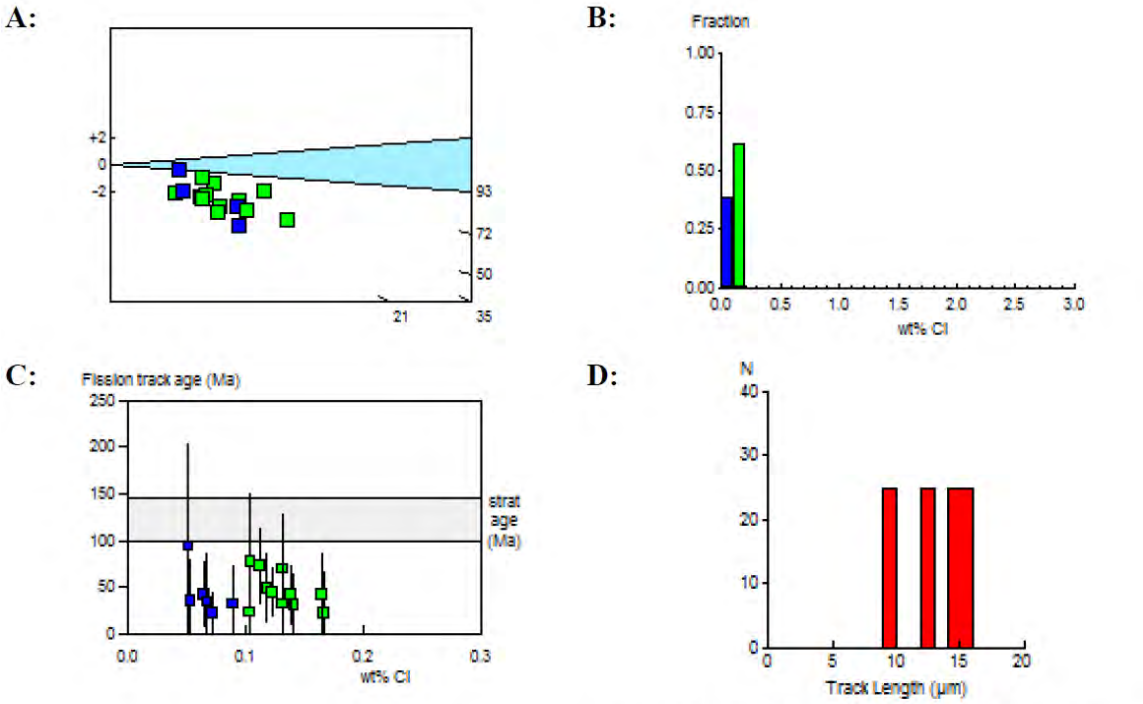
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-11	3	8	43	30	4.238E+05	2.278E+06	0.186	19.7	0.12	48.0 ± 18.5
G1220-11	4	16	57	50	5.085E+05	1.812E+06	0.281	15.7	0.11	72.3 ± 20.6
G1220-11	5	4	11	8	7.945E+05	2.185E+06	0.364	18.9	0.05	93.6 ± 54.7
G1220-11	9	7	43	25	4.449E+05	2.733E+06	0.163	23.6	0.07	42.0 ± 17.2
G1220-11	11	3	24	15	3.178E+05	2.543E+06	0.125	22.0	0.09	32.3 ± 19.8
G1220-11	13	3	22	12	3.973E+05	2.913E+06	0.136	25.2	0.05	35.2 ± 21.7
G1220-11	14	7	26	24	4.635E+05	1.721E+06	0.269	14.9	0.13	69.4 ± 29.6
G1220-11	17	1	12	24	6.621E+04	7.945E+05	0.083	6.9	0.17	21.6 ± 22.4
G1220-11	21	14	82	24	9.270E+05	5.429E+06	0.171	46.9	0.12	44.1 ± 12.8
G1220-11	22	4	48	50	1.271E+05	1.526E+06	0.083	13.2	0.07	21.6 ± 11.2
G1220-11	23	4	33	15	4.238E+05	3.496E+06	0.121	30.2	0.14	31.3 ± 16.6
G1220-11	24	4	25	24	2.648E+05	1.655E+06	0.160	14.3	0.17	41.3 ± 22.3
G1220-11	26	6	20	12	7.945E+05	2.648E+06	0.300	22.9	0.10	77.3 ± 36.0
G1220-11	29	8	50	24	5.297E+05	3.311E+06	0.160	28.6	0.14	41.3 ± 15.8
G1220-11	30	3	33	21	2.270E+05	2.497E+06	0.091	21.6	0.10	23.5 ± 14.2
G1220-11	32	3	24	20	2.384E+05	1.907E+06	0.125	16.5	0.13	32.3 ± 19.8
G1220-11	37	2	15	30	1.059E+05	7.945E+05	0.133	6.9	0.07	34.5 ± 26.0
		97	568		3.778E+05	2.212E+06		19.1		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 12.480 with 16 degrees of freedom
P(χ^2) = 71.0%
Age Dispersion = 1.818% (did not converge)

Ns / Ni = 0.171 ± 0.019
Mean Ratio = 0.174 ± 0.020

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.319E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 44.1 ± 5.0 Ma
CENTRAL AGE = 44.1 ± 5.0 Ma



Mean track length 12.67 ± 1.35 µm Std. Dev. 2.70 µm 4 tracks

Figure S2.38: Apatite Fission Track Age Data of sample 538823

GC1129-68 Apatite 538735
Counted by: MEM

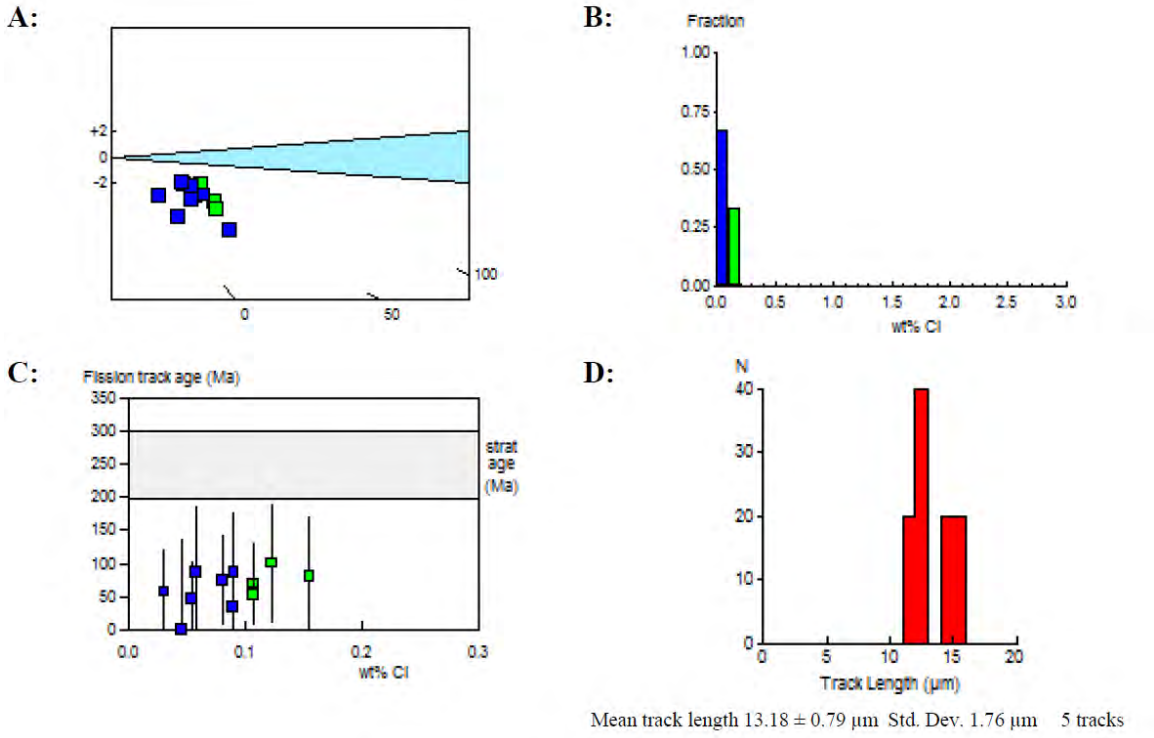
Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-12	6	0	14	24	0.000E+00	9.270E+05	0.000	8.0	0.05	0.0 ± 30.8
G1220-12	13	4	13	12	5.297E+05	1.721E+06	0.308	14.8	0.16	79.5 ± 45.5
G1220-12	14	0	7	16	0.000E+00	6.952E+05	0.000	6.0	0.05	0.0 ± 68.6
G1220-12	15	7	18	49	2.270E+05	5.837E+05	0.389	5.0	0.12	100.3 ± 44.8
G1220-12	16	4	18	35	1.816E+05	8.172E+05	0.222	7.0	0.03	57.5 ± 31.8
G1220-12	19	7	26	24	4.635E+05	1.721E+06	0.269	14.8	0.11	69.6 ± 29.7
G1220-12	20	6	29	24	3.973E+05	1.920E+06	0.207	16.6	0.11	53.5 ± 24.1
G1220-12	21	6	21	48	1.986E+05	6.952E+05	0.286	6.0	0.08	73.8 ± 34.2
G1220-12	23	3	17	24	1.986E+05	1.126E+06	0.176	9.7	0.05	45.7 ± 28.6
G1220-12	30	5	15	49	1.621E+05	4.864E+05	0.333	4.2	0.09	86.0 ± 44.5
G1220-12	31	4	12	24	2.648E+05	7.945E+05	0.333	6.8	0.06	86.0 ± 49.7
G1220-12	32	5	39	36	2.207E+05	1.721E+06	0.128	14.8	0.09	33.2 ± 15.8
		51	229		2.220E+05	9.970E+05		8.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 9.761 with 11 degrees of freedom
P(χ^2) = 55.2%
Age Dispersion = 1.118% (did not converge)

Ns / Ni = 0.223 ± 0.034
Mean Ratio = 0.221 ± 0.037

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.323E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can: ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can: ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 57.6 ± 9.1 Ma
CENTRAL AGE = 57.6 ± 9.1 Ma



Mean track length 13.18 ± 0.79 µm Std. Dev. 1.76 µm 5 tracks

759

760

761 **Figure S2.39:** Apatite Fission Track Age Data of sample 538735

GC1129-69 Apatite 538737
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-13	5	54	232	16	5.363E+06	2.304E+07	0.233	198.1	0.13	60.3 ± 9.3
G1220-13	8	1	5	6	2.648E+05	1.324E+06	0.200	11.4	0.09	51.9 ± 56.9
G1220-13	9	1	24	10	1.589E+05	3.814E+06	0.042	32.8	0.08	10.8 ± 11.1
G1220-13	10	5	42	30	2.648E+05	2.225E+06	0.119	19.1	0.09	30.9 ± 14.7
G1220-13	11	5	19	20	3.973E+05	1.510E+06	0.263	13.0	0.03	68.2 ± 34.3
G1220-13	12	1	11	16	9.932E+04	1.092E+06	0.091	9.4	0.01	23.6 ± 24.7
G1220-13	13	5	11	9	8.828E+05	1.942E+06	0.455	16.7	0.11	117.3 ± 63.4
G1220-13	15	1	14	10	1.589E+05	2.225E+06	0.071	19.1	0.01	18.6 ± 19.2
G1220-13	16	14	39	12	1.854E+06	5.164E+06	0.359	44.4	0.15	92.8 ± 29.0
G1220-14	3	0	13	10	0.000E+00	2.066E+06	0.000	17.7	0.01	0.0 ± 33.7
G1220-14	4	3	24	16	2.980E+05	2.384E+06	0.125	20.4	0.06	32.6 ± 20.0
G1220-14	5	2	15	6	5.297E+05	3.973E+06	0.133	34.1	0.01	34.7 ± 26.2
G1220-14	6	2	5	6	5.297E+05	1.324E+06	0.400	11.4	0.14	103.6 ± 86.7
G1220-14	8	39	207	18	3.443E+06	1.827E+07	0.188	156.7	0.28	49.0 ± 8.7
G1220-14	13	2	16	8	3.973E+05	3.178E+06	0.125	27.3	0.19	32.6 ± 24.4
		135	676		1.112E+06	5.568E+06		47.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 17.900 with 14 degrees of freedom
P(χ^2) = 21.1%
Age Dispersion = 12.855% (did not converge)

Ns / Ni = 0.200 ± 0.019
Mean Ratio = 0.187 ± 0.034

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.326E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can: ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can: ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 51.8 ± 5.1 Ma
CENTRAL AGE = 50.6 ± 5.8 Ma

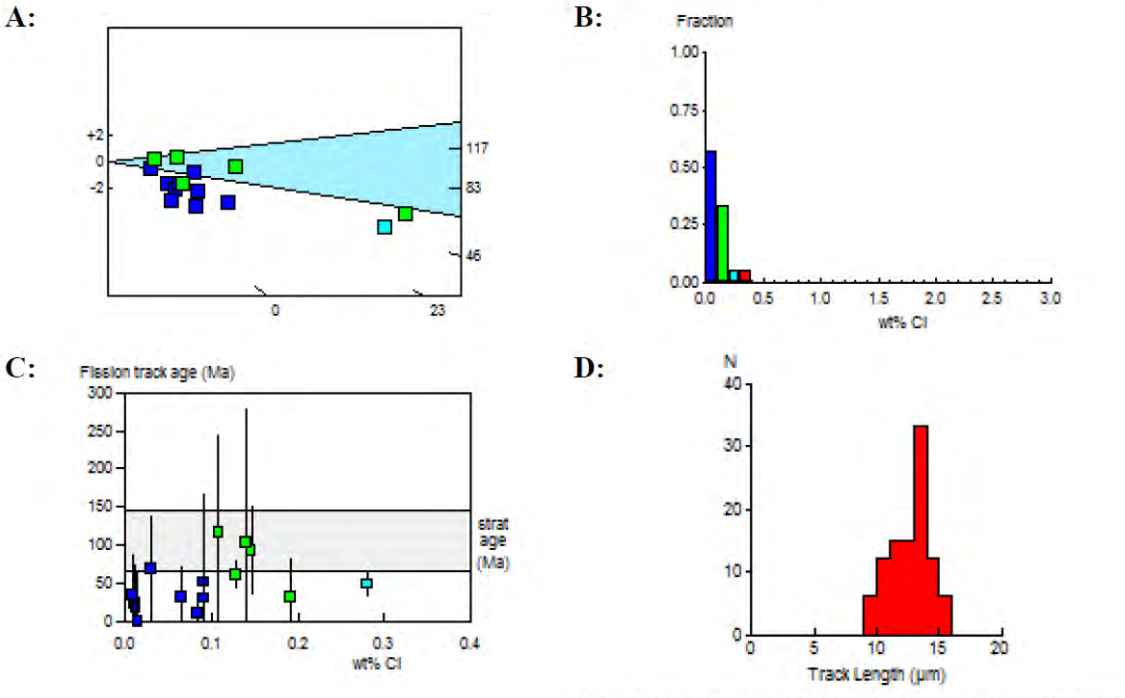


Figure S2.40: Apatite Fission Track Age Data of sample 538737

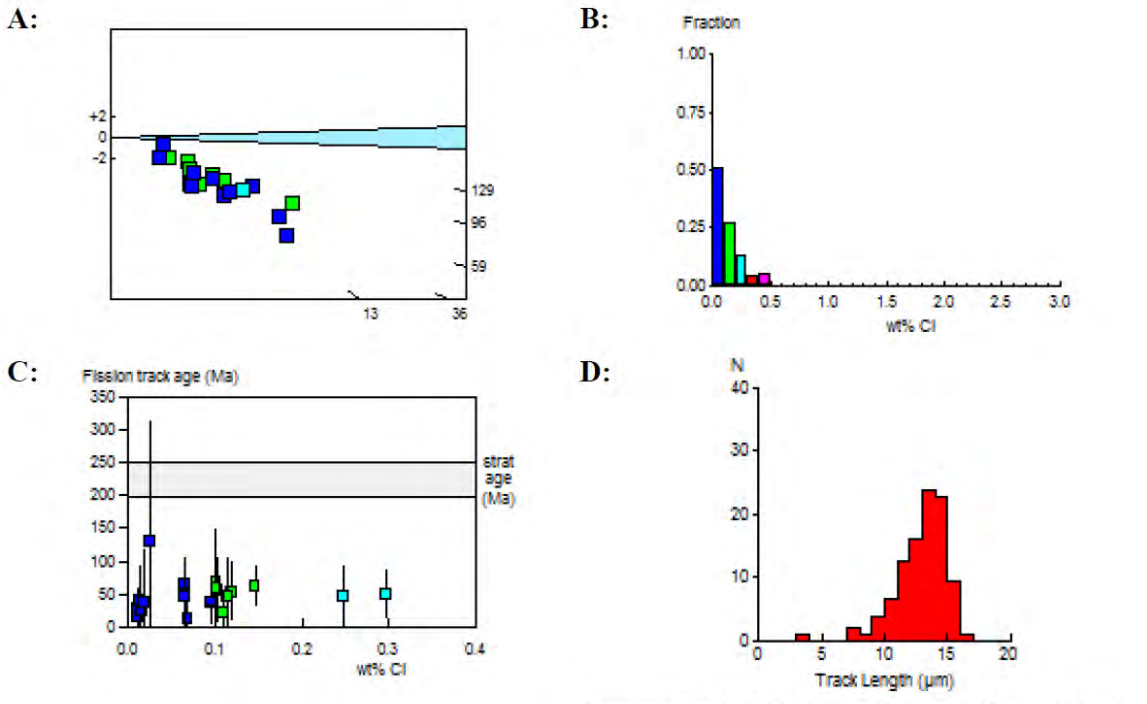
GC1129-70 Apatite 538751
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-15	4	4	15	30	2.119E+05	7.945E+05	0.267	6.8	0.10	69.4 ± 39.1
G1220-15	6	5	28	9	8.828E+05	4.944E+06	0.179	42.3	0.25	46.6 ± 22.7
G1220-15	9	6	28	18	5.297E+05	2.472E+06	0.214	21.1	0.10	55.9 ± 25.2
G1220-15	10	7	34	32	3.476E+05	1.688E+06	0.206	14.4	0.12	53.7 ± 22.3
G1220-15	11	4	37	12	5.297E+05	4.900E+06	0.108	41.9	0.01	28.2 ± 14.9
G1220-15	12	20	85	20	1.589E+06	6.754E+06	0.235	57.8	0.15	61.3 ± 15.3
G1220-15	13	11	79	50	3.496E+05	2.511E+06	0.139	21.5	0.10	36.4 ± 11.7
G1220-15	18	1	19	50	3.178E+04	6.038E+05	0.053	5.2	0.01	13.8 ± 14.1
G1220-15	19	3	6	30	1.589E+05	3.178E+05	0.500	2.7	0.03	129.6 ± 91.7
G1220-15	21	3	17	15	3.178E+05	1.801E+06	0.176	15.4	0.11	46.0 ± 28.9
G1220-15	23	2	23	18	1.766E+05	2.030E+06	0.087	17.4	0.11	22.7 ± 16.8
G1220-15	24	3	19	30	1.589E+05	1.006E+06	0.158	8.6	0.01	41.2 ± 25.6
G1220-15	26	6	40	30	3.178E+05	2.119E+06	0.150	18.1	0.10	39.2 ± 17.2
G1220-15	27	1	20	42	3.783E+04	7.567E+05	0.050	6.5	0.07	13.1 ± 13.4
G1220-15	29	8	91	50	2.543E+05	2.892E+06	0.088	24.7	0.02	23.0 ± 8.5
G1220-15	31	13	52	24	8.607E+05	3.443E+06	0.250	29.5	0.06	65.1 ± 20.3
G1220-15	33	9	47	50	2.860E+05	1.494E+06	0.191	12.8	0.30	49.9 ± 18.2
G1220-15	34	2	9	70	4.540E+04	2.043E+05	0.222	1.7	0.10	57.9 ± 45.3
G1220-15	37	5	28	16	4.966E+05	2.781E+06	0.179	23.8	0.06	46.6 ± 22.7
G1220-15	38	1	7	49	3.243E+04	2.270E+05	0.143	1.9	0.02	37.3 ± 39.9
		114	684		2.809E+05	1.685E+06		14.4		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 15.929 with 19 degrees of freedom
P(χ^2) = 66.2%
Age Dispersion = 8.295% (did not converge)
Ns / Ni = 0.167 ± 0.017
Mean Ratio = 0.180 ± 0.022

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.333E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 43.5 ± 4.6 Ma
CENTRAL AGE = 43.5 ± 4.7 Ma



Mean track length 12.96 ± 0.19 µm Std. Dev. 1.98 µm 105 tracks

766
767 **Figure S2.41:** Apatite Fission Track Age Data of sample 538751

GC1129-72 Apatite 538754

Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-16	3	42	169	20	3.337E+06	1.343E+07	0.249	114.6	0.12	64.9 ± 11.3
G1220-16	4	5	11	20	3.973E+05	8.740E+05	0.455	7.5	0.37	118.2 ± 63.9
G1220-16	6	43	121	6	1.139E+07	3.205E+07	0.355	273.4	0.57	92.6 ± 16.7
G1220-16	8	25	145	16	2.483E+06	1.440E+07	0.172	122.9	0.21	45.1 ± 9.9
G1220-16	12	9	57	64	2.235E+05	1.415E+06	0.158	12.1	0.03	41.3 ± 14.9
G1220-16	13	24	112	15	2.543E+06	1.187E+07	0.214	101.2	0.57	56.0 ± 12.7
G1220-16	15	1	2	24	6.621E+04	1.324E+05	0.500	1.1	0.25	129.9 ± 159.2
G1220-16	16	0	3	16	0.000E+00	2.980E+05	0.000	2.5	0.16	0.0 ± 217.5
G1220-16	19	2	11	24	1.324E+05	7.283E+05	0.182	6.2	0.01	47.5 ± 36.6
G1220-16	21	13	104	50	4.132E+05	3.305E+06	0.125	28.2	0.09	32.7 ± 9.7
G1220-16	22	16	78	12	2.119E+06	1.033E+07	0.205	88.1	0.59	53.6 ± 14.8
G1220-16	24	4	28	18	3.531E+05	2.472E+06	0.143	21.1	0.25	37.4 ± 20.0
G1220-16	28	3	10	35	1.362E+05	4.540E+05	0.300	3.9	0.31	78.3 ± 51.6
G1220-16	31	3	3	12	3.973E+05	3.973E+05	1.000	3.4	0.13	257.3 ± 210.2
G1220-16	32	10	55	40	3.973E+05	2.185E+06	0.182	18.6	0.26	47.5 ± 16.4
G1220-16	34	3	10	36	1.324E+05	4.414E+05	0.300	3.8	0.12	78.3 ± 51.6
G1220-16	35	11	66	25	6.992E+05	4.195E+06	0.167	35.8	0.02	43.6 ± 14.3
G1220-16	36	34	125	16	3.377E+06	1.241E+07	0.272	105.9	0.60	71.0 ± 13.9
G1220-16	41	0	9	49	0.000E+00	2.919E+05	0.000	2.5	0.04	0.0 ± 51.4
G1220-16	42	26	152	25	1.653E+06	9.662E+06	0.171	82.4	0.03	44.7 ± 9.6
		274	1271		8.325E+05	3.862E+06		33.0		

Area of basic unit = 6.293E-07 cm^{-2}

χ^2 = 28.204 with 19 degrees of freedom

P(χ^2) = 8.0%

Age Dispersion = 18.782%

Ns / Ni = 0.216 ± 0.014

Mean Ratio = 0.257 ± 0.048

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass

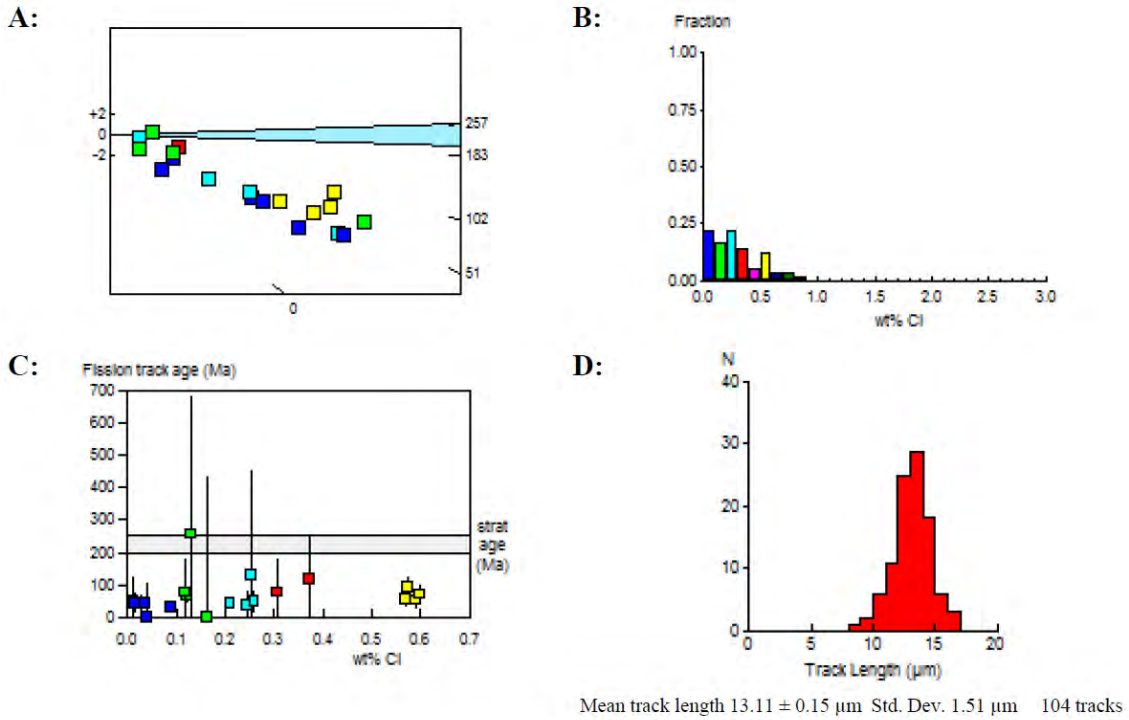
ρ = 1.336E+06cm^{-2} ND = 2070

ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011

bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 56.3 ± 4.1 Ma

CENTRAL AGE = 55.8 ± 5.1 Ma



Mean track length 13.11 ± 0.15 µm Std. Dev. 1.51 µm 104 tracks

Figure S2.42: Apatite Fission Track Age Data of sample 538754

GC1129-73 Apatite 538756
Counted by: MEM

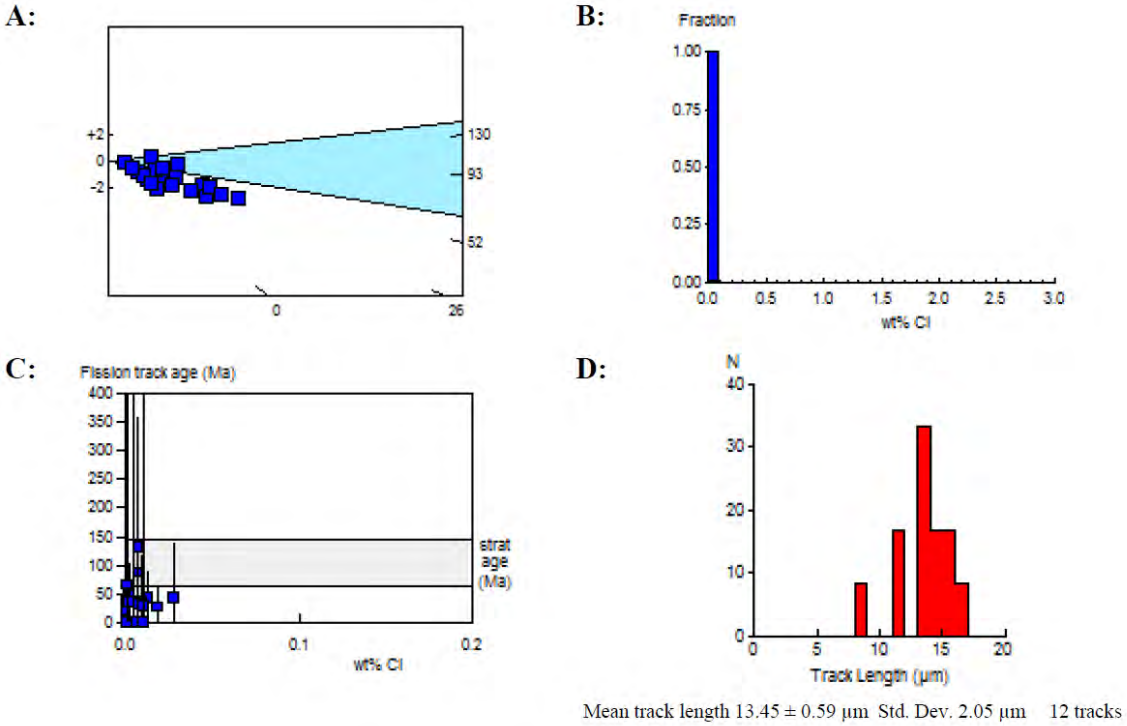
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-17	4	0	5	16	0.000E+00	4.966E+05	0.000	4.2	0.00	0.0 ± 106.2
G1220-17	6	4	25	20	3.178E+05	1.986E+06	0.160	16.9	0.01	42.0 ± 22.6
G1220-17	7	4	12	12	5.297E+05	1.589E+06	0.333	13.5	0.01	87.1 ± 50.4
G1220-17	8	2	20	30	1.059E+05	1.059E+06	0.100	9.0	0.02	26.3 ± 19.5
G1220-17	10	1	7	20	7.945E+04	5.562E+05	0.143	4.7	0.01	37.5 ± 40.1
G1220-17	11	2	13	8	3.973E+05	2.582E+06	0.154	22.0	0.00	40.4 ± 30.7
G1220-17	12	1	8	12	1.324E+05	1.059E+06	0.125	9.0	0.01	32.8 ± 34.8
G1220-17	18	2	4	9	3.531E+05	7.063E+05	0.500	6.0	0.01	130.2 ± 112.9
G1220-17	20	0	3	25	0.000E+00	1.907E+05	0.000	1.6	0.00	0.0 ± 218.0
G1220-17	21	3	28	30	1.589E+05	1.483E+06	0.107	12.6	0.01	28.1 ± 17.1
G1220-17	22	1	6	16	9.932E+04	5.959E+05	0.167	5.1	0.03	43.7 ± 47.2
G1220-17	25	7	47	42	2.648E+05	1.778E+06	0.149	15.1	0.00	39.1 ± 15.9
G1220-17	29	0	8	21	0.000E+00	6.054E+05	0.000	5.2	0.00	0.0 ± 59.2
G1220-17	32	0	1	20	0.000E+00	7.945E+04	0.000	0.7	0.01	0.0 ± 1850.5
G1220-17	33	0	4	25	0.000E+00	2.543E+05	0.000	2.2	0.00	0.0 ± 143.4
G1220-17	34	0	2	6	0.000E+00	5.297E+05	0.000	4.5	0.00	0.0 ± 427.2
G1220-17	35	5	29	49	1.621E+05	9.405E+05	0.172	8.0	0.00	45.2 ± 21.9
G1220-17	36	1	12	10	1.589E+05	1.907E+06	0.083	16.2	0.00	21.9 ± 22.8
G1220-17	37	0	6	16	0.000E+00	5.959E+05	0.000	5.1	0.00	0.0 ± 84.1
G1220-17	39	5	36	30	2.648E+05	1.907E+06	0.139	16.2	0.00	36.4 ± 17.4
G1220-17	42	2	8	16	1.986E+05	7.945E+05	0.250	6.8	0.00	65.5 ± 51.8
		40	284		1.468E+05	1.042E+06		8.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 10.417 with 20 degrees of freedom
P(χ^2) = 96.0%
Age Dispersion = 0.003% (did not converge)

Ns / Ni = 0.141 ± 0.024
Mean Ratio = 0.123 ± 0.028

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.339E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 37.0 ± 6.3 Ma
CENTRAL AGE = 37.0 ± 6.3 Ma



Mean track length 13.45 ± 0.59 µm Std. Dev. 2.05 µm 12 tracks

Figure S2.43: Apatite Fission Track Age Data of sample 538756

GC1129-76 Apatite 538761
Counted by: MEM

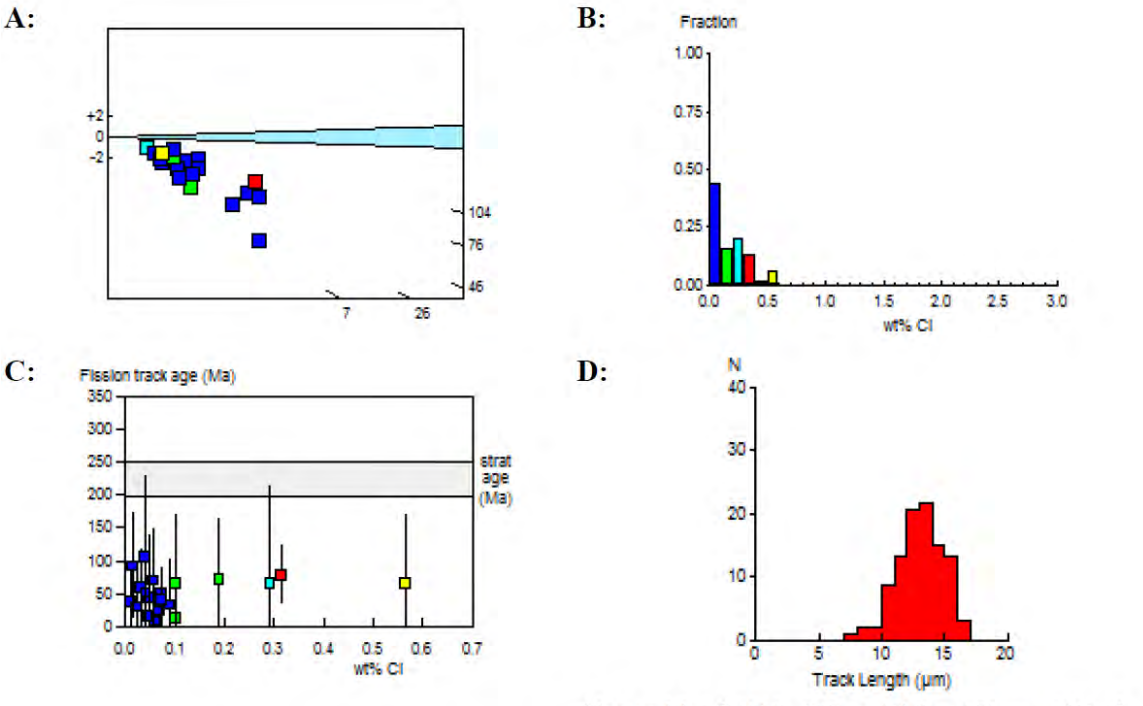
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1220-18	4	4	46	20	3.178E+05	3.655E+06	0.087	31.0	0.07	22.9 ± 12.0
G1220-18	6	1	21	12	1.324E+05	2.781E+06	0.048	23.6	0.10	12.5 ± 12.9
G1220-18	7	2	71	20	1.589E+05	5.641E+06	0.028	47.9	0.06	7.4 ± 5.3
G1220-18	8	1	4	12	1.324E+05	5.297E+05	0.250	4.5	0.29	65.6 ± 73.4
G1220-18	9	10	53	18	8.828E+05	4.679E+06	0.189	39.7	0.07	49.6 ± 17.2
G1220-18	11	4	15	9	7.063E+05	2.648E+06	0.267	22.5	0.06	70.0 ± 39.4
G1220-18	13	2	14	10	3.178E+05	2.225E+06	0.143	18.9	0.01	37.6 ± 28.4
G1220-18	14	1	9	12	1.324E+05	1.192E+06	0.111	10.1	0.02	29.2 ± 30.8
G1220-18	15	16	54	10	2.543E+06	8.581E+06	0.296	72.8	0.32	77.7 ± 22.2
G1220-18	16	7	20	12	9.270E+05	2.648E+06	0.350	22.5	0.01	91.7 ± 40.3
G1220-18	17	2	8	25	1.271E+05	5.085E+05	0.250	4.3	0.10	65.6 ± 51.9
G1220-18	18	1	6	10	1.589E+05	9.534E+05	0.167	8.1	0.05	43.8 ± 47.3
G1220-18	22	1	8	6	2.648E+05	2.119E+06	0.125	18.0	0.09	32.9 ± 34.9
G1220-18	26	5	22	24	3.311E+05	1.457E+06	0.227	12.4	0.03	59.7 ± 29.6
G1220-18	28	3	11	9	5.297E+05	1.942E+06	0.273	16.5	0.19	71.5 ± 46.6
G1220-18	29	3	20	20	2.384E+05	1.589E+06	0.150	13.5	0.07	39.5 ± 24.5
G1220-18	34	12	62	36	5.297E+05	2.737E+06	0.194	23.2	0.04	50.9 ± 16.1
G1220-18	39	1	16	16	9.932E+04	1.589E+06	0.063	13.5	0.05	16.5 ± 17.0
G1220-18	40	4	10	12	5.297E+05	1.324E+06	0.400	11.2	0.04	104.7 ± 62.0
G1220-18	42	2	8	15	2.119E+05	8.475E+05	0.250	7.2	0.57	65.6 ± 51.9
		82	478		4.231E+05	2.466E+06		20.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 24.212 with 19 degrees of freedom
P(χ^2) = 18.8%
Age Dispersion = 34.434%

Ns / Ni = 0.172 ± 0.021
Mean Ratio = 0.193 ± 0.023

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.343E+06cm^{-2} ND = 2070
ρ_D interpolated between top of can; ρ = 1.285E+06cm^{-2} ND = 1011
bottom of can; ρ = 1.346E+06cm^{-2} ND = 1059

POOLED AGE = 45.1 ± 5.5 Ma
CENTRAL AGE = 46.2 ± 7.1 Ma



Mean track length 13.09 ± 0.17 µm Std. Dev. 1.79 µm 106 tracks

Figure S2.44: Apatite Fission Track Age Data of sample 538761

774

GC1129-80 Apatite 53870
Counted by: MEM

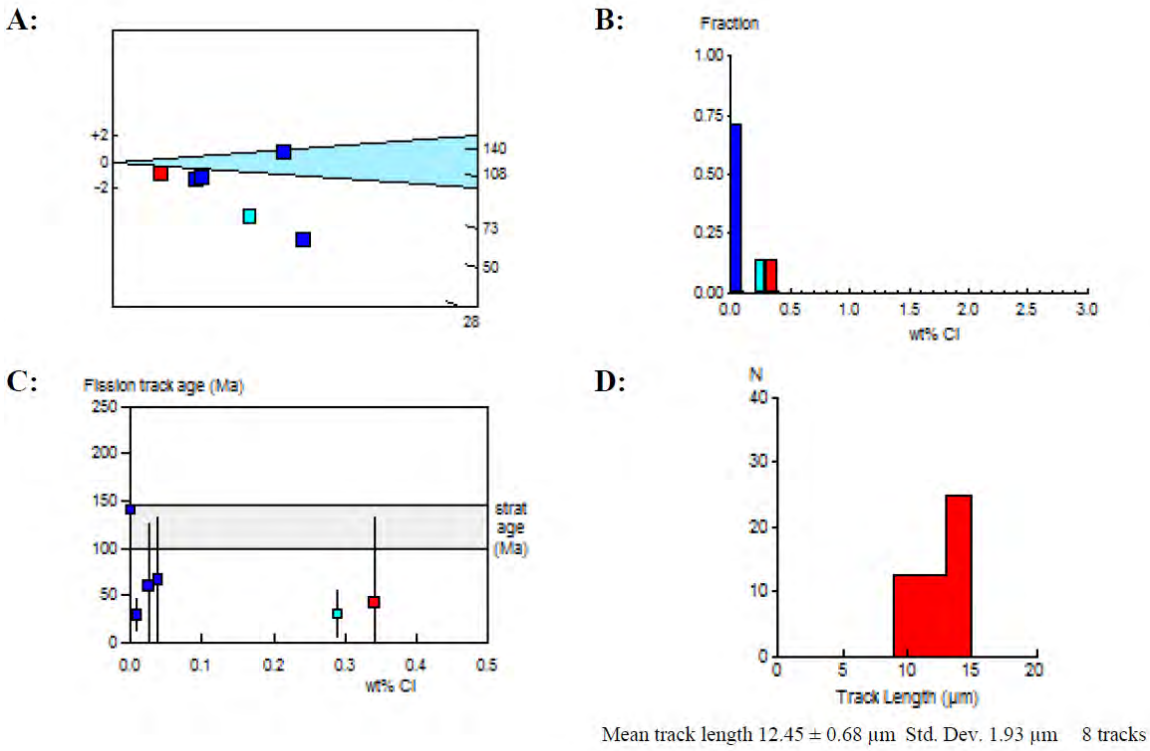
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-8	3	4	17	8	7.945E+05	3.377E+06	0.235	29.9	0.03	59.2 ± 32.9
G1224-8	4	5	19	12	6.621E+05	2.516E+06	0.263	22.3	0.04	66.2 ± 33.3
G1224-8	5	11	98	40	4.370E+05	3.893E+06	0.112	34.5	0.01	28.3 ± 9.0
G1224-8	7	1	6	12	1.324E+05	7.945E+05	0.167	7.0	0.34	42.0 ± 45.4
G1224-8	10	6	51	16	5.959E+05	5.065E+06	0.118	44.9	0.29	29.7 ± 12.8
G1224-8	14	32	57	9	5.650E+06	1.006E+07	0.561	89.2	0.00	140.4 ± 31.3
		59	248		9.665E+05	4.063E+06		36.0		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 24.831 with 5 degrees of freedom
$P(\chi^2)$ = 0.0%
Age Dispersion = 57.691%

Ns / Ni = 0.238 ± 0.034
Mean Ratio = 0.243 ± 0.068

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.287E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can; ρ = 1.239E+06cm^{-2} ND = 975
bottom of can; ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 59.9 ± 8.8 Ma
CENTRAL AGE = 58.8 ± 17.5 Ma



Mean track length 12.45 ± 0.68 µm Std. Dev. 1.93 µm 8 tracks

775
776 **Figure S2.45:** Apatite Fission Track Age Data of sample 538770

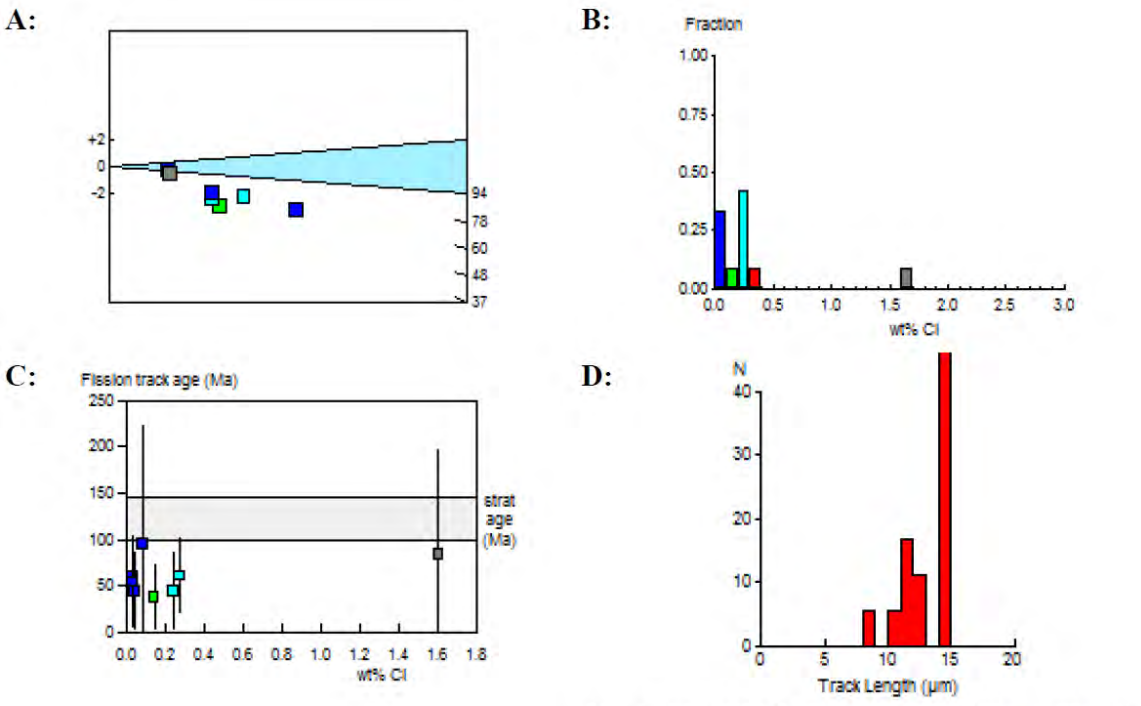
GC1129-81 Apatite 538739
Counted by: MEM

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	Cl (wt%)	F.T. AGE (Ma)
G1224-9	3	21	89	32	1.043E+06	4.420E+06	0.236	39.0	0.03	59.7 ± 14.6
G1224-9	12	5	34	24	3.311E+05	2.251E+06	0.147	19.8	0.14	37.3 ± 17.9
G1224-9	13	5	29	25	3.178E+05	1.843E+06	0.172	16.2	0.04	43.7 ± 21.2
G1224-9	20	5	29	10	7.945E+05	4.608E+06	0.172	40.6	0.24	43.7 ± 21.2
G1224-9	26	6	28	18	5.297E+05	2.472E+06	0.214	21.8	0.03	54.2 ± 24.4
G1224-9	29	3	8	24	1.986E+05	5.297E+05	0.375	4.7	0.08	94.6 ± 64.1
G1224-9	30	3	9	36	1.324E+05	3.973E+05	0.333	3.5	1.61	84.1 ± 56.1
G1224-9	36	11	46	36	4.855E+05	2.030E+06	0.239	17.9	0.27	60.5 ± 20.4
		59	272		4.573E+05	2.108E+06		18.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 2.416 with 7 degrees of freedom
P(χ^2) = 93.3%
Age Dispersion = 0.000% (did not converge)
Ns / Ni = 0.217 ± 0.031
Mean Ratio = 0.236 ± 0.028

Ages calculated using a zeta of 392.9 ± 7.4 for CN5 glass
ρ = 1.293E+06cm^{-2} ND = 2040
ρ_D interpolated between top of can: ρ = 1.239E+06cm^{-2} ND = 975
bottom of can: ρ = 1.354E+06cm^{-2} ND = 1065

POOLED AGE = 54.9 ± 8.0 Ma
CENTRAL AGE = 54.9 ± 8.0 Ma



Mean track length 13.21 ± 0.46 µm Std. Dev. 1.94 µm 18 tracks

Figure S2.46: Apatite Fission Track Age Data of sample 538739

781

GC1017-6 Zircon
Counted by: PFG

453029

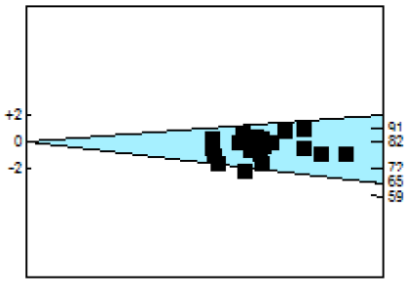
Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1223-3	6	194	115	25	1.233E+07	7.310E+06	1.687	313.7	91.2 ±	11.0
G1223-3	16	237	172	30	1.255E+07	9.111E+06	1.378	391.0	74.6 ±	7.7
G1223-3	21	126	94	15	1.335E+07	9.958E+06	1.340	427.4	72.6 ±	10.0
G1223-3	23	84	54	8	1.669E+07	1.073E+07	1.556	460.3	84.2 ±	14.8
G1223-3	24	122	100	16	1.212E+07	9.932E+06	1.220	426.2	66.1 ±	9.1
G1223-3	27	135	86	15	1.430E+07	9.111E+06	1.570	391.0	84.9 ±	11.9
G1223-3	30	81	58	9	1.430E+07	1.024E+07	1.397	439.5	75.6 ±	13.1
G1223-3	31	125	80	12	1.655E+07	1.059E+07	1.563	454.7	84.5 ±	12.3
G1223-3	35	145	97	25	9.217E+06	6.166E+06	1.495	264.6	80.9 ±	10.8
G1223-3	46	80	63	12	1.059E+07	8.343E+06	1.270	358.0	68.8 ±	11.7
G1223-3	71	133	92	15	1.409E+07	9.746E+06	1.446	418.3	78.3 ±	10.8
G1223-3	72	80	69	10	1.271E+07	1.096E+07	1.159	470.6	62.8 ±	10.4
G1223-3	73	181	127	18	1.598E+07	1.121E+07	1.425	481.2	77.2 ±	9.1
G1223-3	74	101	92	12	1.337E+07	1.218E+07	1.098	522.8	59.5 ±	8.7
G1223-3	77	167	100	18	1.474E+07	8.828E+06	1.670	378.9	90.3 ±	11.6
G1223-3	80	110	73	12	1.457E+07	9.667E+06	1.507	414.9	81.5 ±	12.5
G1223-3	81	117	84	12	1.549E+07	1.112E+07	1.393	477.4	75.4 ±	10.9
G1223-3	82	131	82	12	1.735E+07	1.086E+07	1.598	466.0	86.4 ±	12.3
G1223-3	84	118	71	12	1.563E+07	9.402E+06	1.662	403.5	89.9 ±	13.7
G1223-3	91	202	146	20	1.605E+07	1.160E+07	1.384	497.8	74.9 ±	8.3
		2669	1855		1.377E+07	9.571E+06		410.7		

Area of basic unit = 6.293E-07 cm-²
χ^2 = 13.812 with 19 degrees of freedom
P(χ^2) = 79.5%
Age Dispersion = 0.136% (did not converge)
Ns / Ni = 1.439 ± 0.043
Mean Ratio = 1.441 ± 0.038

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.242E+06cm-² ND = 1960
ρ_D interpolated between top of can; ρ = 1.241E+06cm-² ND = 976
bottom of can; ρ = 1.251E+06cm-² ND = 984

POOLED AGE = 77.9 ± 3.0 Ma
CENTRAL AGE = 77.9 ± 3.0 Ma

A:



782

783

784 **Figure S2.47:** Zircon Fission Track Age Data of sample 453029

GC1129-13 Zircon
Counted by: PFG

538332

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-1	16	100	35	12	1.324E+07	4.635E+06	2.857	183.3	166.7 ±	33.0
G1222-1	29	83	34	15	8.793E+06	3.602E+06	2.441	142.5	142.7 ±	29.2
G1222-1	38	122	41	15	1.292E+07	4.343E+06	2.976	171.8	173.5 ±	31.6
G1222-1	42	125	32	16	1.241E+07	3.178E+06	3.906	125.7	226.8 ±	45.2
G1222-1	43	216	62	20	1.716E+07	4.926E+06	3.484	194.8	202.6 ±	29.6
G1222-1	46	84	32	20	6.674E+06	2.543E+06	2.625	100.6	153.3 ±	32.0
G1222-1	50	147	58	20	1.168E+07	4.608E+06	2.534	182.3	148.0 ±	23.2
G1222-1	53	152	48	15	1.610E+07	5.085E+06	3.167	201.1	184.5 ±	30.8
G1222-1	56	185	45	21	1.400E+07	3.405E+06	4.111	134.7	238.5 ±	40.0
G1222-1	57	138	40	16	1.371E+07	3.973E+06	3.450	157.1	200.7 ±	36.3
G1222-1	62	84	42	18	7.416E+06	3.708E+06	2.000	146.7	117.1 ±	22.3
G1222-1	66	92	30	15	9.746E+06	3.178E+06	3.067	125.7	178.7 ±	37.8
G1222-1	69	184	64	21	1.392E+07	4.843E+06	2.875	191.6	167.7 ±	24.7
G1222-1	76	252	92	30	1.335E+07	4.873E+06	2.739	192.8	159.9 ±	19.8
G1222-1	89	108	47	16	1.073E+07	4.668E+06	2.298	184.6	134.4 ±	23.7
G1222-1	94	79	29	9	1.395E+07	5.120E+06	2.724	202.5	159.0 ±	34.7
G1222-1	95	94	32	10	1.494E+07	5.085E+06	2.938	201.1	171.3 ±	35.3
G1222-1	96	121	49	20	9.614E+06	3.893E+06	2.469	154.0	144.3 ±	24.7
G1222-1	106	166	51	21	1.256E+07	3.859E+06	3.255	152.6	189.5 ±	30.7
G1222-1	109	176	53	24	1.165E+07	3.509E+06	3.321	138.8	193.3 ±	30.6
		2708	916		1.216E+07	4.112E+06		162.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 19.481 with 19 degrees of freedom
P(χ^2) = 42.6%
Age Dispersion = 3.281% (did not converge)

Ns / Ni = 2.956 ± 0.113
Mean Ratio = 2.962 ± 0.118

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 172.4 ± 7.7 Ma
CENTRAL AGE = 172.3 ± 7.9 Ma

A:

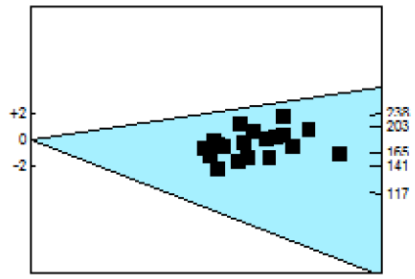


Figure S2.48: Zircon Fission Track Age Data of sample 548332

GC1129-16 Zircon 538342
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-2	2	159	46	18	1.404E+07	4.061E+06	3.457	160.6	201.1 ±	34.0
G1222-2	4	131	33	15	1.388E+07	3.496E+06	3.970	138.3	230.4 ±	45.2
G1222-2	7	74	54	20	5.880E+06	4.290E+06	1.370	169.7	80.5 ±	14.5
G1222-2	12	82	29	10	1.303E+07	4.608E+06	2.828	182.3	165.0 ±	35.9
G1222-2	16	82	37	25	5.212E+06	2.352E+06	2.216	93.0	129.6 ±	25.9
G1222-2	30	192	63	25	1.220E+07	4.004E+06	3.048	158.4	177.6 ±	26.1
G1222-2	35	95	66	25	6.038E+06	4.195E+06	1.439	165.9	84.5 ±	13.7
G1222-2	59	114	55	15	1.208E+07	5.827E+06	2.073	230.5	121.3 ±	20.1
G1222-2	67	99	46	20	7.866E+06	3.655E+06	2.152	144.6	125.9 ±	22.7
G1222-2	69	174	60	25	1.106E+07	3.814E+06	2.900	150.8	169.1 ±	25.6
G1222-2	90	68	28	12	9.005E+06	3.708E+06	2.429	146.7	141.9 ±	32.0
G1222-2	98	175	59	18	1.545E+07	5.209E+06	2.966	206.0	172.9 ±	26.4
G1222-2	101	86	58	18	7.592E+06	5.120E+06	1.483	202.5	87.0 ±	14.9
G1222-2	104	148	51	16	1.470E+07	5.065E+06	2.902	200.3	169.2 ±	27.8
G1222-2	105	69	21	12	9.137E+06	2.781E+06	3.286	110.0	191.3 ±	47.9
G1222-2	111	152	66	25	9.662E+06	4.195E+06	2.303	165.9	134.7 ±	20.1
		1900	772		1.010E+07	4.103E+06		162.3		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 51.741 with 15 degrees of freedom
P(χ^2) = 0.0%
Age Dispersion = 25.974%

Ns / Ni = 2.461 ± 0.105
Mean Ratio = 2.551 ± 0.188

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 143.8 ± 7.0 Ma
CENTRAL AGE = 140.8 ± 11.5 Ma

A:

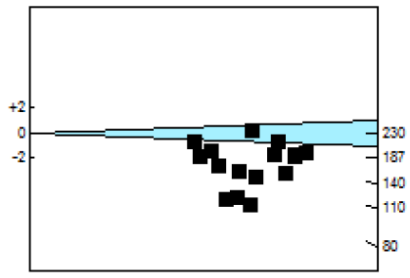


Figure S2.49: Zircon Fission Track Age Data of sample 548342

GC1129-21 Zircon
Counted by: PFG

538356

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-3	1	128	76	25	8.136E+06	4.831E+06	1.684	191.1	98.8 ±	14.5
G1222-3	3	108	87	25	6.865E+06	5.530E+06	1.241	218.7	72.9 ±	10.6
G1222-3	4	62	53	15	6.568E+06	5.615E+06	1.170	222.1	68.8 ±	13.0
G1222-3	7	142	80	25	9.026E+06	5.085E+06	1.775	201.1	104.0 ±	14.8
G1222-3	15	56	33	16	5.562E+06	3.277E+06	1.697	129.6	99.5 ±	22.0
G1222-3	17	111	79	24	7.349E+06	5.231E+06	1.405	206.9	82.5 ±	12.3
G1222-3	21	86	56	16	8.541E+06	5.562E+06	1.536	220.0	90.1 ±	15.6
G1222-3	25	121	64	25	7.691E+06	4.068E+06	1.891	160.9	110.8 ±	17.3
G1222-3	36	92	43	16	9.137E+06	4.271E+06	2.140	168.9	125.2 ±	23.3
G1222-3	37	150	137	50	4.767E+06	4.354E+06	1.095	172.2	64.4 ±	7.8
G1222-3	40	88	71	24	5.827E+06	4.701E+06	1.239	185.9	72.8 ±	11.7
G1222-3	41	82	84	20	6.515E+06	6.674E+06	0.976	264.0	57.4 ±	9.0
G1222-3	42	74	59	15	7.839E+06	6.250E+06	1.254	247.2	73.7 ±	13.0
G1222-3	46	84	60	25	5.339E+06	3.814E+06	1.400	150.8	82.2 ±	14.0
G1222-3	48	55	48	16	5.462E+06	4.767E+06	1.146	188.5	67.4 ±	13.4
G1222-3	49	116	81	30	6.144E+06	4.290E+06	1.432	169.7	84.1 ±	12.3
G1222-3	51	169	154	35	7.673E+06	6.992E+06	1.097	276.5	64.5 ±	7.3
G1222-3	54	130	102	24	8.607E+06	6.754E+06	1.275	267.1	74.9 ±	10.1
G1222-3	59	82	76	24	5.429E+06	5.032E+06	1.079	199.0	63.4 ±	10.2
G1222-3	60	209	116	40	8.303E+06	4.608E+06	1.802	182.3	105.6 ±	12.5
		2145	1559		6.956E+06	5.056E+06		200.0		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 40.594 with 19 degrees of freedom
P(χ^2) = 0.3%
Age Dispersion = 15.072%

Ns / Ni = 1.376 ± 0.046
Mean Ratio = 1.417 ± 0.072

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 80.8 ± 3.3 Ma
CENTRAL AGE = 80.9 ± 4.3 Ma

A:

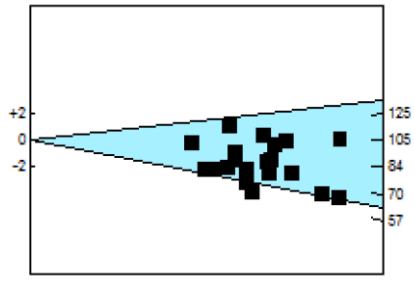


Figure S2.50: Zircon Fission Track Age Data of sample 548356

GC1129-23 Zircon
Counted by: PFG

538802

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-4	6	115	94	24	7.614E+06	6.224E+06	1.223	246.1	71.9 ±	10.1
G1222-4	7	132	97	24	8.740E+06	6.422E+06	1.361	254.0	79.9 ±	10.9
G1222-4	12	125	72	18	1.104E+07	6.356E+06	1.736	251.4	101.8 ±	15.2
G1222-4	17	117	58	15	1.239E+07	6.144E+06	2.017	243.0	118.1 ±	19.2
G1222-4	18	79	88	32	3.923E+06	4.370E+06	0.898	172.8	52.8 ±	8.3
G1222-4	19	95	80	25	6.038E+06	5.085E+06	1.188	201.1	69.8 ±	10.7
G1222-4	20	130	91	25	8.263E+06	5.784E+06	1.429	228.8	83.9 ±	11.6
G1222-4	23	160	94	30	8.475E+06	4.979E+06	1.702	196.9	99.8 ±	13.2
G1222-4	26	61	68	10	9.693E+06	1.081E+07	0.897	427.4	52.8 ±	9.4
G1222-4	27	101	73	25	6.420E+06	4.640E+06	1.384	183.5	81.2 ±	12.6
G1222-4	29	33	33	18	2.913E+06	2.913E+06	1.000	115.2	58.8 ±	14.5
G1222-4	32	48	57	16	4.767E+06	5.661E+06	0.842	223.9	49.6 ±	9.8
G1222-4	39	92	66	18	8.122E+06	5.827E+06	1.394	230.4	81.9 ±	13.3
G1222-4	42	155	100	40	6.158E+06	3.973E+06	1.550	157.1	91.0 ±	11.9
G1222-4	44	110	107	20	8.740E+06	8.502E+06	1.028	336.2	60.5 ±	8.3
G1222-4	48	93	56	16	9.236E+06	5.562E+06	1.661	220.0	97.4 ±	16.6
G1222-4	52	104	58	20	8.263E+06	4.608E+06	1.793	182.3	105.1 ±	17.4
G1222-4	55	161	129	36	7.107E+06	5.694E+06	1.248	225.2	73.3 ±	8.8
G1222-4	58	98	74	40	3.893E+06	2.940E+06	1.324	116.3	77.8 ±	12.1
G1222-4	62	217	137	30	1.149E+07	7.257E+06	1.584	287.0	92.9 ±	10.4
		2226	1632		7.339E+06	5.380E+06		212.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 46.098 with 19 degrees of freedom
P(χ^2) = 0.0%
Age Dispersion = 16.956%

Ns / Ni = 1.364 ± 0.044
Mean Ratio = 1.363 ± 0.074

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can: ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can: ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 80.1 ± 3.2 Ma
CENTRAL AGE = 78.9 ± 4.4 Ma

A:

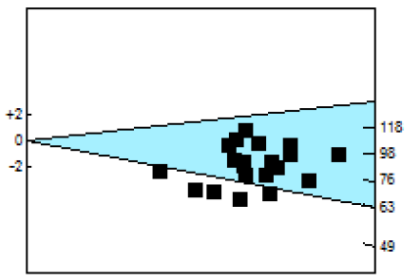


Figure S2.51: Zircon Fission Track Age Data of sample 548802

GC1129-24 Zircon
Counted by: PFG

538804

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-5	1	179	76	30	9.481E+06	4.026E+06	2.355	159.2	137.7 ±	19.1
G1222-5	2	182	118	30	9.640E+06	6.250E+06	1.542	247.2	90.5 ±	10.9
G1222-5	4	156	94	25	9.916E+06	5.975E+06	1.660	236.3	97.3 ±	12.9
G1222-5	5	164	85	25	1.042E+07	5.403E+06	1.929	213.7	113.0 ±	15.3
G1222-5	8	212	132	40	8.422E+06	5.244E+06	1.606	207.4	94.2 ±	10.7
G1222-5	13	147	80	25	9.344E+06	5.085E+06	1.837	201.1	107.7 ±	15.2
G1222-5	14	131	91	20	1.041E+07	7.230E+06	1.440	285.9	84.5 ±	11.7
G1222-5	15	175	111	25	1.112E+07	7.055E+06	1.577	279.0	92.5 ±	11.4
G1222-5	19	205	131	40	8.144E+06	5.204E+06	1.565	205.8	91.8 ±	10.5
G1222-5	20	253	143	40	1.005E+07	5.681E+06	1.769	224.7	103.7 ±	11.1
G1222-5	22	169	108	30	8.952E+06	5.721E+06	1.565	226.2	91.8 ±	11.5
G1222-5	26	144	88	25	9.153E+06	5.594E+06	1.636	221.2	96.0 ±	13.2
G1222-5	32	163	106	25	1.036E+07	6.738E+06	1.538	266.5	90.2 ±	11.5
G1222-5	33	163	86	30	8.634E+06	4.555E+06	1.895	180.2	111.1 ±	15.0
G1222-5	37	272	176	40	1.081E+07	6.992E+06	1.545	276.5	90.7 ±	9.0
G1222-5	41	201	135	40	7.985E+06	5.363E+06	1.489	212.1	87.4 ±	9.9
G1222-5	44	101	79	20	8.025E+06	6.277E+06	1.278	248.2	75.1 ±	11.4
G1222-5	45	287	200	30	1.520E+07	1.059E+07	1.435	419.0	84.3 ±	8.0
G1222-5	46	117	82	24	7.747E+06	5.429E+06	1.427	214.7	83.8 ±	12.2
G1222-5	58	143	81	25	9.089E+06	5.149E+06	1.765	203.6	103.5 ±	14.6
		3564	2202		9.615E+06	5.941E+06		234.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 19.764 with 19 degrees of freedom
P(χ^2) = 40.9%
Age Dispersion = 1.808% (did not converge)

Ns / Ni = 1.619 ± 0.044
Mean Ratio = 1.643 ± 0.053

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can: ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can: ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 95.0 ± 3.4 Ma
CENTRAL AGE = 95.0 ± 3.4 Ma

A:

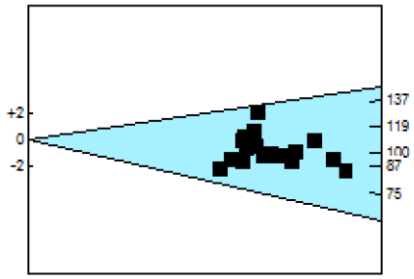


Figure S2.52: Zircon Fission Track Age Data of sample 548804

GC1129-26 Zircon
Counted by: PFG

538808

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-6	1	72	37	21	5.448E+06	2.800E+06	1.946	110.7	114.0 ±	23.2
G1222-6	8	91	28	12	1.205E+07	3.708E+06	3.250	146.6	189.3 ±	41.1
G1222-6	15	65	22	10	1.033E+07	3.496E+06	2.955	138.2	172.3 ±	42.7
G1222-6	17	98	54	12	1.298E+07	7.151E+06	1.815	282.8	106.4 ±	18.2
G1222-6	18	86	31	10	1.367E+07	4.926E+06	2.774	194.8	161.9 ±	34.1
G1222-6	28	117	51	15	1.239E+07	5.403E+06	2.294	213.7	134.2 ±	22.7
G1222-6	45	150	58	16	1.490E+07	5.760E+06	2.586	227.8	151.1 ±	23.6
G1222-6	50	107	54	15	1.134E+07	5.721E+06	1.981	226.2	116.1 ±	19.6
G1222-6	51	102	40	9	1.801E+07	7.063E+06	2.550	279.3	149.0 ±	28.0
G1222-6	66	90	38	9	1.589E+07	6.709E+06	2.368	265.3	138.5 ±	27.0
G1222-6	72	122	48	16	1.212E+07	4.767E+06	2.542	188.5	148.5 ±	25.5
G1222-6	91	158	72	20	1.255E+07	5.721E+06	2.194	226.2	128.4 ±	18.5
G1222-6	94	135	62	18	1.192E+07	5.473E+06	2.177	216.5	127.4 ±	19.8
G1222-6	97	76	33	12	1.006E+07	4.370E+06	2.303	172.8	134.7 ±	28.3
G1222-6	98	184	76	20	1.462E+07	6.038E+06	2.421	238.8	141.5 ±	19.6
G1222-6	111	135	68	15	1.430E+07	7.204E+06	1.985	284.9	116.3 ±	17.5
G1222-6	112	126	67	15	1.335E+07	7.098E+06	1.881	280.7	110.2 ±	16.9
G1222-6	113	162	83	27	9.534E+06	4.885E+06	1.952	193.2	114.3 ±	15.7
G1222-6	121	67	38	9	1.183E+07	6.709E+06	1.763	265.3	103.4 ±	21.1
		2143	960		1.212E+07	5.429E+06		214.7		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 14.701 with 18 degrees of freedom
P(χ^2) = 68.2%
Age Dispersion = 0.325% (did not converge)
Ns / Ni = 2.232 ± 0.087
Mean Ratio = 2.302 ± 0.093

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 130.6 ± 5.9 Ma
CENTRAL AGE = 130.6 ± 5.9 Ma

A:

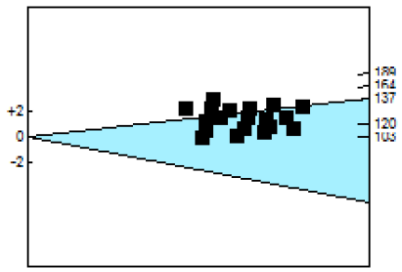


Figure S2.53: Zircon Fission Track Age Data of sample 548808

GC1129-31 Zircon 538364
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-7	3	110	82	16	1.092E+07	8.144E+06	1.341	322.0	78.8 ±	11.6
G1222-7	4	216	134	30	1.144E+07	7.098E+06	1.612	280.7	94.6 ±	10.6
G1222-7	7	106	68	25	6.738E+06	4.322E+06	1.559	170.9	91.5 ±	14.4
G1222-7	8	124	125	30	6.568E+06	6.621E+06	0.992	261.8	58.4 ±	7.5
G1222-7	12	161	105	100	2.558E+06	1.669E+06	1.533	66.0	90.0 ±	11.5
G1222-7	14	100	90	20	7.945E+06	7.151E+06	1.111	282.8	65.3 ±	9.6
G1222-7	18	96	77	30	5.085E+06	4.079E+06	1.247	161.3	73.3 ±	11.3
G1222-7	21	136	104	50	4.322E+06	3.305E+06	1.308	130.7	76.8 ±	10.2
G1222-7	22	156	119	50	4.958E+06	3.782E+06	1.311	149.6	77.0 ±	9.5
G1222-7	26	121	65	30	6.409E+06	3.443E+06	1.862	136.1	109.1 ±	17.0
G1222-7	27	166	115	40	6.595E+06	4.569E+06	1.443	180.7	84.8 ±	10.5
G1222-7	30	195	173	50	6.197E+06	5.498E+06	1.127	217.4	66.3 ±	7.1
G1222-7	33	145	94	30	7.680E+06	4.979E+06	1.543	196.9	90.5 ±	12.2
G1222-7	38	118	48	28	6.697E+06	2.724E+06	2.458	107.7	143.7 ±	24.8
G1222-7	39	115	80	20	9.137E+06	6.356E+06	1.438	251.4	84.4 ±	12.4
G1222-7	48	175	93	25	1.112E+07	5.911E+06	1.882	233.8	110.3 ±	14.4
G1222-7	50	70	54	15	7.416E+06	5.721E+06	1.296	226.2	76.2 ±	13.9
G1222-7	52	156	106	20	1.239E+07	8.422E+06	1.472	333.0	86.4 ±	11.1
G1222-7	59	74	61	20	5.880E+06	4.847E+06	1.213	191.7	71.3 ±	12.4
G1222-7	61	176	128	30	9.323E+06	6.780E+06	1.375	268.1	80.8 ±	9.6
		2716	1921		6.549E+06	4.632E+06		183.2		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 39.755 with 19 degrees of freedom
$P(\chi^2)$ = 0.4%
Age Dispersion = 13.288%

Ns / Ni = 1.414 ± 0.042
Mean Ratio = 1.456 ± 0.073

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 83.0 ± 3.2 Ma
CENTRAL AGE = 83.3 ± 4.0 Ma

A:

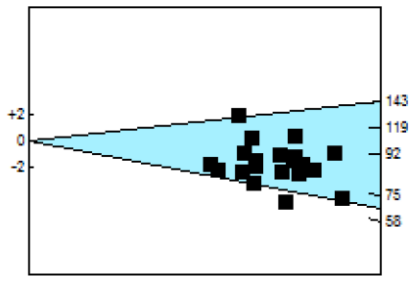


Figure S2.54: Zircon Fission Track Age Data of sample 548364

GC1129-36 Zircon
Counted by: PFG

538374

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-8	3	66	93	40	2.622E+06	3.695E+06	0.710	146.1	41.8 ±	6.8
G1222-8	4	29	32	21	2.194E+06	2.421E+06	0.906	95.7	53.3 ±	13.7
G1222-8	6	147	111	70	3.337E+06	2.520E+06	1.324	99.6	77.8 ±	10.0
G1222-8	13	39	37	30	2.066E+06	1.960E+06	1.054	77.5	62.0 ±	14.3
G1222-8	14	45	50	30	2.384E+06	2.648E+06	0.900	104.7	53.0 ±	11.0
G1222-8	15	67	63	18	5.915E+06	5.562E+06	1.063	219.9	62.6 ±	11.1
G1222-8	16	115	127	50	3.655E+06	4.036E+06	0.906	159.6	53.3 ±	7.0
G1222-8	18	94	51	36	4.149E+06	2.251E+06	1.843	89.0	108.0 ±	19.0
G1222-8	21	99	93	27	5.827E+06	5.473E+06	1.065	216.4	62.6 ±	9.2
G1222-8	23	142	99	40	5.641E+06	3.933E+06	1.434	155.5	84.2 ±	11.2
G1222-8	24	131	109	50	4.163E+06	3.464E+06	1.202	137.0	70.6 ±	9.3
G1222-8	31	70	45	40	2.781E+06	1.788E+06	1.556	70.7	91.3 ±	17.6
G1222-8	33	55	55	25	3.496E+06	3.496E+06	1.000	138.2	58.8 ±	11.3
G1222-8	34	110	111	50	3.496E+06	3.528E+06	0.991	139.5	58.3 ±	8.0
G1222-8	39	112	123	50	3.560E+06	3.909E+06	0.911	154.6	53.6 ±	7.1
G1222-8	44	79	83	20	6.277E+06	6.595E+06	0.952	260.8	56.0 ±	8.9
G1222-8	46	128	126	40	5.085E+06	5.006E+06	1.016	197.9	59.8 ±	7.6
G1222-8	47	163	123	70	3.700E+06	2.792E+06	1.325	110.4	77.9 ±	9.5
G1222-8	50	105	103	40	4.171E+06	4.092E+06	1.019	161.8	60.0 ±	8.4
G1222-8	51	148	129	40	5.880E+06	5.125E+06	1.147	202.6	67.5 ±	8.3
		1944	1763		3.925E+06	3.560E+06		140.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 37.655 with 19 degrees of freedom
P(χ^2) = 0.7%
Age Dispersion = 14.027%

Ns / Ni = 1.103 ± 0.036
Mean Ratio = 1.116 ± 0.060

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 64.8 ± 2.6 Ma
CENTRAL AGE = 64.5 ± 3.3 Ma

A:

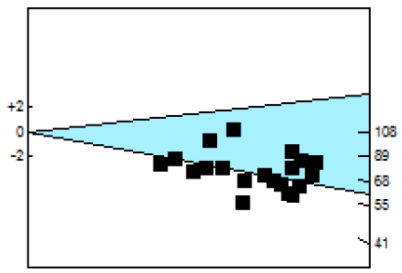


Figure S2.55: Zircon Fission Track Age Data of sample 548374

GC1129-47 Zircon
Counted by: PFG

538709

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-9	3	98	69	15	1.038E+07	7.310E+06	1.420	289.0	83.4 ±	13.3
G1222-9	9	98	38	15	1.038E+07	4.026E+06	2.579	159.2	150.7 ±	29.0
G1222-9	20	115	34	15	1.218E+07	3.602E+06	3.382	142.4	196.9 ±	38.7
G1222-9	29	75	28	12	9.932E+06	3.708E+06	2.679	146.6	156.4 ±	34.8
G1222-9	35	124	70	20	9.852E+06	5.562E+06	1.771	219.9	103.9 ±	15.7
G1222-9	50	114	57	30	6.038E+06	3.019E+06	2.000	119.4	117.1 ±	19.2
G1222-9	56	120	82	24	7.945E+06	5.429E+06	1.463	214.7	85.9 ±	12.5
G1222-9	62	125	72	18	1.104E+07	6.356E+06	1.736	251.3	101.8 ±	15.3
G1222-9	73	150	69	20	1.192E+07	5.482E+06	2.174	216.8	127.2 ±	18.7
G1222-9	86	174	80	21	1.317E+07	6.054E+06	2.175	239.4	127.3 ±	17.5
G1222-9	90	84	50	15	8.899E+06	5.297E+06	1.680	209.4	98.5 ±	17.8
G1222-9	97	130	67	15	1.377E+07	7.098E+06	1.940	280.7	113.7 ±	17.3
		1407	716		1.016E+07	5.172E+06		204.5		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 23.529 with 11 degrees of freedom
P(χ^2) = 1.5%
Age Dispersion = 15.707%

Ns / Ni = 1.965 ± 0.090
Mean Ratio = 2.083 ± 0.164

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 115.1 ± 5.9 Ma
CENTRAL AGE = 115.9 ± 8.0 Ma

A:

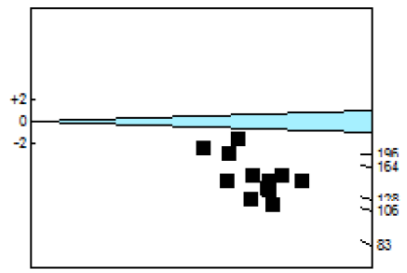


Figure S2.56: Zircon Fission Track Age Data of sample 548709

GC1129-49 Zircon
Counted by: PFG

538713

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-10	4	284	105	30	1.504E+07	5.562E+06	2.705	219.9	157.9 ±	18.4
G1222-10	9	102	36	9	1.801E+07	6.356E+06	2.833	251.3	165.3 ±	32.3
G1222-10	10	309	99	40	1.228E+07	3.933E+06	3.121	155.5	181.9 ±	21.4
G1222-10	11	74	44	20	5.880E+06	3.496E+06	1.682	138.2	98.7 ±	18.9
G1222-10	15	221	75	40	8.780E+06	2.980E+06	2.947	117.8	171.9 ±	23.3
G1222-10	19	82	40	16	8.144E+06	3.973E+06	2.050	157.1	120.1 ±	23.3
G1222-10	22	157	56	20	1.247E+07	4.449E+06	2.804	175.9	163.6 ±	25.8
G1222-10	23	215	75	40	8.541E+06	2.980E+06	2.867	117.8	167.3 ±	22.8
G1222-10	24	126	33	16	1.251E+07	3.277E+06	3.818	129.6	221.8 ±	43.7
G1222-10	25	154	54	20	1.224E+07	4.290E+06	2.852	169.6	166.4 ±	26.6
G1222-10	26	132	38	25	8.390E+06	2.415E+06	3.474	95.5	202.1 ±	37.5
G1222-10	28	161	59	20	1.279E+07	4.688E+06	2.729	185.4	159.3 ±	24.5
G1222-10	30	243	93	36	1.073E+07	4.105E+06	2.613	162.3	152.6 ±	19.0
G1222-10	33	206	52	40	8.184E+06	2.066E+06	3.962	81.7	230.0 ±	36.1
G1222-10	35	74	27	12	9.799E+06	3.575E+06	2.741	141.4	160.0 ±	36.2
G1222-10	44	93	40	12	1.232E+07	5.297E+06	2.325	209.4	136.0 ±	25.9
G1222-10	45	148	76	32	7.349E+06	3.774E+06	1.947	149.2	114.1 ±	16.3
G1222-10	48	114	37	16	1.132E+07	3.675E+06	3.081	145.3	179.6 ±	34.2
G1222-10	56	310	89	30	1.642E+07	4.714E+06	3.483	186.4	202.7 ±	24.8
G1222-10	63	172	47	20	1.367E+07	3.734E+06	3.660	147.7	212.8 ±	35.4
		3377	1175		1.086E+07	3.780E+06		149.4		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 34.175 with 19 degrees of freedom
P(χ^2) = 1.8%
Age Dispersion = 12.165%
Ns / Ni = 2.874 ± 0.097
Mean Ratio = 2.885 ± 0.135

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 167.7 ± 6.9 Ma
CENTRAL AGE = 166.1 ± 8.3 Ma

A:

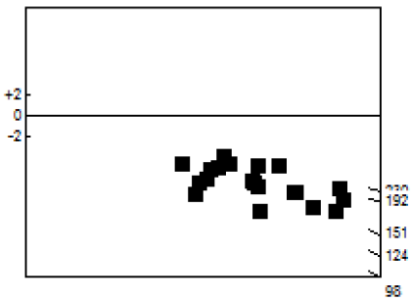


Figure S2.57: Zircon Fission Track Age Data of sample 548713

GC1129-50 Zircon 538719
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-11	6	126	67	24	8.343E+06	4.436E+06	1.881	175.4	110.2 ±	16.9
G1222-11	7	163	67	20	1.295E+07	5.323E+06	2.433	210.5	142.2 ±	20.9
G1222-11	9	76	34	10	1.208E+07	5.403E+06	2.235	213.6	130.8 ±	27.2
G1222-11	10	77	36	8	1.529E+07	7.151E+06	2.139	282.7	125.2 ±	25.5
G1222-11	32	62	47	16	6.158E+06	4.668E+06	1.319	184.6	77.5 ±	15.1
G1222-11	37	249	98	30	1.319E+07	5.191E+06	2.541	205.2	148.5 ±	18.0
G1222-11	48	241	84	30	1.277E+07	4.449E+06	2.869	175.9	167.4 ±	21.6
G1222-11	49	72	26	12	9.534E+06	3.443E+06	2.769	136.1	161.7 ±	37.2
G1222-11	57	75	27	15	7.945E+06	2.860E+06	2.778	113.1	162.1 ±	36.6
G1222-11	58	154	58	15	1.631E+07	6.144E+06	2.655	242.9	155.1 ±	24.2
G1222-11	59	202	80	20	1.605E+07	6.356E+06	2.525	251.3	147.6 ±	19.8
G1222-11	67	148	56	21	1.120E+07	4.238E+06	2.643	167.5	154.4 ±	24.5
G1222-11	68	175	76	30	9.270E+06	4.026E+06	2.303	159.2	134.7 ±	18.8
G1222-11	72	224	108	30	1.187E+07	5.721E+06	2.074	226.2	121.5 ±	14.5
G1222-11	74	113	44	15	1.197E+07	4.661E+06	2.568	184.3	150.1 ±	26.9
G1222-11	77	78	43	15	8.263E+06	4.555E+06	1.814	180.1	106.3 ±	20.4
G1222-11	78	104	56	30	5.509E+06	2.966E+06	1.857	117.3	108.9 ±	18.2
G1222-11	84	71	26	10	1.128E+07	4.132E+06	2.731	163.4	159.4 ±	36.7
G1222-11	85	130	55	15	1.377E+07	5.827E+06	2.364	230.4	138.2 ±	22.5
G1222-11	87	153	68	30	8.104E+06	3.602E+06	2.250	142.4	131.7 ±	19.4
		2693	1156		1.081E+07	4.639E+06		183.4		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 22.837 with 19 degrees of freedom
P(χ^2) = 24.5%
Age Dispersion = 4.020% (did not converge)
Ns / Ni = 2.330 ± 0.082
Mean Ratio = 2.337 ± 0.089

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 136.3 ± 5.8 Ma
CENTRAL AGE = 136.1 ± 5.9 Ma

A:



Figure S2.58: Zircon Fission Track Age Data of sample 548719

GC1129-54 Zircon
Counted by: PFG

538822

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-12	1	280	106	25	1.780E+07	6.738E+06	2.642	266.4	154.3 ±	18.0
G1222-12	9	73	35	10	1.160E+07	5.562E+06	2.086	219.9	122.1 ±	25.3
G1222-12	11	83	58	8	1.649E+07	1.152E+07	1.431	455.5	84.0 ±	14.5
G1222-12	14	76	49	15	8.051E+06	5.191E+06	1.551	205.2	91.0 ±	16.8
G1222-12	15	83	44	12	1.099E+07	5.827E+06	1.886	230.4	110.6 ±	20.8
G1222-12	18	82	46	10	1.303E+07	7.310E+06	1.783	289.0	104.5 ±	19.4
G1222-12	20	50	23	10	7.945E+06	3.655E+06	2.174	144.5	127.2 ±	32.2
G1222-12	26	69	38	8	1.371E+07	7.548E+06	1.816	298.4	106.5 ±	21.7
G1222-12	35	101	60	12	1.337E+07	7.945E+06	1.683	314.1	98.8 ±	16.3
G1222-12	36	113	49	12	1.496E+07	6.489E+06	2.306	256.5	134.9 ±	23.3
G1222-12	37	72	41	8	1.430E+07	8.144E+06	1.756	322.0	103.0 ±	20.3
G1222-12	52	83	30	10	1.319E+07	4.767E+06	2.767	188.5	161.5 ±	34.6
G1222-12	55	132	96	21	9.988E+06	7.264E+06	1.375	287.2	80.8 ±	11.0
G1222-12	56	137	66	12	1.814E+07	8.740E+06	2.076	345.5	121.6 ±	18.4
G1222-12	57	119	66	20	9.455E+06	5.244E+06	1.803	207.3	105.7 ±	16.4
G1222-12	64	115	44	12	1.523E+07	5.827E+06	2.614	230.4	152.7 ±	27.3
		1668	851		1.293E+07	6.597E+06		260.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 27.548 with 15 degrees of freedom
$P(\chi^2)$ = 2.5%
Age Dispersion = 14.504%

Ns / Ni = 1.960 ± 0.083
Mean Ratio = 1.984 ± 0.106

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 114.8 ± 5.5 Ma
CENTRAL AGE = 113.4 ± 6.9 Ma

A:

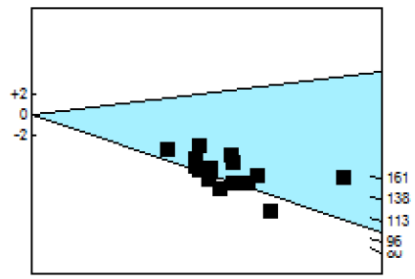


Figure S2.59: Zircon Fission Track Age Data of sample 548822

GC1129-56 Zircon
Counted by: PFG

538722

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)
G1222-13	2	137	57	25	8.708E+06	3.623E+06	2.404	143.2	140.5 ± 22.4
G1222-13	4	164	75	50	5.212E+06	2.384E+06	2.187	94.2	128.0 ± 18.1
G1222-13	5	102	46	16	1.013E+07	4.569E+06	2.217	180.6	129.8 ± 23.2
G1222-13	10	104	33	18	9.181E+06	2.913E+06	3.152	115.2	183.7 ± 37.0
G1222-13	11	110	45	20	8.740E+06	3.575E+06	2.444	141.4	142.9 ± 25.5
G1222-13	14	75	33	20	5.959E+06	2.622E+06	2.273	103.7	133.0 ± 28.0
G1222-13	16	127	54	30	6.727E+06	2.860E+06	2.352	113.1	137.6 ± 22.6
G1222-13	21	215	65	25	1.367E+07	4.132E+06	3.308	163.3	192.6 ± 27.6
G1222-13	23	80	24	20	6.356E+06	1.907E+06	3.333	75.4	194.1 ± 45.4
G1222-13	26	87	35	15	9.217E+06	3.708E+06	2.486	146.6	145.3 ± 29.3
G1222-13	36	96	16	25	6.102E+06	1.017E+06	6.000	40.2	345.3 ± 93.6
G1222-13	60	165	75	40	6.555E+06	2.980E+06	2.200	117.8	128.8 ± 18.2
G1222-13	68	103	48	15	1.091E+07	5.085E+06	2.146	201.0	125.6 ± 22.2
G1222-13	73	103	47	20	8.184E+06	3.734E+06	2.191	147.6	128.3 ± 22.8
G1222-13	88	121	72	25	7.691E+06	4.577E+06	1.681	180.9	98.6 ± 14.9
G1222-13	92	42	16	15	4.449E+06	1.695E+06	2.625	67.0	153.3 ± 45.2
G1222-13	94	146	52	25	9.280E+06	3.305E+06	2.808	130.7	163.9 ± 26.7
G1222-13	96	106	31	40	4.211E+06	1.232E+06	3.419	48.7	199.0 ± 40.9
G1222-13	97	81	45	12	1.073E+07	5.959E+06	1.800	235.6	105.5 ± 19.8
G1222-13	103	181	56	18	1.598E+07	4.944E+06	3.232	195.4	188.3 ± 29.1
		2345	925		7.862E+06	3.101E+06		122.6	

Area of basic unit = 6.293E-07 cm-²
χ^2 = 38.224 with 19 degrees of freedom
P(χ^2) = 0.6%
Age Dispersion = 16.661%

Ns / Ni = 2.535 ± 0.098
Mean Ratio = 2.713 ± 0.207

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm-² ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm-² ND = 1060
bottom of can; ρ = 1.349E+06cm-² ND = 1061

POOLED AGE = 148.2 ± 6.7 Ma
CENTRAL AGE = 148.5 ± 8.8 Ma

A:

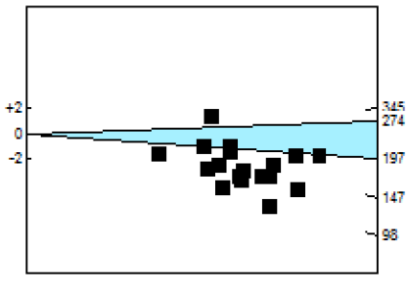


Figure S2.60: Zircon Fission Track Age Data of sample 548722

GC1129-59 Zircon
Counted by: PFG

538726

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)
G1222-14	5	166	48	20	1.319E+07	3.814E+06	3.458	150.8	201.3 ± 33.3
G1222-14	11	156	85	30	8.263E+06	4.502E+06	1.835	178.0	107.6 ± 14.7
G1222-14	13	156	34	16	1.549E+07	3.377E+06	4.588	133.5	265.7 ± 50.7
G1222-14	24	155	144	50	4.926E+06	4.577E+06	1.076	180.9	63.3 ± 7.5
G1222-14	43	67	56	12	8.872E+06	7.416E+06	1.196	293.2	70.3 ± 12.8
G1222-14	46	83	49	12	1.099E+07	6.489E+06	1.694	256.5	99.4 ± 18.1
G1222-14	47	240	132	30	1.271E+07	6.992E+06	1.818	276.4	106.6 ± 11.8
G1222-14	58	121	39	21	9.156E+06	2.951E+06	3.103	116.7	180.9 ± 33.6
G1222-14	61	103	28	20	8.184E+06	2.225E+06	3.679	87.9	213.9 ± 45.9
G1222-14	68	148	82	20	1.176E+07	6.515E+06	1.805	257.6	105.8 ± 14.8
G1222-14	69	202	91	25	1.284E+07	5.784E+06	2.220	228.7	129.9 ± 16.7
G1222-14	77	87	35	40	3.456E+06	1.390E+06	2.486	55.0	145.3 ± 29.3
G1222-14	82	81	78	25	5.149E+06	4.958E+06	1.038	196.0	61.1 ± 9.8
		1765	901		8.737E+06	4.460E+06		176.3	

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 105.019 with 12 degrees of freedom
$P(\chi^2)$ = 0.0%
Age Dispersion = 41.153%

Ns / Ni = 1.959 ± 0.080
Mean Ratio = 2.307 ± 0.305

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 114.8 ± 5.4 Ma
CENTRAL AGE = 117.5 ± 14.6 Ma

A:

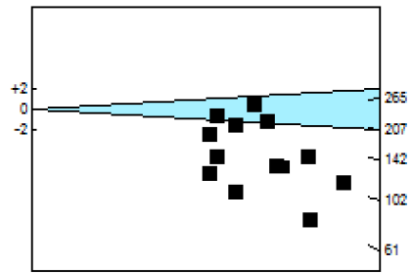


Figure S2.61: Zircon Fission Track Age Data of sample 548726

GC1129-60 Zircon
Counted by: PFG

538728

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-15	1	123	40	12	1.629E+07	5.297E+06	3.075	209.4	179.3 ±	32.9
G1222-15	2	103	53	15	1.091E+07	5.615E+06	1.943	222.0	113.9 ±	19.4
G1222-15	8	169	55	30	8.952E+06	2.913E+06	3.073	115.2	179.2 ±	28.1
G1222-15	9	99	52	18	8.740E+06	4.591E+06	1.904	181.5	111.6 ±	19.3
G1222-15	10	146	84	40	5.800E+06	3.337E+06	1.738	131.9	102.0 ±	14.2
G1222-15	13	116	36	18	1.024E+07	3.178E+06	3.222	125.6	187.7 ±	36.1
G1222-15	17	98	53	16	9.733E+06	5.264E+06	1.849	208.1	108.4 ±	18.7
G1222-15	28	108	38	12	1.430E+07	5.032E+06	2.842	198.9	165.9 ±	31.5
G1222-15	34	91	54	18	8.034E+06	4.767E+06	1.685	188.5	98.9 ±	17.1
G1222-15	39	188	84	25	1.195E+07	5.339E+06	2.238	211.1	131.0 ±	17.5
G1222-15	43	295	128	50	9.375E+06	4.068E+06	2.305	160.8	134.8 ±	14.6
G1222-15	46	146	61	25	9.280E+06	3.877E+06	2.393	153.3	140.0 ±	21.6
G1222-15	47	157	83	25	9.979E+06	5.276E+06	1.892	208.6	110.9 ±	15.3
G1222-15	57	119	53	15	1.261E+07	5.615E+06	2.245	222.0	131.4 ±	21.9
G1222-15	60	199	72	25	1.265E+07	4.577E+06	2.764	180.9	161.4 ±	22.5
G1222-15	61	106	47	20	8.422E+06	3.734E+06	2.255	147.6	132.0 ±	23.3
G1222-15	62	316	140	60	8.369E+06	3.708E+06	2.257	146.6	132.1 ±	13.8
G1222-15	65	216	100	20	1.716E+07	7.945E+06	2.160	314.1	126.5 ±	15.6
G1222-15	67	244	111	40	9.693E+06	4.410E+06	2.198	174.3	128.7 ±	15.0
G1222-15	70	134	49	25	8.517E+06	3.115E+06	2.735	123.1	159.7 ±	26.9
		3173	1393		9.906E+06	4.349E+06		171.9		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 27.226 with 19 degrees of freedom
$P(\chi^2)$ = 10.0%
Age Dispersion = 7.078% (did not converge)

Ns / Ni = 2.278 ± 0.073
Mean Ratio = 2.339 ± 0.104

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can: ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can: ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 133.3 ± 5.3 Ma
CENTRAL AGE = 133.3 ± 5.8 Ma

A:

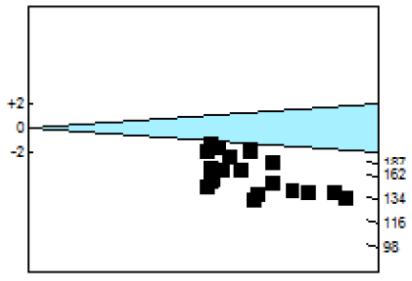


Figure S2.62: Zircon Fission Track Age Data of sample 548728

GC1129-63 Zircon 538732
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)
G1222-16	1	196	61	60	5.191E+06	1.616E+06	3.213	63.9	187.2 ± 27.8
G1222-16	5	112	53	16	1.112E+07	5.264E+06	2.113	208.1	123.7 ± 20.8
G1222-16	15	188	88	32	9.336E+06	4.370E+06	2.136	172.7	125.1 ± 16.4
G1222-16	24	93	30	15	9.852E+06	3.178E+06	3.100	125.6	180.7 ± 38.2
G1222-16	25	135	42	25	8.581E+06	2.670E+06	3.214	105.5	187.3 ± 33.4
G1222-16	26	63	33	25	4.004E+06	2.098E+06	1.909	82.9	111.9 ± 24.2
G1222-16	30	315	128	50	1.001E+07	4.068E+06	2.461	160.8	143.9 ± 15.5
G1222-16	33	143	51	25	9.089E+06	3.242E+06	2.804	128.1	163.7 ± 27.0
G1222-16	36	84	40	15	8.899E+06	4.238E+06	2.100	167.5	123.0 ± 23.8
G1222-16	37	219	73	25	1.392E+07	4.640E+06	3.000	183.4	175.0 ± 24.0
G1222-16	38	114	50	25	7.246E+06	3.178E+06	2.280	125.6	133.4 ± 22.8
G1222-16	48	222	158	40	8.819E+06	6.277E+06	1.405	248.1	82.5 ± 8.8
G1222-16	59	66	47	16	6.555E+06	4.668E+06	1.404	184.5	82.5 ± 15.9
G1222-16	65	153	39	25	9.725E+06	2.479E+06	3.923	98.0	227.9 ± 41.2
G1222-16	76	153	57	28	8.683E+06	3.235E+06	2.684	127.9	156.8 ± 24.6
G1222-16	77	178	51	50	5.657E+06	1.621E+06	3.490	64.1	203.1 ± 32.6
G1222-16	79	87	55	15	9.217E+06	5.827E+06	1.582	230.3	92.9 ± 16.1
G1222-16	82	30	13	25	1.907E+06	8.263E+05	2.308	32.7	135.0 ± 44.9
G1222-16	84	152	87	60	4.026E+06	2.304E+06	1.747	91.1	102.5 ± 14.0
G1222-16	112	51	27	20	4.052E+06	2.145E+06	1.889	84.8	110.7 ± 26.5
		2754	1183		7.392E+06	3.175E+06		125.5	

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 74.926 with 19 degrees of freedom
$P(\chi^2)$ = 0.0%
Age Dispersion = 25.081%

Ns / Ni = 2.328 ± 0.081
Mean Ratio = 2.438 ± 0.160

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can: ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can: ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 136.2 ± 5.7 Ma
CENTRAL AGE = 135.8 ± 9.7 Ma

A:

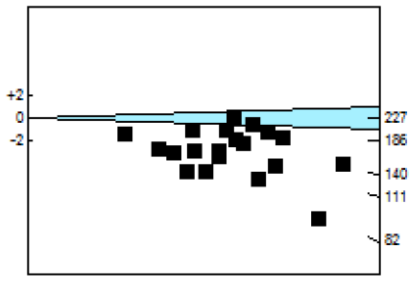


Figure S2.63: Zircon Fission Track Age Data of sample 548732

GC1129-67 Zircon 538733
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-17	3	103	32	20	8.184E+06	2.543E+06	3.219	100.5	187.6 ±	38.2
G1222-17	12	74	38	12	9.799E+06	5.032E+06	1.947	198.9	114.1 ±	22.9
G1222-17	32	90	26	9	1.589E+07	4.591E+06	3.462	181.5	201.5 ±	45.1
G1222-17	56	53	27	12	7.018E+06	3.575E+06	1.963	141.3	115.0 ±	27.3
G1222-17	57	140	76	15	1.483E+07	8.051E+06	1.842	318.3	108.0 ±	15.6
G1222-17	60	60	29	10	9.534E+06	4.608E+06	2.069	182.2	121.2 ±	27.6
G1222-17	65	83	28	14	9.421E+06	3.178E+06	2.964	125.6	172.9 ±	38.0
G1222-17	66	217	149	20	1.724E+07	1.184E+07	1.456	468.0	85.5 ±	9.3
G1222-17	70	143	64	25	9.089E+06	4.068E+06	2.234	160.8	130.8 ±	19.9
G1222-17	83	76	41	10	1.208E+07	6.515E+06	1.854	257.5	108.7 ±	21.2
G1222-17	86	64	37	10	1.017E+07	5.880E+06	1.730	232.4	101.5 ±	21.1
G1222-17	104	113	46	15	1.197E+07	4.873E+06	2.457	192.6	143.6 ±	25.3
		1216	593		1.123E+07	5.479E+06		216.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 27.043 with 11 degrees of freedom
P(χ^2) = 0.5%
Age Dispersion = 19.129%

Ns / Ni = 2.051 ± 0.103
Mean Ratio = 2.266 ± 0.182

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 120.1 ± 6.6 Ma
CENTRAL AGE = 125.0 ± 10.0 Ma

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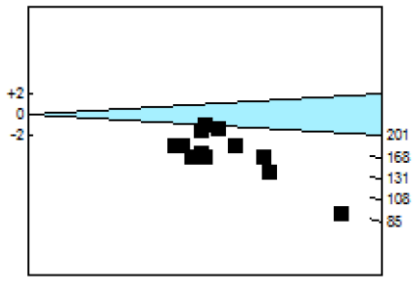


Figure S2.64: Zircon Fission Track Age Data of sample 548733

GC1129-68 Zircon 538735
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)
G1222-18	2	122	69	20	9.693E+06	5.482E+06	1.768	216.7	103.7 ± 15.8
G1222-18	8	76	69	16	7.548E+06	6.853E+06	1.101	270.9	64.8 ± 10.9
G1222-18	13	118	73	20	9.375E+06	5.800E+06	1.616	229.3	94.9 ± 14.3
G1222-18	17	125	75	20	9.932E+06	5.959E+06	1.667	235.5	97.8 ± 14.5
G1222-18	20	184	86	20	1.462E+07	6.833E+06	2.140	270.1	125.3 ± 16.6
G1222-18	44	98	67	20	7.786E+06	5.323E+06	1.463	210.4	85.9 ± 13.8
G1222-18	45	74	73	15	7.839E+06	7.733E+06	1.014	305.7	59.7 ± 9.9
G1222-18	59	126	74	16	1.251E+07	7.349E+06	1.703	290.5	99.9 ± 14.8
G1222-18	70	58	50	15	6.144E+06	5.297E+06	1.160	209.4	68.2 ± 13.3
G1222-18	108	97	90	20	7.707E+06	7.151E+06	1.078	282.6	63.4 ± 9.4
		1078	726		9.412E+06	6.339E+06		250.5	

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 26.239 with 9 degrees of freedom
P(χ^2) = 0.2%
Age Dispersion = 19.125%
Ns / Ni = 1.485 ± 0.071
Mean Ratio = 1.471 ± 0.118

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.348E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 87.2 ± 4.7 Ma
CENTRAL AGE = 84.9 ± 6.9 Ma

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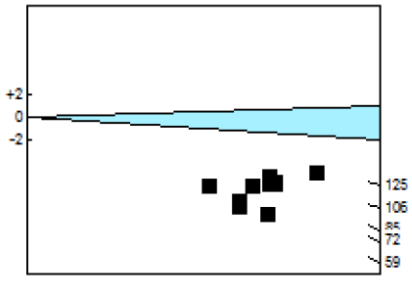


Figure S2.65: Zircon Fission Track Age Data of sample 548735

GC1129-70 Zircon
Counted by: PFG

538751

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-20	2	226	149	60	5.985E+06	3.946E+06	1.517	156.0	89.1 ±	9.6
G1222-20	3	102	79	25	6.483E+06	5.021E+06	1.291	198.5	75.9 ±	11.5
G1222-20	13	54	50	20	4.290E+06	3.973E+06	1.080	157.0	63.6 ±	12.6
G1222-20	16	62	50	20	4.926E+06	3.973E+06	1.240	157.0	72.9 ±	14.0
G1222-20	17	47	38	12	6.224E+06	5.032E+06	1.237	198.9	72.7 ±	16.0
G1222-20	18	47	48	12	6.224E+06	6.356E+06	0.979	251.2	57.6 ±	11.9
G1222-20	20	60	66	30	3.178E+06	3.496E+06	0.909	138.2	53.5 ±	9.6
G1222-20	49	85	55	20	6.754E+06	4.370E+06	1.545	172.7	90.7 ±	15.8
G1222-20	50	103	96	30	5.456E+06	5.085E+06	1.073	201.0	63.1 ±	9.1
G1222-20	52	107	53	24	7.085E+06	3.509E+06	2.019	138.7	118.3 ±	20.1
G1222-20	55	49	41	15	5.191E+06	4.343E+06	1.195	171.7	70.3 ±	15.0
G1222-20	57	81	58	21	6.129E+06	4.389E+06	1.397	173.5	82.1 ±	14.2
G1222-20	64	77	60	15	8.157E+06	6.356E+06	1.283	251.2	75.4 ±	13.1
G1222-20	66	93	46	24	6.158E+06	3.046E+06	2.022	120.4	118.5 ±	21.5
G1222-20	68	83	77	30	4.396E+06	4.079E+06	1.078	161.2	63.4 ±	10.1
G1222-20	70	66	73	25	4.195E+06	4.640E+06	0.904	183.4	53.2 ±	9.1
G1222-20	79	60	46	18	5.297E+06	4.061E+06	1.304	160.5	76.7 ±	15.1
G1222-20	80	103	98	30	5.456E+06	5.191E+06	1.051	205.2	61.9 ±	8.8
G1222-20	86	129	87	30	6.833E+06	4.608E+06	1.483	182.1	87.1 ±	12.3
G1222-20	88	98	71	16	9.733E+06	7.051E+06	1.380	278.7	81.1 ±	12.8
		1732	1341		5.770E+06	4.467E+06		176.6		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 34.683 with 19 degrees of freedom
P(χ^2) = 1.5%
Age Dispersion = 14.228%

Ns / Ni = 1.292 ± 0.047
Mean Ratio = 1.299 ± 0.070

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.349E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can: ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can: ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 75.9 ± 3.3 Ma
CENTRAL AGE = 75.1 ± 4.1 Ma

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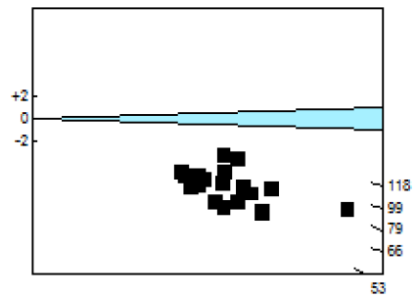


Figure S2.66: Zircon Fission Track Age Data of sample 548751

GC1129-72 Zircon 538754
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-21	1	108	65	25	6.865E+06	4.132E+06	1.662	163.3	97.5 ±	15.5
G1222-21	11	197	91	15	2.087E+07	9.640E+06	2.165	381.0	126.8 ±	16.3
G1222-21	16	269	104	18	2.375E+07	9.181E+06	2.587	362.9	151.2 ±	17.8
G1222-21	23	76	33	12	1.006E+07	4.370E+06	2.303	172.7	134.8 ±	28.3
G1222-21	33	67	35	18	5.915E+06	3.090E+06	1.914	122.1	112.2 ±	23.6
G1222-21	44	179	78	40	7.111E+06	3.099E+06	2.295	122.5	134.3 ±	18.5
G1222-21	48	73	44	30	3.867E+06	2.331E+06	1.659	92.1	97.4 ±	18.7
G1222-21	49	162	120	30	8.581E+06	6.356E+06	1.350	251.2	79.3 ±	9.7
G1222-21	52	62	31	16	6.158E+06	3.079E+06	2.000	121.7	117.2 ±	25.9
G1222-21	53	103	49	16	1.023E+07	4.867E+06	2.102	192.3	123.1 ±	21.6
G1222-21	58	128	50	21	9.686E+06	3.783E+06	2.560	149.5	149.6 ±	25.2
G1222-21	60	160	68	30	8.475E+06	3.602E+06	2.353	142.4	137.7 ±	20.2
G1222-21	62	134	51	40	5.323E+06	2.026E+06	2.627	80.1	153.5 ±	25.5
G1222-21	65	84	40	40	3.337E+06	1.589E+06	2.100	62.8	123.0 ±	23.8
G1222-21	67	149	78	40	5.919E+06	3.099E+06	1.910	122.5	112.0 ±	15.9
G1222-21	71	33	16	9	5.827E+06	2.825E+06	2.063	111.6	120.8 ±	36.9
G1222-21	74	95	55	16	9.435E+06	5.462E+06	1.727	215.9	101.3 ±	17.3
G1222-21	86	88	39	16	8.740E+06	3.873E+06	2.256	153.1	132.1 ±	25.6
G1222-21	89	56	26	15	5.933E+06	2.754E+06	2.154	108.9	126.1 ±	30.1
G1222-21	90	102	53	25	6.483E+06	3.369E+06	1.925	133.1	112.8 ±	19.3
		2325	1126		7.828E+06	3.791E+06		149.8		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 27.372 with 19 degrees of freedom
P(χ^2) = 9.6%
Age Dispersion = 11.678%

Ns / Ni = 2.065 ± 0.075
Mean Ratio = 2.086 ± 0.074

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.349E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 121.0 ± 5.2 Ma
CENTRAL AGE = 120.5 ± 6.2 Ma

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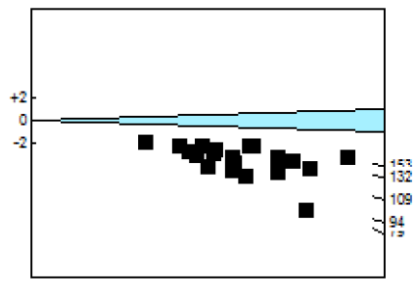


Figure S2.67: Zircon Fission Track Age Data of sample 548754

GC1129-73 Zircon 538756
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-22	1	98	28	15	1.038E+07	2.966E+06	3.500	117.2	203.7 ±	43.9
G1222-22	4	29	9	3	1.536E+07	4.767E+06	3.222	188.4	187.8 ±	71.8
G1222-22	10	78	46	12	1.033E+07	6.091E+06	1.696	240.7	99.5 ±	18.6
G1222-22	11	93	48	12	1.232E+07	6.356E+06	1.938	251.2	113.6 ±	20.4
G1222-22	26	37	14	8	7.349E+06	2.781E+06	2.643	109.9	154.4 ±	48.6
G1222-22	27	41	16	5	1.303E+07	5.085E+06	2.563	201.0	149.8 ±	44.3
G1222-22	28	33	21	4	1.311E+07	8.343E+06	1.571	329.7	92.3 ±	25.8
G1222-22	29	57	31	6	1.510E+07	8.210E+06	1.839	324.5	107.8 ±	24.2
G1222-22	30	73	25	12	9.667E+06	3.311E+06	2.920	130.8	170.4 ±	39.7
		539	238		1.112E+07	4.912E+06		194.1		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 12.504 with 8 degrees of freedom
P(χ^2) = 13.0%
Age Dispersion = 16.040%

Ns / Ni = 2.265 ± 0.176
Mean Ratio = 2.432 ± 0.234

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.349E+06cm^{-2} ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm^{-2} ND = 1060
bottom of can; ρ = 1.349E+06cm^{-2} ND = 1061

POOLED AGE = 132.6 ± 10.8 Ma
CENTRAL AGE = 133.1 ± 13.2 Ma

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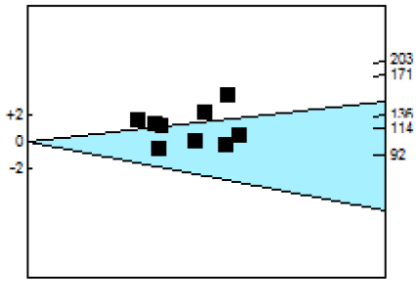


Figure S2.68: Zircon Fission Track Age Data of sample 548756

GC1129-76 Zircon 538761
Counted by: PFG

Slide ref	Current grain no	N_s	N_i	N_a	ρ_S	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1222-23	1	114	62	18	1.006E+07	5.473E+06	1.839	216.3	107.8 ±	17.2
G1222-23	3	77	28	10	1.224E+07	4.449E+06	2.750	175.8	160.6 ±	35.6
G1222-23	6	76	47	16	7.548E+06	4.668E+06	1.617	184.5	94.9 ±	17.8
G1222-23	7	96	41	9	1.695E+07	7.239E+06	2.341	286.1	137.0 ±	25.8
G1222-23	12	86	64	21	6.508E+06	4.843E+06	1.344	191.4	79.0 ±	13.2
G1222-23	17	176	70	60	4.661E+06	1.854E+06	2.514	73.3	147.0 ±	21.1
G1222-23	19	41	20	12	5.429E+06	2.648E+06	2.050	104.7	120.1 ±	32.9
G1222-23	20	101	45	12	1.337E+07	5.959E+06	2.244	235.5	131.4 ±	23.8
G1222-23	21	174	73	40	6.912E+06	2.900E+06	2.384	114.6	139.5 ±	19.7
G1222-23	24	79	34	15	8.369E+06	3.602E+06	2.324	142.3	136.0 ±	28.1
G1222-23	28	75	36	18	6.621E+06	3.178E+06	2.083	125.6	122.1 ±	24.9
G1222-23	29	62	30	18	5.473E+06	2.648E+06	2.067	104.7	121.1 ±	27.1
G1222-23	33	95	57	18	8.387E+06	5.032E+06	1.667	198.9	97.8 ±	16.6
G1222-23	34	53	18	15	5.615E+06	1.907E+06	2.944	75.4	171.8 ±	47.1
G1222-23	37	95	47	20	7.548E+06	3.734E+06	2.021	147.6	118.4 ±	21.3
G1222-23	38	59	29	20	4.688E+06	2.304E+06	2.034	91.1	119.2 ±	27.2
G1222-23	39	198	77	25	1.259E+07	4.894E+06	2.571	193.4	150.3 ±	20.5
G1222-23	61	130	47	30	6.886E+06	2.490E+06	2.766	98.4	161.5 ±	27.8
G1222-23	82	51	24	15	5.403E+06	2.543E+06	2.125	100.5	124.5 ±	30.9
G1222-23	93	89	52	30	4.714E+06	2.754E+06	1.712	108.8	100.4 ±	17.7
		1927	901		7.256E+06	3.393E+06		134.1		

Area of basic unit = 6.293E-07 cm-²
χ^2 = 24.614 with 19 degrees of freedom
$P(\chi^2)$ = 17.4%
Age Dispersion = 10.101%
Ns / Ni = 2.139 ± 0.086
Mean Ratio = 2.170 ± 0.093

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.349E+06cm-² ND = 2121
ρ_D interpolated between top of can; ρ = 1.348E+06cm-² ND = 1060
bottom of can; ρ = 1.349E+06cm-² ND = 1061

POOLED AGE = 125.3 ± 5.9 Ma
CENTRAL AGE = 124.6 ± 6.5 Ma

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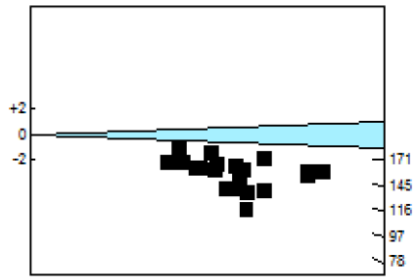


Figure S2.69: Zircon Fission Track Age Data of sample 548761

GC1129-78 Zircon
Counted by: PFG

453010

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1223-1	17	116	86	15	1.229E+07	9.111E+06	1.349	391.4	73.0 ±	10.5
G1223-1	19	74	60	16	7.349E+06	5.959E+06	1.233	256.0	66.8 ±	11.7
G1223-1	21	100	76	15	1.059E+07	8.051E+06	1.316	345.9	71.2 ±	11.0
G1223-1	23	89	85	15	9.428E+06	9.005E+06	1.047	386.8	56.7 ±	8.7
G1223-1	24	77	73	15	8.157E+06	7.733E+06	1.055	332.2	57.1 ±	9.4
G1223-1	25	71	46	12	9.402E+06	6.091E+06	1.543	261.7	83.4 ±	15.9
G1223-1	31	148	105	25	9.407E+06	6.674E+06	1.410	286.7	76.2 ±	9.9
G1223-1	46	218	175	40	8.660E+06	6.952E+06	1.246	298.7	67.4 ±	7.0
G1223-1	51	99	74	25	6.293E+06	4.704E+06	1.338	202.1	72.4 ±	11.3
G1223-1	81	71	75	15	7.522E+06	7.945E+06	0.947	341.3	51.3 ±	8.6
G1223-1	83	90	73	15	9.534E+06	7.733E+06	1.233	332.2	66.7 ±	10.6
G1223-1	84	71	80	16	7.051E+06	7.945E+06	0.887	341.3	48.1 ±	7.9
G1223-1	85	83	86	16	8.243E+06	8.541E+06	0.965	366.9	52.3 ±	8.1
G1223-1	92	62	83	15	6.568E+06	8.793E+06	0.747	377.7	40.5 ±	6.9
G1223-1	93	143	107	24	9.468E+06	7.085E+06	1.336	304.3	72.3 ±	9.4
G1223-1	94	96	60	15	1.017E+07	6.356E+06	1.600	273.1	86.5 ±	14.4
G1223-1	99	154	138	30	8.157E+06	7.310E+06	1.116	314.0	60.4 ±	7.2
G1223-1	106	113	81	15	1.197E+07	8.581E+06	1.395	368.6	75.5 ±	11.1
G1223-1	122	134	127	24	8.872E+06	8.409E+06	1.055	361.2	57.2 ±	7.2
G1223-1	123	52	34	10	8.263E+06	5.403E+06	1.529	232.1	82.7 ±	18.3
		2061	1724		8.780E+06	7.345E+06		315.5		

Area of basic unit = 6.293E-07 cm-²
χ^2 = 29.703 with 19 degrees of freedom
P(χ^2) = 5.6%
Age Dispersion = 9.296% (did not converge)

Ns / Ni = 1.195 ± 0.039
Mean Ratio = 1.217 ± 0.052

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.241E+06cm-² ND = 1960
ρ_D interpolated between top of can; ρ = 1.241E+06cm-² ND = 976
bottom of can; ρ = 1.251E+06cm-² ND = 984

POOLED AGE = 64.7 ± 2.6 Ma
CENTRAL AGE = 64.6 ± 3.0 Ma

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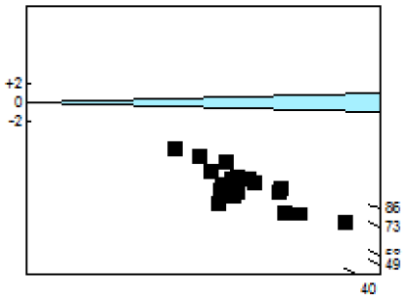


Figure S2.70: Zircon Fission Track Age Data of sample 453010

GC1129-79 Zircon
Counted by: PFG

453032

Slide ref	Current grain no	N_s	N_i	N_a	ρ_s	ρ_i	RATIO	U (ppm)	F.T. AGE (Ma)	
G1223-2	3	36	68	16	3.575E+06	6.754E+06	0.529	290.0	28.8 ±	6.0
G1223-2	5	32	36	20	2.543E+06	2.860E+06	0.889	122.8	48.2 ±	11.8
G1223-2	6	76	55	20	6.038E+06	4.370E+06	1.382	187.6	74.8 ±	13.4
G1223-2	16	54	46	19	4.516E+06	3.847E+06	1.174	165.2	63.6 ±	12.9
G1223-2	19	67	60	20	5.323E+06	4.767E+06	1.117	204.7	60.5 ±	10.9
G1223-2	28	74	62	16	7.349E+06	6.158E+06	1.194	264.4	64.6 ±	11.2
G1223-2	32	206	154	30	1.091E+07	8.157E+06	1.338	350.2	72.4 ±	7.9
G1223-2	34	198	150	36	8.740E+06	6.621E+06	1.320	284.3	71.5 ±	7.9
G1223-2	35	156	84	30	8.263E+06	4.449E+06	1.857	191.0	100.3 ±	13.8
G1223-2	44	67	38	15	7.098E+06	4.026E+06	1.763	172.9	95.3 ±	19.5
G1223-2	49	97	73	25	6.166E+06	4.640E+06	1.329	199.2	71.9 ±	11.3
G1223-2	50	36	79	25	2.288E+06	5.021E+06	0.456	215.6	24.8 ±	5.0
G1223-2	53	89	101	25	5.657E+06	6.420E+06	0.881	275.7	47.8 ±	7.0
G1223-2	55	79	82	30	4.185E+06	4.343E+06	0.963	186.5	52.2 ±	8.3
G1223-2	59	108	84	36	4.767E+06	3.708E+06	1.286	159.2	69.6 ±	10.3
G1223-2	60	127	88	36	5.606E+06	3.884E+06	1.443	166.8	78.1 ±	11.0
G1223-2	61	58	51	18	5.120E+06	4.502E+06	1.137	193.3	61.6 ±	11.9
G1223-2	64	127	113	25	8.072E+06	7.183E+06	1.124	308.4	60.9 ±	8.0
G1223-2	70	116	84	30	6.144E+06	4.449E+06	1.381	191.0	74.7 ±	10.9
G1223-2	72	58	39	15	6.144E+06	4.132E+06	1.487	177.4	80.4 ±	16.8
		1861	1547		6.072E+06	5.048E+06		216.7		

Area of basic unit = 6.293E-07 cm^{-2}
χ^2 = 71.412 with 19 degrees of freedom
P(χ^2) = 0.0%
Age Dispersion = 26.872%

Ns / Ni = 1.203 ± 0.041
Mean Ratio = 1.202 ± 0.078

Ages calculated using a zeta of 87.7 ± 0.8 for U3 glass
ρ = 1.241E+06cm^{-2} ND = 1960
ρ_D interpolated between top of can: ρ = 1.241E+06cm^{-2} ND = 976
bottom of can: ρ = 1.251E+06cm^{-2} ND = 984

POOLED AGE = 65.2 ± 2.7 Ma
CENTRAL AGE = 62.8 ± 4.7 Ma

A:

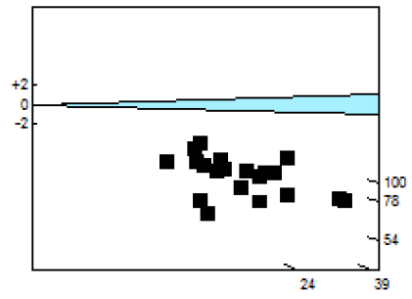


Figure S2.71: Zircon Fission Track Age Data of sample 453032

S3: Principles of Interpretation of AFTA and ZFTA Data in Sedimentary Basins

S3.1 Introduction

In this Appendix, we set out some basic aspects of the processes underlying the AFTA and ZFTA techniques, and explain the principles involved in determining thermal history parameters from AFTA and ZFTA data. As the processes involved in AFTA are understood in considerably more detail than for ZFTA, the discussion is centred on AFTA, with some additional comments on ZFTA provided at the end of the Appendix.

Detrital apatite grains are incorporated into sedimentary rocks from three dominant sources - crystalline basement rocks, older sediments and contemporaneous volcanism. Apatites derived from the first two sources will, in general, contain fission tracks when they are deposited, with AFTA parameters characteristic of the source regions. However, apatites derived from contemporaneous volcanism, or from rapidly uplifted basement, will contain no tracks when they are deposited. For now, we will restrict discussion to this situation, and generalise at a later point to cover the case of apatites which contain tracks that have been inherited from source regions.

S3.2 Basic principles of Apatite Fission Track Analysis

Fission tracks are trails of radiation damage, which are produced within apatite grains at a more or less constant rate through geological time, as a result of the spontaneous fission of 238U impurity atoms. Therefore, the number of fission events which occur within an apatite grain during a fixed time interval depends on the magnitude of the time interval and the uranium content of the grain. Each fission event leads to the formation of a single fission track, and the proportion of tracks which can intersect a polished surface of an apatite grain depends on the length of the tracks. Therefore, the number of tracks which are etched in unit area of the surface of an apatite grain (the "spontaneous track density") depends on three factors - (i) The time over which tracks have been accumulating; (ii) The uranium content of the apatite grain; and, (iii) The distribution of track lengths in the grain. In sedimentary rocks which have not been subjected to temperatures greater than ~50°C since deposition, spontaneous fission tracks have a characteristic distribution of confined track lengths, with a mean length in the range 14-15 μm and a standard deviation of ~1 μm. In such samples, by measuring the spontaneous track density and the uranium content of a collection of apatite grains, a "fission track age" can be calculated which will be equal to the time over which tracks have been accumulating. The technique is calibrated against other isotopic systems using age standards which also have this type of length distribution (see Supporting information chapter S2).

In samples which have been subjected to temperatures greater than ~50°C after deposition, fission tracks are shortened because of the gradual repair of the radiation damage which constitutes the unetched tracks. In effect, the tracks shrink from each end, in a process which is known as fission track "annealing". The final length of each individual track is essentially determined by the maximum temperature which that track has experienced. A time difference of an order of magnitude produces a change in fission track parameters which is equivalent to a temperature change of only ~10°C, so temperature is by far the dominant factor in determining the final fission track parameters. As temperature increases, all existing tracks shorten to a length determined by the prevailing temperature, regardless of when they were formed. After the temperature has subsequently decreased, all tracks formed prior to the thermal maximum are "frozen" at the degree of length reduction they attained at that time. Thus, the length of each track can be thought of as a maximum-reading thermometer, recording the maximum temperature to which it has been subjected. Therefore, in samples for which the present temperature is maximum, all tracks have much the same length, resulting in a narrow, symmetric distribution. The degree of shortening will depend on the temperature, with the mean track

length falling progressively from ~14 μm at 50°C, to zero at around 110°-120°C - the precise temperature depending on the timescale of heating and the composition of the apatites present in the sample (see below). Values quoted here relate to times of the order of 107 years (heating rates around 1 to 10°C/Ma) and average apatite composition. If the effective timescale of heating is shorter than 107 years, the temperature responsible for a given degree of track shortening will be higher, depending in detail on the kinetics of the annealing process (Green et al., 1986; Laslett et al., 1987; Duddy et al., 1988; Green et al., 1989b). Shortening of tracks produces an accompanying reduction in the fission track age, because of the reduced proportion of tracks which can intersect the polished surface. Therefore, the fission track age is also highly temperature dependent, falling to zero at around 120°C due to total erasure of all tracks. Samples which have been heated to a maximum paleotemperature less than ~120°C at some time in the past and subsequently cooled will contain two populations of tracks, and will show a more complex distribution of lengths and ages. If the maximum paleotemperature was less than ~50°C then the two components will not be resolvable, but for maximum paleotemperatures between ~50° and 120°C the presence of two components can readily be identified. Tracks formed prior to the thermal maximum will all be shortened to approximately the same degree (the precise value depending on the maximum paleotemperature), while those formed during and after cooling will be longer, due to the lower prevailing temperatures. The length distribution in such samples will be broader than in the simple case, consisting of a shorter and a longer component, and the fission track age will reflect the amount of length reduction shown by the shorter component (determined by the maximum paleotemperature). If the maximum paleotemperature was sufficient to shorten tracks to between 9 and 11 μm, and cooling to temperatures of ~50°C or less was sufficiently rapid, tracks formed after cooling will have lengths of 14-15 μm and the resulting track length distribution will show a characteristic bimodal form. If the maximum paleotemperature was greater than ~110 to 120°C, all pre-existing tracks will be erased, and all tracks now present will have formed after the onset of cooling. The fission track age in such samples relates directly to the time of cooling. In thermal history scenarios in which a heating episode is followed by cooling and then temperature increases again, the tracks formed during the second heating phase will undergo progressive shortening. The tracks formed prior to the initial cooling, which were shortened in the first heating episode, will not undergo further shortening until the temperature exceeds the maximum temperature reached in the earlier heating episode.

In practice, differences in timescale of heating can complicate this simple description. In detail, it is the integrated time-temperature effect of the two heating episodes which should be considered.) If the maximum and peak paleotemperatures in the two episodes are sufficiently different (>~10°C), and the later peak paleotemperature is less than the earlier maximum value, then the AFTA parameters allow determination of both episodes. As the peak paleotemperature in the later episode approaches the earlier maximum, the two generations of tracks become increasingly more difficult to resolve, and when the two paleotemperatures are the same, both components are shortened to an identical degree and all information on the earlier heating phase will be lost. No information is preserved on the approach to maximum paleotemperature because the great majority of tracks formed up to that time have the same mean track length. Only those tracks formed in the last few per cent of the history prior to the onset of cooling are not shortened to the same degree (because temperature dominates over time in the annealing kinetics). These form a very small proportion of the total number of tracks and therefore cannot be resolved within the length distribution because of the inherent spread of several μm in the length distribution. To summarise, AFTA allows determination of the magnitude of the maximum temperature and the time at which cooling from that maximum began. In some circumstances, determination of a subsequent peak paleotemperature and the time of cooling is also possible.

S3.3 Quantitative understanding of fission track annealing in apatite

Annealing kinetics and modelling the development of AFTA parameters

Our understanding of the behaviour of fission tracks in apatite during geological thermal histories is based on study of the response of fission tracks to elevated temperatures in the laboratory (Green et al., 1986; Laslett et al., 1987; Duddy et al., 1988; Green et al., 1989b), in geological situations (Green et al.

1989a), observations of the lengths of spontaneous tracks in apatites from a wide variety of geological environments (Gleadow et al. 1986), and the relationship between track length reduction and reduction in fission track age observed in controlled laboratory experiments (Green, 1988).

These studies resulted in the capability to simulate the development of AFTA parameters resulting from geological thermal histories for an apatite of average composition (Durango apatite, ~0.43 wt% Cl). Full details of this modelling procedure have been explained in Green et al. (1989b). The following discussion presents a brief explanation of the approach.

Geological thermal histories involving temperatures varying through time are broken down into a series of isothermal steps. The progressive shortening of track length through sequential intervals is calculated using the extrapolated predictions of an empirical kinetic model fitted to laboratory annealing data. Contributions from tracks generated throughout the history (remembering that new tracks are continuously generated through time as new fissions occur) are summed to produce the final distribution of track lengths expected to result from the input history. In summing these components, care is taken to allow for various biases which affect revelation of confined tracks (Laslett et al. 1982). The final length reduction of each component of tracks is converted to a contribution of fission track age, using the relationship between track length and density reduction determined by Green (1988). These age contributions are summed to generate the final predicted fission track age.

This approach depends critically on the assumption that extrapolation of the laboratorybased kinetic model to geological timescales, over many orders of magnitude in time, is valid. This was assessed critically by Green et al. (1989b), who showed that predictions from this approach agree well with observed AFTA parameters in apatites of the appropriate composition in samples from a series of reference wells in the Otway Basin of south-east Australia (Gleadow and Duddy, 1981; Gleadow et al., 1983; Green et al., 1989a). This point is illustrated in Figure C.1. Green et al. (1989b) also quantitatively assessed the errors associated with extrapolation of the Laslett et al. (1987) model from laboratory to geological timescales (i.e. precision, as opposed to accuracy). Typical levels of precision are ~0.5 µm for mean lengths <~10 µm, and ~0.3 µm for lengths >~10 µm. These figures are equivalent to an uncertainty in estimates of maximum paleotemperature derived using this approach of ~10°C. Precision is largely independent of thermal history for any reasonable geological history. Accuracy of prediction from this model is limited principally by the effect of apatite composition on annealing kinetics, as explained in the next section.

Compositional effects

Natural apatites essentially have the composition Ca5(PO4)3(F,OH,Cl). Most common detrital and accessory apatites are predominantly Fluor-apatites, but may contain appreciable amounts of chlorine. The amount of chlorine in the apatite lattice exerts a subtle compositional control on the degree of annealing, with apatites richer in fluorine being more easily annealed than those richer in chlorine. The result of this effect is that in a single sample, individual apatite grains may show a spread in the degree of annealing (i.e. length reduction and fission track age reduction). This effect becomes most pronounced in the temperature range 90 - 120°C (assuming a heating timescale of ~10 Ma), and can be useful in identifying samples exposed to paleotemperatures in this range. At temperatures below ~80°C, the difference in annealing sensitivity is less marked, and compositional effects can largely be ignored.

Our original quantitative understanding of the kinetics of fission track annealing, as described above, relates to a single apatite (Durango apatite) with ~0.43 wt% Cl, on which most of our original experimental studies were carried out. Recently, we have extended this quantitative understanding to apatites with Cl contents up to ~3 wt%. This new, multi-compositional kinetic model is based both on new laboratory annealing studies on a range of apatites with different F-Cl compositions (Figure C.2), and on observations of geological annealing in apatites from a series of samples from exploration wells in which the section is currently at maximum temperature since deposition. A composite model for Durango apatite composition was first created by fitting a common model to the old laboratory data (from Green et al., 1986) and the new geological data for a similar composition. This was then extended to other compositions on the basis of the multi-compositional laboratory and geological data sets.

Details of the multi-compositional model are contained in a Technical Note, available from Geotrack in Melbourne.
The multi-compositional model allows prediction of AFTA parameters for any Cl content between 0 and 3 wt%, using a similar approach to that used in our original single composition modelling, as outlined above. Then, for an assumed or measured distribution of Cl contents within a sample, the composite parameters for the sample can be predicted. The range of Cl contents from 0 to 3 wt% spans the range of compositions commonly encountered, as discussed in the next section.
Predictions of the new multi-compositional model are in good agreement with the geological constraints on annealing rates provided by the Otway Basin reference wells, as shown in Figure C.3. However, note that the AFTA data from these Otway Basin wells were among those used in construction of the new model, so this should not be viewed as independent verification, but rather as a demonstration of the overall consistency of the model.

Distributions of Cl content in common AFTA samples

Figure C.4a shows a histogram of Cl contents, measured by electron microprobe, in apatite grains from more than 100 samples of various types. Most grains have Cl contents less than ~0.5 wt%. The majority of grains with Cl contents greater than this come from volcanic sources and basic intrusives, and contain up to ~2 wt% Cl. Figure s3.4b shows the distribution of Cl contents measured in randomly selected apatite grains from 61 samples of "typical" quartzo-feldspathic sandstone. This distribution is similar to that in Figure S3.4a, except for a more rapid fall-off as Cl content increases. Apatites from most common sandstones give distributions of Cl content which are very similar to that in Figure C.4b. Volcanogenic sandstones typically contain apatites with higher Cl contents, with a much flatter distribution for Cl contents up to ~1.5%, falling to zero at ~2.5 to 3 wt%, as shown in Figure S3.4c. Cl contents in granitic basement samples and high-level intrusives are typically much more dominated by compositions close to end-member Fluorapatite, although many exceptions occur to this general rule. Information about the spread of Cl contents in samples analyzed in this report can befound in S1.

Alternative kinetic models

Recently, both Carlson (1990) and Crowley et al. (1991) have published alternative kinetic models for fission track annealing in apatite. Carlson's model is based on our laboratory annealing data for Durango apatite (Green et al., 1986) and other (unpublished) data. In his abstract, Carlson claims that because his model is "based on explicit physical mechanisms, extrapolations of annealing rates to the lower temperatures and longer timescales required for the interpretation of natural fission track length distributions can be made with greater confidence than is the case for purely empirical relationships fitted to the experimental annealing data". As explained in detail by Green et al. (1993), all aspects of Carlson's model are in fact purely empirical, and his model is inherently no "better" for the interpretation of data than any other. In fact, detailed inspection shows that Carlson's model does not fit the laboratory data set at all well. Therefore, we recommend against use of this model to interpret AFTA data. The approach taken by Crowley et al. (1991) is very similar to that taken by Laslett et al., (1987). They have fitted models to new annealing data in two apatites of different composition - one close to end-member Fluorapatite (B-5) and one having a relatively high Sr content (113855). The model developed by Crowley et al. (1991) from their own annealing data for the B-5 apatite gives predictions in geological conditions which are consistently higher than measured values, as shown in Figure C.5. Corrigan (1992) reported a similar observation in volcanogenic apatites in samples from a series of West Texas wells. Since the B-5 apatite is close to end-member Fluor-apatite, while the Otway Group apatites contain apatites with Cl contents from zero up to ~3 wt% (and the West Texas apatites have up to 1 wt%), the fluorapatites should have mean lengths rather less than the measured values, which should represent a mean over the range of Cl contents present. Therefore, the predictions of the Crowley et al. (1991) B-5 model appear to be consistently high.
We attribute this to the rather restricted temperature-time conditions covered by the experiments of Crowley et al. (1991), with annealing times between one and 1000 hours, in contrast to times between 20 minutes and 500 days in the experiments of Green et al. (1986). In addition, few of the measured

length values in Crowley et al.'s study fall below 11 μm (in only five out of 60 runs in which lengths were measured in apatite B-5) and their model is particularly poorly defined in this region.
Crowley et al. (1991) also fitted a new model to the annealing data for Durango apatite published by Green et al. (1986). Predictions of their fit to our data are not very much different to those from the Laslett et al. (1987) model (Figure C.6). We have not pursued the differences between their model and ours in detail because the advent of our multi-compositional model has rendered the single compositional approach obsolete.

S3.4 Evidence for elevated paleotemperatures from AFTA

The basic principle involved in the interpretation of AFTA data in sedimentary basins is to determine whether the degree of annealing shown by tracks in apatite from a particular sample could have been produced if the sample has never been hotter than its present temperature at any time since deposition. To do this, the burial history derived from the stratigraphy of the preserved sedimentary section is used to calculate a thermal history for each sample using the present geothermal gradient and surface temperature (i.e. assuming these have not changed through time). This is termed the "Default Thermal History". For each sample, the AFTA parameters predicted as a result of the Default Thermal History are then compared to the measured data. If the data show a greater degree of annealing than calculated on the basis of this history, the sample must have been hotter at some time in the past. In this case, the AFTA data are analyzed to provide estimates of the magnitude of the maximum paleotemperature in that sample, and the time at which cooling commenced from the thermal maximum.
The degree of annealing is assessed in two ways - from fission track age and track length data. The stratigraphic age provides a basic reference point for the interpretation of fission track age, because reduction of the fission track age below the stratigraphic age unequivocally reveals that appreciable annealing has taken place after deposition of the host sediment. Large degrees of fission track age reduction, with the pooled or central fission track age very much less than the stratigraphic age, indicate severe annealing, which requires paleotemperatures of at least ~100°C for any reasonable geological time-scale of heating (>~1 Ma). Note that this applies even when apatites contain tracks inherited from source areas. More moderate degrees of annealing can be detected by inspection of the single grain age data, as the most sensitive (fluorine-rich) grains will begin to give fission track ages significantly less than the stratigraphic age before the central or pooled age has been reduced sufficiently to give a noticeable signal. Note that this aspect of the single grain age data can also be used for apatites which have tracks inherited from source areas. If signs of moderate annealing (from single grain age reduction) or severe annealing (from the reduction in pooled or central age) are seen in samples in which the Default Thermal History predicts little or no effect, the sample must have been subjected to elevated paleotemperatures at some time in the past. Figure C.7 shows how increasing degrees of annealing are observable in radial plots of the single grain fission track age data.
Similarly, the present temperature from which a sample is taken, and the way in which this has been approached (as inferred from the preserved sedimentary section), forms a basic point of reference for track length data. The observed mean track length is compared with the mean length predicted from the Default Thermal History. If the observed degree of track shortening in a sample is greater than that expected from the Default Thermal History (i.e. the mean length is significantly less than the predicted value), either the sample must have been subjected to higher paleotemperatures at some time after deposition, or the sample contains shorter tracks which were inherited from sediment source areas at the time the sediment was deposited. If shorter tracks were inherited from source areas, the sample should still contain a component of longer tracks corresponding to the tracks formed after deposition. In general, the fission track age should be greater than the stratigraphic age. This can be assessed quantitatively using the computer models for the development of AFTA parameters described in an earlier section. If the presence of shorter tracks cannot be explained by their inheritance from source areas, the sample must have been hotter in the past.

S3.5 Quantitative determination of the magnitude of maximum paleotemperature and the timing of cooling using AFTA

Values of maximum paleotemperature and timing of cooling in each sample are determined using a forward modelling approach based on the quantitative description of fission track annealing described in earlier sections. The Default Thermal History described above is used as the basis for this forward modelling, but with the addition of episodes of elevated paleotemperatures as required to explain the data. AFTA parameters are modelled iteratively through successive thermal history scenarios in order to identify thermal histories that can account for observed parameters. The range of values of maximum paleotemperature and timing of cooling which can account for the measured AFTA parameters (fission track age and track length distribution) are defined using a maximum likelihood-based approach. In this way, best estimates ("maximum likelihood values") can be defined together with ±95% confidence limits.

In samples in which all tracks have been totally annealed at some time in the past, only a minimum estimate of maximum paleotemperature is possible. In such cases, AFTA data provide most control on the time at which the sample cooled to temperatures at which tracks could be retained. The time at which cooling began could be earlier than this time, and therefore the timing also constitutes a minimum estimate. Comparison of the AFTA parameters predicted by the multi-compositional model with measured values in samples which are currently at their maximum temperatures since deposition shows a good degree of consistency, suggesting the uncertainty in application of the model should be less than ±10°C. This constitutes a significant improvement over earlier approaches, since the kinetic models used are constrained in both laboratory and geological conditions. It should be appreciated that relative differences in maximum paleotemperature can be identified with greater precision than absolute paleotemperatures, and it is only the estimation of absolute paleotemperature values to which the ±10°C uncertainty relates.

Cooling history

If the data are of high quality and provided that cooling from maximum paleotemperatures began sufficiently long ago (so that the history after this time is represented by a significant proportion of the total tracks in the sample), determination of the magnitude of a subsequent peak paleotemperature and the timing of cooling from that peak may also be possible (as explained in Section C.2). A similar approach to that outlined above provides best estimates and corresponding ±95% confidence limits for this episode. Such estimates may simply represent part of a protracted cooling history, and evidence for a later discrete cooling episode can only be accepted if this scenario provides a significantly improved fit to the data. Geological evidence and consistency of estimates between a series of samples can also be used to verify evidence for a second episode.

In practice, most typical AFTA datasets are only sufficient to resolve two discrete episodes of heating and cooling. One notable exception to this is when a sample has been totally annealed in an early episode, and has then undergone two (or more) subsequent episodes with progressively lower peak paleotemperatures in each. But in general, complex cooling histories involving a series of episodes of heating and cooling will allow resolution of only two episodes, and the results will depend on which episodes dominate the data. Typically this will be the earliest and latest episodes, but if multiple cooling episodes occur within a narrow time interval the result will represent an approximation to the actual history.

S3.6 Qualitative assessment of AFTA parameters

Various aspects of thermal history can often be assessed by qualitative assessment of AFTA parameters. For example, samples which have reached maximum paleotemperatures sufficient to produce total annealing, and which only contain tracks formed after the onset of cooling, can be identified from a number of lines of evidence.

In a vertical sequence of samples showing increasing degrees of annealing, the transition from rapidly decreasing fission track age with increasing depth to more or less the same age over a range of depth denotes the transition from partial to total annealing of all tracks formed prior to the thermal maximum. In samples in which all tracks have been totally annealed, the single grain age data should show that none of the individual grain fission track ages are significantly older than the time of cooling, and grains in all compositional groups should give the same fission track age unless the sample has been further

disturbed by a later episode. If the sample cooled rapidly to sufficiently low temperatures, little annealing will have taken place since cooling, and all grains will give ages which are compatible with a single population around the time of cooling, as shown in Figure S3.7.
Inspection of the distribution of single grain ages in partially annealed samples can often yield useful information on the time of cooling, as the most easily annealed grains (those richest in fluorine) may have been totally annealed prior to cooling, while more retentive (Cl-rich) compositions were only partially annealed (as in Figure S3.7, centre).
The form of the track length distribution can also provide information, from the relative proportions of tracks with different lengths. All of these aspects of the data can be used to reach a preliminary thermal history interpretation.

S3.7 Allowing for tracks inherited from source areas

The effect of tracks inherited from source areas, and present at the time the apatite is deposited in the host sediment, is often posed as a potential problem for AFTA. However, this can readily be allowed for in analyzing both the fission track age and length data.
In assessing fission track age data to determine the degree of annealing, the only criterion used is the comparison of fission track age with the value expected on the basis of the Default Thermal History. From this point of view, inherited tracks do not affect the conclusion: if a grain or a sample gives a fission track age which is significantly less than expected, the grain or sample has clearly undergone a higher degree of annealing than can be accounted for by the Default Thermal History, and therefore must have been hotter in the past, whether the sample contained tracks when it was deposited or not.
The presence of inherited tracks does impose a limit on our ability to detect postdepositional annealing from age data alone, as in samples which contain a fair proportion of inherited tracks, moderate degrees of annealing may reduce the fission track age from the original value, but not to a value which is significantly less than the stratigraphic age. This is particularly noticeable in the case of Tertiary samples containing apatites derived from Paleozoic basement. In such cases, although fission track age data may show no evidence of post-depositional annealing, track length data may well show such evidence quite clearly.
The influence of track lengths inherited from source areas can be allowed for by comparison of the fission track age with the value predicted by the Default Thermal History combined with inspection of the track length distribution. If the mean length is much less than the length predicted by the Default Thermal History, either the sample has been subjected to elevated paleotemperatures, sufficient to produce the observed degree of length reduction, or else the sample contains a large proportion of shorter tracks inherited from source areas. However, in the latter case, the sample should give a pooled or central fission track age correspondingly older than the stratigraphic age, while the length distribution should contain a component of longer track lengths corresponding to the value predicted by the Default Thermal History. It is important in this regard that the length of a track depends primarily on the maximum temperature to which it has been subjected, whether in the source regions or after deposition in the sedimentary basin. Thus, any tracks retaining a provenance signature will have lengths towards the shorter end of the distribution where track lengths will not have "equilibrated" with the temperatures attained since deposition.
In general, it is only in extreme cases that inherited tracks render track length data insensitive to post-depositional annealing. For example, if practically all the tracks in a particular sample were formed prior to deposition, perhaps in a Pliocene sediment in which apatites were derived from a stable Paleozoic shield with fission track ages of ~300 Ma or more, the track length distribution will, in general, be dominated by inheritance, as only ~2% of tracks would have formed after deposition. Postdepositional heating will not be detectable as long as the maximum paleotemperature is insufficient to cause greater shortening than that which occurred in the source terrain.
Even in such extreme cases, once a sample is exposed to temperatures sufficient to produce greater shortening than that inherited from source areas, the inherited tracks and those formed after deposition will all undergo the same degree of shortening, and the effects of post-depositional annealing can be recognised. In such cases, the presence of tracks inherited from source areas is actually very useful,

because the number of tracks formed after deposition is so small that little or no information would beavailable without the inherited tracks.

S3.8 Plots of fission track age and mean track length vs depth and temperature

AFTA data from well sequences are usually plotted as shown in Figure S3.8. This figure shows AFTA data for two scenarios: one in which deposition has been essentially continuous from the Carboniferous to the present and all samples are presently at their maximum paleotemperature since deposition (Figure C.8a); and, one in which the section was exposed to elevated paleotemperatures prior to cooling in the Early Tertiary (Figure S3.8b).

In both figures, fission track age and mean track length are plotted against depth and present temperature. Presentation of AFTA data in this way often provides insight into the thermal history interpretation, following principles outlined earlier in this supporting information chapter.

In Figure S3.8a**,** for samples at temperatures below ~70°C, the fission track age is either greater than or close to the stratigraphic age, and little fission track age reduction has affected these samples. Track lengths in these samples are all greater than ~13 µm. In progressively deeper samples, both the fission track age and mean track length are progressively reduced to zero at a present temperature of around 110°C, with the precise value depending on the spread of apatite compositions present in the sample. Track length distributions in the shallowest samples would be a mixture of tracks retaining information on the thermal history of source regions, while in deeper samples, all tracks would be shortened to a length determined by the prevailing temperature. This pattern of AFTA parameters is characteristic of a sequence which is currently at maximum temperatures.

The data in Figure S3.8b show a very different pattern. The fission track age data show a rapid decrease in age, with values significantly less than the stratigraphic age at temperatures of ~40 to 50°C, at which such a degree of age reduction could not be produced in any geological timescale. Below this rapid fall, the fission track ages do not change much over ~1 km (30°C). This transition from rapid fall to consistent ages is diagnostic of the transition from partial to total annealing. Samples above the "breakin slope" contain two generations of tracks: those formed prior to the thermal maximum, which have been partially annealed (shortened) to a degree which depends on the maximum paleotemperature; and, those formed after cooling, which will be longer. Samples below the break-in slope contain only one generation of tracks, formed after cooling to lower temperatures at which tracks can be retained. At greater depths, where temperatures increase to ~90°C and above, the effect of present temperatures begins to reduce the fission track ages towards zero, as in the "maximum temperatures now" case.

The track length data also reflect the changes seen in the fission track age data. At shallow depths, the presence of the partially annealed tracks shortened prior to cooling causes the mean track length to decrease progressively as the fission track age decreases. However, at depths below the break in slope in the age profile, the track length increases again as the shorter component is totally annealed and so does not contribute to the measured distribution of track lengths. At greater depths, the mean track lengths decrease progressively to zero once more due to the effects of the present temperature regime. Examples of such data have been presented, e.g. by Green (1989) and Kamp and Green (1990).

S3.9 Determining paleogeothermal gradients and amount of section removed on unconformities

Estimates of maximum paleotemperatures in samples over a range of depths in a vertical sequence provides the capability of determining the paleogeothermal gradient immediately prior to the onset of cooling from those maximum paleotemperatures. The degree to which the paleogeothermal gradient can be constrained depends on a number of factors, particularly the depth range over which samples are analyzed. If samples are only analyzed over ~1 km, then the paleotemperature difference over that range may be only ~20 to 30°C. Since maximum paleotemperatures can often only be determined within a ~10°C range, this introduces considerable uncertainty into the final estimate of paleogeothermal gradient (see Figure S3.9).

Another important factor is the difference between maximum paleotemperatures and present temperatures (“net cooling”). If this is only ~10°C, which is similar to the uncertainty in absolute paleotemperature determination, only broad limits can be established on the paleogeothermal gradient. In general, the control on the paleogeothermal gradient improves as the amount of net

cooling increases. However, if the net cooling becomes so great that many samples were totally annealed prior to the onset of cooling - so that only minimum estimates of maximum paleotemperatures are possible - constraints on the paleogeothermal gradient from AFTA come only from that part of the section in which samples were not totally annealed. In this case, integration of AFTA data with VR measurements can be particularly useful in constraining the paleo-gradient.

Having constrained the paleogeothermal gradient at the time cooling from maximum paleotemperatures began, if we assume a value for surface temperature at that time, the amount of section subsequently removed by uplift and erosion can be calculated as shown in Figure S3.10. The *net* amount of section removed is obtained by dividing the difference between the paleo-surface temperature (Ts) and the intercept of the paleotemperature profile at the present ground surface (Ti) by the estimated paleogeothermal gradient. The *total* amount of section removed is obtained by adding the thickness of section subsequently redeposited above the unconformity to the *net* amount estimated as in Figure S3.10. If the analysis is performed using depths from the appropriate unconformity, then the analysis will directly yield the *total* amount of section removed.

Geotrack have developed a method of deriving estimates of both the paleogeothermal gradient and the net amount of section removed using estimated paleotemperatures derived from AFTA and VR. Perhaps more importantly, this method also provides rigorous values for upper and lower 95% confidence limits on each parameter. The method is based on maximum likelihood estimation of the paleogeothermal gradient and the surface intercept, from a table of paleotemperature and depth values. The method is able to accept ranges for paleotemperature estimates (e.g. where the maximum paleotemperature can only be constrained to between, for example, 60 and 90°C), as well as upper and lower limits (e.g. <60°C for samples which show no detectable annealing; >110°C in samples which were totally annealed). Estimates of paleotemperature from AFTA and VR may be combined or analyzed separately. Some results from this method have been reported by Bray et al. (1992). Full details of the methods employed are presented in a confidential, in-house, Geotrack research report, copies of which are available on request from the Melbourne office.

Results are presented in two forms. Likelihood profiles, plotting the log-likelihood as a function of either gradient or section removed, portray the probability of a given value of gradient or section removed. The best estimate is given by the value of gradient or section removed for which the log-likelihood is maximised. Ideally, the likelihood profiles should show a quadratic form, and values of gradient or section removed at which the log-likelihood has fallen by two from the maximum value define the upper and lower 95% confidence limits on the estimates. An alternative method of portraying this information is a crossplot of gradient against section removed, in which values which fall within 95% confidence limits (in two dimensions) are contoured. Note that the confidence limits defined by this method are rather tighter than those from the likelihood profiles, as the latter only reflect variation in one parameter, whereas the contoured crossplot takes variation of both parameters into account. It must be emphasised that this method relies on the assumption that the paleotemperature profile was linear both throughout the section analyzed and through the overlying section which has been removed. While the second part of this assumption can never be confirmed independently, visual inspection of the paleotemperature estimates as a function of depth should be sufficient to verify or deny the linearity of the paleotemperature profile through the preserved section.

Results of this procedure are shown in this report if the data allow sufficiently welldefined paleotemperature estimates to justify use of the method. Where the AFTA data suggest that the section is currently at maximum temperature since deposition, or that the paleotemperature profile was non-linear, or where data are of insufficient quality to allow rigorous paleotemperature estimation, the method is not used.

S3.10 Thermal history interpretation of Zircon Fission Track Age data

The understanding of the annealing characteristics of fission tracks in zircon is not as well developed as for apatite. From both laboratory and geological evidence, fission tracks in zircon are known to be less sensitive to annealing than fission tracks in apatite. Zircon fission track ages are conventionally interpreted in terms of a closure temperature, defined as the nominal temperature below which a radiogenic product, in this case fission tracks, is effectively retained. Although the closure temperature

in zircon is not well constrained, geological evidence suggests values between 175° and 300°C. Based on comparison of zircon fission track ages with ages from other radiometric systems, Hurford (1986) proposed a closure temperature for fission track retention in zircon of 240° ± 50°C. A value around 240°C or possibly higher is broadly consistent with all available evidence, although the ±50°C quoted uncertainty would allow unrealistically low values, and our experience suggests a value between 220 and 300°C would be most appropriate.

In contrast, tracks in apatite begin to be retained when the mineral cools below a temperature of the order of ~110°-120°C (depending on apatite composition). Where fission track ages are available for apatite and zircon, the cooling history of a sample can be assessed because of the differing thermal sensitivities of the tracks in the minerals. In samples which have cooled rapidly from high temperatures of >240°C to less than ~50°C, the apatite and zircon fission track ages are usually concordant, as tracks in both minerals began to be retained at roughly the same time.

The type of analysis presented for apatite data is not possible in zircon, because it is not usually possible to measure track lengths in zircon due to the etching properties of the mineral. This makes it difficult to determine whether a thermally affected zircon fission track age represents partial or total annealing, and, thus, whether the fission track age can be interpreted in terms of a specific "event". Another problem with the interpretation of zircon data is posed by the lack of constraint on the temperatures required for a given degree of partial annealing. Therefore, unlike the situation with apatite, it is not possible to rigorously estimate maximum paleotemperatures on the basis of zircon data alone. On the basis of experience in a small number of sedimentary basins, it appears that significant age reduction in zircon does not normally occur below temperatures of ~200°C. However, the effects of composition on annealing rates in zircon are not well understood, and it is possible that some particularly sensitive zircons may exist which undergo annealing at lower temperatures.

Investigation of zircon data in samples with different levels of vitrinite reflectance suggests that no significant age reduction occurs in zircon at VR levels below ~4%. VR values in excess of 5% are required in order to produce any significant age reduction. Some evidence suggests that the rock needs to reach the stage of incipient recrystallisation in order to produce observable effects in zircon.

Based on all evidence currently available, we define the following broad paleotemperature ranges which may be defined on the basis of ZFTA data: Some age reduction 200-300°C

Appreciable age reduction ≥240°C Total annealing of most grains >300°C

It should be borne in mind that these paleotemperatures are only very loosely constrained, and any conclusions based thereon should be treated with caution.

S3 References

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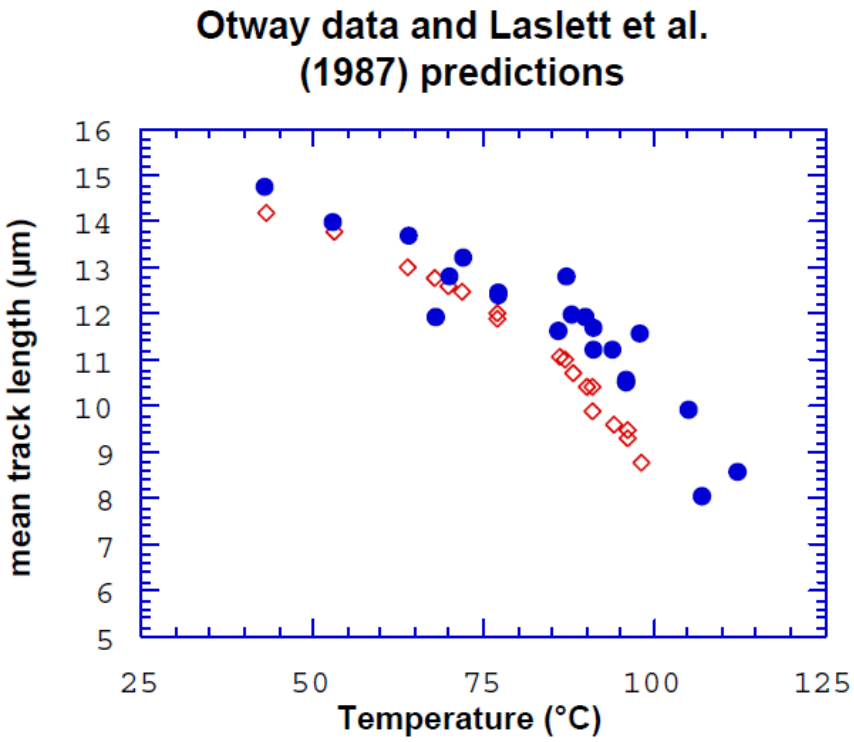
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1576 **Figure S3.1a** Comparison of mean track length (solid circles) measured in samples from four Otway
1577 Basin reference wells (from Green et al, 1989a) and predicted mean track lengths (open diamonds) from
1578 the kinetic model of fission track annealing from Laslett et al. (1987). The predictions underestimate the
1579 measured values, but they refer to an apatite composition that is more easily annealed than the
1580 majority of apatites in these samples, so this is expected.

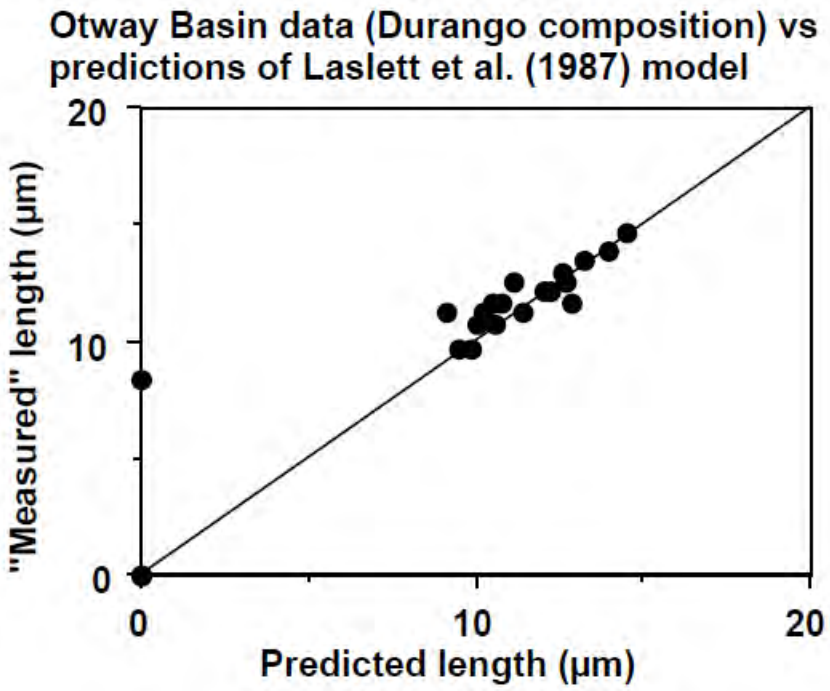


Figure S3.1b Comparison of the mean track length in apatites of the same Cl content as Durango apatite from the Otway Group samples illustrated in figure C.1a, with values predicted for apatite of the same composition by the model of Laslett et al. (1987). The agreement is clearly very good except possibly at lengths below ~10 µm.

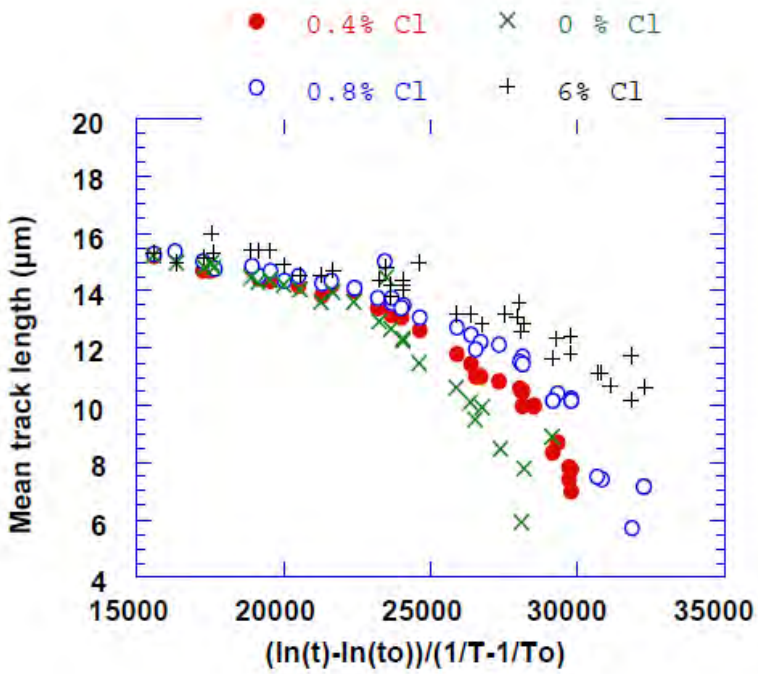


Figure S3.2 Mean track length in apatites with four different chlorine contents, as a combined function of temperature and time, to reduce the data to a single scale. Fluorapatites are more easily annealed than chlorapatites, and the annealing kinetics show a progressive change with increasing Cl content.

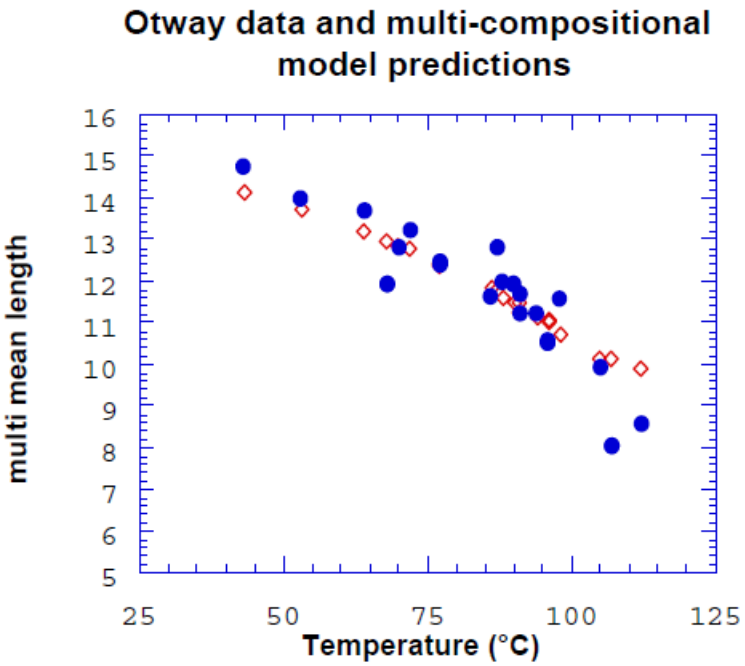


Figure S3.3 Comparison of measured mean track length (solid circles) in samples from four Otway Basin reference wells (from Green et al, 1989a) and predicted mean track lengths (open diamonds) from the new multi-compositional kinetic model of fission track annealing described in Section C.3. This model takes into account the spread of Cl contents in apatites from the Otway Group samples and the influence of Cl content on annealing rate. The agreement is clearly very good over the range of the data.

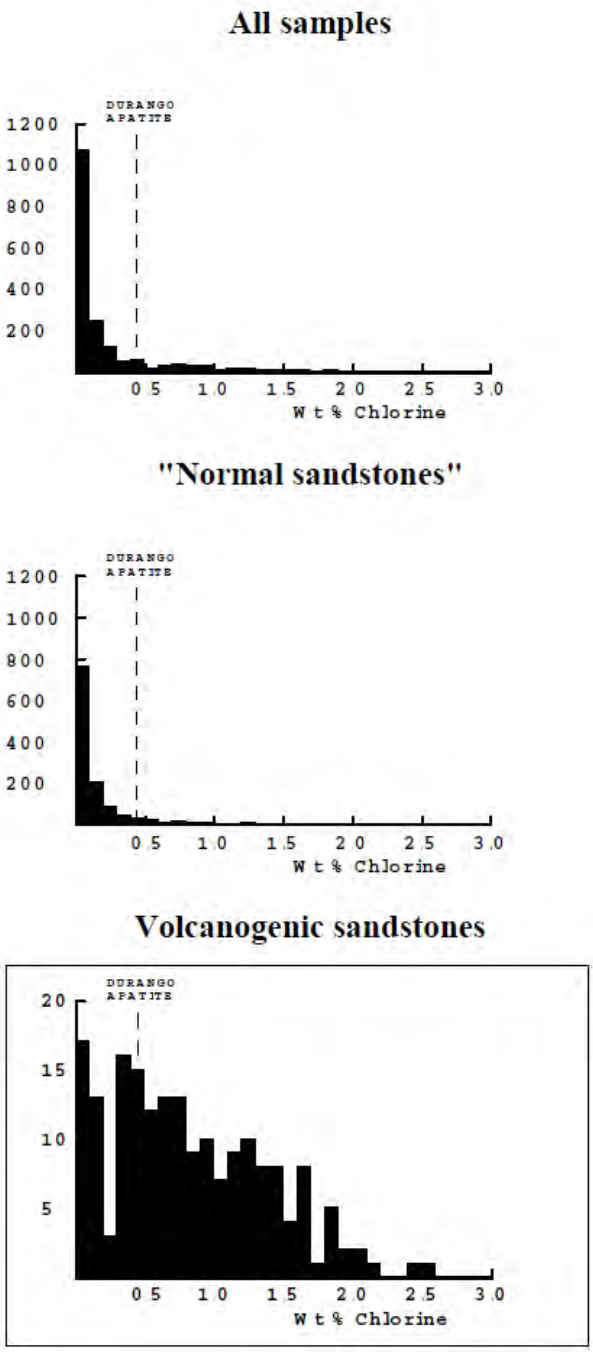


Figure C.4 a: Histogram of Cl contents (wt%) in over 1750 apatite grains from over 100 samples of various sedimentary and igneous rocks. Most samples give Cl contents below ~0.5 wt %, while those apatites giving higher Cl contents are characteristic of volcanogenic sandstones and basic igneous sources.
b: Histogram of Cl contents (wt%) in 1168 apatite grains from 61 samples which can loosely be characterised as "normal sandstone". The distribution is similar to that in the upper figure, except for a lower number of grains with Cl contents greater than ~1%.
c: Histogram of Cl contents (wt%) in 188 apatite grains from 15 samples of volcanogenic sandstone. The distribution is much flatter than the other two, with much higher proportion of Cl-rich grains.

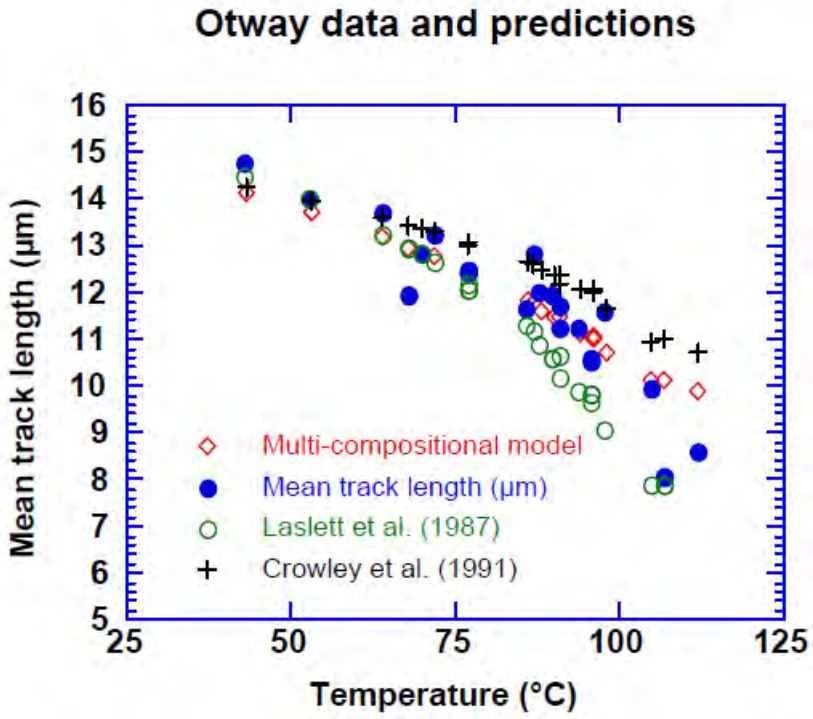


Figure S3.5 Comparison of mean track length in samples from four Otway Basin reference wells (from Green et al, 1989a) and predicted mean track lengths from three kinetic models for fission track annealing. The Crowley et al. (1991) model relates to almost pure Fluorapatite (B-5), yet overpredicts mean lengths in the Otway Group samples which are dominated by Cl-rich apatites. The predictions of that model are therefore not reliable.

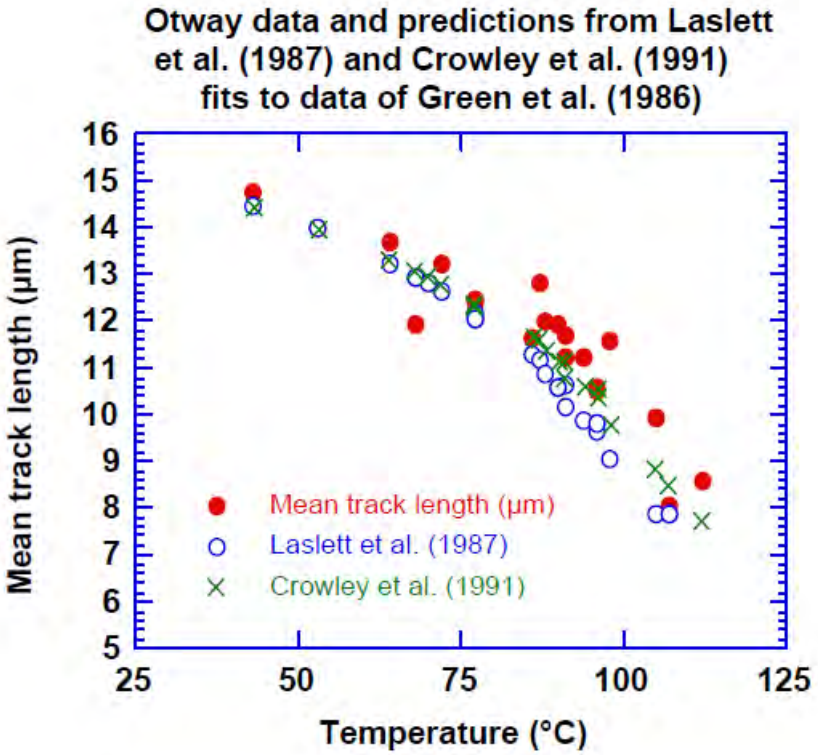


Figure S3.6 Comparison of mean track length in samples from four Otway Basin reference wells with values predicted from Laslett et al. (1987) and the model fitted to the annealing data of Green et al. (1986) by Crowley et al. (1991). The predictions of the two models are not very different.

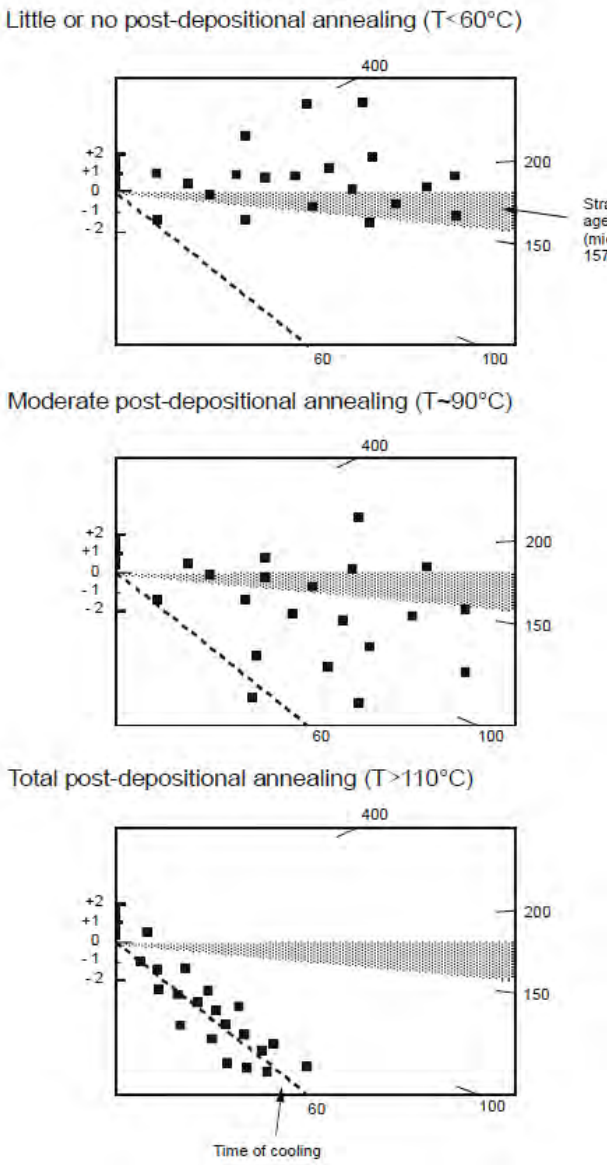


Figure S3.7 Radial plots of single grain age data in three samples of mid-Jurassic sandstone that have been subjected to varying degrees of post-depositional annealing prior to cooling at ~60 Ma. The id-point of the stratigraphic age range has been taken as the reference value (corresponding to the horizontal).

The upper diagram represents a sample which has remained at paleotemperatures less than ~60°C, and has therefore undergone little or no post-depositional annealing. All single grain ages are either compatible with the stratigraphic age (within y = ±2 in the radial plot) or older than the stratigraphic age (yi > 2).

The centre diagram represents a sample which has undergone a moderate degree of postdepositional annealing, having reached a maximum paleotemperature of around ~90°C prior to cooling. While some of the individual grain ages are compatible with the stratigraphic age (-2 < yi < +2) and some may be significantly greater than the stratigraphic age (yi > 2), a number of grains give ages which are significantly less than the stratigraphic age (y < 2).

The lower diagram represents a sample in which all apatite grains were totally annealed, at paleotemperatures greater than ~110°C, prior to rapid cooling at ~60 Ma. All grains give fission track ages compatible with a fission track age of ~60 Ma (i.e., all data plot within ±2 of the radial line corresponding to an age of ~60 Ma), and most are significantly younger than the stratigraphic age.

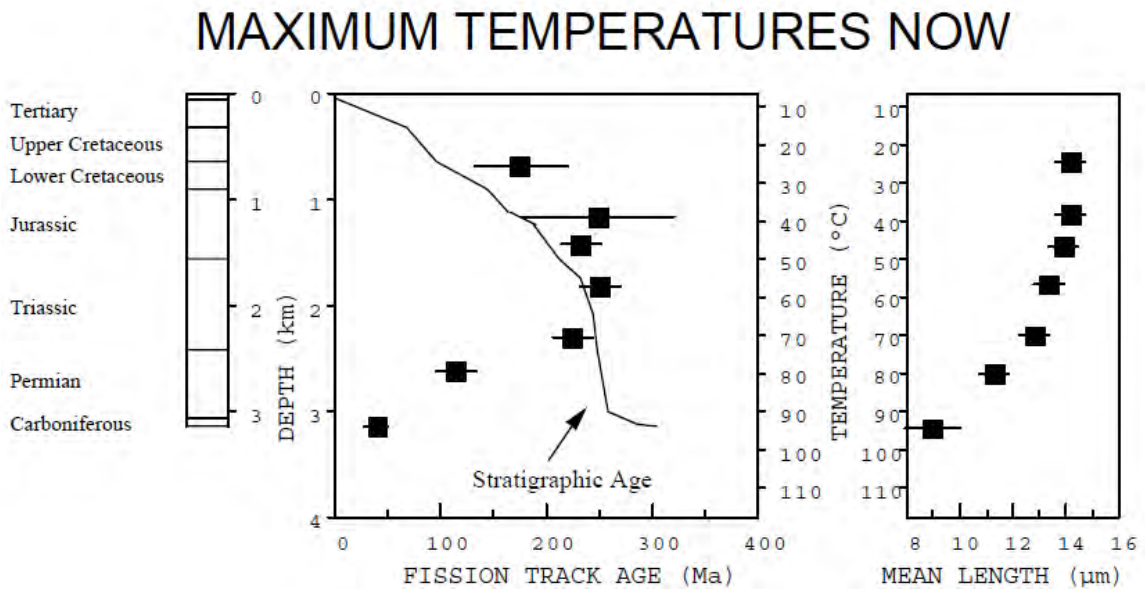


Figure S3.8a Typical pattern of AFTA parameters in a well in which samples throughout the entire section are currently at their maximum temperatures since deposition. Both the fission track age and mean track length undergo progressive reduction to zero at temperatures of ~100 - 110°C, the actual value depending on the range of apatite compositions present.

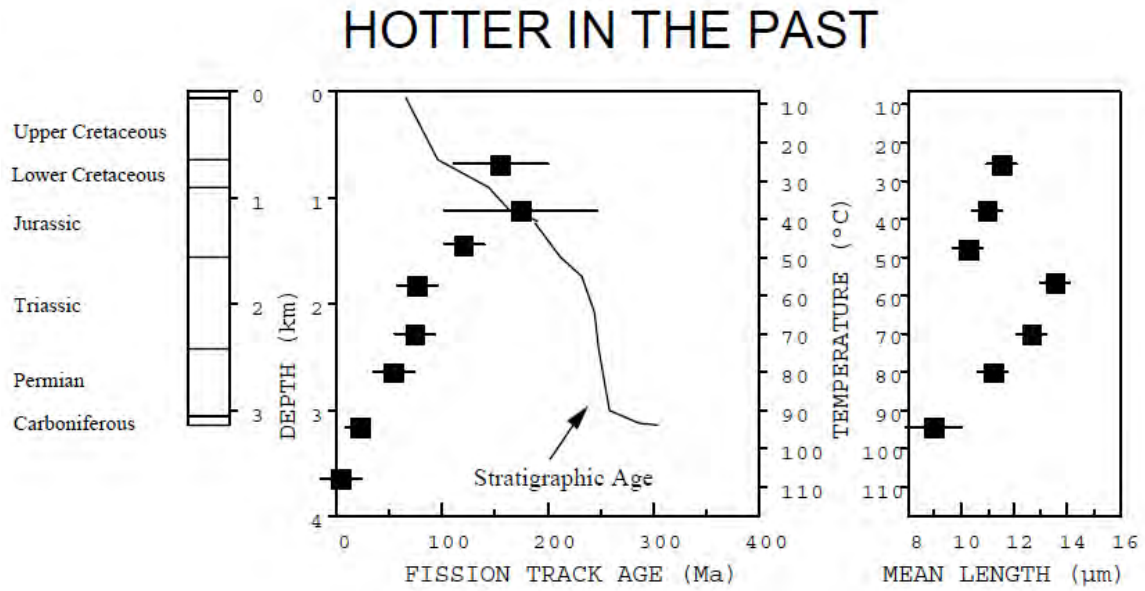


Figure S3.8b Typical pattern of AFTA parameters in a well in which samples throughout the section were exposed to elevated paleotemperatures after deposition (prior to cooling in the Early Tertiary, in this case). Both the fission track age and mean track length show more reduction at temperatures of ~40 to 50°C than would be expected at such temperatures. At greater depths (higher temperatures), the constancy of fission track age and the increase in track length are both diagnostic of exposure to elevated paleotemperatures. See Supporting information chapter S3 for further discussion

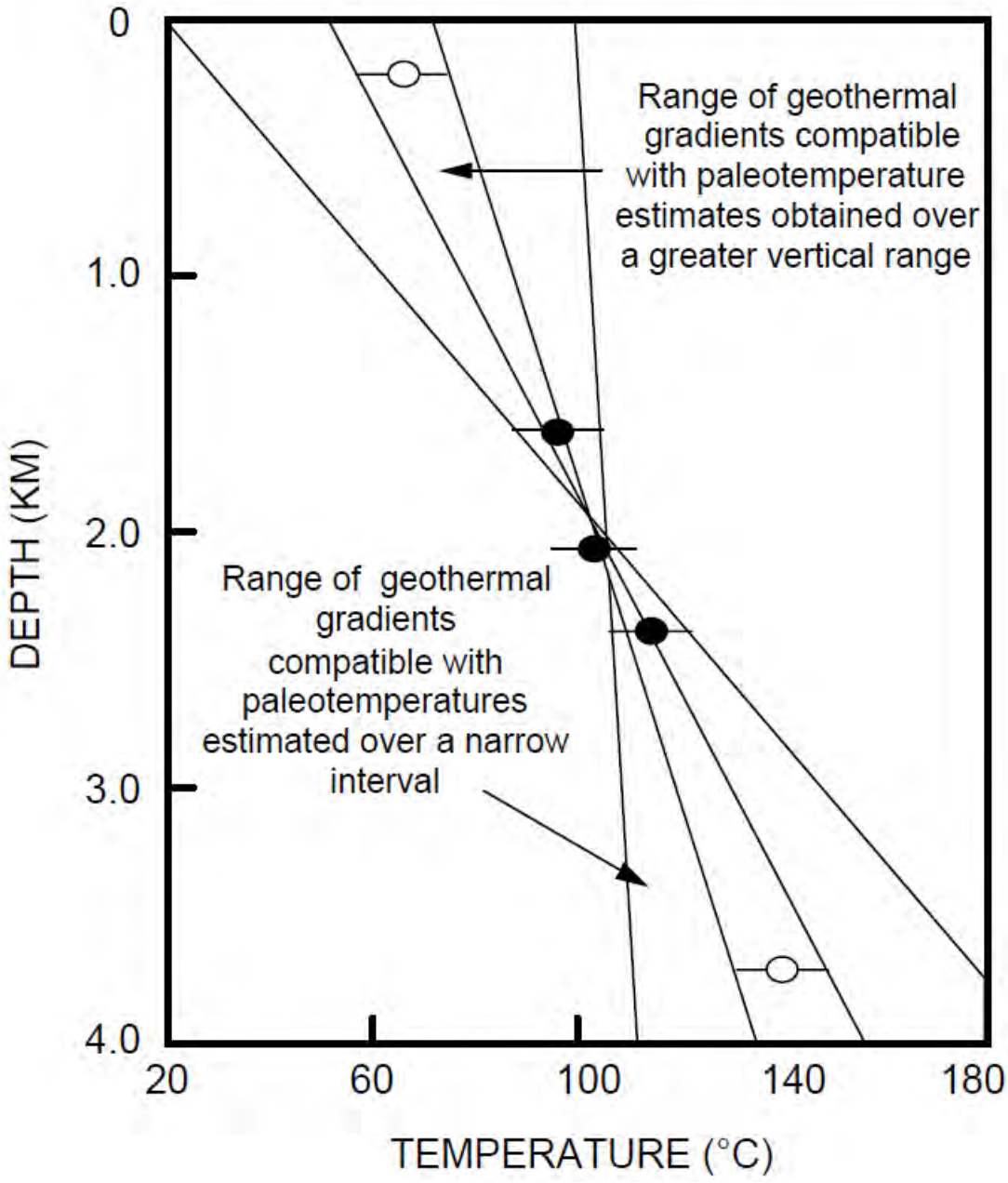


Figure S3.9 It is important to obtain paleotemperature constraints over as great a range of depths as possible in order to provide a reliable estimate of paleogeothermal gradient. If paleotemperatures are only available over a narrow depth range, then the paleogeothermal gradient can only be very loosely constrained.

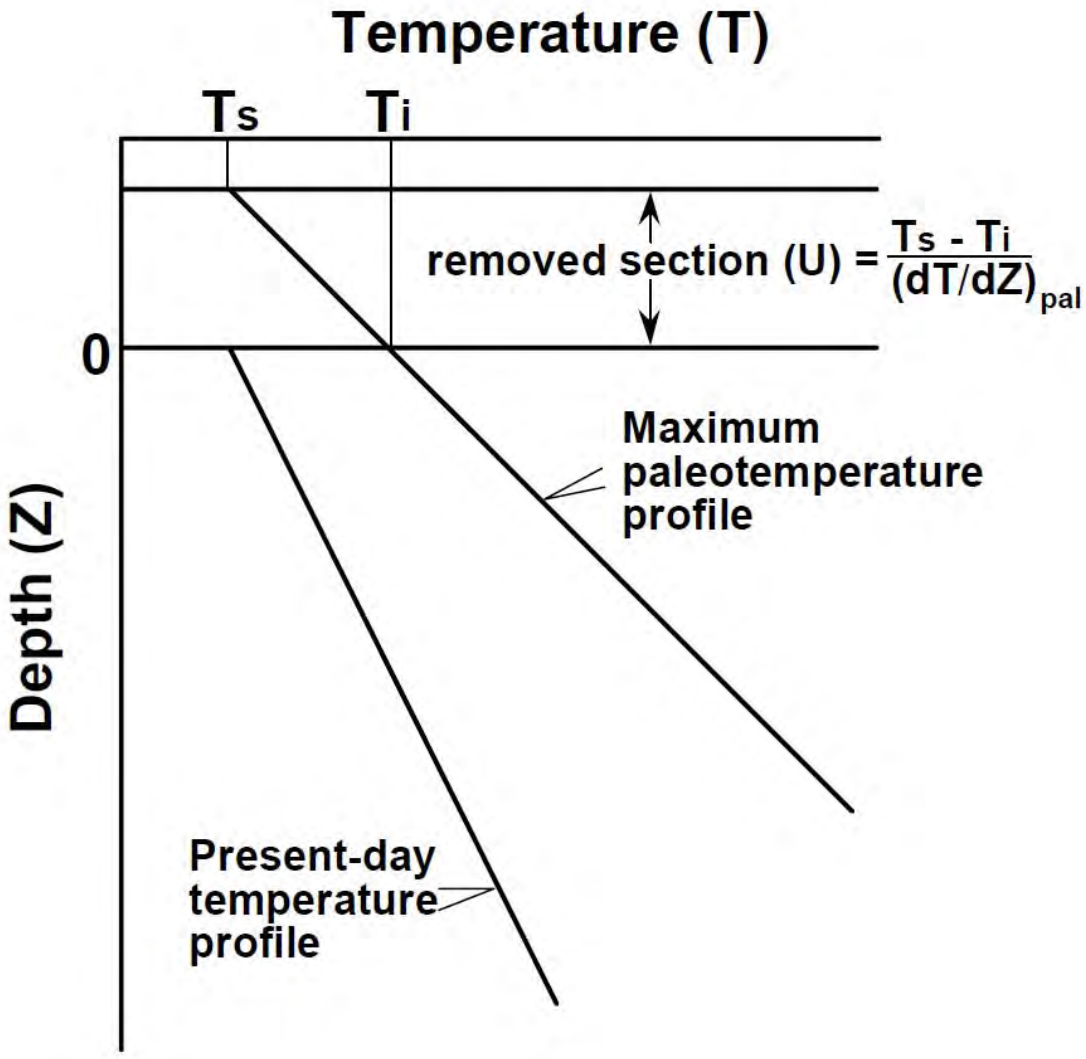


Figure S3.10 If the paleogeothermal gradient can be constrained by AFTA and VR, as explained in the text, then for an assumed value of surface temperature, Ts, the amount of section removed can be estimated, as shown.

S4: Vitrinite Reflectance Data

S4.1 Integration of Vitrinite Reflectance Data with AFTA

Vitrinite reflectance is a time-temperature indicator governed by a kinetic response in a similar manner to the annealing of fission tracks in apatite as described in supporting information chapter S3.

In this study, vitrinite reflectance data are interpreted on the basis of the distributed activation energy model describing the evolution of VR with temperature and time described by Burnham and Sweeney (1989), as implemented in the BasinModTM software package of Platte River Associates. In a considerable number of wells from around the world, in which AFTA has been used to constrain the thermal history, we have found that the Burnham and Sweeney (1989) model gives good agreement between predicted and observed VR data, in a variety of settings.

As in the case of fission track annealing, it is clear from the chemical kinetic description embodied in equation 2 of Burnham and Sweeney (1989) that temperature is more important than time in controlling the increase of vitrinite reflectance. If the Burnham and Sweeney (1989) distributed activation energy model is expressed in the form of an Arrhenius plot (a plot of the logarithm of time versus inverse absolute temperature), then the slopes of lines defining contours of equal vitrinite reflectance in such a plot are very similar to those describing the kinetic description of annealing of fission tracks in Durango apatite developed by Laslett et al. (1987), which is used to interpret the AFTA data in this report. This feature of the two quite independent approaches to thermal history analysis means that for a particular sample, a given degree of fission track annealing in apatite of Durango composition will be associated with the same value of vitrinite reflectance regardless of the heating rate experienced by a sample. Thus paleotemperature estimates based on either AFTA or VR data sets should be equivalent, regardless of the duration of heating. As a guide, Table S4.1 gives paleotemperature estimates for various values of VR for two different heating times.

One practical consequence of this relationship between AFTA and VR is, for example, that a VR value of 0.7% is associated with total annealing of all fission tracks in apatite of Durango composition, and that total annealing of all fission tracks in apatites of more Chlorine-rich composition is accomplished between VR values of 0.7 and ~0.9%.

Furthermore, because vitrinite reflectance continues to increase progressively with increasing temperature, VR data allow direct estimation of maximum paleotemperatures in the range where fission tracks in apatite are totally annealed (generally above ~110°C) and where therefore AFTA only provides minimum estimates. Maximum paleotemperature estimates based on vitrinite reflectance data from a well in which most AFTA samples were totally annealed will allow constraints on the paleogeothermal gradient that would not be possible from AFTA alone. In such cases the AFTA data should allow tight constraints to be placed on the time of cooling and also the cooling history, since AFTA parameters will be dominated by the effects of tracks formed after cooling from maximum paleotemperatures. Even in situations where AFTA samples were not totally annealed, integration of AFTA and VR can allow paleotemperature control over a greater range of depth, e.g. by combining AFTA from sand-dominated units with VR from other parts of the section, thereby providing tighter constraint on the paleogeothermal gradient.

S4.2 Supplied vitrinite reflectance values

Vitrinite reflectance values supplied by the GEUS are summarized in Table S4.2. No information was provided in regard to analytical techniques, but based on the numbers of measurements for each sample the mean values are regarded as well-defined and reliable.

S4. References

Burnham, A.K. and Sweeney, J.J. (1989) A chemical kinetic model of vitrinite reflectance maturation. *Geochim. et Cosmochim. Acta*, 53, 2649-2657.

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Paleotemperature (°C / °F)	Vitrinite Reflectance (%) 1 Ma Duration of heating	10 Ma Duration of heating
40 / *104*	0.29	0.32
50 / *122*	0.31	0.35
60 / *140*	0.35	0.40
70 / *158*	0.39	0.45
80 / *176*	0.43	0.52
90 / *194*	0.49	0.58
100 / *212*	0.55	0.64
110 / *230*	0.61	0.70
120 / *248*	0.66	0.78
130 / *266*	0.72	0.89
140 / *284*	0.81	1.04
150 / *302*	0.92	1.20
160 / *320*	1.07	1.35
170 / *338*	1.23	1.55
180 / *356*	1.42	1.80
190 / *374*	1.63	2.05
200 / *392*	1.86	2.33
210 / *410*	2.13	2.65
220 / *428*	2.40	2.94
230 / *446*	2.70	3.23

Table S4.1: Paleotemperature - vitrinite reflectance nomogram based on Equation 2 of Burnham and Sweeney (1989)

Source #	Location	Stratigraphic Subdivision	Stratigraphic age (Ma)	VR (Range) %	N
outcrop					
538306	Pq. Khu Tuong N, Vietnam 10.29380556 103.9815	Cretaceous	145-65	0.61	76
538312	Pq. Khu Tuong N, Vietnam 10.29380556 103.9815	Cretaceous	145-65	0.62	77
538382	Sanlong, Cambodia 10.79911111 104.7533722	Triassic?	251-199	-	
538714	Phnom Kbal Remeas, Cambodia 10.61730556 104.2433611	Permian	299-251	4.27	85
538716	Phnom Kbal Remeas, Cambodia 10.61730556 104.2433611	Permian	299-251	4.47	77
538740	Vealrinh, Cambodia 10.72597222 103.7972778	Cretaceous	145-65	0.88	105
538741	Vealrinh, Cambodia 10.72597222 103.7972778	Cretaceous	145-65	0.76	92
538742	Vealrinh, Cambodia 10.72597222 103.7972778	Cretaceous	145-65	0.78	103
538744	Vealrinh, Cambodia 10.72597222 103.7972778	Cretaceous	145-65	0.79	75
538750	Rang Pein, Cambodia 11.37808333 104.3515556	Triassic	251-199	3.29	35
538752	Prak Sambo, Cambodia 11.39877778 104.3871389	Triassic	251-199	3.39	10
538755	Phnom Cheal, Cambodia 11.38938889 104.5029722	Triassic	251-199	4.48	88
538759	Phnom Srong, Cambodia 11.26541667 104.6096944	Triassic	251-199	3.98	60

Note: Some samples may contain both vitrinite and inertinite. Only vitrinite data is shown.

*1 See chapter S1 for discussion of present temperature data.

Table S4.2: Vitrinite reflectance sample details and results supplied by GEUS - Vietnam and Cambodia

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